

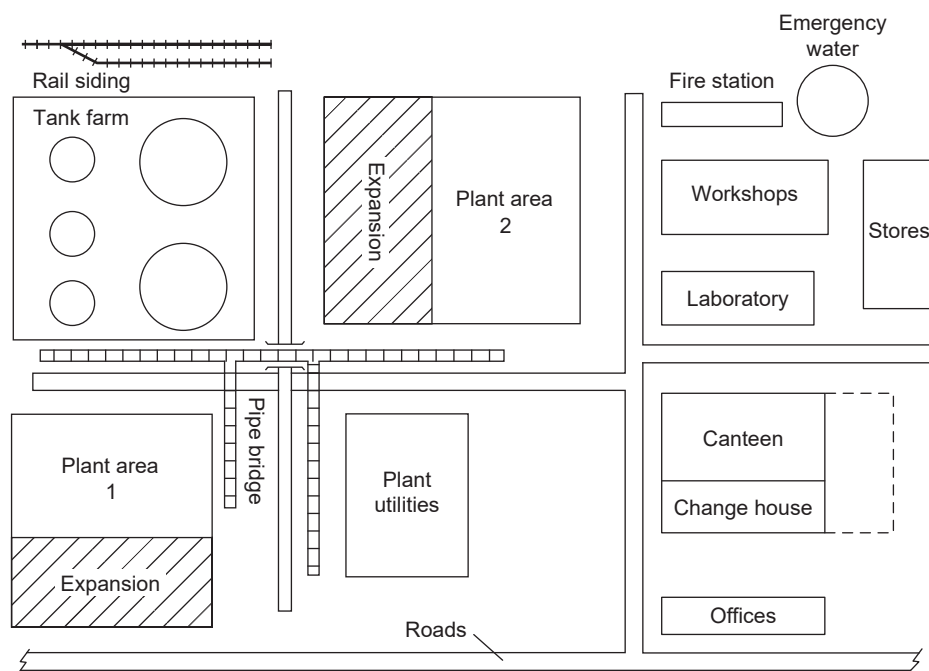


FIG. 11.1 Typical site plan.

processes. Control rooms are normally located adjacent to the processing units, but with potentially hazardous processes may have to be sited at a safer distance.

The siting of the main process units determines the layout of the plant roads, pipe alleys, and drains. Access roads to each building are needed for construction and for operation and maintenance.

Utility buildings should be sited to give the most economical run of pipes to and from the process units.

Cooling towers should be sited so that under the prevailing wind the plume of condensate spray drifts away from the plant area and adjacent properties.

The main storage areas should be placed between the loading and unloading facilities and the process units they serve. Storage tanks containing hazardous materials should be sited at least 70 m (200 ft) from the site boundary.

A typical plot plan is shown in Fig. 11.1.

A comprehensive discussion of site layout is given by Mecklenburgh (1985); see also House (1969), Kaess (1970), and Meissner and Shelton (1992).

11.4 Plant layout

The economic construction and efficient operation of a process unit will depend on how well the plant and equipment specified on the process flowsheet is laid out.

A detailed account of plant layout techniques cannot be given in this short section. A fuller discussion can be found in the book edited by Mecklenburgh (1985) and in articles by Kern (1977a–f, 1978a–f), Meissner and Shelton (1992), Brandt et al. (1992), and Russo and Tortorella (1992).

The principal factors to be considered are:

1. Economic considerations: construction and operating costs
2. The process requirements
3. Convenience of operation
4. Convenience of maintenance
5. Safety
6. Future expansion
7. Modular construction

Costs

The cost of construction can be minimized by adopting a layout that gives the shortest run of connecting pipe between equipment and the least amount of structural steel work; however, this will not necessarily be the best arrangement for operation and maintenance.

Process requirements

An example of the need to take into account process considerations is the need to elevate the base of columns to provide the necessary net positive suction head to a pump (see [Chapter 20](#)) or the operating head for a thermosiphon reboiler (see [Chapter 19](#)).

Operation

Equipment that needs to have frequent operator attention should be located convenient to the control room. Valves and instruments should be located at convenient positions and heights for operator access. Sample points must be convenient for operator access and must also be near to a drain point so that sample lines can be flushed. Sufficient working space and headroom must be provided to allow easy access to equipment. If it is anticipated that equipment will need replacement, then sufficient space must be allowed to permit access for lifting equipment.

Maintenance

Heat exchangers need to be sited so that the tube bundles can be easily withdrawn for cleaning and tube replacement. Vessels that require frequent replacement of catalyst or packing should be located on the outside of buildings. Equipment that requires dismantling for maintenance, such as compressors and large pumps, should be placed under cover.

Safety

Blast walls may be needed to isolate potentially hazardous equipment and confine the effects of an explosion. At least two escape routes for operators must be provided from each level in process buildings.

Plant expansion

Equipment should be located so that it can be conveniently tied in with any future expansion of the process. Space should be left on pipe racks for future needs, and service pipes should be oversized to allow for future requirements.

Modular construction

In recent years there has been a move to assemble sections of plant at the plant manufacturer's site. These modules include the equipment, structural steel, piping, and instrumentation. The modules are then transported to the plant site by road or by sea.

The advantages of modular construction are:

1. Improved quality control
2. Reduced construction cost
3. Less need for skilled labor on site
4. Less need for skilled personnel on overseas sites

Some of the disadvantages are:

1. Higher design costs
2. More structural steel work
3. More flanged connections
4. Possible problems with on-site assembly

A fuller discussion of techniques and applications of modular construction is given by [Shelley \(1990\)](#), [Hesler \(1990\)](#), and [Whittaker \(1984\)](#).

General considerations

Open structural steelwork buildings are normally used for process equipment. Closed buildings are used for process operations that require protection from the weather, for small plants, or for processes that require ventilation with scrubbing of the vent gas.

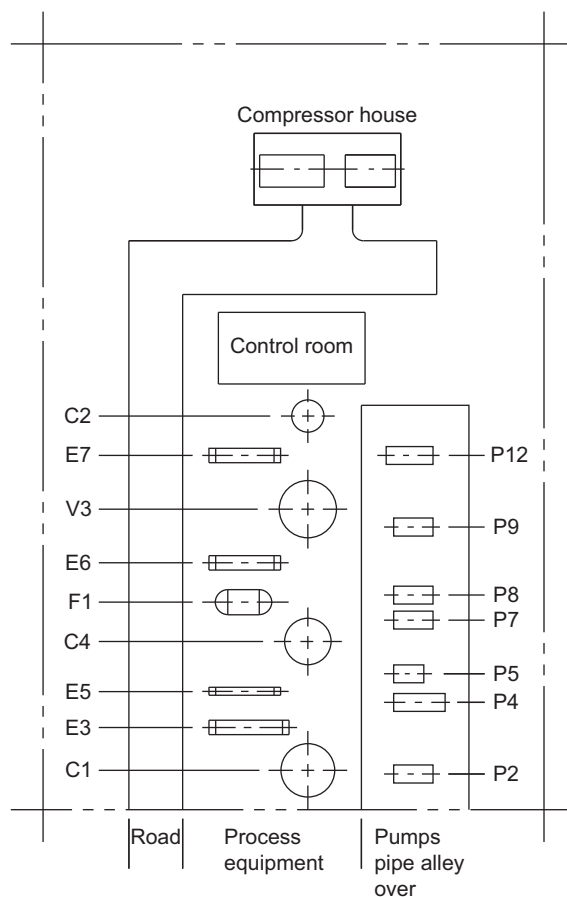


FIG. 11.2 A typical plant layout.

The arrangement of the major items of equipment often follows the sequence given on the process flowsheet, with the columns and vessels arranged in rows and the ancillary equipment, such as heat exchangers and pumps, positioned along the outside. A typical preliminary layout is shown in [Fig. 11.2](#).

11.4.1 Techniques used in site and plant layout

Cardboard cutouts of the equipment outlines can be used to make trial plant layouts. Simple models, made up from rectangular and cylindrical blocks, can be used to study alternative layouts in plan and elevation. Cutouts and simple block models can also be used for site layout studies. Once the layout of the major pieces of equipment has been decided, the plan and elevation drawings can be made and the design of the structural steelwork and foundations undertaken.

Large-scale models, to a scale of at least 1:30, are sometimes still made for major projects. These models are used for piping design and to decide the detailed arrangement of small items of equipment, such as valves, instruments, and sample points. Piping isometric diagrams can be taken from the finished models. The models are also useful on the construction site and for operator training. Proprietary kits of parts are available for the construction of plant models.

Computer-aided design (CAD) tools are being increasingly used for plant layout studies, and computer models are complementing, if not yet completely replacing, physical models. Several proprietary programs are available for the generation of three-dimensional models of plant layout and piping. Present systems allow designers to zoom in on a section of plant and view it from various angles. Developments of computer technology will soon enable engineers to virtually walk through the plant. A typical computer-generated model is shown in [Fig. 11.3](#).

Some of the advantages of computer graphics modeling compared with actual scale models are:

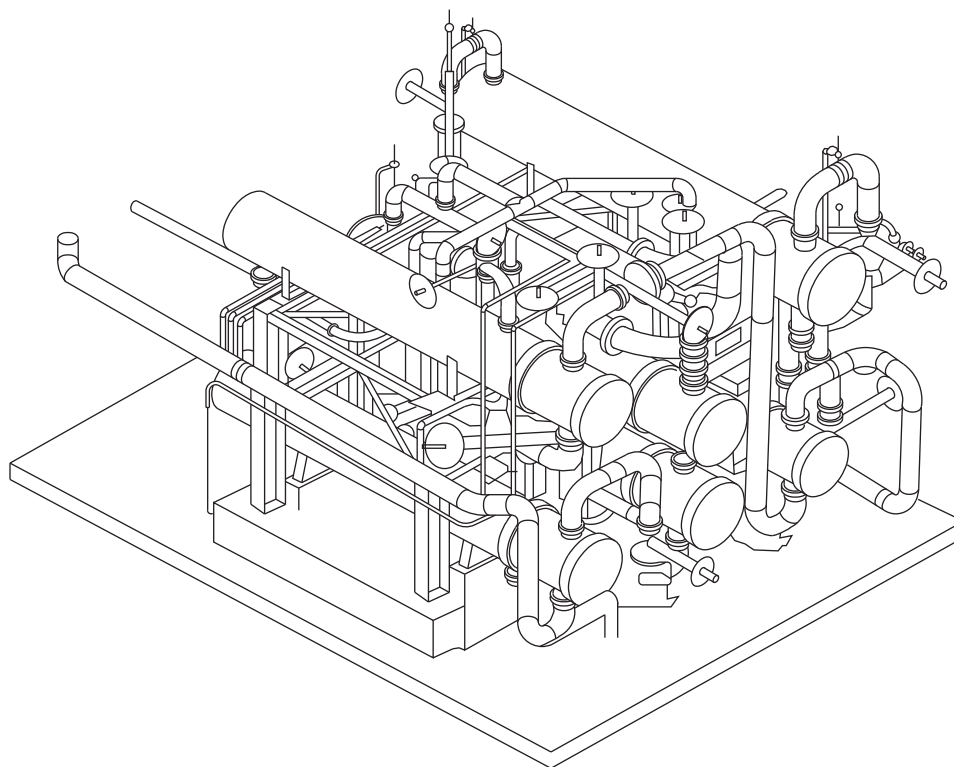


FIG. 11.3 Computer-generated layout "model." Courtesy: Babcock Construction Ltd.

1. The ease of electronic transfer of information. Piping drawings can be generated directly from the layout model. Bills of quantities: materials, valves, instruments, etc., are generated automatically.
2. The computer model can be part of an integrated project information system, covering all aspects of the project from conception to operation.
3. It is easy to detect interference between pipe runs, and pipes and structural steel that occupy the same space.
4. The physical model has to be transported to the plant site for use in the plant construction and operator training. A computer model can be instantly available in the design office, the customer's offices, and at the plant site.
5. Expert systems and optimization programs can be incorporated in the package to assist the designer to find the best practical layout; see [Madden et al. \(1990\)](#).

11.5 Environmental considerations

All individuals and companies have a duty of care to their neighbors and to the environment in general. In addition to this moral duty, most countries have enacted strict laws to protect the environment and preserve the quality of air, water, and land.

Vigilance is required in both the design and operation of process plants to ensure that legal standards are met and that no harm is done to the environment.

Consideration must be given to:

1. All emissions to land, air, and water
2. Waste management
3. Smells
4. Noise
5. Visual impact
6. Any other nuisances
7. The environmental friendliness of the products

11.5.1 Environmental legislation

It is not feasible to review the entire body of legislation that has been enacted to protect the environment in this chapter. States, provinces, and municipalities often pass local legislation that is stricter than the national laws. For example, the South Coast Air Quality Management District (SCAQMD), which sets air quality standards for the Los Angeles basin, has consistently advocated air quality limits that exceed the U.S. national standards. The interpretation of statutes can change over time as a result of court decisions and executive orders, and environmental legislation is also revised frequently, so the design engineer should always check with the local, regional, and federal authorities to ensure that the correct standards are being applied in the design and to ensure that the correct information is being generated for permit applications. This section provides a brief overview of some of the main environmental laws in North America. More information on U.S. federal laws can be found on the U.S. Environmental Protection Agency (EPA) website at www.epa.gov/laws-regulations. Information on Canadian laws is given on the Environment Canada website at www.canada.ca/en/environment-climate-change. Full citations for all of the laws are given in the references in Section 11.6.

The National Environmental Policy Act of 1969

The National Environmental Policy Act (NEPA) requires that all legislation must consider environmental impact. Government agencies must prepare environmental impact assessments. The president must make an annual Environmental Quality Report to Congress, discussing current status and recent trends in the air, water, and terrestrial environments; natural resources; and government programs that have impacts on the environment.

The act also established the Council on Environmental Quality in the Executive Office of the President, with goals of assisting in the preparation of the Environmental Quality Report, reviewing government programs, carrying out research, and recommending national policies to improve environmental protection.

The Clean Air Act (1970)

The Clean Air Act (CAA) was enacted to improve air quality. The CAA was amended and strengthened in 1990. Under the CAA, the EPA is empowered to set National Ambient Air Quality Standards (NAAQS), which are allowable ambient levels for seven contaminants:

1. Ozone
2. Carbon monoxide
3. Lead
4. Nitrogen dioxide
5. Sulfur dioxide
6. PM₁₀: particulate matter with mean diameter less than 10 μm
7. PM_{2.5}: particulate matter with mean diameter less than 2.5 μm

Areas that fail to meet the NAAQS levels are classified as “nonattainment” areas and must take remediation steps, such as mandating the use of cleaner-burning, federal reformulated gasoline.

The CAA also mandated the EPA to set the U.S. National Emission Standards for Hazardous Air Pollutants (NESHAP). These standards regulate air emissions of 189 listed hazardous air pollutants. The standards were set based on the best 12% of controlled facilities in each industry. A similarly high technical standard was set by EPA in the Maximum Available Control Technology rules (MACT) that have been applied to emissions of volatile organics (1995) and sulfur and other inorganic compounds (2002) from oil refineries.

The CAA has had a significant impact on ambient levels of the pollutants that it regulates. Acid precipitation and surface water acidity have been reduced, particularly in the Midwest and Northeast. Ozone nonattainment has not been so successful, as ozone formation is closely linked to emissions of nitrogen oxides, which are not as tightly regulated. Charts showing nonattainment regions can be downloaded from www.epa.gov.

The Federal Water Pollution Control Act (“The Clean Water Act,” 1972)

The initial goal of the Clean Water Act (CWA) was to achieve clean water for swimming, boating, and protecting fish and wildlife by 1983. The act was amended in 1977 and 1987 to strengthen the focus on water quality and emissions of toxic compounds.

The CWA empowers the EPA to set water quality standards for pollutants in surface waters. Under the CWA, effluent guidelines are established for each industrial sector. It is unlawful to discharge any pollutant into navigable waters without a permit from the EPA.

The Safe Drinking Water Act (1974)

The Safe Drinking Water Act (SDWA) authorized the EPA to set standards on the required purity of surface or subterranean water that could potentially be used for drinking. Owners and operators of public water systems must comply with these EPA standards. The SDWA established the Office of Ground Water and Drinking Water (OGDW) to oversee implementation of the act and protect public health by ensuring safe drinking water.

The Resource Conservation and Recovery Act (1976)

The ultimate goal of the Resource Conservation and Recovery Act (RCRA) is to protect groundwater from contamination. The legislation addresses only current and future facilities, as abandoned or historic sites are covered by CERCLA and SARA, described later.

Under RCRA, waste producers are required to take a “cradle to grave” approach to waste management. The producer of the waste is legally liable from the moment that waste is produced until its final disposal. A waste generator must identify waste material as “hazardous waste” if it is on a regulatory list or has a characteristic of flammability, toxicity, corrosivity, or reactivity. Once a material has been identified as hazardous waste, it must be clearly labeled and tracked when in transport. The waste must be treated in special facilities to low levels of contaminants. The final residual solid material, for example, incinerator ash, must be disposed into a registered hazardous waste landfill.

Additional regulations to aid in the identification of hazardous waste are given in the Code of Federal Regulations (40 CFR 261.3, 1999).

The Comprehensive Environmental Response, Compensation and Liability Act (or Superfund, 1980)

The Superfund act addressed hazardous waste from abandoned operations and historic production sites. The act created a tax on the chemical and petroleum industries to pay for remediation of uncontrolled or abandoned hazardous waste sites. Prohibitions and other requirements were established for abandoned sites, including liability for anyone responsible for a release of hazardous material at such a site.

The Comprehensive Environmental Response, Compensation and Liability Act (CERCLA) legislation also authorized the EPA to take action to remediate sites, either by short-term removal of material if there was an immediate threat of release or longer-term remediation actions if there was no immediate threat to life.

The Superfund Amendments and Reauthorization Act (1986)

The Superfund Amendments and Reauthorization Act (SARA) of 1986 amended CERCLA and made additions to the program. The act stressed the importance of permanent remediation and the use of innovative cleanup technologies. Greater participation of citizens, local communities, and state governments in all steps of the Superfund program was encouraged. The EPA was required to revise the Hazard Ranking System (HRS) to ensure that it accurately assessed the relative degree of risk to human health and the environment caused by uncontrolled hazardous waste sites. The size of the trust fund for site restoration was increased to \$8.5 billion. Title III of SARA authorized the Emergency Planning and Community Right-to-Know Act (EPCRA) cited in [Chapter 10](#).

The Pollution Prevention Act (1990)

The Pollution Prevention Act (PPA) was passed to encourage waste minimization and reduction of pollution at the source by more efficient processing and raw materials use. The EPA was authorized to promote source reduction and recycling through grants, technical assistance, and dissemination of information.

The Oil Pollution Act of 1990 (1990)

The Oil Pollution Act (OPA) was passed in the aftermath of the Exxon Valdez oil spill in the Prince William Sound in Alaska. It created a tax on oil to cover the cost of responding to major spills when the responsible party is unwilling or unable to do so. The limits on liability for causing oil spills are clearly laid out in the law, and oil shippers and storage facilities are required to submit plans for responding to a large spill to the EPA.

The major environmental laws in Canada are discussed next.

The Department of the Environment Act (E-10, 1985)

This act established the Department of the Environment and defined the responsibilities of the Minister of the Environment. The minister was charged with establishing programs to protect the environment and reduce pollution, ensuring that new federal programs are assessed for environmental impact, and reporting environmental information to the Canadian public.

The Canadian Environmental Protection Act (C-33, 1999)

The Canadian Environmental Protection Act (CEPA) charges Environment Canada with controlling toxic substances, reducing pollution, and eliminating persistent bio-accumulating toxic substances. Enforcement officers are given the power to issue environmental protection compliance orders on the spot to stop illegal activity and correct any violations of the act.

CEPA is also the preferred legislation that Canada will use to enforce regulations to meet its commitments to reduce greenhouse gas emissions as a signatory of the Kyoto Protocol.

The Canada Water Act (C-11, 1985)

The Canada Water Act empowers the Minister of the Environment to conclude agreements with the provinces on water resource management and water quality management. Drinking water and recreational water quality standards are set by Health Canada.

11.5.2 Waste minimization

Waste arises mainly as by-products or unused reactants from the process, or as off-specification product produced through misoperation. There will also be fugitive emissions from leaking seals and flanges and inadvertent spills and discharges through misoperation. In emergency situations, material may be discharged to the atmosphere through vents normally protected by bursting disks and relief valves.

Before considering “end-of-pipe” approaches for treating and managing waste products, the design engineer should always try to minimize production of waste at the source. The hierarchy of waste management approaches is:

1. Source reduction: Don't make the waste in the first place. This is the best practice.
2. Recycle: Find a use for the waste stream.
3. Treatment: Reduce the severity of the environmental impact.
4. Disposal: Meet the requirements of the law.

Source reduction is accomplished during process design. Some of the strategies that can be considered include:

1. *Purification of feeds.* Reducing the concentration of impurities in the feed usually leads to reduced side reactions and less waste formation. This approach can also reduce the need for purges and vent streams. Feed impurities also often lead to degradation of solvents and catalysts. Care must be taken to select a purification approach that does not itself lead to more waste formation.
2. *Protect catalysts and adsorbents.* Deactivated catalysts and adsorbents are solid wastes from the process. In some cases, relatively small amounts of contaminants can cause a load of catalyst or adsorbent to become useless. The catalyst or adsorbent should be protected by using a guard bed of suitable material to adsorb or filter out contaminants before they can damage the catalyst.
3. *Eliminate use of extraneous materials.* When different solvent or mass separating agents are used, this leads to waste formation when the solvents become degraded. If the plant or site uses relatively few solvents, it may be economical to build a solvent-recovery plant. Liquid wastes from spent solvents are common in fine chemicals and pharmaceuticals manufacture.
4. *Increase recovery from separations.* Higher product recovery leads to lower concentrations of product in the waste streams. Higher purity recycle streams usually lead to less waste formation. These benefits must always be traded off against the extra capital and energy costs involved in driving the separation processes to higher recovery or purity.
5. *Improve fuel quality.* Switching to a cleaner-burning fuel, such as natural gas, reduces the emissions from fired heaters. In some locations this must be traded off against the higher cost of natural gas relative to heating oil and coal.

Unused reactants can be recycled and off-specification product reprocessed. Integrated processes can be selected, with the waste from one process becoming the raw material for another. For example, the otherwise waste hydrogen chloride produced in a chlorination process can be used for chlorination using a different reaction, as in the balanced, chlorination-oxyhydrochlorination process for vinyl chloride production. It may be possible to sell waste to another company for use as raw material in their manufacturing processes, for example, the use of off-specification and recycled plastics in the production of lower-grade products, such as the ubiquitous black plastic bucket.

Processes and equipment should be designed to reduce the chances of misoperation by providing tight control systems, alarms, and interlocks. Sample points, process equipment drains, and pumps should be sited so that any leaks flow into the plant effluent collection system, not directly to sewers. Hold-up systems, tanks, and ponds should be provided to retain spills for treatment. Flanged joints should be kept to the minimum needed for the assembly and maintenance of equipment. Fugitive emissions from packings and seals can be reduced by specifying dual seals, dry-gas seals, or seal-less pumps.

A technique that is sometimes used for waste minimization is the five-step review:

1. Identify waste *components* for regulatory impact.
2. Identify waste *streams* for size and economic impact.
3. List the *root causes* of the waste streams.
4. List and analyze *modifications* to address the root causes.
5. Prioritize and *implement* the best solutions.

The information gathered in the first two steps is often collected in an effluent summary worksheet. An effluent summary lists the regulated pollutants produced by the process and summarizes the quantities produced and where they originate. The effluent summary can be used to focus waste minimization efforts and as a basis for the design of effluent treatment processes. The information in the effluent summary may also be required for obtaining permits to operate the plant or for preparing more formal environmental impact analyses to convince investors or insurers that environmental impact has been properly addressed. An example effluent summary sheet is given in Appendix G. A template is available in Microsoft Excel format in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

The American Petroleum Institute Publication 302 (1991) discusses source reduction, recycle, treatment, and disposal of wastes. Other source reduction techniques are given by [Smith and Petela \(1991\)](#) and [El-Halwagi \(2017\)](#). The UK Institution of Chemical Engineers has published a guide to waste minimization ([ICHEME 1997](#)).

11.5.3 Waste management

When waste is produced, processes must be incorporated in the design for its treatment and safe disposal. The following techniques can be considered:

1. Dilution and dispersion
2. Discharge to foul-water sewer (with the agreement of the appropriate authority)
3. Physical treatments: scrubbing, settling, absorption, and adsorption; see [Chapter 16](#)
4. Chemical treatment: precipitation (for example, of heavy metals), neutralization
5. Biological treatment: activated sludge and other processes
6. Incineration on land or at sea
7. Landfill at controlled sites
8. Sea dumping (now subject to tight international control)

Several standards have been written to address waste management systems. In the United States, EPA standard 40 CFR 260 (2006) provides general guidelines, while EPA 40 CFR 264 (2006) gives standards for waste treatment, storage, and disposal. Standards for the petroleum industry are given in API publications 300 (1991) and 303 (1992). The main international standard for waste management systems is ISO 140001 (2004), which has been adopted as the national standard in the countries of the EU. ASTM standard 11.04 (2006) should also be consulted.

Gaseous wastes

Gaseous effluents that contain toxic or noxious substances need treatment before discharge into the atmosphere. The practice of relying on dispersion from tall stacks is seldom entirely satisfactory. Gaseous pollutants can be removed by absorption or adsorption. Absorption by scrubbing with water or a suitable solvent or base is probably the most widely used method for high-volume gas streams, while adsorption onto activated carbon or a zeolitic adsorbent is used for smaller gas streams. The design of adsorption plants is addressed in [Chapter 16](#), and the design of scrubbing towers is described in [Chapter 17](#). Finely dispersed solids can be removed by scrubbing or using electrostatic precipitators; see [Chapter 18](#). Flammable gases can be burnt. The sources of air pollution and their control are covered in several books: [Walk et al. \(1997\)](#), [Heumann \(1997\)](#), [Davies \(2000\)](#), [Cooper and Ally \(2010\)](#), and [Schnelle et al. \(2017\)](#). [McGowan and Santoleri \(2007\)](#) discuss methods for reducing emissions of volatile organic compounds (VOCs).

Liquid wastes

The waste liquids from a chemical process, other than aqueous effluent, will usually be flammable and can be disposed of by burning in suitably designed incinerators. Care must be taken to ensure that the temperatures attained in the incinerator are high enough to completely destroy any harmful compounds that may be formed, such as the possible formation of dioxins when burning chlorinated compounds. The gases leaving an incinerator may be scrubbed and acid gases neutralized. A typical incinerator for burning gaseous or liquid wastes is shown in [Chapter 3](#), Fig. 3.13. The design of incinerators for hazardous waste and the problems inherent in the disposal of waste by incineration are discussed by [Butcher \(1990\)](#) and [Baker-Counsell \(1987\)](#).

In the past, small quantities of liquid waste in drums were disposed of by dumping at sea or in landfill sites. This is not an environmentally acceptable method and is now subject to stringent controls.

Solid wastes

Solid waste can be burnt in suitable incinerators or disposed by burial at licensed landfill sites. As with liquid wastes, the dumping of toxic solid waste at sea is no longer acceptable.

Aqueous wastes

Aqueous waste streams include process water, utility wastewater, and site run-off. Water that is used or formed in the process must be sent to effluent treatment. Common process water effluents include:

- Water contaminated with ammonia or hydrogen sulfide from gas scrubbers
- Salt waters from deionizers, softeners, neutralization steps, and washing operations
- Water contaminated with hydrocarbons, for example, from condensers
- Biologically contaminated water, for example, fermentation broths or water from washing operations
- Spent acid and caustic streams

The site utility systems produce large wastewater flows. A purge known as a "blowdown" is taken from both the cooling water and the boiler feed water to prevent the accumulation of solids in either recirculating system. Cooling water blowdown is often the largest contributor to the site wastewater. The blowdown streams can be high in minerals content and also contain chemicals such as biocides and corrosion inhibitors that have been added to the boiler feed water or cooling water.

It is also a best practice to collect run-off water from the plant area and treat it in the site wastewater plant before discharging it to the environment. Run-off water can come from rain, fire hydrant flushing, and equipment washing. As the water flows over the ground around the plant, it can become contaminated with organic chemicals that have leaked from the plant. Most plants are designed so that all the run-off is collected into local sewers or ditches that are routed to the site wastewater treatment plant.

The principal factors that determine the nature of an aqueous industrial effluent and on which strict controls will be placed by the responsible authority are:

1. pH
2. Suspended solids
3. Toxicity
4. Biological oxygen demand

The pH can be adjusted by the addition of acid or alkali. Spent acid or alkali solutions must usually be neutralized before they can be sent to water treatment plants. Lime (calcium oxide) is frequently used to neutralize acidic effluents. In the case of sulfuric acid, the use of lime leads to formation of calcium sulfate. Calcium sulfate that is potentially contaminated with trace organic material has low value and can be used as road fill. An alternative approach is to neutralize with more expensive ammonia, forming ammonium sulfate, which can be sold as a fertilizer.

Suspended solids can be removed by settling, using clarifiers (see [Chapter 18](#)).

For some effluents, it is possible to reduce the toxicity to acceptable levels by dilution. Other effluents need chemical treatment.

The oxygen concentration in a water course must be maintained at a level sufficient to support aquatic life. For this reason, the biological oxygen demand of an effluent is of utmost importance. It is measured by a standard test: the BOD5 (5-day biological oxygen demand). This test measures the quantity of oxygen that a given volume of the effluent (when diluted with water containing suitable bacteria, essential inorganic salts, and saturated with oxygen) will absorb in 5 days, at a constant temperature of 20 °C. The results are reported as parts of oxygen

absorbed per million parts effluent (ppm). The BOD5 test is a rough measure of the strength of the effluent: the organic matter present. It does not measure the total oxygen demand, as any nitrogen compounds present will not be completely oxidized in 5 days. The ultimate oxygen demand (UOD) can be determined by conducting the test over a longer period, up to 90 days. If the chemical composition of the effluent is known, or can be predicted from the process flow-sheet, the UOD can be estimated by assuming complete oxidation of the carbon present to carbon dioxide and the nitrogen present to nitrate:

$$\text{UOD} = 2.67C + 4.57N$$

where C and N are the concentrations of carbon and nitrogen in ppm by weight.

Activated sludge processes are usually used to reduce the biological oxygen demand of an aqueous effluent before discharge. Where wastewater is discharged into the sewers with the agreement of the local water authorities, a charge will normally be made according to the BOD value and any treatment required. Where treated effluent is discharged to water courses with the agreement of the appropriate regulatory authority, the BOD5 limit will typically be set at 20 ppm. A full discussion of aqueous effluent treatment is given by [Eckenfelder et al. \(1985\)](#); see also [Eckenfelder \(1999\)](#).

11.5.4 Noise

Noise can cause a serious nuisance in the neighborhood of a process plant. Care must be taken when selecting and specifying equipment such as compressors, air-cooler fans, induced and forced draught fans for furnaces, conveyors, grinders, dryers, and other noisy plant. Excessive noise can also be generated when venting through steam and other relief valves and from flare stacks. Such equipment should be fitted with silencers. Vendors' specifications should be checked to ensure that equipment complies with statutory noise levels, both for the protection of employees (see [Chapter 10](#)) and for noise pollution considerations. Noisy equipment should, as far as practicable, be sited well away from the site boundary. Earth banks and screens of trees can be used to reduce the noise level perceived outside the site.

11.5.5 Visual impact

The appearance of the plant should be considered at the design stage. There is little that can be done to change the appearance of a modern-style plant, where most of the equipment and piping are outside and in full view, but some steps can be taken to minimize the visual impact. Large equipment, such as storage tanks, can be painted to blend in with, or even contrast with, the surroundings. For example, the Richmond oil refinery in the San Francisco Bay Area has its storage tanks painted to blend in with the surrounding hills. Landscaping and screening by belts of trees can also help improve the overall appearance of the site.

11.5.6 Environmental auditing

An environmental audit is a systematic examination of how a business operation affects the environment. It will include all emissions to air, land, and water and cover the legal constraints, the effect on the community, the landscape, and the ecology. Products will be considered, as well as processes. When applied at the design stage of a new development, it is more correctly called an *environmental impact assessment*.

The aims of the audit or assessment are to:

1. Identify environmental problems associated with manufacturing process and the use of the products before they become liabilities
2. Develop standards for good working practices
3. Provide a basis for company policy
4. Ensure compliance with environmental legislation
5. Satisfy requirements of insurers
6. Be seen to be concerned with environmental questions, which is important for public relations
7. Minimize the production of waste, which is an economic factor

Environmental auditing is discussed in the books by [Grayson \(1992\)](#), [McGaw \(1993\)](#), and [Pain \(2018\)](#).

Life cycle assessment

Life cycle assessment is a more exhaustive procedure than environmental auditing and is used to compare the long-term sustainability of alternative designs. A life cycle assessment considers all the environmental costs and impacts of the process, its feed stocks, and the physical plant itself, from initial construction through to final decommissioning and site remediation. The methods for carrying out a life cycle assessment are given in ISO standards BS EN ISO 14040 and BS EN ISO 14044 (which have superseded the older 14041, 14042, and 14043 standards). A good introduction to life cycle assessment is given by [Clift \(2001\)](#). Many examples of life cycle assessments can be found in the journals *Environmental Science and Technology*, *Environmental Progress*, and *The International Journal of Life Cycle Assessment*.

11.7 References

- Baker-Counsell, J. (1987). *Hazardous wastes: the future for incineration*. *Process Engineering*, (April), 26.
- Brandt, D., George, W., Hathaway, C., & McClintock, N. (1992). Plant layout, Part 2: The impact of codes, standards and regulations. *Chemical Engineering*, New York, 99(4), 97.
- Butcher, C. (1990). *Incinerating hazardous waste*, 471 p. 27). London: Chemical Engineering. April 12.
- CCPS. (2003). *Guidelines for facility siting and layout*. Wiley-AIChE.
- Clift, R. (2001). Clean technology and industrial ecology. In R. M. Harrison (Ed.), *Pollution: Causes, effects and control* (4th ed.). Royal Society of Chemistry.
- Cooper, C. D., & Ally, F. C. (2010). *Air pollution control: A design approach* (4th ed.). Waveland Press.
- Davies, W. T. (Ed.). (2000). *Air pollution engineering manual* (2nd ed.). Wiley International.
- Eckenfelder, W. W. (1999). *Industrial water pollution control* (2nd ed.). McGraw-Hill.
- Eckenfelder, W. W., Patoczka, J., & Watkin, A. T. (1985). Wastewater treatment. *Chemical Engineering*, New York, 92(9), 60.
- El-Halwagi, M. M. (2017). *Sustainable design through process integration: Fundamentals and application to industrial pollution prevention, resource conservation, and profitability enhancement* (2nd ed.). Elsevier.
- Grayson, L. (Ed.). (1992). *Environmental auditing: A guide to best practices in the UK and Europe*. Technical Communications.
- Hesler, W. E. (1990). Modular design: Where it fits. *Chemical Engineering Progress*, 86(10), 76.
- Heumann, W. L. (1997). *Industrial air pollution control systems*. McGraw-Hill.
- House, F. F. (1969). Engineers guide to plant layout. *Chemical Engineering*, New York, 76(7), 120.
- ICHEME. (1997). *Waste minimization, a practical guide*. Institution of Chemical Engineers.
- Kaess, D. (1970). Guide to trouble free plant layouts. *Chemical Engineering*. New York, 77(6), 122.
- Kern, R. (1977a). How to manage plant design to obtain minimum costs. *Chemical Engineering*, New York, 84(May 23), 130.
- Kern, R. (1977b). Specifications are the key to successful plant design. *Chemical Engineering*, New York, 8(July 4), 123.
- Kern, R. (1977c). Layout arrangements for distillation columns. *Chemical Engineering*, New York, 8(Aug. 15), 153.
- Kern, R. (1977d). How to find optimum layout for heat exchangers. *Chemical Engineering*, New York, 8(Sept. 12), 169.
- Kern, R. (1977e). Arrangement of process and storage vessels. *Chemical Engineering*, New York, 8(Nov. 7), 93.
- Kern, R. (1977f). How to get the best process-plant layouts for pumps and compressors. *Chemical Engineering*, New York, 8(Dec. 5), 131.
- Kern, R. (1978a). Pipework design for process plants. *Chemical Engineering*, New York, 85(Jan. 30), 105.
- Kern, R. (1978b). Space requirements and layout for process furnaces. *Chemical Engineering*, New York, 85(Feb. 27), 117.
- Kern, R. (1978c). Instrument arrangements for ease of maintenance and convenient operation. *Chemical Engineering*, New York, 85(April 10), 127.
- Kern, R. (1978d). How to arrange plot plans for process plants. *Chemical Engineering*, New York, 85(May 8), 19.
- Kern, R. (1978e). Arranging the housed chemical process plant. *Chemical Engineering*, New York, 85(July 17), 123.
- Kern, R. (1978f). Controlling the cost factor in plant design. *Chemical Engineering*, New York, 85(Aug. 14), 141.
- Madden, J., Pulford, C., & Shadbolt, N. (1990). Plant layout—untouched by human hand? *Chemical Engineering*, London, 474(May 24), 32.
- McGaw, D. (1993). *Environmental auditing and compliance manual*. Van Nostrand Reinhold.
- McGowan, T. F., & Santoleri, J. J. (2007). VOC emission controls for the CPI. *Chemical Engineering*, New York, 114(2), 34.
- Mecklenburgh, J. C. (Ed.). (1985). *Process plant layout*. Godwin/Longmans.
- Meissner, R. E., & Shelton, D. C. (1992). Plant layout, Part 1: Minimizing problems in plant layout. *Chemical Engineering*, New York, 99(4), 97.
- Merims, R. (1966). Plant location and site considerations. In R. Landau (Ed.), *The chemical plant*. Reinhold.
- Pain, S. W. (2018). *Safety, health and environmental auditing: A practical guide* (2nd ed.). CRC Press.
- Russo, T. J., & Tortorella, A. J. (1992). Plant layout, Part 3: The contribution of CAD. *Chemical Engineering*, New York, 99(4), 97.
- Schnelle, K. B., Dunn, R. F., & Ternes, M. E. (2017). *Air pollution control technology handbook* (2nd ed.). CRC Press.
- Shelley, S. (1990). Making inroads with modular construction. *Chemical Engineering*, New York, 97(8), 30.
- Smith, R., & Petela, E. (1991). *Waste minimization in the process industries: 3. Separation and recycle systems*, 513 p. 13). London: Chemical Engineering.
- Walk, K., Warner, C. F., & Davis, W. T. (1997). *Air pollution: Its origin and control* (3rd ed.).
- Whittaker, R. (1984). Onshore modular construction. *Chemical Engineering*, New York, 91(5), 80.

American standards

API Publication 300. (1991). *Generation and management of wastes and secondary materials in the petroleum refining industry: 1987-1988*. American Petroleum Institute.

- API Publication 302. (1991). *Waste minimization in the petroleum industry: Source reduction, recycle, treatment, disposal: A compendium of practices*. American Petroleum Institute.
- API Publication 303. (1992). *Generation and management of wastes and secondary materials*. American Petroleum Institute.
- ASTM 11.04. (2020). *Waste management*. American Society for Testing Materials.
- EPA 40 CFR 260. (2019). *Hazardous waste management systems: General*. U.S. Environmental Protection Agency.
- EPA 40 CFR 261. (2019). *Identification and listing of hazardous waste*. U.S. Environmental Protection Agency.
- EPA 40 CFR 262. (2019). *Standards applicable to generators of hazardous waste*. U.S. Environmental Protection Agency.
- EPA 40 CFR 264. (2019). *Standards for owners and operators of hazardous waste treatment, storage and disposal facilities*. U.S. Environmental Protection Agency.
- NFPA 70. (2019). *National electrical code*. National Fire Protection Association.

International standards

- BS EN ISO 14001:2015. (2015). *Environmental management systems. Requirements with guidance for use*.
- BS EN ISO 14040:2006. (2006). *Environmental management – Life cycle assessment – Principles and framework* (2nd ed).
- BS EN ISO 14044:2006. (2006). *Environmental management – Life cycle assessment – Requirements and guidelines*.

U.S. laws

- The Clean Air Act. (1970). 42 U.S.C. s/s 7401 et seq.
- The Clean Water Act. (1977). 33 U.S.C. s/s 1251 et seq.
- The Comprehensive Environmental Response, Compensation and Liability Act. (1980). 42 U.S.C. s/s 9601 et seq.
- The National Environmental Policy Act of 1969 (1969) 42 U.S.C. 4321-4347.
- The Oil Pollution Act of 1990 (1990) 33 U.S.C. 2702-2761.
- The Pollution Prevention Act. (1990). 42 U.S.C. 13101 and 13102 et seq.
- The Resource Conservation and Recovery Act. (1976). U.S.C. s/s 312 et seq (p. 42).
- The Safe Drinking Water Act. (1974). 42 U.S.C. s/s 300f et seq.
- The Superfund Amendments and Reauthorization Act. (1986). 42 U.S.C. 9601 et seq.

Canadian laws

- The Canada Water Act (R.S.C., 1985, c. C-11).
- The Canadian Environmental Protection Act (S.C. 1999, c. 33).
- The Department of the Environment Act (R.S.C., 1985, c. E-10).

Optimization in design

KEY LEARNING OBJECTIVES

- How to improve a design using optimization methods
- Understand the role of constraints in limiting designs
- How to recognize design trade-offs
- Awareness of the methods used for solving optimization problems
- Why experienced design engineers very rarely use rigorous optimization methods in industrial practice

12.1 Introduction

Optimization is an intrinsic part of design: the designer seeks the best, or optimum, solution to a problem.

Many design decisions can be made without formally setting up and solving a mathematical optimization problem. The design engineer will often rely on a combination of experience and judgment, and in some cases the best design will be immediately obvious. Other design decisions have such a trivial impact on process costs that it makes more sense to make a close guess at the answer than to properly set up and solve the optimization problem. In every design, though, there will be several problems that require more rigorous optimization. This chapter introduces the techniques for formulating and solving optimization problems, as well as some of the pitfalls that are commonly encountered in optimization.

In this book, the discussion of optimization will, of necessity, be limited to a brief overview of the main techniques used in process and equipment design. Chemical engineers working in industry use optimization methods for process operations far more than they do for design, as discussed in Section 12.12. Chemical engineering students would benefit greatly from more classes in operations research methods, which are generally part of the industrial engineering curriculum. These methods are used in almost every industry for planning, scheduling, and supply chain management, all of which are critical operations for plant operation and management. There is an extensive literature on operations research methods and several good books on the application of optimization methods in chemical engineering design and operations. A good overview of operations research methods is given in the classic introductory text by [Hillier and Lieberman \(2014\)](#). Applications of optimization methods in chemical engineering are discussed by [Rudd and Watson \(1968\)](#), [Stoecker \(1989\)](#), [Floudas \(1995\)](#), [Biegler et al. \(1997\)](#), [Edgar and Himmelblau \(2001\)](#), and [Diwekar \(2008\)](#).

12.2 The design objective

An optimization problem is always stated as the maximization or minimization of a quantity called the objective. For chemical engineering design projects, the objective should be a measure of how effectively the design meets the customer's needs. This will usually be a measure of economic performance. Some typical objectives are given in Table 12.1.

The overall corporate objective is usually to maximize operating income, cash flow, or earnings before interest and taxes (EBIT), but the design engineer will often find it more convenient to use other objectives when working on subcomponents of the design. The optimization of subsystems is discussed in more detail in Section 12.5.

The first step in formulating the optimization problem is to state the objective as a function of a finite set of variables, sometimes referred to as the decision variables:

$$z = f(x_1, x_2, x_3, \dots, x_n) \quad (12.1)$$

where:

z = objective

$x_1, x_2, x_3, \dots, x_n$ = decision variables

This function is called the objective function. The decision variables may be independent, but they will usually be related to each other by many constraint equations. The optimization problem can then be stated as a maximization or minimization of the objective function subject to the set of constraints. Constraint equations are discussed in the next section.

Design engineers often face difficulties in formulating the objective function. Some of the economic objectives that are widely used in making investment decisions lead to intrinsically difficult optimization problems. For example, discounted cash flow rate of return (DCFROR) is difficult to express as a simple function and is highly nonlinear, whereas net present value (NPV) increases with project size and is unbounded unless a constraint is set on plant size or available capital. Optimization is therefore often carried out using simple objectives such as "minimize cost of production."

Health, safety, environmental, and societal impact costs and benefits are difficult to quantify and relate to economic benefit. These factors can be introduced as constraints, but few engineers would advocate building a plant in which every piece of equipment was designed for the minimum legally permissible safety and environmental performance. An experienced designer would usually select a design that was marginally worse in economic performance if it was clearly significantly safer.

An additional complication in formulating the objective function is the quantification of uncertainty. Economic objective functions are generally highly sensitive to the prices used for feeds, raw materials, and energy and to estimates of project capital cost. These costs and prices are forecasts or estimates and are usually subject to substantial error. Cost estimation and price forecasting are discussed in Chapters 7 and 8. There may also be uncertainty in the decision variables, either from variation in the plant inputs, variations introduced by unsteady plant operation, or imprecision in the design data and the constraint equations. Optimization under uncertainty is a specialized subject in its own right and is beyond the scope of this book. See Chapter 5 of Diwekar (2008) for a good introduction to the subject.

TABLE 12.1 Typical design optimization objectives

Maximize	Minimize
Project net present value	Project expense
Return on investment	Cost of production
Reactor productivity per unit volume	Total annualized cost
Plant availability (time on stream)	Plant inventory (for safety reasons)
Process yield of main product	Formation of waste products

12.3 Constraints and degrees of freedom

12.3.1 Constraints

The constraints on the optimization are the set of equations that bound the decision variables and relate them to each other.

If we write $\mathbf{x}$ as a vector of n decision variables, then we can state the optimization problem as:

$$\begin{aligned} &\text{Optimize (Max. or Min.) } z = f(\mathbf{x}) \\ &\text{subject to (s.t.):} \quad \mathbf{g}(\mathbf{x}) \leq 0 \\ &\quad \quad \quad \mathbf{h}(\mathbf{x}) = 0 \end{aligned} \tag{12.2}$$

where: z = the scalar objective

$f(\mathbf{x})$ = the objective function

$\mathbf{g}(\mathbf{x})$ = a m_i vector of inequality constraints

$\mathbf{h}(\mathbf{x})$ = a m_e vector of equality constraints

The total number of constraints is $m = m_i + m_e$

Equality constraints arise from conservation equations (mass, mole, energy, and momentum balances) and constitutive relations (the laws of chemistry and physics, correlations of experimental data, design equations, etc.). Any equation that is introduced into the optimization model that contains an “=” sign will become an equality constraint. Many examples of such equations can be found throughout this book.

Inequality constraints generally arise from the external constraints discussed in Section 1.2: safety limits, legal limits, market and economic limits, technical limits set by design codes and standards, feed and product specifications, availability of resources, etc. Some examples of inequality constraints might include:

Main product purity ≥ 99.99 wt%

Feed water concentration ≤ 20 ppmw

NO_x emissions ≤ 50 kg/yr

Production rate $\leq 400,000$ metric tons per year

Maximum design temperature for ASME Boiler and Pressure Vessel Code Section VIII Division 2 ≤ 900 °F

Investment capital $\leq \$50$ MM (50 million dollars)

The effect of constraints is to limit the parameter space. This can be illustrated using a simple two-parameter problem:

$$\begin{aligned} &\text{Max. } z = x_1^2 + 2x_2^2 \\ &\text{s.t. } x_1 + x_2 = 5 \\ &\quad \quad \quad x_2 \leq 3 \end{aligned}$$

The two constraints can be plotted on a graph of x_1 vs. x_2 , as in [Fig. 12.1](#).

In the case of this example, it is clear by inspection that the set of constraints does not bound the problem. In the limit $x_1 \rightarrow \infty$, the solution to the equality constraint is $x_2 \rightarrow -\infty$, and the objective function gives $z \rightarrow \infty$, so no maximum can be found. Problems of this kind are referred to as unbounded. For this problem to have a solution, we need an additional constraint of the form:

$$\begin{aligned} &x_1 \leq a \text{ (where } a > 2) \\ &x_2 \geq b \text{ (where } b < 3) \\ &\text{or } h(x_1, x_2) = 0 \end{aligned}$$

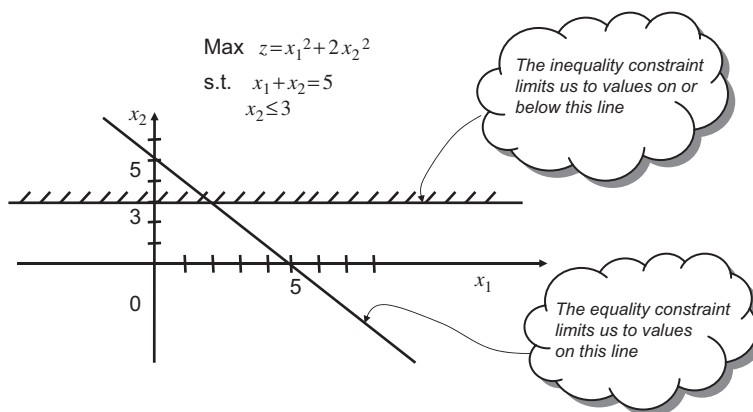


FIG. 12.1 Constraints on a simple optimization problem.

to define a closed search space.

It is also possible to overconstrain the problem. For example, if we set the problem:

$$\begin{aligned} \text{Max. } z &= x_1^2 + 2x_2^2 \\ \text{s.t. } x_1 + x_2 &= 5 \\ x_2 &\leq 3 \\ x_1 &\leq 1 \end{aligned}$$

In this case, it can be seen from Fig. 12.2 that the feasible region defined by the inequality constraints does not contain any solution to the equality constraint. The problem is therefore infeasible as stated.

12.3.2 Degrees of freedom

If the problem has n variables and m_e equality constraints, then it has $n - m_e$ degrees of freedom. If $n = m_e$ then there are no degrees of freedom and the set of m_e equations can be solved for the n variables. If $m_e > n$ then the problem is over-specified. In most cases, however, $m_e < n$ and $n - m_e$ is the number of parameters that can be independently adjusted to find the optimum.

When inequality constraints are introduced into the problem, they generally set bounds on the range over which parameters can be varied and hence reduce the space in which the search for the optimum is carried out. Very often, the optimum solution to a constrained problem is found to be at the edge of the search space (i.e., at one of the inequality constraint boundaries). In such cases, that inequality constraint becomes equal to zero (as written in Equation 12.2) and is said to be *active*. It is often possible to use engineering insight and understanding of chemistry and physics to simplify the optimization problem. If the behavior of a system is well understood, the design engineer can

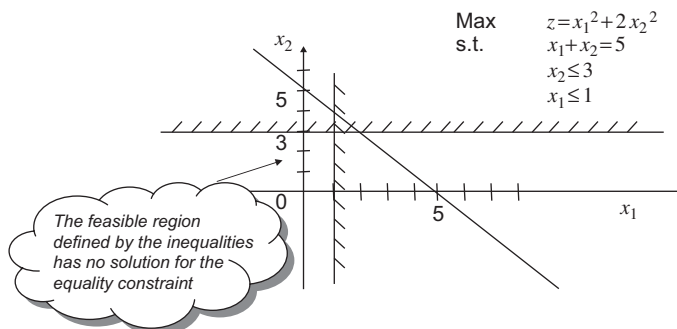


FIG. 12.2 An overconstrained problem.

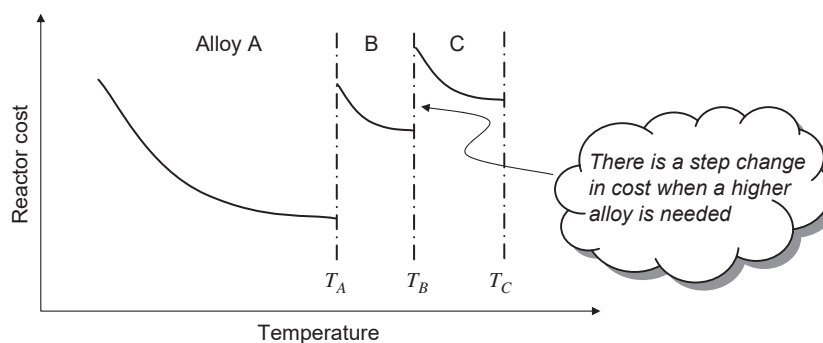


FIG. 12.3 Variation of reactor cost with temperature.

decide that an inequality constraint is likely to be active. By converting the inequality constraint into an equality constraint, the number of degrees of freedom is reduced by one and the problem is made simpler.

This can be illustrated by a simple reactor optimization example. The size and cost of a reactor are proportional to residence time, which decreases as temperature is increased. The optimal temperature is usually a trade-off between reactor cost and the formation of by-products in side reactions. But if there were no side reactions, then the next constraint would be the maximum temperature allowed by the pressure vessel design code. More expensive alloys might allow for operation at higher temperatures. The variation of reactor cost with temperature will look something like Fig. 12.3, where T_A , T_B , and T_C are the maximum temperatures allowed by the vessel design code for alloys A, B, and C, respectively.

The design engineer could formulate this problem in several ways. It could be solved as three separate problems, one corresponding to each alloy, each with a constraint on temperature $T < T_{\text{alloy}}$. The design engineer would then pick the solution that gave the best value of the objective function. The problem could also be formulated as a mixed-integer, nonlinear program with integer variables to determine the selection of alloy and set the appropriate constraint (see Section 12.11). The design engineer could also recognize that alloy A costs a lot less than alloy B, and the higher alloys only give a relatively small extension in the allowable temperature range. It is clear that cost decreases with temperature, so the optimum temperature will be T_A for alloy A and T_B for alloy B. Unless the design engineer is aware of some other effect that has an impact on cost as temperature is increased, it is safe to write $T = T_A$ as an equality constraint and solve the resulting problem. If the cost of alloy B is not excessive, then it would be prudent to also solve the problem with $T = T_B$, using the cost of alloy B.

The correct formulation of constraints is the most important step in setting up an optimization problem. Inexperienced engineers are often unaware of many constraints and consequently find “optimal” designs that are dismissed as unfeasible by more experienced designers.

12.4 Trade-offs

If the optimal value of the objective is not at a constraint limit, it will usually be determined by a trade-off between two or more effects. Trade-offs are common in design, because better performance in terms of increased purity, increased recovery, or reduced energy or raw materials use usually comes at the expense of higher capital expense, operating expense, or both. The optimization problem must capture the trade-off between cost and benefit.

A well-known example of a trade-off is the optimization of process heat recovery. A high degree of heat recovery requires close temperature approaches in the heat exchangers (see Section 3.5), which leads to high capital cost as the exchangers require more surface area. If the minimum temperature approach is increased, the capital cost is reduced but less energy is recovered. We can plot the capital cost and energy cost against the minimum approach temperature, as shown schematically in Fig. 12.4. If the capital cost is annualized (see Section 9.7) then the two costs can be added to give a total cost. The optimum value of the approach temperature, $\Delta T_{\text{optimum}}$, is then given by the minimum point in the total cost curve.

Some common trade-offs encountered in the design of chemical plants include:

- More separations equipment and operating cost vs. lower product purity
- More recycle costs vs. increased feed use and waste formation

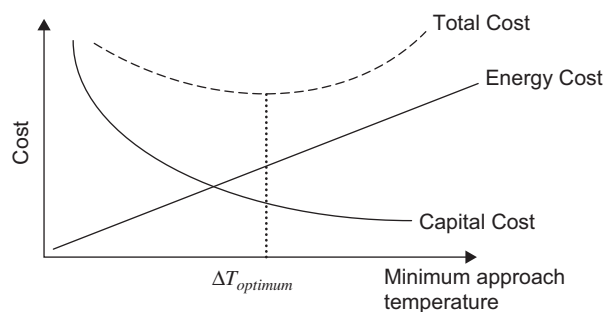


FIG. 12.4 The capital–energy trade-off in process heat recovery.

- More heat recovery vs. cheaper heat exchange network
- Higher reactivity at high pressure vs. more expensive reactors and higher compression costs
- Fast reactions at high temperature vs. product degradation
- Marketable by-products vs. more plant expense
- Cheaper steam and electricity vs. more off-site capital cost

Stating an optimization problem as a trade-off between two effects is often useful in conceptualizing the problem and interpreting the optimal solution. For example, in the case of process heat recovery, it is usually found that the shape of the total cost curve in Fig. 12.4 is relatively flat over the range $15^{\circ}\text{C} < \Delta T_{\text{optimum}} < 40^{\circ}\text{C}$. Knowing this, most experienced designers would not worry about finding the value of $\Delta T_{\text{optimum}}$, but would instead select a value for the minimum temperature approach within the range 15°C to 40°C , based on knowledge of the customer's preference for high energy efficiency or low capital expense.

12.5 Problem decomposition

The task of formally optimizing the design of a complex processing plant involving several hundred variables, usually with highly nonlinear relationships between the variables, is formidable, if not impossible. The task can be reduced by dividing the process into more manageable units, identifying the key variables, and concentrating work where the effort will give the greatest benefit. Some caution is needed when optimizing subproblems.

Subdivision, and optimization of the subunits rather than the whole, will not necessarily give the optimum design for the whole process. The optimization of one unit may be at the expense of another. For example, it will usually be satisfactory to optimize the reflux ratio for a distillation column independently of the rest of the plant, but if the column is part of a separation stage following a reactor, in which the product is separated from the unreacted materials, then the design of the column will interact with, and may well determine, the optimization of the reactor design. Care must always be taken to ensure that subcomponents are not optimized at the expense of other parts of the plant.

Equipment optimization is usually treated as a subproblem that is solved after the main process variables such as reactor conversion, recycle ratios, and product recoveries have been optimized. For example, the detailed design of heat exchangers is usually a trade-off between pressure drop and heat transfer. Higher shell- or tube-side velocities will give a higher heat-transfer coefficient, leading to a lower area and cheaper exchanger, but will also cause a higher pressure drop. A common practice is to make an allowance for exchanger pressure drop when solving the process flowsheet and then optimize the heat exchanger design subject to not exceeding the constraint of allowable pressure drop during detailed design. If heat exchanger costs are a significant fraction of total capital cost, this approach can lead to poor overall optimization, as the arbitrary assignment of pressure drops and inaccurate estimation of heat transfer coefficients in the process-level model will probably not lead to the optimal design.

Another example of a problem decomposition that is often applied is the use of the pinch design method in heat exchanger network design, described in Sections 3.5.1 and 3.5.3. If we choose to follow the pinch design rule, then no heat should be transferred across the pinch, and the heat exchanger network design problem is decomposed into two separate, smaller problems above and below the pinch. This is convenient, particularly when solving relatively small problems as hand calculations. Unfortunately, this approach has the drawback that we might miss opportunities to match the same streams above and below the pinch and hence reduce the number of exchangers needed by combining an exchanger from the above-pinch problem with one from the below-pinch problem. When designing

large networks involving many process streams and multiple utility streams, the imposition of utility pinches, as well as process pinches, can lead to the formation of impractical networks with many small heat exchangers.

12.6 Optimization of a single decision variable

If the objective is a function of a single variable, x , the objective function $f(x)$ can be differentiated with respect to x to give $f'(x)$. Any stationary points in $f(x)$ can then be found as the solutions of $f'(x) = 0$. If the second derivative of the objective function is greater than zero at a stationary point, then the stationary point is a local minimum. If the second derivative is less than zero, then the stationary point is a local maximum, and if it is equal to zero, then it is a saddle point. If x is bounded by constraints, then we must also check the values of the objective function at the upper and lower limiting constraints. Similarly, if $f(x)$ is discontinuous, then the value of $f(x)$ on either side of the discontinuity should also be checked.

This procedure can be summarized as the following algorithm:

$$\begin{aligned} \text{Min. } z &= f(x) \\ \text{s.t. } x &\geq x_L \\ x &\leq x_U \end{aligned} \quad (12.3)$$

1. Solve $f' = \frac{df(x)}{dx} = 0$ to find values of x_S .
2. Evaluate $f'' = \frac{d^2f(x)}{dx^2}$ for each value of x_S . If $f'' > 0$ then x_S corresponds to a local minimum in $f(x)$.
3. Evaluate $f(x_S)$, $f(x_L)$, and $f(x_U)$.
4. If the objective function is discontinuous, then evaluate $f(x)$ on either side of the discontinuity, x_{D1} and x_{D2} .
5. The overall optimum is the value from the set $(x_L, x_S, x_{D1}, x_{D2}, x_U)$ that gives the lowest value of $f(x)$.

This is illustrated graphically in Fig. 12.5(a) for a continuous objective function. In Fig. 12.5(a), x_L is the optimum point, even though there is a local minimum at x_{S1} . Fig. 12.5(b) illustrates the case of a discontinuous objective function. Discontinuous functions are quite common in engineering design, arising, for example, when changes in temperature or pH cause a change in metallurgy. In Fig. 12.5(b) the optimum is at x_{D1} , even though there is a local minimum at x_S .

If the objective function can be expressed as a differentiable equation, then it is usually also easy to plot a graph like those in Fig. 12.5 and quickly determine whether the optimum lies at a stationary point or a constraint.

12.7 Search methods

In design problems, the objective function very often cannot be written as a simple equation that is easily differentiated. This is particularly true when the objective function requires solving large computer models, possibly using several different programs and requiring several minutes, hours, or days to converge a single solution. In such cases, the optimum is found using a search method. The concept of search methods is most easily explained for single-variable problems, but search methods are at the core of the solution algorithms for multivariable optimization as well.

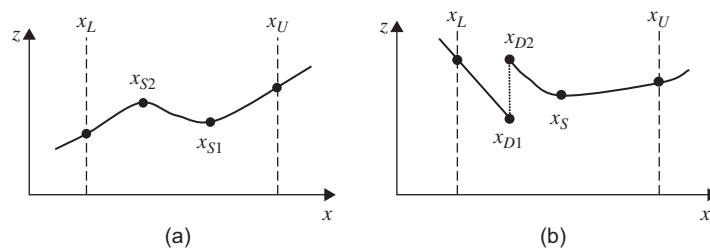


FIG. 12.5 Optimization of a single variable between bounds.

12.7.1 Unrestricted search

If the decision variable is not bounded by constraints, then the first step is to determine a range in which the optimum lies. In an unrestricted search, we make an initial guess of x and assume a step size, h . We then calculate $z_1 = f(x)$, $z_2 = f(x + h)$, and $z_3 = f(x - h)$. From the values of z_1 , z_2 , and z_3 we determine the direction of search that leads to improvement in the value of the objective, depending on whether we wish to minimize or maximize z . We then continue increasing (or decreasing) x by successive steps of h until the optimum is passed.

In some cases, it may be desirable to accelerate the search procedure, in which case the step size can be doubled at each step. This gives the sequence $f(x + h)$, $f(x + 3h)$, $f(x + 7h)$, $f(x + 15h)$, etc.

Unrestricted searching is a relatively simple method of bounding the optimum for problems that are not constrained. In engineering design problems, it is almost always possible to state upper and lower bounds for every parameter, so unrestricted search methods are not widely used in design.

Once a restricted range that contains the optimum has been established, restricted range search methods can be used. These can be broadly classified as direct methods that find the optimum by eliminating regions in which it does not lie and indirect methods that find the optimum by making an approximate estimate of $f'(x)$.

12.7.2 Regular search (three-point interval search)

The three-point interval search starts by evaluating $f(x)$ at the upper and lower bounds, x_L and x_U , and at the center point $(x_L + x_U)/2$. Two new points are then added in the midpoints between the bounds and the center point, at $(3x_L + x_U)/4$ and $(x_L + 3x_U)/4$, as shown in Fig. 12.6. The three adjacent points with the lowest values of $f(x)$ (or the highest values for a maximization problem) are then used to define the next search range.

By eliminating two of the four quarters of the range at each step, this procedure reduces the range by half each cycle. To reduce the range to a fraction ϵ of the initial range therefore takes n cycles, where $\epsilon = 0.5^n$. Because each cycle requires calculating $f(x)$ for two additional points, the total number of calculations is $2n = 2 \log \epsilon / \log 0.5$.

The procedure is terminated when the range has been reduced sufficiently to give the desired precision in the optimum. For design problems, it is usually not necessary to specify the optimal value of the decision variables to high precision, so ϵ is usually not a very small number.

12.7.3 Golden-section search

The golden-section search, sometimes called the golden mean search, is as simple to implement as the regular search but is more computationally efficient if $\epsilon < 0.29$. In the golden-section search only one new point is added at each cycle.

The golden-section method is illustrated in Fig. 12.7. We start by evaluating $f(x_L)$ and $f(x_U)$ corresponding to the upper and lower bounds of the range, labeled A and B in the figure. We then add two new points, labeled C and D, each located a distance ωAB from the bounds A and B, that is, located at $x_L + \omega(x_U - x_L)$ and $x_U - \omega(x_U - x_L)$. For a minimization problem, the point that gives the highest value of $f(x)$ is eliminated. In Fig. 12.7, this is point B. A single new point, E, is added, such that the new set of points AECD is symmetric with the old set of points ACDB.

For the new set of points to be symmetric with the old set of points, $AE = CD = \omega AD$.

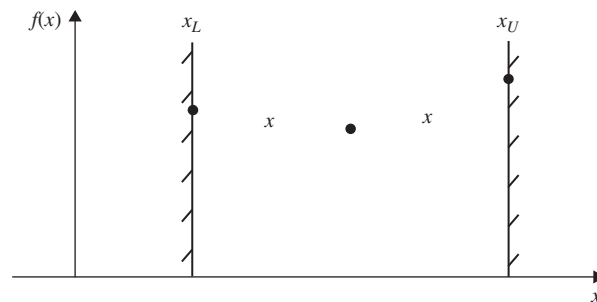


FIG. 12.6 Regular search.

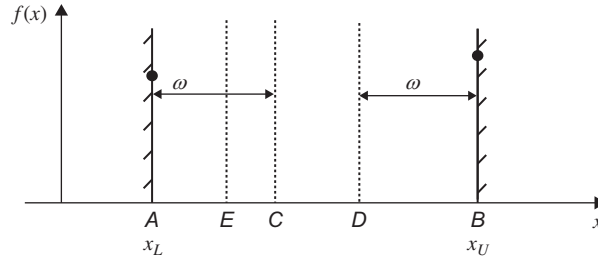


FIG. 12.7 Golden section search.

But we know $DB = \omega AB$, so $AD = (1 - \omega)AB$ and $CD = (1 - 2\omega)AB$
 so $(1 - 2\omega) = \omega(1 - \omega)$

$$\omega = \frac{3 \pm \sqrt{5}}{2}$$

Each new point reduces the range to a fraction $(1 - \omega) = 0.618$ of the original range. To reduce the range to a fraction ϵ of the initial range therefore requires $n = \log \epsilon / \log 0.618$ function evaluations.

The number $(1 - \omega)$ is known as the *golden mean*. The significance of this number has been known since ancient times. Livio (2002) gives a very entertaining account of its history and occurrence in art, architecture, music, and nature.

12.7.4 Quasi-Newton method

Newton's method is a super-linear, indirect search method that seeks the optimum by solving $f'(x)$ and $f''(x)$ and searching for where $f'(x) = 0$. The value of x at step $k + 1$ is calculated from the value of x at step k using:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} \quad (12.4)$$

and the procedure is repeated until $(x_{k+1} - x_k)$ is less than a convergence criterion or tolerance, ϵ .

If we do not have explicit formulae for $f'(x)$ and $f''(x)$, then we can make a finite difference approximation about a point, in which case:

$$x_{k+1} = x_k - \frac{[f(x_k + h) - f(x_k - h)]/2h}{[f(x_k + h) - 2f(x_k) + f(x_k - h)]/h^2} \quad (12.5)$$

Care is needed in setting the step size, h , and the tolerance for convergence, ϵ . The quasi-Newton method generally gives fast convergence unless $f''(x)$ is close to zero, in which case convergence is poor.

All of the methods discussed in this section are best suited for unimodal functions, that is, functions with no more than one maximum or minimum within the bounded range.

12.8 Optimization of two or more decision variables

A two-variable optimization problem can be stated as:

$$\begin{aligned} \text{Min. } z &= f(x_1, x_2) \\ \text{s.t. } h(x_1, x_2) &= 0 \\ g(x_1, x_2) &\leq 0 \end{aligned} \quad (12.6)$$

For simplicity, all problems will be stated as minimization problems from here on. A maximization problem can be rewritten as $\text{Min. } z = -f(x_1, x_2)$.

With two parameters, we can plot contour lines of z on a graph of x_1 vs. x_2 and hence get a visual representation of the behavior of z . For example, Fig. 12.8 shows a schematic of a contour plot for a function that exhibits a local minimum of <30 at about $(4,13)$ and a global minimum of <10 at about $(15,19)$. Contour plots are useful for understanding some of the key features of multivariable optimization that become apparent as soon as we consider more than one decision variable.

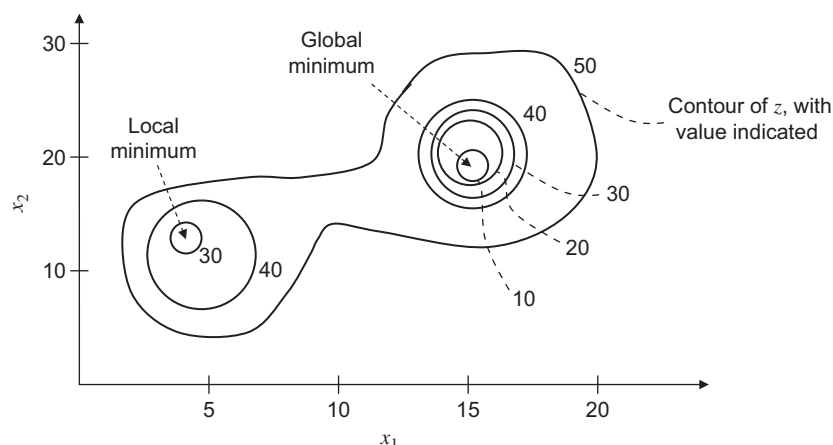


FIG. 12.8 Optimization of two decision variables.

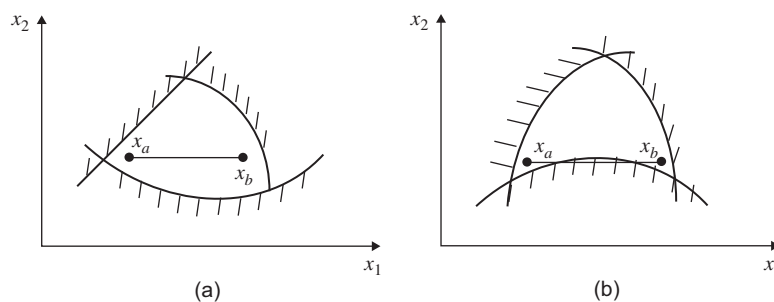


FIG. 12.9 Convexity for a two-variable problem. (a) Convex feasible region. (b) Nonconvex feasible region

12.8.1 Convexity

Constraint boundaries can also be plotted in the (x_1, x_2) parameter space, as illustrated in Fig. 12.9. If the constraints are not linear, there is a possibility that the feasible region may not be convex. A convex feasible region, illustrated in Fig. 12.9(a), is one in which any point on a straight line between any two points inside the feasible region also lies within the feasible region. This can be stated mathematically as:

$$\begin{aligned} x &= \alpha x_a + (1 - \alpha)x_b \in \text{FR} \\ \forall x_a, x_b \in \text{FR}, 0 < \alpha < 1 \end{aligned} \quad (12.7)$$

where

x_a, x_b = any two points belonging to the feasible region

FR = the set of points inside the feasible region bounded by the constraints

α = a constant

If any two points in the feasible region can be found such that some point on a straight line between them lies outside of the feasible region, then the feasible region is nonconvex, as illustrated in Fig. 12.9(b).

The importance of convexity is that problems with a convex feasible region are more easily solved to a global optimum. Problems with nonconvex feasible regions are prone to convergence to local minima. Nonconvexity is common in chemical engineering problems due to the nonlinear nature of many of the equality constraint equations.

12.8.2 Searching in two dimensions

The procedures for searching in two dimensions are mostly extensions of the methods used for single-variable line searches:

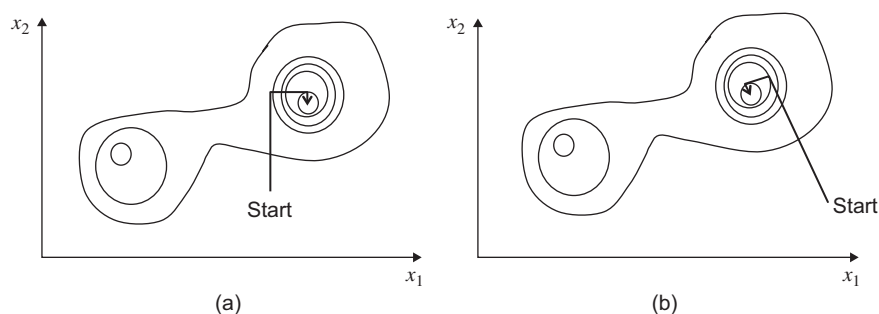


FIG. 12.10 Search methods. (a) Univariate search. (b) Steepest descent.

1. Find an initial solution (x_1, x_2) inside the feasible region.
2. Determine a search direction.
3. Determine step lengths δx_1 and δx_2 .
4. Evaluate $z = f(x_1 + \delta x_1, x_2 + \delta x_2)$.
5. Repeat steps 2 to 4 until convergence.

If x_1 and x_2 are varied one at a time, then the method is known as a univariate search and is the same as carrying out successive line searches in each parameter. If the step length is determined so as to find the minimum with respect to the variable searched, then the calculation steps towards the optimum, as shown in Fig. 12.10(a). This method is simple to implement, but can be very slow to converge. Other direct methods include pattern searches, such as the factorial designs used in statistical design of experiments (see, for example, Montgomery, 2012), the EVOP method (Box, 1957), and the sequential simplex method (Spendley et al., 1962).

Indirect methods can also be applied to problems with two or more decision variables. In the steepest descent method (also known as the gradient method), the search direction is along the gradient at point (x_1, x_2) , that is, orthogonal to the contours of $f(x_1, x_2)$. A line search is then carried out to establish a new minimum point where the gradient is reevaluated. This procedure is repeated until the convergence criterion is met, as shown in Fig. 12.10(b).

12.8.3 Problems in multivariable optimization

Some common problems that are encountered in multivariable optimization can be described for a two-variable problem and are illustrated in Fig. 12.11. In Fig. 12.11(a), the shape of the contours is such that a univariate search would be very slow to converge. Using an indirect method such as steepest descent would be more appropriate in this case. Fig. 12.11(b) shows the problem of convergence to a local optimum. In this scenario, different answers are obtained for different initial solutions. This problem can be overcome by using pattern searches with a larger grid or by using probabilistic methods such as simulated annealing or genetic algorithms that introduce some possibility of moving away from a local optimum. An introduction to probabilistic methods is given in Diwekar (2008). Probabilistic methods are also useful when faced with a nonconvex feasible region, as pictured in Fig. 12.11(c).

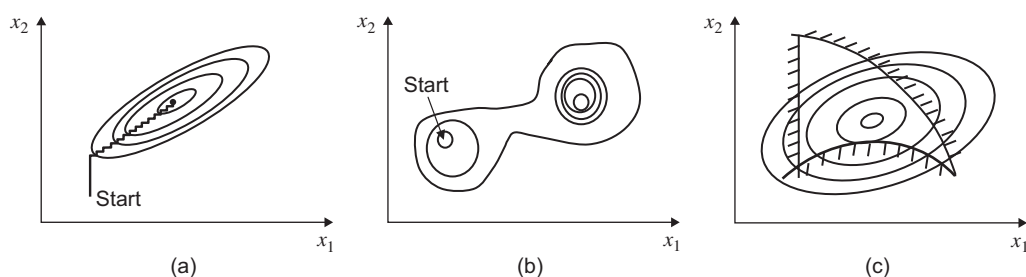


FIG. 12.11 Common problems in multivariable optimization. (a) Slow convergence. (b) Convergence to local optimum. (c) Nonconvex feasible region.

12.8.4 Multivariable optimization

When there are more than two decision variables, it is much harder to visualize the parameter space, but the same issues of initialization, convergence, convexity, and local optima are faced. The solution of large multivariable optimization problems is at the core of the field of operations research. Operations research methods are widely used in industry, particularly in manufacturing facilities, as discussed in [Section 12.12](#).

The following sections give only a cursory overview of this fascinating subject. Readers who are interested in learning more should refer to [Hillier and Lieberman \(2014\)](#) and the other references cited in this chapter.

12.9 Linear programming

A set of continuous linear constraints always defines a convex feasible region. If the objective function is also linear and $x_i > 0$ for all x_i , then the problem can be written as a linear program (LP). A simple two-variable illustration of an LP is given in [Fig. 12.12](#).

Linear programs always solve to a global optimum. The optimum must lie on the boundary at an intersection between constraints, which is known as a *vertex* of the feasible region. The inequality constraints that intersect at the optimum are said to be active and have $h(\mathbf{x}) = 0$, where $\mathbf{x}$ is the vector of decision variables.

Many algorithms have been developed to solve LPs, of which the most widely used are based on the SIMPLEX algorithm developed by [Dantzig \(1963\)](#). The SIMPLEX method introduces slack and surplus variables to transform the inequality constraints into equalities. For example, if:

$$x_1 + x_2 - 30 \leq 0$$

we can introduce a slack variable, S_1 , and write:

$$x_1 + x_2 - 30 + S_1 = 0$$

The resulting set of equalities is solved to obtain a feasible solution, in which some of the slack and surplus variables will be zero, corresponding to active constraints. The algorithm then searches the vertices of the feasible region, decreasing the objective at each step until the optimum is reached. Details of the SIMPLEX method are given in most optimization or operations research textbooks. See, for example, [Hillier and Lieberman \(2014\)](#) or [Edgar and Himmelblau \(2001\)](#). There have been many improvements to the SIMPLEX algorithm over the years, but it is still the method used in most commercial solvers.

Some problems that can occur in solving LPs are illustrated in [Fig. 12.13](#). In [Fig. 12.13\(a\)](#), the contours of the objective function are exactly parallel to one of the constraints. The problem is said to be degenerate and has an infinite number of solutions along the line of that constraint. [Fig. 12.13\(b\)](#) shows a problem where the feasible region is unbounded. This situation does not usually occur in engineering design unless the problem has been badly formulated. The situation in [Fig. 12.13\(c\)](#) is more common, in which the problem is overconstrained and there is no feasible region.

Linear programming can be used to solve very large problems, with thousands of variables and constraints. The method is widely used in operations, particularly in optimization of oil refineries and petrochemical plants. It is used a lot less in design, as design problems almost inevitably contain many nonlinear equations.

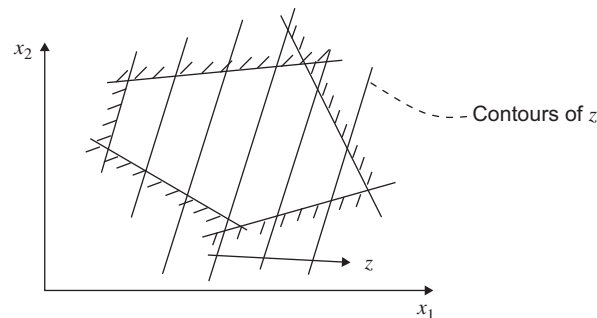


FIG. 12.12 A linear program.

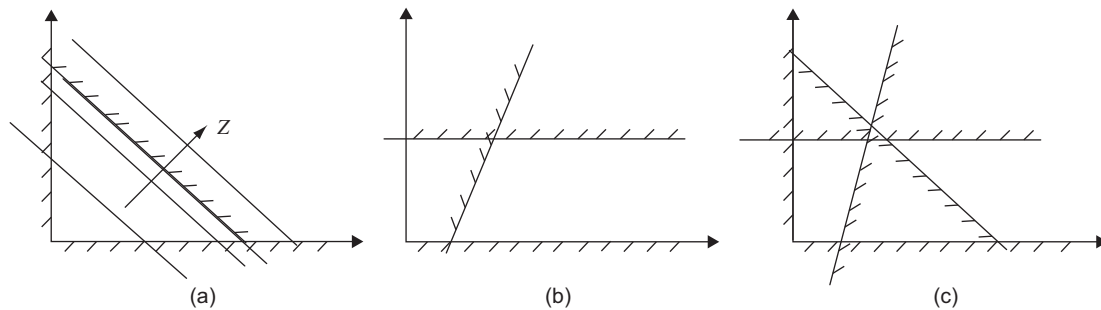


FIG. 12.13 Problems in linear programming. (a) Objective function parallel to a constraint (degenerate problem). (b) Feasible region unbounded. (c) No feasible region.

12.10 Nonlinear programming

When the objective function and/or the constraints are nonlinear, the optimization must be solved as a nonlinear program (NLP). Three main methods are used for solving an NLP.

12.10.1 Successive linear programming

In successive linear programming (SLP), $f(x)$, $g(x)$, and $h(x)$ are linearized at an initial point. The resulting LP is solved to give an initial solution, and $f(x)$, $g(x)$, and $h(x)$ are linearized again at the new point. The procedure is then repeated until convergence. If the new point is outside the feasible region, the nearest point lying inside the feasible region is used.

With SLP there is no guarantee of convergence or global optimality. The method is widely used, nonetheless, as it is a simple extension of linear programming. It should be noted that whenever discontinuous linear functions are used to approximate a nonlinear function, the problem behaves like an SLP. There is no guarantee of convexity or convergence to the optimal solution.

12.10.2 Successive quadratic programming

The successive quadratic programming (SQP) algorithm is similar to SLP, but instead approximates $f(x)$ as a quadratic function and uses quadratic programming methods that give faster convergence than SLP. SQP works well for highly NLPs with relatively few variables, for example, optimizing a process simulation or the design of a single piece of equipment. Biegler et al. (1997) suggest SQP is the best method for problems with fewer than 50 variables and where the gradients must be found numerically.

12.10.3 Reduced gradient method

Reduced gradient methods are related to the SIMPLEX algorithm. The method linearizes the constraints and introduces slack and surplus variables to transform the inequalities into equalities. The n -dimensional vector x is then partitioned into $n - m$ independent variables, where m is the number of constraints. A search direction is determined in the space of the independent variables, and a quasi-Newton method is used to determine an improved solution of $f(x)$ that still satisfies the nonlinear constraints. If all the equations are linear, this reduces to the SIMPLEX method (Wolfe, 1962). Various algorithms have been proposed, using different methods for carrying out the search and returning to a feasible solution, for example, the generalized reduced gradient (GRG) algorithm (Abadie & Guigou, 1969) and the MINOS algorithm (Murtagh & Saunders, 1978, 1982).

Reduced gradient methods are particularly effective for sparse problems with a large number of variables. A problem is said to be sparse if each constraint involves only a few of the variables. This is a common situation in design problems, where many of the constraints are written in terms of only one or two variables. Reduced gradient methods also work better when many of the constraints are linear, as less computational time is spent linearizing constraints and returning the solution to the feasible region. Because of the decomposition of the problem, fewer calculations are required per iteration, particularly if analytical expressions for the gradients are known (which is usually not the case in design). The reduced gradient method is often used in optimizing large spreadsheet models that contain many linear constraints.

All of the nonlinear programming algorithms can suffer from the convergence and local optima problems described in Section 12.8.3. Probabilistic methods such as simulated annealing and genetic algorithms can be used if it is suspected that the feasible region is nonconvex or multiple local optima are present.

12.11 Mixed-integer programming

Many of the decisions faced in operations involve discrete variables. For example, if we calculate that we need to ship 3.25 trucks of product from plant A to plant B each week, we could send 3 trucks for 3 weeks and then 4 trucks in the fourth week, or we could send 4 trucks each week, with the fourth truck only one-quarter filled, but we cannot send 3.25 trucks every week. Some common operational problems involving discrete variables include:

- Production scheduling: Determine the production schedule and inventory to minimize the cost of meeting demand. This is particularly important for batch plants, when the plant can make different products.
- Transshipment problems and supply chain management: Satisfy demands at different producing plants and sales destinations from different supply points, warehouses, and production facilities. Shipping quantities are often constrained to certain size lots corresponding to rail tankers, road tankers, drums, etc.
- Assignment problems: Schedule workers to different tasks.

Discrete variables are also sometimes used in process design, for example, the number of trays or the feed tray of a distillation column, and in process synthesis, to allow selection between flowsheet options, as described later.

Discrete decisions are addressed in operations research by introducing integer variables. When integer variables are introduced, an LP becomes a mixed-integer linear program (MILP) and a nonlinear program becomes a mixed-integer nonlinear program (MINLP). Binary integer variables are particularly useful, as they can be used to formulate rules that enable the optimization program to choose between options. For example, if we define y as a binary integer variable such that:

if $y = 1$ a feature exists in the optimal solution, and
if $y = 0$ the feature does not exist in the optimal solution,

then we can formulate constraint equations such as:

$$\sum_{i=1}^n y_i = 1 \text{ choose only one of } n \text{ options}$$

$$\sum_{i=1}^n y_i \leq m \text{ choose at most } m \text{ of } n \text{ options}$$

$$\sum_{i=1}^n y_i \geq m \text{ choose at least } m \text{ of } n \text{ options}$$

$$y_k - y_j \leq 0 \text{ if item } k \text{ is selected, item } j \text{ must be selected, but not vice versa}$$

$$\left. \begin{array}{l} g_1(x) - M y \leq 0 \\ g_2(x) - M(1 - y) \leq 0 \\ M \text{ is a large scalar value} \end{array} \right\} \text{ either } g_1(x) \leq 0 \text{ or } g_2(x) \leq 0$$

The last rule listed can be used to select between alternative constraints.

12.11.1 Mixed-integer programming algorithms

Although integer variables are convenient for problem formulation, if too many integer variables are used, the number of options explodes in a combinatorial manner and solution becomes difficult. Mixed-integer problems can be solved efficiently using methods such as the “branch and bound” algorithm. The branch and bound method starts by treating all integer variables as continuous and solving the resulting LP or NLP to give a first approximation. All integer variables are then rounded to the nearest integer to give a second approximation. The problem is then partitioned into two new integer problems for each integer variable that had a nonintegral solution in the first

approximation. In one branch a constraint is added that forces the integer variable to be greater than or equal to the next highest integer, whereas in the other branch a constraint is added that forces the variable to be equal to or less than the next lowest integer. For example, if a variable was found to have an optimal value $y = 4.4$ in the first approximation, then the new constraints would be $y \geq 5$ in one branch and $y \leq 4$ in the other. The branched problems are then solved to give new first approximations, and the branching procedure is repeated until an integer solution is found.

When an integer solution is found, it is used to set a bound on the value of the objective. For example, in a minimization problem, the optimal solution must be less than or equal to the bound set by this integral solution. Consequently, all branches with greater values of the objective can be discarded, as forcing the variables in these branches to integer values will lead to deterioration in the objective rather than improvement. The procedure then continues branching on all the nonintegral integer variables from each first approximation, setting new bounds each time an improved integer solution is found, until all of the branches have been bounded and the optimal solution has been obtained. See Hillier and Lieberman (2014) or Edgar and Himmelblau (2001) for details of the algorithm and examples of its application.

The branch and bound method can be used for MINLP problems, but it requires solving a large number of NLP problems and is therefore computationally intensive. Instead, methods such as the Generalized Benders' Decomposition and Outer Approximation algorithms are usually preferred. These methods solve a master MILP problem to initialize the discrete variables at each stage and then solve an NLP subproblem to optimize the continuous variables. Details of these methods are given in Floudas (1995), Biegler et al. (1997), and Diwekar (2008).

12.11.2 Superstructure optimization

Binary integer variables can be used to formulate optimization problems that choose between flowsheet options. For example, consider the problem of selecting a reactor. We can set up a unit cell consisting of a well-mixed reactor, a plug-flow reactor (PFR), and a bypass in parallel, each with a valve upstream, as illustrated in Fig. 12.14(a). If a binary variable is used to describe whether the valve is open or closed and a constraint is introduced such that only one of the valves is open, then the optimization will select the best option. A set of such unit cells can be built into a superstructure, incorporating additional features such as recycles, as shown schematically in Fig. 12.14(b). A more rigorous superstructure that encompasses other options such as side-stream feeds to the plug-flow reactor (PFR) was developed by Kokossis and Floudas (1990).

The optimization of such a superstructure can identify reactor networks or mixing arrangements that would not be intuitively obvious to the design engineer. Similar superstructure formulations have been proposed for other process synthesis problems such as distillation column sequencing, design of heat-exchange networks, and design of site utility systems. Biegler et al. (1997) give an excellent overview of the use of superstructure-based methods in process synthesis.

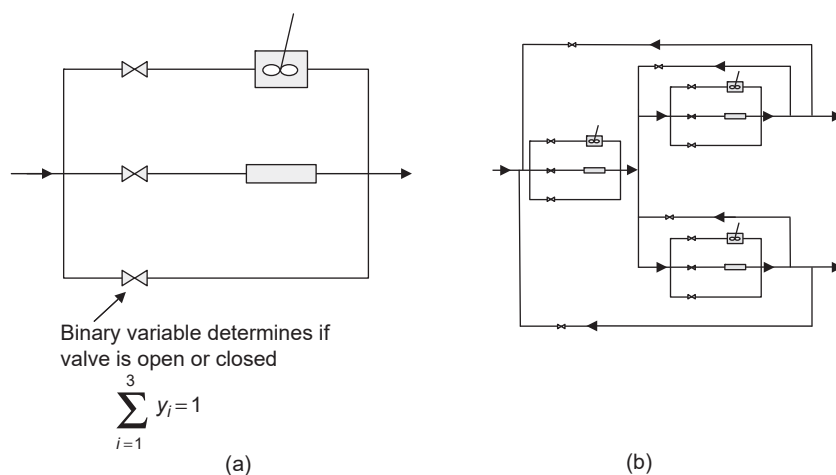


FIG. 12.14 Application of integer programming to reactor design. (a) Unit cell of reactor options. (b) Superstructure of unit cells and recycles.

12.12 Optimization in industrial practice

12.12.1 Optimization of process operations

Perhaps not surprisingly, operations research methods are widely used in process operations. Few manufacturing plants do not use LP or MILP tools for planning and scheduling. Supply chain management is very important to economic performance, and is usually carried out using large MILP models. The models used in industry for these purposes are often not very sophisticated, but proper formulation of constraints and the ability to solve robustly with a large number of variables are usually more important features of tools for these applications.

The use of operations research methods for plant and supply chain applications is taught as part of the industrial engineering curriculum in most universities. Chemical engineers considering a career in manufacturing would be well advised to develop a solid grounding in these methods.

12.12.2 Optimization of batch and semicontinuous processes

In batch operation, there will be periods when product is being produced, followed by nonproductive periods when the product is discharged and the equipment prepared for the next batch. The rate of production will be determined by the total batch time, productive plus nonproductive periods.

$$\text{Batches per year} = \frac{8760 \times \text{plant attainment}}{\text{batch cycle time}} \quad (12.8)$$

where the “plant attainment” is the fraction of the total hours in a year (8760) that the plant is in operation.

$$\text{Annual production} = \text{quantity produced per batch} \times \text{batches per year.}$$

$$\text{Cost per unit of production} = \frac{\text{annual cost of production}}{\text{annual production rate}} \quad (12.9)$$

With many batch operations, the production rate decreases during the production period; for example, batch reactors and plate and frame filter presses. There is then an optimum batch size, or optimum cycle time, that gives the minimum cost per unit of production.

For some continuous processes, the period of continuous production will be limited by gradual changes in process conditions. Examples include the deactivation of catalysts or the fouling of heat-exchange surfaces. Production is lost during the periods when the plant is shut down for catalyst renewal or equipment cleaning. As with batch processes, there is an optimum cycle time to give the minimum production cost. The optimum time between shut-downs can be found by determining the relationship between cycle time and cost per unit of production (the objective function) and using one of the optimization techniques outlined in this section to find the minimum.

With discontinuous processes, the period between shut-downs will usually be a function of equipment size. Increasing the size of critical equipment will extend the production period, but at the expense of increased capital cost. The designer must strike a balance between the savings gained by reducing the nonproductive period and the increased investment required.

In some batch plants several trains of identical equipment are operated in a sequence that allows some degree of heat recovery or enables downstream equipment to operate continuously. In this type of plant the time allowed for each operation in the sequence is optimized so that an overall schedule for the plant can be developed. Scheduling of batch processes is described in [Biegler et al. \(1997\)](#).

12.12.3 Optimization in process design

Few, if any, industrial designs are rigorously optimized. This is because:

1. The cost of building rigorous models of reactor kinetics and hydraulics that give accurate prediction of by-product yields is usually not justified. The amount of time available for the project is usually insufficient for such models to be built. The errors introduced by uncertainty in the process models may be much larger than the differences in performance predicted for different designs.
2. The uncertainty in the forecasts of future prices is usually so large that it dominates most differences between design alternatives.

3. Regardless of the quality of the tools used or the experience of the estimator, it is usually impossible to make a capital cost estimate within $\pm 15\%$ without completing a substantial amount of design work (see [Chapter 7](#)). Many design decisions are thus made on the basis of sketchy cost estimates. The cost of going back and revisiting these design decisions at a later stage in the project when more design detail is available is usually not justified.
4. Criteria such as safety, operability, reliability, and flexibility are of vital importance in process design. These features make the design more robust to variations in the design assumptions and operating requirements. A safe, operable, and reliable plant will often require more expense above the cost of the economically “optimal” design. This extra expense is difficult to trade off against the nonfinancial benefits of having a process that is easier to run.
5. In most cases there are several “near-optimal” designs. The difference between the values of the objective obtained for each of these is often not statistically significant, given the errors in prices, cost estimates, and yields.

In industrial process design, optimization usually involves carrying out sufficient analysis to be confident that the design is reasonably close to the optimum. The most important things for the design engineer to understand are:

1. What are the constraints on the design?
2. Which constraints are hard (inviolable) and which are soft (can be modified)?
3. Where are the discontinuities in cost? For example, what conditions cause a change to a more costly metallurgy or a different design code?
4. What are the main design trade-offs?
5. How does the objective function vary with the main process parameters?
6. What are the major cost components of the process (both capital and operating costs), and what radical changes could be made to the process to reduce these costs?

Experienced design engineers usually think through these questions carefully to satisfy themselves that their design is “good enough.” Only very occasionally do they formulate an optimization problem and solve it rigorously.

Example 12.1

Optimize the design of a distillation column to separate 225 metric tons per hour of an equimolar mixture of benzene, toluene, ethylbenzene, paraxylene, and orthoxylene with minimum total annualized cost. The feed is a saturated liquid at 330 kPa. The recovery of toluene in the distillate should be greater than 99%, and the recovery of ethylbenzene in the bottoms should be greater than 99%.

Solution

The first step is to determine the design factor. If we assume a design factor of 10%, then the equipment should be designed for a flow rate of 248 metric tons per hour (te/h). This flow rate is used in simulating the process for the purpose of sizing equipment, but energy consumption must be based on the reboiler and condenser duties expected for a 225 te/h feed rate.

This is a single distillation column, which is easy to model in any of the commercial simulation programs. UniSim (Honeywell, Inc.) was used for the purpose of this example. The simulation was set up using the component recoveries of toluene and ethylbenzene as column specifications, which gave rapid convergence. Tray sizing calculations were run using the UniSim tray sizing utility. A tray spacing of 0.61 m (2 ft) was assumed, and other tray parameters were left at the UniSim default values. Two meters were added to the column height to allow space for a sump and demister. Sieve trays were used, and the stage efficiency was assumed to be 80%. Details of the column simulation and design are given in Examples 4.6 and 4.7.

To optimize the design, we need to formulate an objective function. The distillation column has the following cost components:

- Capital costs: column shell, internals, condenser, receiver drum, reboiler, pumps, piping, instrumentation, structure, foundations, etc.
- Operating costs: cost of heating for the reboiler and cost of cooling for the condenser

The purchased equipment costs can be estimated based on information from the process simulation using the cost correlations given in [Chapter 7](#). The column shell is a pressure vessel, and the design can be completed using the methods given in [Chapter 14](#). The details of how to complete these calculations are not important here, but Example 14.2 and Example 7.3 provide detailed explanations of the method followed. Carbon steel construction was assumed. The purchased equipment costs can be converted into an installed capital cost by multiplying by an installation factor. For the purposes of this example, the installation factor can be assumed to be 4.0; see Section 7.6. The installed capital costs can be converted into an annual capital charge by dividing by 3, using a rule of thumb that is developed in Section 9.7.

TABLE 12.2 Optimization results

Iteration number	1	2	3	4	5	6	7	8	9
Number of trays	40	90	120	70	80	76	84	80	80
Feed tray	20	45	60	35	40	38	42	27	53
Column height (m)	26.4	56.9	75.2	44.7	50.8	48.4	53.2	50.8	50.8
Column diameter (m)	5.49	4.42	4.42	4.42	4.42	4.42	4.42	4.42	4.57
Reflux ratio	3.34	2.50	2.48	2.57	2.52	2.54	2.51	2.48	2.78
Reboiler duty, Q_r (MW)	34.9	28.3	28.2	28.8	28.5	28.6	28.4	28.2	30.4
Condenser duty, Q_c (MW)	33.9	27.3	27.2	27.8	27.5	27.6	27.4	27.2	29.4
Annualized capital cost (MM\$/y)	0.82	0.95	1.25	0.83	0.89	0.87	0.91	0.89	0.94
Annual energy cost (MM\$/y)	8.59	6.96	6.93	7.10	7.01	7.04	6.99	6.93	7.50
Total annualized cost (MM\$/y)	9.41	7.91	8.18	7.93	7.900	7.905	7.904	7.82	8.44

The operating costs are simple to estimate from the condenser and reboiler duties if the cost of energy is known. For this example, the cost of heat is taken as \$7.5/GJ and the cost of cooling is \$0.2/GJ.

The objective function can then be written as:

$$\begin{aligned} \text{Min.: Total annualized cost (TAC)} &= \text{cost of heating} + \text{cost of cooling} + \text{annualized capital cost} \\ &= 5.5Q_r + 0.2Q_c + (4/3) (\Sigma \text{ purchased equipment costs}) \end{aligned}$$

where:

Q_r = annual reboiler energy consumption (GJ/yr)

Q_c = annual condenser energy consumption (GJ/yr)

The optimization problem is strictly a MINLP, as we need to consider discrete variables (number of trays, feed tray) as well as continuous variables (reflux ratio, reboiler duty, etc.). This problem is actually relatively easy to formulate and solve rigorously, but instead we will step through the calculation to illustrate how an experienced industrial designer would approach the problem.

Table 12.2 gives the results of several iterations of the optimization.

1. To begin, we need to find a feasible solution. As an initial guess we can use 40 trays with the feed on tray 20. The column converges with a reflux ratio of 3.34 and diameter 5.49 m. This is large, but not unreasonable for such a high flow rate. Looking at the components of the total annualized cost, the capital is contributing \$0.8 MM/yr and energy is contributing \$8.6 MM/yr, so the costs are dominated by energy cost. It is clear that adding more stages and reducing the reflux ratio will reduce the total cost. (If capital costs were dominating, then we would reduce the number of stages.) There is no upper hard constraint on column height, but there is a soft constraint. At the time of writing there are only 14 cranes in the world that can lift a column taller than 80 m. There are 48 cranes that can lift a column up to 60 m. We can therefore expect that the cost of lifting a column >60 m height will go up as it becomes more expensive to rent the necessary equipment for installation. We can start by assuming a soft constraint that the maximum height must be less than 60 m.
2. Using 90 trays with feed on tray 45 gives a reflux ratio of 2.5 and diameter 4.42 m. The column height is 56 m, which allows some space for vessel supports and clearance for piping at the column base and still comes in under the 60-m target. The capital cost increases to \$0.95 MM/yr, while energy cost is reduced to \$6.96 MM/yr, giving a total annualized cost of \$7.91 MM/yr and savings of \$1.5 MM/yr relative to the initial design.
3. We should explore whether going to an even taller column would make sense. We can investigate this by increasing the installation factor from 4 to 5 for the column shell to allow for the higher cost of using one of the larger cranes. If we increase the number of trays to 120, the column height is 75 m, which will give a total height of close to 80 m when installed. The total annualized cost increases to \$8.2 MM/yr, so we can conclude that it is probably not economical to go to a total height above 60 m. We can notice, though, that the reflux ratio didn't change much

when we added extra trays. This suggests that we are getting close to minimum reflux. It might therefore be worth backing off from the maximum column height constraint to see if there is an optimum number of trays.

4. Adding a design with 70 trays and feed on tray 35 (roughly halfway between 40 and 90) gives reflux ratio 2.57 and total annualized cost \$7.94 MM/yr. This is not an improvement on the 90-tray design, so the optimum must be between 70 and 90 trays.
5. A design with 80 trays and feed on tray 40 (halfway between 70 and 90) gives reflux ratio 2.52 and total annualized cost \$7.900 MM/yr. This is better than 70 or 90 trays. If we wanted to proceed further to establish the optimum, we could continue reducing the search space using a regular search until we get to the optimum number of trays. Instead, an experienced designer would note that the difference in cost within the range examined (\$0.03 MM/yr) is relatively small compared with the error in the capital cost estimate ($\pm 30\%$, or \$0.29 MM/yr). Because the optimum appears to be fairly flat with respect to the number of trays over the range 70 to 90, it is reasonable to take the optimum as 80 trays. (As a confirmation, iterations 6 and 7, with 76 and 84 trays, indicate that the optimum indeed lies at 80 ± 2 trays).
6. Having fixed the number of trays at 80, we should now optimize the feed tray. We start by adding two new points, with the feed tray at trays 27 and 53. These give total annualized costs of \$7.82 MM/yr and \$8.43 MM/yr, respectively. The minimum cost is given by the lower bound on feed tray location. If we try a higher feed tray (say, tray 26) the UniSim tray sizing utility gives a warning "head loss under downcomers is too large." We could overcome this warning by modifying the tray design, but once again we can notice that the annualized cost savings that we have gained by optimizing feed tray (\$0.08 MM/yr) is small compared with the error in the capital cost, so the design with feed tray 27 is close enough to optimum.

The column design is thus set at 80 trays, with feed on tray 27, giving a column 50.8 m high and 4.42 m diameter.

The solution obtained is "good enough" but is not rigorously optimal. Several possible variations in flow scheme were not considered. For example, we could have examined use of feed pre-heat, intermediate-stage condensers or reboilers, or more efficient column internals such as high-efficiency trays or structured packing. The column cost may also be reduced if different diameters or different internals were used in the rectifying and stripping sections. In the broader context of the process, it may be possible to supply the heat required for the reboiler using heat recovered from elsewhere in the process, in which case the cost of energy will be reduced and the capital energy trade-off will be altered. In the overall process context, we could also question whether the column needs such high recoveries of toluene and ethylbenzene, because the high recoveries clearly lead to a high reflux rate and column energy cost.

12.13 References

- Abadie, J., & Guigou, J. (1969). *Gradient réduit généralisé*, Électricité de France Note HI 069/02.
- Biegler, L. T., Grossman, I. E., & Westerberg, A. W. (1997). *Systematic methods of chemical process design*. Prentice Hall.
- Box, G. E. P. (1957). Evolutionary operation: A method for increasing industrial productivity. *Applied Statistics*, 6, 81.
- Dantzig, G. B. (1963). *Linear programming and extensions*. Princeton University Press.
- Diwekar, U. (2008). *Introduction to applied optimization* (2nd ed.). Springer.
- Edgar, T. E., & Himmelblau, D. M. (2001). *Optimization of chemical processes* (2nd ed.). McGraw-Hill.
- Floudas, C. A. (1995). *Nonlinear and mixed-integer Optimization: Fundamentals and applications*. Oxford University Press.
- Hillier, F. S., & Lieberman, G. J. (2014). *Introduction to operations research* (10th ed.). McGraw-Hill.
- Kokossis, A. C., & Floudas, C. A. (1990). Optimization of complex reactor networks – 1. Isothermal operation. *Chemical Engineering Science*, 45(3), 595.
- Livio, M. (2002). *The golden ratio*. Random House.
- Montgomery, D. C. (2012). *Design and analysis of experiments* (8th ed.). Wiley.
- Murtagh, B. A., & Saunders, M. A. (1978). Large-scale linearly constrained optimization. *Mathematical Programming*, 14, 41.
- Murtagh, B. A., & Saunders, M. A. (1982). A projected Lagrangian algorithm and its implementation for sparse non-linear constraints. *Mathematical Programming Study*, 16, 84.
- Rudd, D. F., & Watson, C. C. (1968). *Strategy of process design*. Wiley.
- Spendley, W., Hext, G. R., & Himsforth, F. R. (1962). *Technometrics*, 4, 44.
- Stoecker, W. F. (1989). *Design of thermal systems* (3rd ed.). McGraw-Hill.
- Wolfe, P. (1962). Methods of non-linear programming. *Notices of the American Mathematical Society*, 9, 308.

12.14 Nomenclature

		Dimensions in \$MLT\theta\$
A	A constant	—
B	A constant	—
$f(x)$	General function of x	—
$f'(x)$	First derivative of function of x with respect to x	—
$f''(x)$	Second derivative of function of x with respect to x	—
FR	The set of points contained in a feasible region	—
$g(x)$	A m_i vector of inequality constraints	—
$g(x)$	General inequality constraint equation in x	—
$h(x)$	A m_e vector of equality constraints	—
$h(x)$	General equality constraint equation in x	—
h	Step length in a search algorithm	—
M	A large scalar constant	—
m	Number of constraints	—
m_e	Number of equality constraints	—
m_i	Number of inequality constraints	—
n	Number of variables	—
Q_c	Condenser duty in distillation	ML^2T^{-3}
Q_r	Reboiler duty in distillation	ML^2T^{-3}
$S_1, S_2 \dots$	Slack and surplus variables	—
T	Temperature	θ
T_{alloy}	Maximum allowed temperature for an alloy	θ
T_A, T_B, T_C	Maximum allowed temperature for alloys A, B, and C	θ
U	Overall heat transfer coefficient	$MT^{-3}\theta^{-1}$
X	A vector of n decision variables	—
$x_1, x_2 \dots$	Continuous variables	—
$y_1, y_2 \dots$	Integer (discrete) variables	—
z	The objective (in optimization)	—
α	A constant between 0.0 and 1.0	—
ϵ	Fraction of search range or tolerance for convergence	—
ΔT	Temperature difference	θ
$\Delta T_{optimum}$	The optimal minimum temperature approach in heat recovery	θ

		Dimensions in \$MLT θ
$\delta x_1, \delta x_2$	Small increments in x_1 and x_2	—
ω	Ratio used in golden-section search (= 0.381966)	—
Suffixes		
$D1$	lower side of a discontinuity	—
$D2$	upper side of a discontinuity	—
i	i^{th} variable	—
j	j^{th} variable	—
k	k^{th} iteration	—
L	lower bound	—
S	stationary point	—
U	upper bound	—

12.15 Problems

- 12.1.** A separator divides a process stream into three phases: a liquid organic stream, a liquid aqueous stream, and a gas stream. The feed stream contains three components, all of which are present to some extent in the separated streams. The composition and flow rate of the feed stream are known. All the streams will be at the same temperature and pressure. The phase equilibrium constants for the three components are available.
- How many design variables must be specified to calculate the output stream compositions and flow rates?
 - How would you optimize these variables if the objective of the separator were to maximize recovery of condensable components into the organic liquid stream? What constraints might limit the attainable recovery?
- 12.2.** A rectangular tank with a square base is constructed from 5-mm steel plates. If the capacity required is 8 cubic meters, determine the optimum dimensions if the tank has:
- a closed top
 - an open top
- 12.3.** Estimate the optimum thickness of insulation for the roof of a house given the following information. The insulation will be installed flat on the attic floor.

Overall heat transfer coefficient for the insulation as a function of thickness, U values (see [Chapter 19](#)):

thickness, mm	0	25	50	100	150	200	250
$U, \text{Wm}^{-2}\text{K}^{-1}$	20	0.9	0.7	0.3	0.25	0.20	0.15

The cost of insulation, including installation, is \$120/m³. Capital charges (see [Chapter 9](#)) are 20% per year. The cost of fuel, allowing for the efficiency of the heating system, is \$8/GJ. The cost of cooling is \$5/GJ. Average temperatures for any region of the United States or Canada can be found online at www.weather.com (under the averages tab). Assume the house is heated or cooled to maintain an internal temperature in the range 70 to 80 °F.

Note: The rate at which heat is being lost or gained is given by $U \times \Delta T, \text{W/m}^2$, where U is the overall coefficient and ΔT the temperature difference; see [Chapter 19](#).

- 12.4.** What is the optimum practical shape for an above-ground dwelling to minimize the heat losses through the building fabric? When is (or was) this optimum shape used? Why is this optimum shape seldom used in richer societies?
- 12.5.** Hydrogen is manufactured from methane by either steam reforming (reaction with steam) or partial oxidation (reaction with oxygen). Both processes are endothermic. What reactor temperature and pressure would you expect to be optimal for these processes? What constraints might apply?

- 12.6. Ethylene and propylene are valuable monomers. A key step in the recovery of these materials is fractionation of the olefin from the corresponding paraffin (ethane or propane). These fractionation steps require refrigeration of the overhead condenser and very large distillation columns with many stages. Raising the pressure at which the column operates improves the performance of the refrigeration system, but increases the number of stages needed. Formulate the objective function for optimizing the recovery of ethylene from an ethylene–ethane mixture. What are the key constraints? What will be the main trade-offs?
- 12.7. If you had to design a plant for pasteurizing milk, what constraints would you place on the design?
- 12.8. A catalytic process was designed to make 150,000 metric tons per year of product with a net profit of \$0.25/lb of product. The catalyst for the process costs \$10/lb, and it takes 2 months to shut down the process, empty the old catalyst, reload fresh catalyst, and restart the process. The feed and product recovery and purification sections can be pushed to make as much as 120% of design basis capacity. The reactor section is sized with sufficient catalyst to make 100% of design basis when operated with fresh catalyst at 500 °F. The reactor can only be operated at temperatures up to 620 °F for safety reasons. The reactor weight hourly space velocity (lb of product per hour per lb of catalyst) is given by the equation:

$$\text{WHSV} = 4.0 \times 10^6 \exp\left(\frac{-8000}{T}\right) \exp(-8.0 \times 10^{-5} \times t \times T)$$

where

t = time on stream in months

T = temperature

Find the optimal temperature versus time profile for the reactor and determine how long the process should be operated before the catalyst is changed out. (Hint: The initial temperature does not have to be 500 °F).

- 12.9. The portfolio of investment projects shown here has been proposed for a company for next year:

Project	Net present value (MM\$)	Cost (MM\$)
A	100	61
B	60	28
C	70	33
D	65	30
E	50	25
F	50	17
G	45	25
H	40	12
I	40	16
J	30	10

- Develop a spreadsheet optimization program to select the optimal portfolio of projects to maximize total portfolio net present value (NPV), given a total budget of \$100 million. (This is a simple MILP.)
- How would the portfolio and NPV change if the budget were increased to \$110 million?
- Because of corporate cost-cutting, the budget is reduced to \$80 million. Which projects are now funded, and what is the new NPV?
- Based on your answers to parts a to c, can you draw any conclusions on which projects are likely to be funded regardless of the financial situation?
- Can you see any problems with this project selection strategy? If so, how would you recommend they should be addressed?

Equipment selection, specification, and design

KEY LEARNING OBJECTIVES

- Where to find information on process equipment
- How to obtain equipment information from vendors

13.1 Introduction

Part I of this book covered process design: the synthesis of the complete process as an assembly of units, each carrying out a specific process operation. In Part II, the selection, specification, and design of the equipment required to carry out these process operations is considered in more detail.

In practice, plant design and process design cannot be separated. The selection and specification of one piece of equipment will often require the use of additional equipment and thus have implications on the process flow diagram. For example, if a continuous dryer is selected for drying a solid product, it may be necessary to add a heater to pre-heat the drying gas, a cyclone or filter to recover solid fines from the off-gas, a cooler and flash drum to cool the off-gas and recover solvent, a vent scrubber to prevent solvent emissions, etc. The design team must understand all the flowsheet implications of equipment selection and design to arrive at an accurate cost estimate and process optimization.

This chapter gives a short introduction to the selection and design of process equipment and provides a guide to the following chapters. Most process operations are carried out in closed pressure vessels, which are addressed in [Chapter 14](#). [Chapter 15](#) discusses the design of chemical and biochemical reactors. Separation processes are covered in [Chapters 16](#) and [17](#). [Chapter 18](#) addresses operations that involve solids handling. [Chapter 19](#) describes the design of equipment for heat transfer and [Chapter 20](#) covers the transport and storage of fluids.

Each chapter and section of Part II is intended to be a stand-alone guide to the design of a particular operation, but in some cases cross-references to sections of other chapters are given to avoid duplication. Throughout Part II the emphasis is on selection and sizing of equipment, and it is assumed that the reader is familiar with the fundamentals of kinetics, thermodynamics, and transport processes. Further details on the scientific principles and theory that underlie the design and operation of process equipment can be found in the numerous textbooks cited in each section and in general books on unit operations, such as [Chhabra and Gurappa \(2019\)](#) and [McCabe et al. \(2004\)](#).

Capital cost correlations for each type of equipment are not given in Part II, as the determination of capital costs for all types of equipment was discussed in [Chapter 7](#). Similarly, materials selection was covered in [Chapter 6](#). Although the role of safety in design was described in [Chapter 10](#), some additional safety issues that are specific to certain unit operations are covered in the relevant sections of Part II.

13.2 Sources of equipment design information

13.2.1 Proprietary and nonproprietary equipment

The equipment used in the chemical process industries can be divided into two classes: proprietary and nonproprietary. Proprietary equipment, such as pumps, compressors, filters, centrifuges, and dryers, is designed and sold as standard catalogue items by specialist manufacturers. Nonproprietary equipment is designed as special, one-off, items for particular processes; for example, reactors, distillation columns and heat exchangers, and is custom-built by specialist fabricators.

Unless employed by one of the specialist equipment manufacturers, the chemical engineer is not normally involved in the detailed design of proprietary equipment. The chemical engineer's job will be to specify the process duty (flow rate, heat load, temperature, pressure, etc.) and then select an appropriate piece of equipment to meet that duty, consulting with the vendors to ensure that the equipment supplied is suitable. Proprietary equipment is often only made in certain standard sizes, and the design engineer must determine which size is best suited for the application or whether to use multiple units in parallel to accommodate the desired flow. Chemical engineers may be involved with the vendor's designers in modifying standard equipment for particular applications, for example, a standard tunnel dryer designed to handle particulate solids may be adapted to dry synthetic fibers. As was pointed out in [Chapter 1](#), the use of standard off-the-shelf equipment, whenever possible, will reduce costs.

Reactors, columns, flash drums, decanters, and other vessels are usually designed as special items for a given project. In particular, reactor designs are usually unique, except where more or less standard equipment is used, such as an agitated, jacketed vessel. Distillation columns, vessels, and tubular heat exchangers, though nonproprietary items, will be designed to conform to recognized standards and codes; this reduces the amount of design work involved.

The chemical engineer's part in the design of "nonproprietary" equipment is usually limited to selecting and "sizing" the equipment. For example, in the design of a distillation column, the design engineer will typically determine the number of plates; the type and design of plate; diameter of the column; and the position of the inlet, outlet, and instrument nozzles. This information would then be transmitted, in the form of sketches and specification sheets, to the specialist mechanical design group, or the fabricator's design team, for detailed design.

It must be emphasized that companies that are engaged in the manufacture of chemicals, fuels, polymers, foods, and pharmaceuticals almost never build their own process equipment. The design engineers from the operating company usually provide specifications to detailed design groups at an engineering, procurement, and construction (EPC) company, who then subcontract the equipment manufacture to specialist equipment fabricators. Even one-of-a-kind items such as reactors, distillation columns, and heat exchangers are built by specialist manufacturers. The accurate transmission of design details is therefore very important, and the process industries have developed many standard specifications to facilitate information exchange with vendors. Standard specifications should be used whenever possible, as these lead to cheaper designs and reduce the risk of needing rework during construction.

13.2.2 Published information on process equipment

Technical literature

Descriptions and illustrations of most types of process equipment can be found in various handbooks: [Green and Southard \(2018\)](#), [Schweitzer \(1997\)](#), and [Crouper et al. \(2012\)](#). *Perry's Chemical Engineers' Handbook* remains the most comprehensive compilation of chemical engineering information. The online version provided by Knovel is the most accessible format. Many specialized books have been written on individual unit operations; these are cited throughout the following chapters.

Equipment manufacturers often write articles in the trade journals. Although these are primarily promotional, they can be quite informative. The trade journals also contain advertisements that can help identify manufacturers. Articles by equipment vendors are common in *Chemical Engineering* and *The Chemical Engineer* and appear somewhat less frequently in *Chemical Engineering Progress* and *Hydrocarbon Processing*. The journals usually contain a reader response card that can be faxed or mailed in to receive advertisers' brochures and sales literature. These can be used to build up a library of vendors' catalogues.

Every year the journal *Chemical Engineering* publishes a buyers' guide. The *Chemical Engineering Buyers' Guide* lists over 500 manufacturers and provides indexes by product type, company name, and trade name, as well as listing websites and contact information for industry associations. It can be used as a "yellow pages" of chemical industry suppliers, but like other directories, it is not fully comprehensive, as not all manufacturers will pay to be listed.

In the United Kingdom, a commercial organization, Technical Indexes Ltd., publishes the Process Engineering Index, which contains information from over 3000 manufacturers and suppliers of process equipment globally.

Online information

All equipment vendors now maintain an online presence, but there is a wide variation in the quality of the websites and the amount of information provided.

Several directory sites have been set up to serve the chemical and process industries. Of these, the best at time of writing is www.chemindustry.com, which has links to many vendors. More limited information is also available at www.chemengg.com and www.cheresources.com. A good site for finding new and used equipment for sale is www.equipnet.com.

Manufacturer's websites are usually easily located using online search engines and often provide details of equipment construction, standard sizes, available metallurgies, specification sheets, and performance information. The *Chemical Engineering Buyers' Guide* can also be used to identify vendor websites for specific equipment types. Manufacturers' association websites usually provide the most comprehensive listings of vendors; see, for example, the websites of the Valve Manufacturers' Association (www.vma.org), Tubular Exchanger Manufacturers' Association (<http://tema.org>), and Conveyor Equipment Manufacturers' Association (<http://cemanet.org>). Other manufacturers' associations are usually easy to find by searching the Internet.

Some equipment types are relatively easy to find using search engines ("crystallizer," "rotary agglomerator," "bioreactor," etc.), but locating the vendors of industrial plant can be more difficult when the equipment name is in common usage ("furnace," "dryer," "filter," "pump," etc.). In such cases, the best approach is to begin at one of the chemical engineering directory sites listed earlier.

13.3 Guide to equipment selection and design

Table 13.1 is a guide to the design of the most common types of process equipment. The numbers refer to the section of this book that provides design guidelines. Table 13.2 is a similar guide for separation processes, which have been grouped based on the phases that are separated. Capital cost correlations for most of the equipment listed in these tables can be found in Table 7.2.

TABLE 13.1 Guide to equipment design

Equipment type	Basic sizing	Detailed design
Reactors		
Basic reactors	15.2, 15.5	As pressure vessels,
Bioreactors	15.9	Chapter 14
Catalytic reactors	15.8	
Multiphase reactors	15.7	
Nonisothermal reactors	15.6	
Separation columns		
Absorption	16.2.4, 17.14	Shells as pressure vessels:
Distillation	17.2–17.13	Chapter 14
Extraction	17.16	Internals:
Single-stage flash	16.3, 17.3.3	Trays 17.12–17.13
Stripping	16.2.4, 17.14	Packing 17.14
Other separation processes		See Table 13.2
Heat exchange equipment		
Air coolers		19.16
Boilers, reboilers, vaporizers		19.11
Condensers		19.10
Fired heaters		19.17
Plate heat exchangers		19.12
Shell and tube exchangers		19.1–19.9
Transport equipment		
Compression of gases		20.6
Conveying of solids		18.3
Pumping of liquids		20.7
Solids handling equipment		
Size reduction (grinding)		18.9
Size enlargement (forming)		18.8
Heating and cooling solids		18.10

Numbers refer to the sections in this book.

TABLE 13.2 Separation processes						
		Minor Component				
		Solid		Liquid		Gas/Vapor
Major component	Solid	Sorting	18.4	Pressing	18.6.5	
		Screening	18.4.1	Drying	18.7	
		Hydrocyclones	18.4.2			
		Classifiers	18.4.3			
		Jigs	18.4.4			
		Tables	18.4.5			
		Centrifuges	18.4.6			
		Dense media	18.4.7			
		Flotation	18.4.8			
		Magnetic	18.4.9			
		Electrostatic	18.4.10			
	Liquid	Thickeners	18.6.1	Decanters	16.4.1	Stripping 16.2.4
		Clarifiers	18.6.1	Coalescers	16.4.3	17.14
		Hydrocyclones	18.6.4	Solvent extraction	16.5.6	
		Filtration	18.6.2	Leaching	16.5.6	
		Centrifuges	18.6.3	Chromatography	16.5.7	
		Crystallizers	16.5.2	Distillation	Chapter 17	
		Evaporators	16.5.1			
		Precipitation	16.5.3			
		Membranes	16.5.4			
		Reverse osmosis	16.5.4			
		Ion exchange	16.5.5			
		Adsorption	16.5.7			
	Gas/Vapor	Gravity Settlers	18.5.1	Separating vessels	16.3	Adsorption 16.2.1
		Impingement	18.5.2	Demisting pads	16.3	Absorption 16.2.4
		Settlers		Cyclones	18.5.3	17.14
		Cyclones	18.5.3	Wet scrubbers	18.5.5	Membranes 16.2.2
		Filters	18.5.4	Electrostatic	18.5.6	Cryogenic 16.2.3
		Wet scrubbers	18.5.5	precipitators		distillation Chapter 17
		Electrostatic precipitators	18.5.6			Condensation 16.2.5

Numbers refer to the sections in this book. The terms major and minor component only apply where different phases are to be separated (i.e., not to those on the diagonal). Note that separation processes include processes for separating phases, as well as for recovering one or more components from a mixture.

13.4 References

- Chhabra, R. P., & Gurappa, B. (Eds.). (2019) (6th ed.) *Coulson and Richardson's chemical engineering: Volume 2B. Separation processes*. Butterworth-Heinemann.
- Couper, J. R., Penney, W. R., & Fair, J. R. (2012). *Chemical process equipment* (3rd ed.). Butterworth-Heinemann.
- Green, D. W., & Southard, M. Z. (Eds.). (2018). *Perry's chemical engineers' handbook* (9th ed.). McGraw-Hill.
- McCabe, W. L., Smith, J. C., & Harriott, P. (2004). *Unit operations of chemical engineering* (7th ed.). McGraw-Hill.
- Schweitzer, P. A. (Ed.). (1997). *Handbook of separation techniques for chemical engineers* (3rd ed.). McGraw Hill.

Design of pressure vessels

KEY LEARNING OBJECTIVES

- What factors a process engineer must consider when setting specifications for a pressure vessel
- How pressure vessels are designed and built and what determines the vessel wall thickness
- How to design and size vessels for use as reactors, columns, separators, and other process uses
- How codes and standards are used in pressure vessel design

14.1 Introduction

This chapter covers those aspects of the mechanical design of chemical plants that are of particular interest to chemical engineers. The main topic considered is the design of pressure vessels. The design of storage tanks is also discussed briefly. Most reactors, separation columns, flash drums, heat exchangers, surge tanks, and other vessels in a chemical plant will need to be designed as pressure vessels, so this topic is relevant to a broad range of process equipment.

The chemical engineer will not usually be called on to undertake the detailed mechanical design of a pressure vessel. Vessel design is a specialized subject and will be carried out by mechanical engineers who are conversant with the current design codes and methods of stress analysis. However, the chemical engineer will be responsible for developing and specifying the basic design information for a particular vessel and needs to have a general appreciation of pressure vessel design to work effectively with the specialist designer.

Another reason why the process engineer must have an appreciation of methods of fabrication, design codes, and other constraints on pressure vessel design is because these constraints often dictate limits on the process conditions. Mechanical constraints can cause significant cost thresholds in design, for example, when a costlier grade of alloy is required above a certain temperature.

The basic data needed by the specialist designer will be:

1. Vessel function
2. Process materials and services
3. Operating and design temperature and pressure
4. Materials of construction
5. Vessel dimensions and orientation
6. Type of vessel heads to be used

7. Openings and connections required
8. Specification of heating and cooling jackets or coils
9. Type of agitator
10. Specification of internal fittings

An elementary understanding of pressure vessel design is needed in the preliminary stages of design, as most correlations for pressure vessel costs are based on the weight of metal required and hence require an estimate of the vessel wall thickness, as well as its volume. In many cases, the required wall thickness will be determined by the combination of loads acting on the vessel rather than internal pressure alone.

A data sheet for pressure vessel design is given in Appendix G, available online at www.elsevier.com/books-and-journals/book-companion/9780128211793. Pressure vessel information is also included in the data sheets for fixed-bed reactors, vapor–liquid contactors, and heat exchangers.

There is no strict definition of what constitutes a pressure vessel, and different codes and regulations apply in different countries; however, it is generally accepted that any closed vessel over 150 mm diameter subject to a pressure difference of more than 0.5 bar should be designed as a pressure vessel.

It is not possible to give a completely comprehensive account of vessel design in one chapter. The design methods and data given should be sufficient for the preliminary design of conventional vessels, for the chemical engineer to check the feasibility of a proposed equipment design, to estimate the vessel cost for an economic analysis, and to determine the vessel's general proportions and weight for plant layout purposes. For a more detailed account of pressure vessel design, the reader should refer to the books by Singh and Soler (1992), Escoe (1994) and Moss and Basic (2013). Other useful books on the mechanical design of process equipment are listed in the bibliography at the end of this chapter.

An elementary understanding of the principles of the “strength of materials” (mechanics of solids) will be needed to follow this chapter. Readers who are not familiar with the subject should consult one of the many textbooks available, such as those by Case et al. (1999), Mott and Untener (2016), Seed (2001), Hibbeler (2016), Beer et al. (2014), and Goodno and Gere (2017).

14.1.1 Classification of pressure vessels

For the purposes of design and analysis, pressure vessels are subdivided into two classes, depending on the ratio of the wall thickness to vessel diameter: thin-walled vessels, with a thickness ratio of less than 1:10, and thick-walled above this ratio.

The principal stresses (see Section 14.3.1) acting at a point in the wall of a vessel, due to a pressure load, are shown in Fig. 14.1. If the wall is thin, the radial stress σ_3 will be small and can be neglected in comparison with the other stresses, and the longitudinal and circumferential stresses σ_1 and σ_2 can be taken as constant over the wall thickness. In a thick wall, the magnitude of the radial stress will be significant, and the circumferential stress will vary across the wall. The majority of the vessels used in the chemical and allied industries are classified as thin-walled vessels. Thick-walled vessels are used for high pressures and are discussed in Section 14.14.

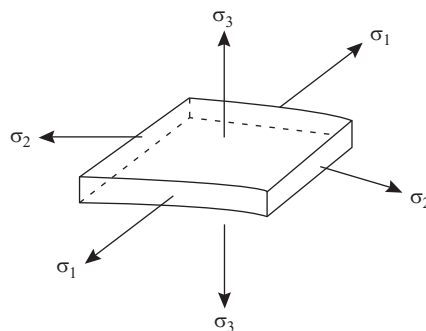


FIG. 14.1 Principal stresses in pressure-vessel wall.

14.2 Pressure vessel codes and standards

In all the major industrialized countries, the design and fabrication of pressure vessels are covered by national standards and codes of practice. In most countries it is a legal requirement that pressure vessels must be designed, constructed, and tested in accordance with part or all of the design code. The primary purpose of the design codes is to establish rules of safety relating to the pressure integrity of vessels and provide guidance on design, materials of construction, fabrication, inspection, and testing. They form a basis of agreement between the manufacturer and customer and the customer's insurance company.

The standard used in North America (and most commonly referenced internationally) is the ASME Boiler and Pressure Vessel Code (the ASME BPV Code). The 12 sections of the ASME BPV Code are listed in [Table 14.1](#). Most chemical plant and refinery vessels fall within the scope of Section VIII of the ASME BPV Code. Section VIII contains three subdivisions:

Division 1: Contains general rules and is most commonly followed, particularly for low-pressure vessels.

Division 2: Contains alternative rules that are more restrictive on materials, design temperatures, design details, fabrication methods, and inspection but allow higher design stresses and hence thinner vessel walls. Division 2 rules are usually chosen for large, high-pressure vessels where the savings in metal cost and fabrication complexity offset the higher engineering and construction costs.

Division 3: Contains alternative rules intended for vessels with design pressures greater than 10,000 psig. It does not establish a maximum pressure for vessels designed in accordance with Division 1 or Division 2, but provides alternative rules that can be followed for thicker-walled vessels.

In the following sections reference will normally be made to the BPV Code Sec. VIII D.1. The scope of the BPV Code Sec. VIII D.1 covers vessels made from iron, steels, and nonferrous metals. It specifically excludes:

1. Vessels within the scope of other sections of the BPV code. For example, power boilers (Sec. I), fiber-reinforced plastic vessels (Sec. X), and transport tanks (Sec. XIII).
2. Fired process tubular heaters.
3. Pressure containers that are integral parts of rotating or reciprocating devices, such as pumps, compressors, turbines, or engines.
4. Piping systems (which are covered by ASME B31.3—see [Chapter 20](#)).
5. Piping components and accessories such as valves, strainers, in-line mixers, and spargers.
6. Vessels containing water at less than 300 psi (2 MPa) and less than 210 °F (99 °C).
7. Hot water storage tanks heated by steam with heat rate less than 0.2 MMBTU/h (58.6 kW), water temperature less than 210 °F (99 °C), and volume less than 120 gal (450 liters).
8. Vessels having internal pressure less than 15 psi (100 kPa) or greater than 3000 psi (20 MPa).
9. Vessels of internal diameter or height less than 6 in (152 mm).
10. Pressure vessels for human occupancy.

TABLE 14.1 The 2004 ASME Boiler and Pressure Vessel Code

SECTIONS	
I	Rules for construction of power boilers
II	Materials
	Part A Ferrous metal specifications
	Part B Nonferrous metal specifications
	Part C Specifications for welding rods, electrodes, and filler metals
	Part D Properties (customary or metric versions)
III	Nuclear power plant components
	NCA General requirements
	Division 1
	Division 2 Code for concrete containments
	Division 3 Containments for transport and storage of spent nuclear fuel and high level radioactive material and waste
IV	Rules for construction of heating boilers
V	Nondestructive examination
VI	Recommended rules for the care and operation of heating boilers

Continued

TABLE 14.1 The 2004 ASME Boiler and Pressure Vessel Code—cont'd

VII	Recommended guidelines for the care of power boilers
VIII	Rules for the construction of pressure vessels
	Division 1
	Division 2 Alternative rules
	Division 3 Alternative rules for the construction of high pressure vessels
IX	Welding and brazing qualifications
X	Fiber-reinforced plastic vessels
XI	Rules for in service inspection of nuclear power plant components
XII	Rules for construction and continued service of transport tanks

The ASME BPV Code can be ordered from ASME and is also available online (for example at ihsmarkit.com). The most recent edition of the code should always be consulted during detailed design.

In addition to the BPV Code Sec. VIII, the process design engineer will frequently need to consult Section II Part D, which lists maximum allowable stress values under Sec. VIII D.1 and D.2, as well as other materials properties. Comprehensive reviews of the ASME code are given by [Ellenberger et al. \(2004\)](#), [Farr and Jawad \(2010\)](#), and [Yokell \(1986\)](#); see also [Green and Southard \(2018\)](#).

In the European Union the design, manufacture, and use of pressure systems is covered by the Pressure Equipment Directive (Council Directive 97/23/EC) whose use became mandatory in May 2002. European standard EN 13445 provides similar rules and guidelines to the ASME BPV Code. The design of fiber-reinforced plastic vessels is covered by European standard EN 13923. The European standards can be obtained from any of the European Union member country national standards agencies; for example, BS EN 13445 can be ordered from www.bsigroup.com. Where national codes are not available, the ASME or European codes would normally be used.

Information and guidance on the pressure vessel codes can be found on the Internet at ihsmarkit.com or www.bsigroup.com.

The national codes and standards dictate the minimum requirements and give general guidance for design and construction; any extension beyond the minimum code requirement will be determined by agreement between the manufacturer and customer.

The codes and standards are drawn up by committees of engineers experienced in vessel design and manufacturing techniques and are a blend of theory, experiment, and experience. They are periodically reviewed, and revisions are issued to keep abreast of developments in design, stress analysis, fabrication, and testing. The latest version of the appropriate national code or standard should always be consulted before undertaking the design of any pressure vessel.

Several commercial computer programs to aid in the design of vessels to the ASME code and other international codes are available. These programs will normally be used by the specialist mechanical engineers who carry out the detailed vessel design. Some examples include:

Pressure Vessel Suite (Computer Engineering, Inc.)
 PVElite and CodeCalc (COADE, Inc.)
 TEMA/ASME and COMPRESS (Codeware, Inc.)

14.3 Fundamentals of strength of materials

This section has been included to provide a basic understanding of the fundamental principles that underlie the design equations given in the sections that follow. The derivation of the equations is given in outline only. A detailed knowledge of the material in this section is not required for preliminary vessel design, but the equations derived here will be referenced and applied in subsequent sections. A full discussion of the topics covered can be found in any text on the “strength of materials” (mechanics of solids).

14.3.1 Principal stresses

The state of stress at a point in a structural member under a complex system of loading is described by the magnitude and direction of the principal stresses. The principal stresses are the maximum values of the normal stresses at

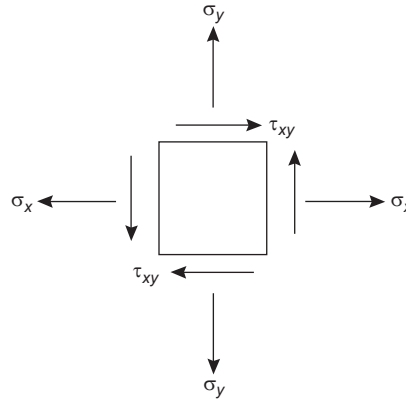


FIG. 14.2 Two-dimensional stress system.

the point, which act on planes on which the shear stress is zero. In a two-dimensional stress system (Fig. 14.2), the principal stresses at any point are related to the normal stresses in the x and y directions σ_x and σ_y and the shear stress τ_{xy} at the point by the following equation:

$$\text{Principal stresses, } \sigma_1, \sigma_2 = \frac{1}{2} (\sigma_y + \sigma_x) \pm \frac{1}{2} \sqrt{(\sigma_y - \sigma_x)^2 + 4\tau_{xy}^2} \quad (14.1)$$

The maximum shear stress at the point is equal to half the algebraic difference between the principal stresses:

$$\text{Maximum shear stress} = \frac{1}{2} (\sigma_1 - \sigma_2) \quad (14.2)$$

Compressive stresses are conventionally taken as negative; tensile as positive.

14.3.2 Theories of failure

The failure of a simple structural element under unidirectional stress (tensile or compressive) is easy to relate to the tensile strength of the material, as determined in a standard tensile test, but for components subjected to combined stresses (normal and shear stress), the position is not so simple, and several theories of failure have been proposed. The three theories most commonly used are described next.

Maximum principal stress theory postulates that a member will fail when one of the principal stresses reaches the failure value in simple tension, σ_e . The failure point in a simple tension is taken as the yield-point stress, or the tensile strength of the material, divided by a suitable factor of safety.

Maximum shear stress theory postulates that failure will occur in a complex stress system when the maximum shear stress reaches the value of the shear stress at failure in simple tension.

For a system of combined stresses, there are three shear stress maxima:

$$\tau_1 = \frac{\sigma_1 - \sigma_2}{2} \quad (14.3a)$$

$$\tau_2 = \frac{\sigma_2 - \sigma_3}{2} \quad (14.3b)$$

$$\tau_3 = \frac{\sigma_3 - \sigma_1}{2} \quad (14.3c)$$

In the tensile test,

$$\tau_e = \frac{\sigma_e}{2} \quad (14.4)$$

The maximum shear stress will depend on the sign of the principal stresses as well as their magnitude, and in a two-dimensional stress system, such as that in the wall of a thin-walled pressure vessel, the maximum value of the shear stress may be that given by putting $\sigma_3 = 0$ in Equations 14.3b and c.

The maximum shear stress theory is often called Tresca's, or Guest's, theory.

Maximum strain energy theory postulates that failure will occur in a complex stress system when the total strain energy per unit volume reaches the value at which failure occurs in simple tension.

The maximum shear-stress theory has been found to be suitable for predicting the failure of ductile materials under complex loading and is the criterion normally used in pressure vessel design.

14.3.3 Elastic stability

Under certain loading conditions, failure of a structure can occur not through gross yielding or plastic failure, but by buckling or wrinkling. Buckling leads to a gross and sudden change of shape of the structure, unlike failure by plastic yielding, where the structure retains the same basic shape. This mode of failure will occur when the structure is not elastically stable: when it lacks sufficient stiffness, or rigidity, to withstand the load. The stiffness of a structural member is dependent not on the basic strength of the material but on its elastic properties (E_Y and ν) and the cross-sectional shape of the member.

The classic example of failure due to elastic instability is the buckling of tall thin columns (struts), which is described in any elementary text on the "strength of materials." For a structure that is likely to fail by buckling, there will be a certain critical value of load below which the structure is stable; if this value is exceeded, catastrophic failure through buckling can occur.

The walls of pressure vessels are usually relatively thin compared with the other dimensions and can fail by buckling under compressive loads. This is particularly important for tall, wide vessels such as distillation columns that can experience compressive loads from wind loads.

Elastic buckling is the decisive criterion in the design of thin-walled vessels under external pressure.

14.3.4 Secondary stresses

In the stress analysis of pressure vessels and pressure vessel components, stresses are classified as primary or secondary. Primary stresses can be defined as those stresses that are necessary to satisfy the conditions of static equilibrium. The membrane stresses induced by the applied pressure and the bending stresses due to wind loads are examples of primary stresses. Primary stresses are not self-limiting; if they exceed the yield point of the material, gross distortion, and in the extreme situation, failure of the vessel will occur.

Secondary stresses are those stresses that arise from the constraint of adjacent parts of the vessel. Secondary stresses are self-limiting; local yielding or slight distortion will satisfy the conditions causing the stress, and failure would not be expected to occur in one application of the loading. The "thermal stress" set up by the differential expansion of parts of the vessel, due to different temperatures or the use of different materials, is an example of a secondary stress. The discontinuity that occurs between the head and the cylindrical section of a vessel is a major source of secondary stress. If free, the dilation of the head would be different from that of the cylindrical section; they are constrained to the same dilation by the welded joint between the two parts. The induced bending moment and shear force due to the constraint give rise to secondary bending and shear stresses at the junction. The magnitude of these discontinuity stresses can be estimated by analogy with the behavior of beams on elastic foundations; see [Hetenyi \(1958\)](#) and [Harvey \(1974\)](#). The estimation of the stresses arising from discontinuities is covered in the books by [Bednar \(1990\)](#) and [Jawad and Farr \(2018\)](#). Other sources of secondary stresses are the constraints arising at flanges, supports, and the change of section due to reinforcement at a nozzle or opening (see Section 14.6).

Though secondary stresses do not affect the "bursting strength" of the vessel, they are an important consideration when the vessel is subject to repeated pressure loading. If local yielding has occurred, residual stress will remain when the pressure load is removed, and repeated pressure cycling can lead to fatigue failure.

14.4 General design considerations for pressure vessels

This section describes general pressure vessel design specifications, most of which would normally be specified by a process engineer.

14.4.1 Design pressure

A vessel must be designed to withstand the maximum pressure to which it is likely to be subjected in operation.

For vessels under internal pressure, the design pressure (sometimes called maximum allowable working pressure [MAWP]) is taken as the pressure at which the relief device is set. This will normally be 5% to 10% above the normal working pressure to avoid spurious operation during minor process upsets. For example, API STD 520 recommended practice sets a 10% margin between the normal operating pressure and the design pressure. When deciding the design pressure, the hydrostatic pressure in the base of the column should be added to the operating pressure, if significant.

Vessels subject to external pressure should be designed to resist the maximum differential pressure that is likely to occur in service. Vessels likely to be subjected to vacuum should be designed for a full negative pressure of 1 bar, unless fitted with an effective, and reliable, vacuum breaker.

14.4.2 Design temperature

The strength of metals decreases with increasing temperature (see [Chapter 6](#)), so the maximum allowable stress will depend on the material temperature. The maximum design temperature at which the maximum allowable stress is evaluated should be taken as the maximum working temperature of the material, with due allowance for any uncertainty involved in predicting vessel wall temperatures. Additional rules apply for welded vessels, as described in ASME BPV Code Sec. VIII D.1 part UW. The minimum design metal temperature (MDMT) should be taken as the lowest temperature expected in service. The designer should consider the lowest operating temperature, ambient temperature, auto-refrigeration, process upsets, and other sources of cooling in determining the minimum.

14.4.3 Materials

Pressure vessels are constructed from plain carbon steels, low- and high-alloy steels, other alloys, clad plate, and reinforced plastics.

Selection of a suitable material must take into account the suitability of the material for fabrication (particularly welding), as well as the compatibility of the material with the process environment; see [Chapter 6](#).

The pressure vessel design codes and standards include lists of acceptable materials in accordance with the appropriate material standards. The ASME BPV Code Sec. II Part D gives maximum allowable stresses as a function of temperature and maximum temperatures permitted under Sections I, III, VIII, and XII of the BPV code for ferrous and nonferrous metals. The design of pressure vessels using reinforced plastics is described in ASME BPV Code Sec. X.

14.4.4 Maximum allowable stress (nominal design strength)

For design purposes, it is necessary to decide a value for the maximum allowable stress (nominal design strength) that can be accepted in the material of construction.

This is determined by applying a suitable safety factor to the maximum stress that the material could be expected to withstand without failure under standard test conditions. The safety factor allows for any uncertainty in the design methods, the loading, the quality of the materials, and the workmanship.

The basis for establishing the maximum allowable stress values in the ASME BPV Code is given in ASME BPV Code Sec. II Part D, Mandatory Appendix 1. At temperatures where creep and stress rupture strength do not govern the selection of stresses, the maximum allowable stress is the lowest of:

1. The specified minimum tensile strength at room temperature divided by 3.5
2. The tensile strength at temperature divided by 3.5
3. The specified minimum yield strength at room temperature divided by 1.5
4. The yield strength at temperature divided by 1.5

At temperatures where creep and stress rupture strength govern, the maximum allowable stress is the lowest of:

1. The average stress to produce a creep rate of 0.01%/1000 hr
2. F times the average stress to cause rupture at the end of 100,000 hr, where $F = 0.67$ for temperatures below 1500 °F (815 °C)—see the code for higher temperatures
3. 0.8 times the minimum stress to cause rupture after 100,000 hr

In some cases where short-time tensile properties govern and slightly greater deformation is acceptable, higher stress values are allowed under ASME BPV Code Sec. VIII D.1. These exceed 67% but do not exceed 90% of the yield strength at temperature. These cases are indicated with a note (G5) in the BPV Code tables. Use of these higher values can result in deformation and changes in the vessel dimensions. They are not recommended for flanges or other applications where changes in dimensions could lead to leaks or vessel malfunction.

The maximum allowable stress values for ASME BPV Code Sec. VIII D.1 are given in ASME BPV Code Sec II Part D Table 1A for ferrous metals and Table 1B for nonferrous metals. Maximum allowable stress values for Sec. VIII D.2 are given in Sec. II Part D Table 2A for ferrous metals and Table 2B for nonferrous metals. Different values are given for plate, tubes, castings, forgings, bar, pipe, and small sections, as well as for different grades of each metal.

Typical maximum allowable stress values for some common materials are shown in Table 14.2. These may be used for preliminary designs. The ASME BPV Code should be consulted for the values to be used for detailed vessel design.

TABLE 14.2 Typical maximum allowable stresses for plates under ASME BPV Code Sec. VIII D.1 (The appropriate material standards should be consulted for particular grades and plate thicknesses.)

Material	Grade	Min. tensile strength (ksi)	Min. yield strength (ksi)	Maximum temperature (°F)	Maximum allowable stress at temperature °F (ksi = 1000 psi)				
					100	300	500	700	900
Carbon steel	A285 Gr A	45	24	900	12.9	12.9	12.9	11.5	5.9
Killed carbon steel	A515 Gr 60	60	32	1000	17.1	17.1	17.1	14.3	5.9
Low-alloy steel 1¼ Cr, ½ Mo, Si	A387 Gr 22	60	30	1200	17.1	16.6	16.6	16.6	13.6
Stainless steel 13 Cr	410	65	30	1200	18.6	17.8	17.2	16.2	12.3
Stainless steel 18 Cr, 8 Ni	304	75	30	1500	20.0	15.0	12.9	11.7	10.8
Stainless steel 18 Cr, 10 Ni, Cb	347	75	30	1500	20.0	17.1	15.0	13.8	13.4
Stainless steel 18 Cr, 10 Ni, Ti	321	75	30	1500	20.0	16.5	14.3	13.0	12.3
Stainless steel 16 Cr, 12 Ni, 2 Mo	316	75	30	1500	20.0	15.6	13.3	12.1	11.5

Note:

1. The stress values for type 304 stainless steel are not the same as those given for stainless steel 304L in Table 7.8.

2. 1 ksi = 1000 psi = 6.8948 N/mm²

14.4.5 Welded joint efficiency and construction categories

The strength of a welded joint will depend on the type of joint and the quality of the welding. The ASME BPV Code Sec. VIII D.1 defines four categories of weld (Part UW-3):

1. Longitudinal or spiral welds in the main shell, necks, or nozzles or circumferential welds connecting hemispherical heads to the main shell, necks, or nozzles.
2. Circumferential welds in the main shell, necks, or nozzles or connecting a formed head other than hemispherical.
3. Welds connecting flanges, tubesheets, or flat heads to the main shell, a formed head, neck, or nozzle.
4. Welds connecting communicating chambers or nozzles to the main shell, to heads, or to necks.

Details of the different types of welds used in pressure vessel construction are given in Section 14.11.

The soundness of welds is checked by visual inspection and by nondestructive testing (radiography).

The possible lower strength of a welded joint compared with the virgin plate is usually allowed for in design by multiplying the allowable design stress for the material by a joint efficiency E . The value of the joint efficiency used in design will depend on the type of joint and amount of radiography required by the design code. Typical values are shown in Table 14.3. A joint efficiency of 1.0 is only permitted for butt joints formed by double welding and subjected to full radiographic examination. Taking the factor as 1.0 implies that the joint is equally as strong as the virgin plate; this is achieved by radiographing the complete weld length and cutting out and remaking any defects. The use of lower joint efficiencies in design, though saving costs on radiography, will result in a thicker, heavier vessel, and the designer must balance any cost savings on inspection and fabrication against the increased cost of materials.

The ASME BPV Code Sec. VIII D.1 Part UW describes the requirements for pressure vessels fabricated by welding. Limiting plate thicknesses are specified for each type of weld with the exception of double-welded butt joints. Requirements for radiographic examination of welds are also specified. Section UW-13 of the code specifies the types of welds that can be used to attach heads and tubesheets to shells. Section UW-16 gives rules for attachment of nozzles to vessels.

The BPV Code should be consulted to determine the allowed joint types for a particular vessel. Any pressure vessel containing lethal substances will require full radiographic testing of all butt welds.

TABLE 14.3 Maximum allowable joint efficiency

Joint description	Joint category	Degree of radiographic examination		
		Full	Spot	None
Double-welded butt joint or equivalent	A, B, C, D	1.0	0.85	0.70
Single-welded butt joint with backing strip	A, B, C, D	0.9	0.8	0.65
Single-welded butt joint without backing strip	A, B, C	NA	NA	0.60
Double full-fillet lap joint	A, B, C	NA	NA	0.55
Single full-fillet lap joint with plug welds	B, C	NA	NA	0.50
Single full-fillet lap joint without plug welds	A, B	NA	NA	0.45

14.4.6 Corrosion allowance

The “corrosion allowance” is the additional thickness of metal added to allow for material lost by corrosion and erosion or scaling (see [Chapter 6](#)). The ASME BPV Code Sec. VIII D.1 states that the vessel user shall specify corrosion allowances (Part UG-25). Minimum wall thicknesses calculated using the rules given in the code are in the fully corroded condition (Part UG-16). Corrosion is a complex phenomenon, and it is not possible to give specific rules for the estimation of the corrosion allowance required for all circumstances. The allowance should be based on experience with the material of construction under similar service conditions to those for the proposed design. For carbon and low-alloy steels, where severe corrosion is not expected, a minimum allowance of 2.0 mm should be used; where more severe conditions are anticipated, this should be increased to 4.0 mm. Most design codes and standards specify a minimum allowance of 1.0 mm, but under the ASME BPV Code Sec. VIII no corrosion allowance is needed when past experience indicates that corrosion is only superficial or does not occur.

14.4.7 Design loads

A structure must be designed to resist gross plastic deformation and collapse under all the conditions of loading. The loads to which a process vessel will be subject in service are listed next. They can be classified as major loads that must always be considered in vessel design and subsidiary loads. Formal stress analysis to determine the effect of the subsidiary loads is only required in the codes and standards where it is not possible to demonstrate the adequacy of the proposed design by other means, such as by comparison with the known behavior of existing vessels.

Major loads

1. Design pressure, including any significant static head of liquid
2. Maximum weight of the vessel and contents under operating conditions
3. Maximum weight of the vessel and contents under the hydraulic test conditions
4. Wind loads
5. Earthquake (seismic) loads
6. Loads supported by, or reacting on, the vessel

Subsidiary loads

1. Local stresses caused by supports, internal structures, and connecting pipes
2. Shock loads caused by water hammer or by surging of the vessel contents
3. Bending moments caused by eccentricity of the center of the working pressure relative to the neutral axis of the vessel
4. Stresses due to temperature differences and differences in the coefficient of expansion of materials
5. Loads caused by fluctuations in temperature and pressure

A vessel will not be subject to all these loads simultaneously. The designer must determine what combination of possible loads gives the worst situation (the “governing case”) and design for that loading condition.

14.4.8 Minimum practical wall thickness

There will be a minimum wall thickness required to ensure that any vessel is sufficiently rigid to withstand its own weight, and any incidental loads. The ASME BPV Code Sec. VIII D.1 specifies a minimum wall thickness of $\{1/16\}$ in (1.5 mm), not including corrosion allowance, and regardless of vessel dimensions and material of construction. As a general guide, the wall thickness of any vessel should not be less than the values given here; the values include a corrosion allowance of 2 mm:

Vessel diameter (m)	Minimum thickness (mm)
1	5
1–2	7
2–2.5	9
2.5–3.0	10
3.0–3.5	12

14.5 The design of thin-walled vessels under internal pressure

14.5.1 Cylinders and spherical shells

The walls of thin vessels can be considered to be “membranes”; supporting loads without significant bending or shear stresses; similar to the walls of a balloon. The analysis of the membrane stresses induced in the wall by internal pressure gives a basis for determining the minimum wall thickness required for vessel shells. The actual thickness required will also depend on the stresses arising from the other loads to which the vessel is subjected.

For a cylindrical shell, the stresses in the walls can be determined from simple force balances, as the wall stresses must balance the forces due to pressure. If we consider a horizontal section through the vessel (Fig. 14.3a), then the force due to pressure on the cross-section is:

$$F_L = \frac{P_i \pi D^2}{4} \quad (14.5)$$

Where P_i = internal pressure

D = mean diameter

F_L = force in longitudinal direction

This force must be balanced by the longitudinal stress in the wall of the cylinder, which acts only on the sectioned area of the wall:

$$F_L = \sigma_L \pi D t \quad (14.6)$$

Where σ_L = longitudinal stress

t = wall thickness

Equating Equations 14.5 and 14.6:

$$\sigma_L = \frac{P_i D}{4t} \quad (14.7)$$

Similarly, if we consider a vertical section in an infinite cylinder (Fig. 14.3b), the force due to pressure on a vertical section of length L is:

$$F_v = P_i D L \quad (14.8)$$

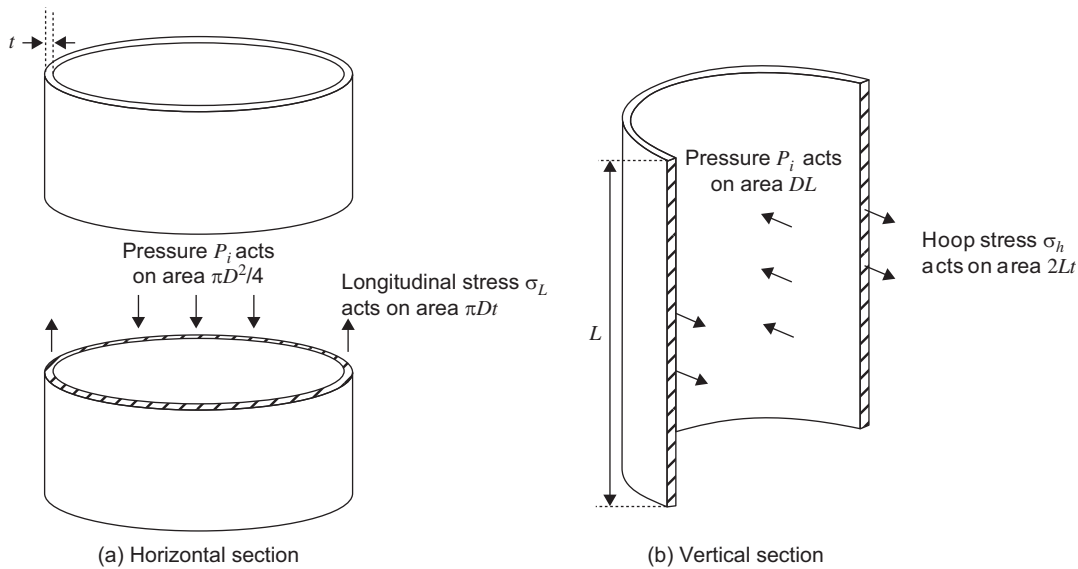


FIG. 14.3 Stresses in the walls of cylindrical vessels due to internal pressure.

Where F_v = force in horizontal direction

L = length

This force is balanced by the circumferential or hoop stress in the wall of the cylinder, which acts only on the sectioned area of the cylinder:

$$F_v = \sigma_h (2 L t) \quad (14.9)$$

Where σ_h = hoop stress

Equating Equations 14.8 and 14.9:

$$\sigma_h = \frac{P_i D}{2t} \quad (14.10)$$

The minimum wall thickness that is required to contain the internal pressure can be determined using Equations 14.7 and 14.10.

If D_i is internal diameter and t the minimum thickness required, the mean diameter will be $(D_i + t)$; substituting this for D in Equation 14.10 gives:

$$t = \frac{P_i(D_i + t)}{2S}$$

where S is the maximum allowable stress and P_i the internal pressure. Rearranging gives:

$$t = \frac{P_i D_i}{2S - P_i} \quad (14.11)$$

If we allow for the welded joint efficiency, E , this becomes:

$$t = \frac{P_i D_i}{2SE - P_i} \quad (14.12)$$

The equation specified by the ASME BPV Code (Sec. VIII D.1 Part UG-27) is:

$$t = \frac{P_i D_i}{2SE - 1.2P_i} \quad (14.13)$$

This differs slightly from Equation 14.12, as it is derived from the formula for thick-walled vessels.

Similarly, for longitudinal stress the code specifies:

$$t = \frac{P_i D_i}{4SE + 0.8P_i} \quad (14.14)$$

The ASME BPV Code specifies that the minimum thickness shall be the greater value determined from Equations 14.13 and 14.14. If these equations are rearranged and used to calculate the MAWP for a vessel of a given thickness, then the MAWP is the lower value predicted by the two equations.

For a spherical shell, the code specifies:

$$t = \frac{P_i D_i}{4SE - 0.4P_i} \quad (14.15)$$

Any consistent set of units can be used for Equations 14.13 to 14.15.

14.5.2 Heads and closures

The ends of a cylindrical vessel are closed by heads of various shapes. The principal types used are:

1. Flat plates and formed flat heads; [Fig. 14.4](#)
2. Hemispherical heads; [Fig. 14.5\(a\)](#)
3. Ellipsoidal heads; [Fig. 14.5\(b\)](#)
4. Torispherical heads; [Fig. 14.5\(c\)](#)

Hemispherical, ellipsoidal, and torispherical heads are collectively referred to as domed heads. They are formed by pressing or spinning; large diameters are fabricated from formed sections. Torispherical heads are often referred

to as dished ends. The preferred proportions of domed heads are given in the standards and codes. Vessel heads can be made in any size, but standard sizes (which come in 6-inch increments) will usually be cheaper.

Choice of closure

Flat plates are used as covers for manways and as the channel covers of heat exchangers. Formed flat ends, known as “flange-only” ends, are manufactured by turning over a flange with a small radius on a flat plate (see Fig. 14.4a). The corner radius reduces the abrupt change of shape at the junction with the cylindrical section, which reduces the local stresses to some extent. “Flange-only” heads are the cheapest type of formed head to manufacture, but their use is limited to low-pressure and small-diameter vessels.

Standard torispherical heads (dished ends) are the most commonly used end closure for vessels up to operating pressures of 15 bar. They can be used for higher pressures, but above 10 bar their cost should be compared with that of an equivalent ellipsoidal head. Above 15 bar an ellipsoidal head will usually prove to be the most economical closure to use.

A hemispherical head is the strongest shape, capable of resisting about twice the pressure of a torispherical head of the same thickness. The cost of forming a hemispherical head will, however, be higher than that for a shallow torispherical head. Hemispherical heads are used for high pressures.

14.5.3 Design of flat ends

Though the fabrication cost is low, flat ends are not a structurally efficient form, and very thick plates would be required for high pressures or large diameters.

The design equations used to determine the thickness of flat ends are based on the analysis of stresses in flat plates. The thickness required will depend on the degree of constraint at the plate periphery.

The ASME BPV Code specifies the minimum thickness as:

$$t = D_e \sqrt{\frac{C P_i}{S E}} \quad (14.16)$$

where C = a design constant, dependent on the edge constraint

D_e = nominal plate diameter

S = maximum allowable stress

E = joint efficiency

Any consistent set of units can be used.

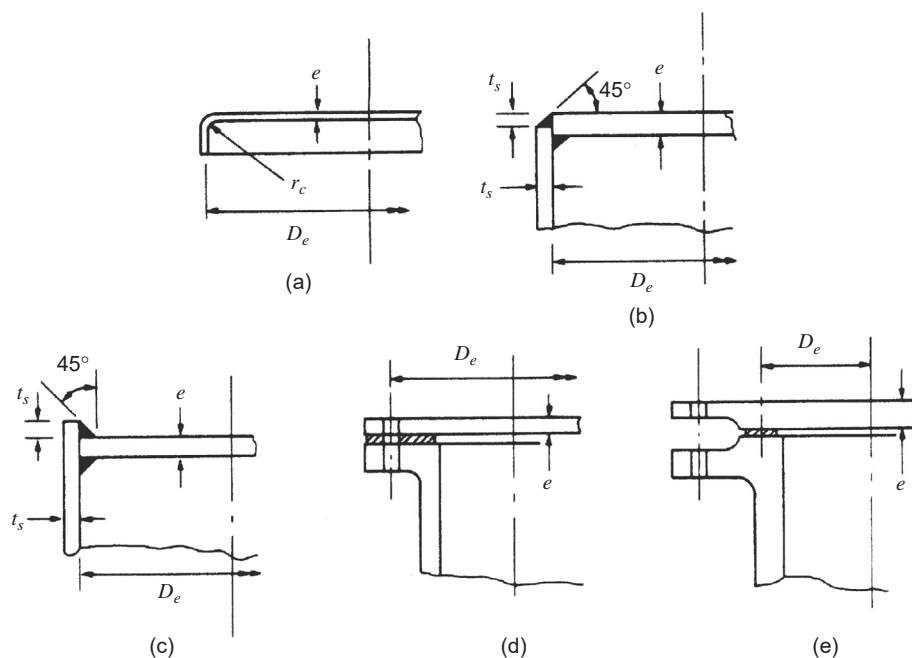


FIG. 14.4 Flat-end closures. (a) Flanged plate. (b) Welded plate. (c) Welded plate. (d) Bolted cover. (e) Bolted cover.

Values for the design constant C and the nominal plate diameter D_e are given in the ASME BPV Code for various arrangements of flat end closures (Sec. VIII D.1 Part UG-34).

Some typical values of the design constant and nominal diameter for the designs shown in Fig. 14.4 are given here. For detailed design, the ASME BPV Code should be consulted.

- (a) Flanged-only end, $C = 0.17$ if corner radius is not more than $3t$; otherwise, $C = 0.1$; D_e is equal to D_i .
- (b and c) Plates welded to the end of the shell with a fillet weld, angle of fillet 45 degrees and weld depth 70% of the thickness of the shell, $C = 0.33 t/t_s$, where t_s is the shell thickness. $D_e = D_i$.
- (d) Bolted cover with a full face gasket (see Section 14.10), $C = 0.25$ and D_e is the bolt circle diameter (the diameter of a circle connecting the centers of the bolt holes).
- (e) Bolted end cover with a narrow-face gasket, $C = 0.3$ and D_e should be taken as the mean diameter of the gasket.

14.5.4 Design of domed ends

Design equations and charts for the various types of domed heads are given in the ASME BPV Code and should be used for detailed design. The code covers both unpierced and pierced heads. Pierced heads are those with openings or connections. The head thickness must be increased to compensate for the weakening effect of the holes where the opening or branch is not locally reinforced (see Section 14.6).

For convenience, simplified design equations are given in this section. These are suitable for the preliminary sizing of unpierced heads and for heads with fully compensated openings or branches.

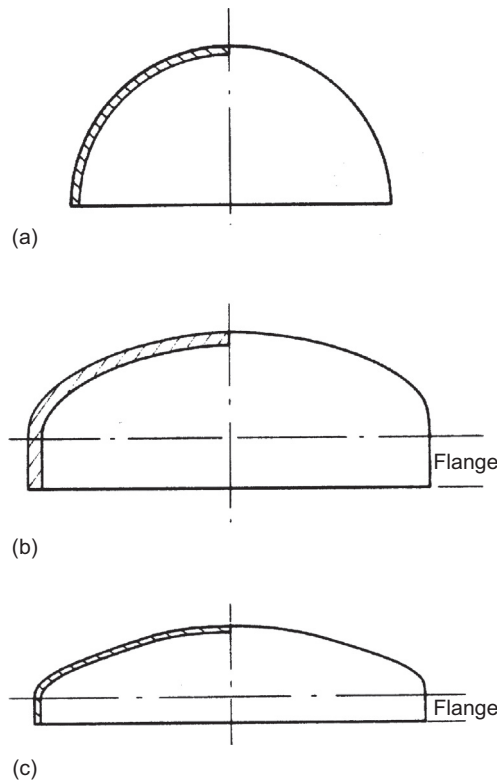


FIG. 14.5 Domed heads. (a) Hemispherical. (b) Ellipsoidal. (c) Torispherical.

Hemispherical heads

For equal stress in the cylindrical section and hemispherical head of a vessel, the thickness of the head need only be half that of the cylinder; however, as the dilation of the two parts would then be different, discontinuity stresses would be set up at the head and cylinder junction. For no difference in dilation between the two parts (equal diametrical strain), it can be shown that for steels (Poisson ratio = 0.3), the ratio of the hemispherical head thickness to cylinder thickness should be 7/17. However, the stress in the head would then be greater than that in the cylindrical section, and the optimum thickness ratio is normally taken as 0.6; see [Brownell and Young \(1959\)](#).

In the ASME BPV Code Sec. VIII D.1, the equation specified is the same as for a spherical shell:

$$t = \frac{P_i D_i}{4SE - 0.4P_i} \quad (14.17)$$

Ellipsoidal heads

Most standard ellipsoidal heads are manufactured with a major and minor axis ratio of 2:1. For this ratio, the following equation can be used to calculate the minimum thickness required (ASME BPV Code Sec. VIII D.1 Part UG-32):

$$t = \frac{P_i D_i}{2SE - 0.2P_i} \quad (14.18)$$

Torispherical heads

A torispherical shape is formed from part of a torus and part of a sphere ([Fig. 14.5c](#)). The shape is close to that of an ellipse but is easier and cheaper to fabricate.

In [Fig. 14.6](#), R_k is the knuckle radius (the radius of the torus) and R_c the crown radius (the radius of the sphere). The stress will be higher in the torus section than the spherical section.

There are two junctions in a torispherical end closure: that between the cylindrical section and the head and that at the junction of the crown and the knuckle radii. The bending and shear stresses caused by the differential dilation that will occur at these points must be taken into account in the design of the heads. The ASME BPV Code gives the design equation (Sec. VIII D.1 Part UG-32):

$$t = \frac{0.885 P_i R_c}{SE - 0.1P_i} \quad (14.19)$$

The ratio of the knuckle to crown radii should not be less than 0.06, to avoid buckling; and the crown radius should not be greater than the diameter of the cylindrical section. Any consistent set of units can be used with Equations 14.17 to 14.19. For formed heads (no welds or joints in the head), the joint efficiency E is taken as 1.0.

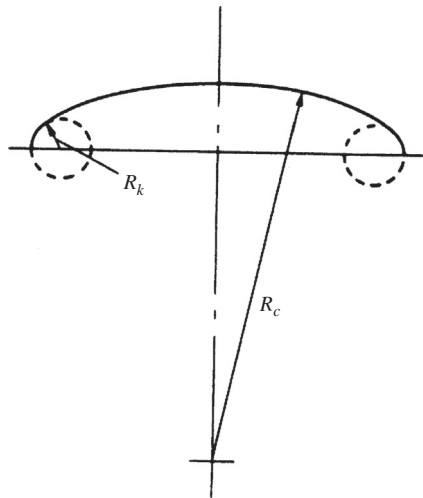


FIG. 14.6 Torisphere.

Flanges (skirts) on domed heads

Formed domed heads are made with a short, straight cylindrical section, called a flange or skirt (Fig. 14.5). This ensures that the weld line is away from the point of discontinuity between the head and the cylindrical section of the vessel.

14.5.5 Conical sections and end closures

Conical sections (reducers) are used to make a gradual reduction in diameter from one cylindrical section to another of smaller diameter.

Conical ends are used to facilitate the smooth flow and removal of solids from process equipment such as hoppers, spray dryers, and crystallizers.

The thickness required at any point on a cone is related to the diameter by the following expression:

$$t = \frac{P_i D_c}{2SE - P_i} \cdot \frac{1}{\cos \alpha} \quad (14.20)$$

where D_c = the diameter of the cone at the point

α = half the cone apex angle

The equation given in the ASME BPV Code is:

$$t = \frac{P_i D_i}{2 \cos \alpha (SE - 0.6P_i)} \quad (14.21)$$

This equation will only apply at points away from the cone-to-cylinder junction. Bending and shear stresses will be caused by the different dilations of the conical and cylindrical sections. A formed section would normally be used for the transition between a cylindrical section and conical section; except for vessels operating at low pressures, or under hydrostatic pressure only. The transition section would be made thicker than the conical or cylindrical section and formed with a knuckle radius to reduce the stress concentration at the transition (Fig. 14.7). The thickness for the conical section away from the transition can be calculated from Equation 14.21.

The code should be consulted for details of how to size the knuckle zone.

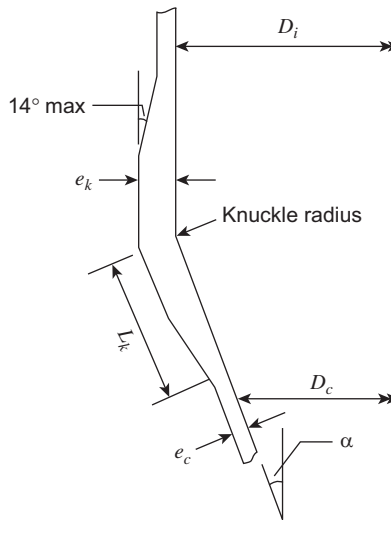
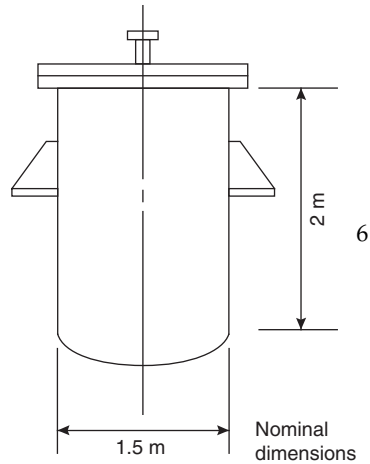


FIG. 14.7 Conical transition section.

Example 14.1

Estimate the thickness required for the component parts of the vessel shown in the diagram. The vessel is to operate at a pressure of 14 bar (absolute) and temperature of 260 °C. The material of construction will be plain carbon steel. Welds will be fully radiographed. A corrosion allowance of 2 mm should be used.

**Solution**

Design pressure, take as 10% above operating gauge pressure:

$$\begin{aligned} &= (14 - 1) \times 1.1 \\ &= 14.3 \text{ bar} \\ &= 1.43 \text{ N/mm}^2 \end{aligned}$$

Design temperature 260 °C (500 °F).

From Table 14.2, maximum allowable stress = $12.9 \times 10^3 \text{ psi} = 88.9 \text{ N/mm}^2$.

Cylindrical section

$$t = \frac{1.43 \times 1.5 \times 10^3}{(2 \times 89 \times 1) - (1.2 \times 1.43)} = 12.2 \text{ mm}$$

$$\text{add corrosion allowance } 12.2 + 2 = 14.2 \text{ mm} \quad (14.13)$$

$$\text{say } \underline{\underline{15 \text{ mm plate}}} \quad \text{or} \quad \underline{\underline{9/16 \text{ inch plate}}}$$

Domed head

1. Try a standard dished head (torisphere):

$$\text{crown radius } R_c = D_i = 1.5 \text{ m}$$

$$\text{knuckle radius} = 6\% R_c = 0.09 \text{ m}$$

A head of this size would be formed by pressing: no joints, so $E = 1$.

$$t = \frac{0.885 \times 1.43 \times 1.5 \times 10^3}{(89 \times 1) - (0.1 \times 1.43)} = \underline{\underline{21.4 \text{ mm}}} \quad (14.19)$$

2. Try a “standard” ellipsoidal head, ratio major: minor axes = 2:1

$$t = \frac{1.43 \times 1.5 \times 10^3}{(2 \times 89 \times 1) - (0.2 \times 1.43)} = \underline{\underline{12.1 \text{ mm}}} \quad (14.18)$$

So an ellipsoidal head would probably be the most economical. Take as same thickness as wall 15 mm or 9/16 inch.

Flat head

Use a full face gasket $C = 0.25$

D_e = bolt circle diameter, take as approx. 1.7 m.

$$t = 1.7 \times 10^3 \sqrt{\frac{0.25 \times 1.43}{89 \times 1}} = \underline{\underline{107.7 \text{ mm}}} \quad (14.16)$$

Add corrosion allowance and round off to 111 mm (4 $\frac{3}{8}$ in).

This shows the inefficiency of a flat cover. It would be better to use a flanged domed head.

14.6 Compensation for openings and branches

All process vessels will have openings for connections, manways, and instrument fittings. The presence of an opening weakens the shell and gives rise to stress concentrations. The stress at the edge of a hole will be considerably higher than the average stress in the surrounding plate. To compensate for the effect of an opening, the wall thickness is increased in the region adjacent to the opening. Sufficient reinforcement must be provided to compensate for the weakening effect of the opening without significantly altering the general dilation pattern of the vessel at the opening. Overreinforcement will reduce the flexibility of the wall, causing a “hard spot,” and giving rise to secondary stresses; typical arrangements are shown in Fig. 14.8.

The simplest method of providing compensation is to weld a pad or collar around the opening (see Fig. 14.8a). The outer diameter of the pad is usually between 1½ and 2 times the diameter of the hole or branch. This method, however, does not give the best disposition of the reinforcing material about the opening, and in some circumstances high thermal stress can arise due to the poor thermal conductivity of the pad-to-shell junction.

At a branch, the reinforcement required can be provided, with or without a pad, by allowing the branch to protrude into the vessel (see Fig. 14.8b). This arrangement should be used with caution for process vessels, as the protrusion will act as a trap for crud, and crevices are created in which localized corrosion can occur. Forged reinforcing rings (see Fig. 14.8c) provide the most effective method of compensation, but are expensive. They would be used for any large openings and branches in vessels operating under severe conditions.

The rules for calculating the minimum amount of reinforcement required are complex. For design purposes, consult the ASME BPV Code Sec. VIII D.1 Part UG-37.

14.7 Design of vessels subject to external pressure

Two types of process vessels are likely to be subjected to external pressure: those operated under vacuum, where the maximum pressure will be 1 bar (1 atm); and jacketed vessels, where the inner vessel will be under the jacket pressure. For jacketed vessels, the maximum pressure difference should be taken as the full jacket pressure, as a situation may arise in which the pressure in the inner vessel is lost. Thin-walled vessels subject to external pressure are vulnerable to failure through elastic instability (buckling), and it is this mode of failure that determines the wall thickness required.

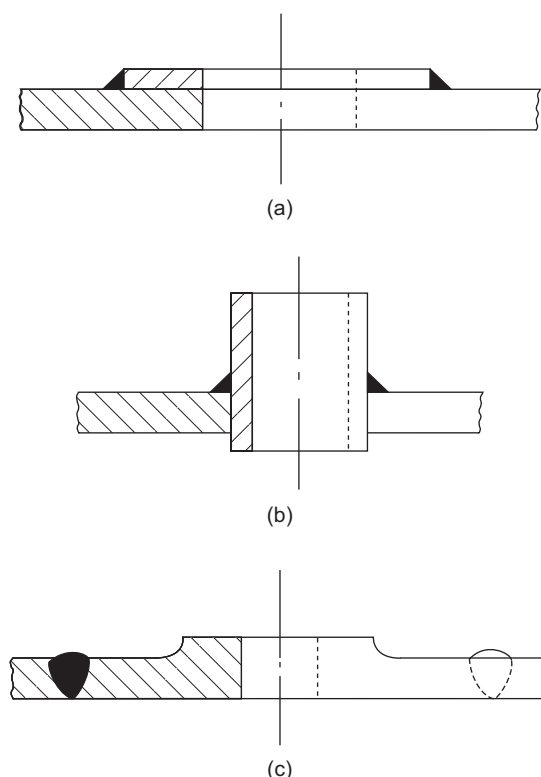


FIG. 14.8 Types of compensation for openings. (a) Welded pad. (b) Inset nozzle. (c) Forged ring.

The method recommended by the BPV Code for vessels subject to compressive stresses is substantially more complex than the method used for tensile stresses and takes into account the fact that the maximum allowable stress in compression is different from that in tension. The ASME BPV Code Sec. VIII D.1 Part UG-28 should be consulted for the approved method for detailed design of cylindrical vessels subject to external pressure. For detailed design of hemispherical vessel heads subject to external pressure, the method given in ASME BPV Code Sec. VIII D.1 Part UG-33 must be followed. Design methods for different shaped heads under external pressure are also given in the standards and codes.

Vessels that are subject to external pressure are often reinforced with internal stiffening rings. Methods for sizing the stiffening rings and determining their spacing are given in the BPV Code.

14.8 Design of vessels subject to combined loading

Pressure vessels are subjected to other loads in addition to pressure (see Section 14.4.7) and must be designed to withstand the worst combination of loading without failure. It is not practical to give an explicit relationship for the vessel thickness to resist combined loads. A trial thickness must be assumed (based on that calculated for pressure alone) and the resultant stress from all loads determined to ensure that the maximum allowable stress intensity is not exceeded at any point. When combined loads are analyzed, the maximum compressive stress must be considered, as well as the maximum tensile stress. The maximum allowable stress in compression is different from the maximum allowable stress in tension and is determined using the method given in ASME BPV Code Sec. VIII D.1 Part UG-23.

The main sources of load to consider are:

1. Pressure
2. Dead weight of vessel and contents

3. Wind
4. Earthquake (seismic)
5. External loads imposed by piping and attached equipment

The primary stresses arising from these loads are considered in the following paragraphs, for cylindrical vessels (Fig. 14.9).

Primary stresses

1. The longitudinal and circumferential stresses due to pressure (internal or external) are given by:

$$\sigma_L = \frac{PD_i}{4t} \quad (14.7)$$

$$\sigma_h = \frac{PD_i}{2t} \quad (14.10)$$

2. The direct stress σ_w due to the weight of the vessel, its contents, and any attachments. The stress will be tensile (positive) for points below the plane of the vessel supports and compressive (negative) for points above the supports (Fig. 14.10). The dead-weight stress will normally only be significant, compared with the magnitude of the other stresses, in tall vessels.

$$\sigma_w = \frac{W_z}{\pi(D_i + t)t} \quad (14.22)$$

where W_z is the total weight that is supported by the vessel wall at the plane considered (see Section 14.8.1).

3. Bending stresses resulting from the bending moments to which the vessel is subjected. Bending moments will be caused by the following loading conditions:
 - a. The wind loads on tall self-supported vessels (Section 14.8.2)
 - b. Seismic (earthquake) loads on tall vessels (Section 14.8.3)
 - c. The dead weight and wind loads on piping and equipment that are attached to the vessel but offset from the vessel center line (Section 14.8.4)
 - d. For horizontal vessels with saddle supports, from the disposition of dead-weight load (see Section 14.9.1)
 The bending stresses will be compressive or tensile, depending on location, and are given by:

$$\sigma_b = \pm \frac{M}{I_v} \left(\frac{D_i}{2} + t \right) \quad (14.23)$$

where M is the total bending moment at the plane being considered and I_v the second moment of area of the vessel about the plane of bending.

$$I_v = \frac{\pi}{64} (D_o^4 - D_i^4) \quad (14.24)$$

4. Torsional shear stresses τ resulting from torque caused by loads offset from the vessel axis. These loads will normally be small and need not be considered in preliminary vessel designs.

The torsional shear stress is given by:

$$\tau = \frac{T}{I_p} \left(\frac{D_i}{2} + t \right) \quad (14.25)$$

where T = the applied torque

$$I_p = \text{polar second moment of area } I_p = (\pi/32)(D_o^4 - D_i^4)$$

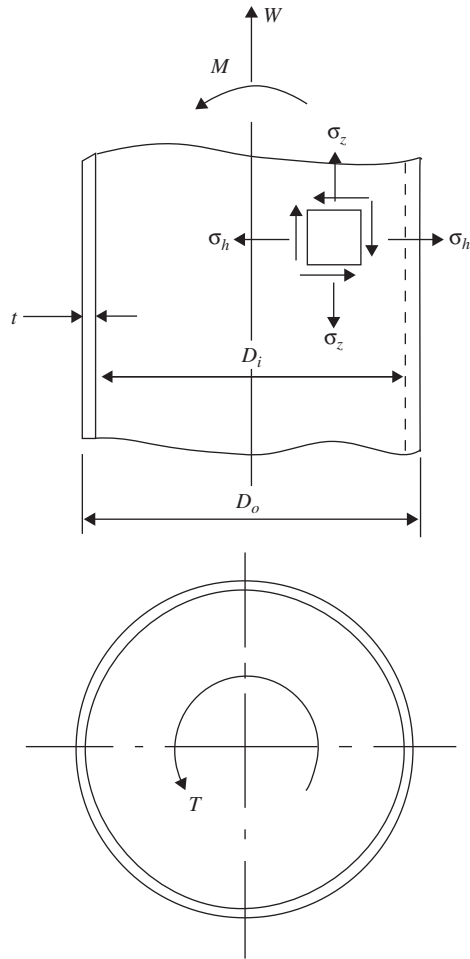


FIG. 14.9 Stresses in a cylindrical shell under combined loading.

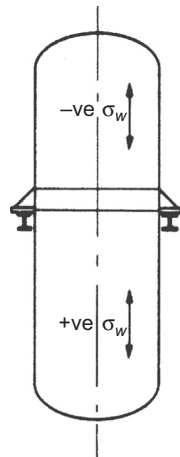


FIG. 14.10 Stresses due to dead-weight loads.

Principal stresses

The principal stresses will be given by:

$$\sigma_1 = \frac{1}{2} \left[\sigma_h + \sigma_z + \sqrt{(\sigma_h - \sigma_z)^2 + 4\tau^2} \right] \quad (14.26)$$

$$\sigma_2 = \frac{1}{2} \left[\sigma_h + \sigma_z - \sqrt{(\sigma_h - \sigma_z)^2 + 4\tau^2} \right] \quad (14.27)$$

where

σ_z = total longitudinal stress

$$= \sigma_L + \sigma_w \pm \sigma_b$$

σ_w should be counted as positive if tension and negative if compressive.

τ is not usually significant.

The third principal stress, that in the radial direction σ_3 , will usually be negligible for thin-walled vessels (see Section 14.1.1). As an approximation, it can be taken as equal to one-half the pressure loading

$$\sigma_3 = 0.5P \quad (14.28)$$

σ_3 will be compressive (negative).

Allowable stress intensity

The maximum intensity of stress allowed will depend on the particular theory of failure adopted in the design method (see Section 14.3.2). The maximum shear-stress theory is normally used for pressure vessel design.

Using this criterion, the maximum stress intensity at any point is taken for design purposes as the numerically greatest value of the following:

$$(\sigma_1 - \sigma_2)$$

$$(\sigma_1 - \sigma_3)$$

$$(\sigma_2 - \sigma_3)$$

The vessel wall thickness must be sufficient to ensure the maximum stress intensity does not exceed the maximum allowable stress (nominal design strength) for the material of construction at any point. The ASME BPV Code Sec. II Part D should be consulted for the maximum allowable stress values in tension or in compression.

Compressive stresses and elastic stability

Under conditions where the resultant axial stress σ_z due to the combined loading is compressive, the vessel may fail by elastic instability (buckling) (see Section 14.3.3). Failure can occur in a thin-walled process column under an axial compressive load by buckling of the complete vessel, as with a strut (Euler buckling); or by local buckling, or wrinkling, of the shell plates. Local buckling will normally occur at a stress lower than that required to buckle the complete vessel. A column design must be checked to ensure that the maximum value of the resultant axial stress does not exceed the critical value at which buckling will occur.

For a curved plate subjected to an axial compressive load, the critical buckling stress σ_c is given by (see [Timoshenko & Gere, 2009](#)):

$$\sigma_c = \frac{E_Y}{\sqrt{3(1 - \nu^2)}} \left(\frac{t}{R_p} \right) \quad (14.29)$$

where R_p is the radius of curvature.

Taking Poisson's ratio as 0.3 gives:

$$\sigma_c = 0.60 E_Y \left(\frac{t}{R_p} \right) \quad (14.30)$$

By applying a suitable factor of safety, Equation 14.30 can be used to predict the maximum allowable compressive stress to avoid failure by buckling. A large factor of safety is required, as experimental work has shown that

cylindrical vessels will buckle at values well below that given by Equation 14.29. For steels at ambient temperature $E_Y = 200,000 \text{ N/mm}^2$ and Equation 14.30 with a factor of safety of 12 gives:

$$\sigma_c = 2 \times 10^4 \left(\frac{t}{D_o} \right) \text{ N/mm}^2 \quad (14.31)$$

The maximum compressive stress in a vessel wall should not exceed that given by Equation 14.31 or the maximum allowable design stress for the material, whichever is the least. For detailed design, the ASME BPV Code Sec. VIII should be consulted and the recommended procedure in the code should be followed.

Stiffening

As with vessels under external pressure, the resistance to failure by buckling can be increased significantly by the use of stiffening rings or longitudinal strips. Methods for estimating the critical buckling stress for stiffened vessels are given in the standards and codes.

Loading

The loads to which a vessel may be subjected will not all occur at the same time. For example, it is the usual practice to assume that the maximum wind load will not occur simultaneously with a major earthquake.

The vessel must be designed to withstand the worst combination of the loads likely to occur in the following situations:

1. During erection (or dismantling) of the vessel.
2. With the vessel erected but not operating.
3. During testing (the hydraulic pressure test).
4. During normal operation.

14.8.1 Weight loads

The major sources of dead weight loads are:

1. The vessel shell.
2. The vessel fittings: manways, nozzles.
3. Internal fittings: plates (plus the fluid on the plates); heating and cooling coils.
4. External fittings: ladders, platforms, piping.
5. Auxiliary equipment that is not self-supported; condensers, agitators.
6. Insulation.
7. The weight of liquid to fill the vessel. The vessel will be filled with water for the hydraulic pressure test and may fill with process liquid due to misoperation.

Note: For vessels on a skirt support (see Section 14.9.2), the weight of the liquid to fill the vessel will be transferred directly to the skirt.

The weight of the vessel and fittings can be calculated from the preliminary design sketches. The weights of standard vessel components: heads, shell plates, manways, branches and nozzles, are given in various handbooks: [Megyesy \(2008\)](#) and [Brownell and Young \(1959\)](#).

For preliminary calculations, the approximate weight of a cylindrical vessel with domed ends, and uniform wall thickness, can be estimated from the following equation:

$$W_v = C_w \pi \rho_m D_m g (H_v + 0.8D_m) t \times 10^{-3} \quad (14.32)$$

where

W_v = total weight of the shell, excluding internal fittings, such as plates, N ,

C_w = a factor to account for the weight of nozzles, manways, internal supports, etc., which can be taken as
 = 1.08 for vessels with only a few internal fittings,
 = 1.15 for distillation columns, or similar vessels, with several manways, and with plate support rings or equivalent fittings,

H_v = height, or length, between tangent lines (the length of the cylindrical section), m,

g = gravitational acceleration, 9.81 m/s^2

t = wall thickness, mm,

ρ_m = density of vessel material, kg/m^3 (see Table 6.2),

D_m = mean diameter of vessel = $(D_i + t \times 10^{-3})$, m.

For a steel vessel, Equation 14.32 reduces to:

$$W_v = 240 C_w D_m (H_v + 0.8 D_m) t \quad (14.33)$$

The following values can be used as a rough guide to the weight of fittings; see Nelson (1963):

1. Caged ladders, steel, 360 N/m length,
2. Plain ladders, steel, 150 N/m length,
3. Platforms, steel, for vertical columns, 1.7 kN/m^2 area,
4. Contacting plates, steel, including typical liquid loading, 1.2 kN/m^2 plate area.

Typical values for the density of insulating materials are (all kg/m^3):

Foam glass	150
Mineral wool	130
Fiberglass	100
Calcium silicate	200

These densities should be doubled to allow for attachment fittings, sealing, and moisture absorption.

14.8.2 Wind loads (tall vessels)

Wind loading will only be important on tall columns installed in the open. Columns and chimney stacks are usually free standing, mounted on skirt supports, and not attached to structural steel work. Under these conditions the vessel under wind loading acts as a cantilever beam (Fig. 14.11). For a uniformly loaded cantilever, the bending moment at any plane is given by:

$$M_x = \frac{W x^2}{2} \quad (14.34)$$

where x is the distance measured from the free end and W the load per unit length (Newtons per meter run).

So the bending moment, and hence the bending stress, will vary parabolically from zero at the top of the column to a maximum value at the base. For tall columns, the bending stress due to wind loading will often be greater than direct stress due to pressure and will determine the plate thickness required. The most economical design will be one in which the plate thickness is progressively increased from the top to the base of the column, with the thickness at the top being sufficient for the pressure load and that at the base sufficient for the pressure plus the maximum bending moment.

Any local increase in the column area presented to the wind will give rise to a local, concentrated load (Fig. 14.12). The bending moment at the column base caused by a concentrated load is given by:

$$M_p = F_p H_p \quad (14.35)$$

where F_p = local, concentrated load,

H_p = the height of the concentrated load above the column base.

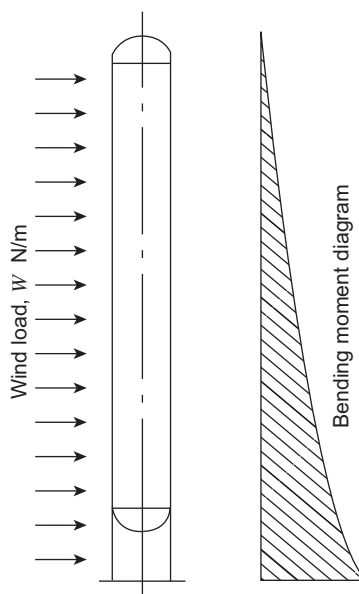


FIG. 14.11 Wind loading on a tall column.

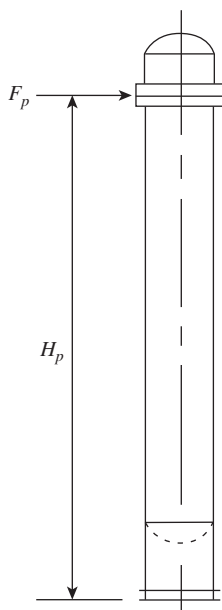


FIG. 14.12 Local wind loading.

Dynamic wind pressure

The load imposed on any structure by the action of the wind will depend on the shape of the structure and the wind velocity.

$$P_w = \frac{1}{2} C_d \rho_a u_w^2 \quad (14.36)$$

where P_w = wind pressure (load per unit area),

C_d = drag coefficient (shape factor),

ρ_a = density of air,

u_w = wind velocity.

The drag coefficient is a function of the shape of the structure and the wind velocity (Reynolds number).

For a smooth cylindrical column or stack, the following semi-empirical equation can be used to estimate the wind pressure:

$$P_w = 0.05u_w^2 \quad (14.37)$$

where P_w = wind pressure, N/m²
 u_w = wind speed, km/h

If the column outline is broken up by attachments, such as ladders or pipe work, the factor of 0.05 in Equation 14.37 should be increased to 0.07 to allow for the increased drag.

A column must be designed to withstand the highest wind speed that is likely to be encountered at the site during the life of the plant. The probability of a given wind speed occurring can be predicted by studying meteorological records for the site location. Data and design methods for wind loading are given in the Engineering Sciences Data Unit (ESDU) Wind Engineering Series (www.esdu.com). Design loadings for locations in the United States are given by Moss and Basic (2013), Megyesy (2008), and Escoe (1994). A wind speed of 160 km/h (100 mph) can be used for preliminary design studies, equivalent to a wind pressure of 1280 N/m² (25 lb/ft²).

At any site, the wind velocity near the ground will be lower than that higher up (due to the boundary layer), and in some design methods a lower wind pressure is used at heights below about 20 m, typically taken as one-half of the pressure above this height.

The loading per unit length of the column can be obtained from the wind pressure by multiplying by the effective column diameter: the outside diameter plus an allowance for the thermal insulation and attachments such as pipes and ladders.

$$W = P_w D_{eff} \quad (14.38)$$

An allowance of 0.4 m should be added for a caged ladder. The calculation of the wind load on a tall column, and the induced bending stresses, is illustrated in Example 14.2. Further examples of the design of tall columns are given by Brownell (1963), Henry (1973), Bednar (1990), Escoe (1994), and Jawad and Farr (2018).

Deflection of tall columns

Tall columns sway in the wind. The allowable deflection will normally be specified as less than 150 mm per 30 m of height (6 in per 100 ft).

For a column with a uniform cross-section, the deflection can be calculated using the formula for the deflection of a uniformly loaded cantilever. A method for calculating the deflection of a column where the wall thickness is not constant is given by Tang (1968).

Wind-induced vibrations

Vortex shedding from tall thin columns and stacks can induce vibrations which, if the frequency of shedding of eddies matches the natural frequency of the column, can be severe enough to cause premature failure of the vessel by fatigue. The effect of vortex shedding should be investigated for free-standing columns with height-to-diameter ratios greater than 10. Methods for estimating the natural frequency of columns are given by Freese (1959) and DeGhetto and Long (1966).

Helical strakes (strips) are fitted to the tops of tall, smooth chimneys to change the pattern of vortex shedding and so prevent resonant oscillation. The same effect will be achieved on a tall column by distributing any attachments (ladders, pipes, and platforms) around the column.

14.8.3 Earthquake loading

The movement of the earth's surface during an earthquake produces horizontal shear forces on tall, self-supported vessels, the magnitude of which increases from the base upward. The total shear force on the vessel will be given by:

$$F_s = a_e \left(\frac{W_v}{g} \right) \quad (14.39)$$

where a_e = the acceleration of the vessel due to the earthquake,
 g = the acceleration due to gravity,
 W_v = total weight of the vessel and contents.

The term (a_e/g) is called the seismic constant C_e , and is a function of the natural period of vibration of the vessel and the severity of the earthquake. Values of the seismic constant have been determined empirically from studies of the damage caused by earthquakes and are available for those geographical locations that are subject to earthquake activity. Values for sites in the United States and procedures for determining the stresses induced in tall columns are given by Megyesy (2008), Escoe (1994), and Moss and Basic (2013).

14.8.4 Eccentric loads (tall vessels)

Ancillary equipment attached to a tall vessel will subject the vessel to a bending moment if the center of gravity of the equipment does not coincide with the center line of the vessel (Fig. 14.13). The moment produced by small fittings, such as ladders, pipes, and manways, will be small and can be neglected. That produced by heavy equipment, such as reflux condensers and side platforms, can be significant and should be considered. The moment is given by:

$$M_e = W_e L_o \quad (14.40)$$

where W_e = dead weight of the equipment,

L_o = distance between the center of gravity of the equipment and the column center line.

To avoid putting undue stress on the column walls, equipment (such as reflux condensers and overhead receiving drums) is usually not attached to the top of a column, but is instead located adjacent to the column in the plant structure. Condensers and receiving vessels are often placed above grade level to provide net positive suction head for reflux and overhead pumps sited at grade.

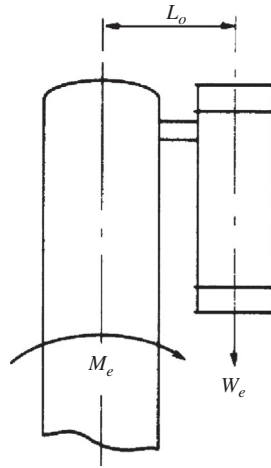


FIG. 14.13 Bending moment due to offset equipment.

14.8.5 Torque

Any horizontal force imposed on the vessel by ancillary equipment, the line of thrust of which does not pass through the center line of the vessel, will produce a torque on the vessel. Such loads can arise through wind pressure on piping and other attachments; however, the torque will normally be small and usually can be disregarded. The pipe work and the connections for any ancillary equipment will be designed so as not to impose a significant load on the vessel.

Example 14.2

Make a preliminary estimate of the plate thickness required for the distillation column specified here:

Height, between tangent lines	50 m
Diameter	2 m
Hemispherical head	
Skirt support, height	3 m
100 sieve plates, equally spaced	
Insulation, mineral wool	75 mm thick
Material of construction, stainless steel, maximum allowable stress	135 N/mm ² at design temperature 200 °C
Operating pressure	10 bar (absolute)
Vessel to be fully radiographed (joint efficiency 1)	
Process service	Gasoline debutanizer

Solution

Design pressure; take as 10% above operating pressure:

$$= (10 - 1) \times 1.1 = 9.9 \text{ bar, say } 10 \text{ bar}$$

$$= 1.0 \text{ N/mm}^2$$

Minimum thickness required for pressure loading:

$$t = \frac{1 \times 2 \times 10^3}{(2 \times 135 \times 1) - (1.2 \times 1)} = 7.4 \text{ mm} \quad (14.13)$$

A much thicker wall will be needed at the column base to withstand the wind and dead weight loads.

As a first trial, divide the column into five sections (courses), with the thickness increasing by 2 mm per section. Try 10, 12, 14, 16, 18 mm.

Dead weight of vessel

Though Equation 14.33 only applies strictly to vessels with uniform thickness, it can be used to get a rough estimate of the weight of this vessel by using the average thickness in the equation, 14 mm.

Take $C_w = 1.15$, vessel with plates

$$D_m = 2 + 14 \times 10^{-3} = 2.014 \text{ m}$$

$$H_v = 50 \text{ m}$$

$$t = 14 \text{ mm}$$

$$W_v = 240 \times 1.15 \times 2.014(50 + 0.8 \times 2.014)14$$

$$= 401,643 \text{ N} \quad (14.33)$$

$$= 402 \text{ kN}$$

Weight of plates:

$$\text{plate area} = \pi/4 \times 2^2 = 3.14 \text{ m}^2$$

$$\text{weight of a plate including liquid on it (see Section 14.8.1)} \approx 1.2 \times 3.14 = 3.8 \text{ kN}$$

$$100 \text{ plates} = 100 \times 3.8 = 380 \text{ kN}$$

Weight of insulation:

$$\text{mineral wool density} = 130 \text{ kg/m}^3$$

$$\begin{aligned} \text{approximate volume of insulation} &= \pi \times 2 \times 50 \times 75 \times 10^{-3} \\ &= 23.6 \text{ m}^3 \end{aligned}$$

$$\text{weight} = 23.6 \times 130 \times 9.81 = 30,049 \text{ N}$$

$$\text{double this to allow for fittings, etc.} = 60 \text{ kN}$$

Total weight:

Shell	402
plates & contents	380
Insulation	60
	842 kN

Note that the weight of the contents of the column would be substantially greater if the column were flooded or entirely filled with liquid. This is the case during hydraulic testing, which should be examined as a different loading scenario.

Wind loading

Take dynamic wind pressure as 1280 N/m^2 , corresponding to 160 kph (100 mph).

$$\begin{aligned} \text{Mean diameter, including insulation} &= 2 + 2(14 + 75) \times 10^{-3} \\ &= 2.18 \text{ m} \end{aligned}$$

$$\text{Loading (per linear meter)} W = 1280 \times 2.18 = 2790 \text{ N/m} \quad (14.38)$$

Bending moment at bottom tangent line:

$$M_x = \frac{2790}{2} \times 50^2 = 3,487,500 \text{ Nm} \quad (14.34)$$

Analysis of stresses

At bottom tangent line:

Pressure stresses:

$$\sigma_L = \frac{1.0 \times 2 \times 10^3}{4 \times 18} = 27.8 \text{ N/mm}^2 \quad (14.7)$$

$$\sigma_h = \frac{1 \times 2 \times 10^3}{2 \times 18} = 55.6 \text{ N/mm}^2 \quad (14.10)$$

Dead weight stress:

$$\begin{aligned} \sigma_w &= \frac{W_v}{\pi(D_i + t)t} = \frac{842 \times 10^3}{\pi(200 + 18)18} \\ &= 7.4 \text{ N/mm}^2 (\text{compressive}) \end{aligned} \quad (14.22)$$

Bending stresses:

$$\begin{aligned} D_o &= 2000 + 2 \times 18 = 2036 \text{ mm} \\ I_v &= \frac{\pi}{64} (2036^4 - 2000^4) = 5.81 \times 10^{10} \text{ mm}^4 \end{aligned} \quad (14.24)$$

$$\begin{aligned}\sigma_b &= \pm \frac{3,487,500 \times 10^3}{5.81 \times 10^{10}} \left(\frac{2000}{2} + 18 \right) \\ &= \pm 61.11 \text{ N/mm}^2\end{aligned}\quad (14.23)$$

The resultant longitudinal stress is:

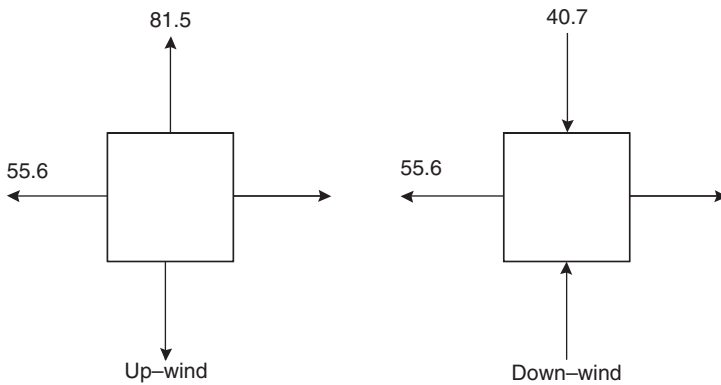
$$\sigma_z = \sigma_L + \sigma_w \pm \sigma_b$$

σ_w is compressive and therefore negative.

$$\sigma_z \text{ (upwind)} = 27.8 - 7.4 + 61.1 = +81.5 \text{ N/mm}^2.$$

$$\sigma_z \text{ (downwind)} = 27.8 - 7.4 - 61.1 = -40.7 \text{ N/mm}^2.$$

As there is no torsional shear stress, the principal stresses will be σ_z and σ_h . The radial stress is negligible: $(P_i/2) = 0.5 \text{ N/mm}^2$.



The greatest difference between the principal stresses will be on the downwind side:

$$(55.6 - (-40.7)) = \underline{\underline{95.5 \text{ N/mm}^2}}$$

well below the maximum allowable design stress.

Note that the bending stress due to wind loading is much larger than the dead weight stress. The hydraulic testing case will have a greater dead weight when the vessel is filled with water, but a simple calculation shows that the maximum weight of water in the vessel (neglecting volume of vessel internals) is $\pi/12 \times \rho \times g (3D_i^2 L + 2D_i^3) = 1582 \text{ kN}$. If this is added to the total weight calculated earlier, then the dead weight stress will increase by about a factor 3. This is still a lot less than the bending stress due to wind load, so the wind load case is the governing case. The hydraulic test will obviously not be scheduled for a day on which 100-mph winds may occur.

Check elastic stability (buckling)

Critical buckling stress:

$$\sigma_c = 2 \times 10^4 \left(\frac{18}{2036} \right) = \underline{\underline{176.8 \text{ N/mm}^2}} \quad (14.31)$$

The maximum compressive stress will occur when the vessel is not under pressure $= 7.4 + 61.1 = 68.5$, well below the critical buckling stress.

So the design is satisfactory. The designer could reduce the plate thickness and recalculate.

14.9 Vessel supports

The method used to support a vessel will depend on the size, shape, and weight of the vessel; the design temperature and pressure; the vessel location and arrangement; and the internal and external fittings and attachments. Horizontal vessels are usually mounted on two saddle supports (Fig. 14.14). Skirt supports are used for tall, vertical columns (Fig. 14.15). Brackets, or lugs, are used for all types of vessels (Fig. 14.16). The supports must be designed to

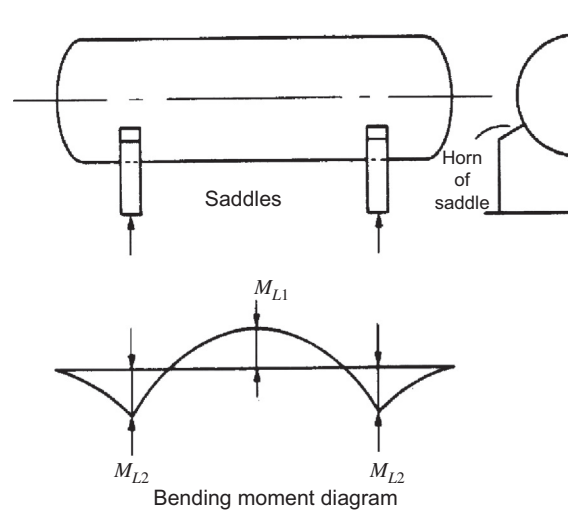


FIG. 14.14 Horizontal cylindrical vessel on saddle supports.

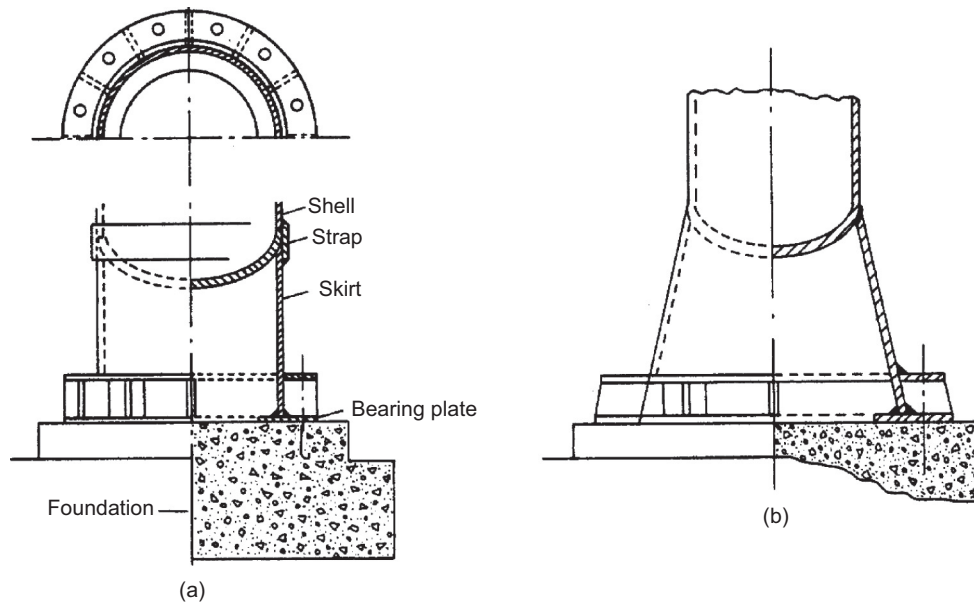


FIG. 14.15 Typical skirt-support designs. (a) Straight skirt. (b) Conical skirt.

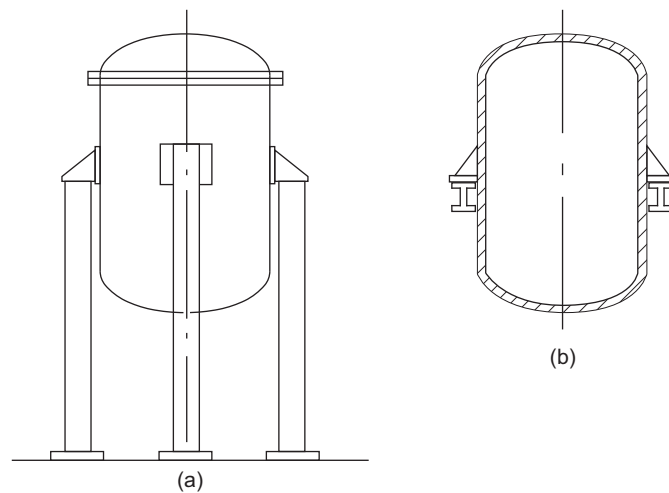


FIG. 14.16 Bracket supports. (a) Supported on legs. (b) Supported from steel work.

carry the weight of the vessel and contents and any superimposed loads, such as wind loads. Supports will impose localized loads on the vessel wall, and the design must be checked to ensure that the resulting stress concentrations are below the maximum allowable design stress. Supports should be designed to allow easy access to the vessel and fittings for inspection and maintenance.

14.9.1 Saddle supports

Though saddles are the most commonly used support for horizontal cylindrical vessels, legs can be used for small vessels. A horizontal vessel will normally be supported at two cross-sections; if more than two saddles are used, the distribution of the loading is uncertain.

A vessel supported on two saddles can be considered a simply supported beam, with an essentially uniform load, and the distribution of longitudinal axial bending moment will be as shown in Fig. 14.14. Maxima occur at the supports and at mid-span. The theoretical optimum position of the supports to give the least maximum bending moment will be the position at which the maxima at the supports and at mid-span are equal in magnitude. For a uniformly loaded beam, the position will be at 21% of the span in from each end. The saddle supports for a vessel will usually be located nearer the ends than this value to make use of the stiffening effect of the ends.

In addition to the longitudinal bending stress, a vessel supported on saddles will be subjected to tangential shear stresses, which transfer the load from the unsupported sections of the vessel to the supports and to circumferential bending stresses. All these stresses need to be considered in the design of large, thin-walled, vessels to ensure that the resultant stress does not exceed the maximum allowable design stress or the critical buckling stress for the material. A detailed stress analysis is beyond the scope of this book. A complete analysis of the stress induced in the shell by the supports is given by Zick (1951). Zick's method forms the basis of the design methods given in the national codes and standards. The method is also given by Brownell and Young (1959), Escoe (1994), and Megyesy (2008).

Design of saddles

The saddles must be designed to withstand the load imposed by the weight of the vessel and contents. They are constructed of bricks or concrete or are fabricated from steel plate. The contact angle should not be less than 120 degrees, and will not normally be greater than 150 degrees. Wear plates are often welded to the shell wall to reinforce the wall over the area of contact with the saddle.

The dimensions of typical "standard" saddle designs are given in Fig. 14.17. To take up any thermal expansion of the vessel, such as that in heat exchangers, the anchor bolt holes in one saddle can be slotted.

Procedures for the design of saddle supports are given by Brownell and Young (1959), Megyesy (2008), Escoe (1994), and Moss and Basic (2013).

14.9.2 Skirt supports

A skirt support consists of a cylindrical or conical shell welded to the base of the vessel. A flange at the bottom of the skirt transmits the load to the foundations. Typical designs are shown in Fig. 14.15. Openings must be provided in the skirt for access and for any connecting pipes; the openings are normally reinforced. The skirt may be welded to the bottom head of the vessel (Fig. 14.18a) or welded flush with the shell (Fig. 14.18b) or welded to the outside of the vessel shell (Fig. 14.18c). The arrangement shown in Fig. 14.18(b) is usually preferred.

Skirt supports are recommended for vertical vessels, as they do not impose concentrated loads on the vessel shell; they are particularly suitable for use with tall columns subject to wind loading, as the support is equally strong in all directions.

Skirt thickness

The skirt thickness must be sufficient to withstand the dead-weight loads and bending moments imposed on it by the vessel; it will not be under the vessel pressure.

The resultant stresses in the skirt will be:

$$\sigma_s \text{ (tensile)} = \sigma_{bs} - \sigma_{ws} \quad (14.41)$$

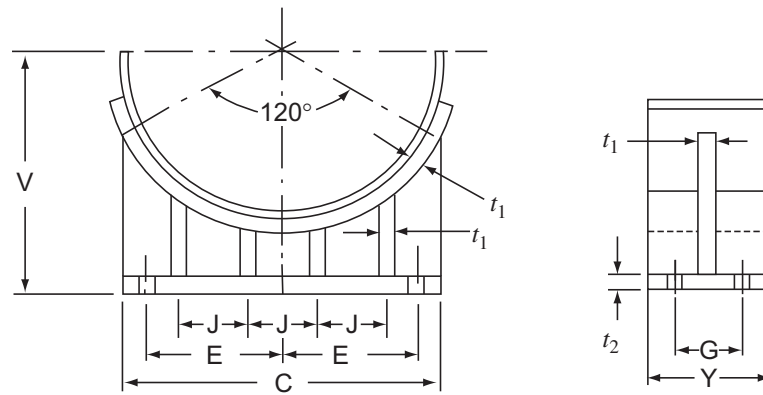
and

$$\sigma_s \text{ (compressive)} = \sigma_{bs} + \sigma_{ws} \quad (14.42)$$

where

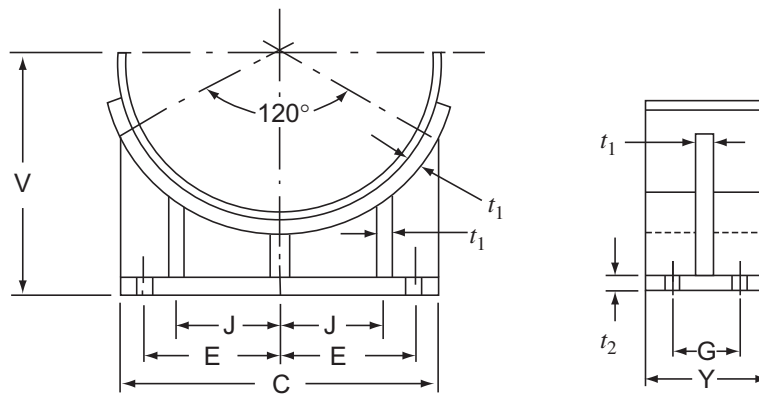
σ_{bs} = bending stress in the skirt

$$= \frac{4 M_s}{\pi (D_s + t_{sk}) t_{sk} D_s}, \quad (14.43)$$



Vessel Diam. (m)	Maximum Weight (kN)	Dimensions (m)						mm			
		V	Y	C	E	J	G	t ₂	t ₁	Bolt Diam.	Bolt Holes
0.6	35	0.48	0.15	0.55	0.24	0.190	0.095	6	5	20	25
0.8	50	0.58	0.15	0.70	0.29	0.225	0.095	8	5	20	25
0.9	65	0.63	0.15	0.81	0.34	0.275	0.095	10	6	20	25
1.0	90	0.68	0.15	0.91	0.39	0.310	0.095	11	8	20	25
1.2	180	0.78	0.20	1.09	0.45	0.360	0.140	12	10	24	30

(a) All contacting edges fillet welded



Vessel Diam. (m)	Maximum Weight (kN)	Dimensions (m)						mm			
		V	Y	C	E	J	G	t ₂	t ₁	Bolt Diam.	Bolt Holes
1.4	230	0.88	0.20	1.24	0.53	0.305	0.140	12	10	24	30
1.6	330	0.98	0.20	1.41	0.62	0.350	0.140	12	10	24	30
1.8	380	1.08	0.20	1.59	0.71	0.405	0.140	12	10	24	30
2.0	460	1.18	0.20	1.77	0.80	0.450	0.140	12	10	24	30
2.2	750	1.28	0.225	1.95	0.89	0.520	0.150	16	12	24	30
2.4	900	1.38	0.225	2.13	0.98	0.565	0.150	16	12	27	33
2.6	1000	1.48	0.225	2.30	1.03	0.590	0.150	16	12	27	33
2.8	1350	1.58	0.25	2.50	1.10	0.625	0.150	16	12	27	33
3.0	1750	1.68	0.25	2.64	1.18	0.665	0.150	16	12	27	33
3.2	2000	1.78	0.25	2.82	1.26	0.730	0.150	16	12	27	33
3.6	2500	1.98	0.25	3.20	1.40	0.815	0.150	16	12	27	33

(b) All contacting edges fillet welded.

FIG. 14.17 Standard steel saddles. (a) For vessels up to 1.2 m. (b) For vessels greater than 1.2 m. Adapted from Bhattacharyya 2011.

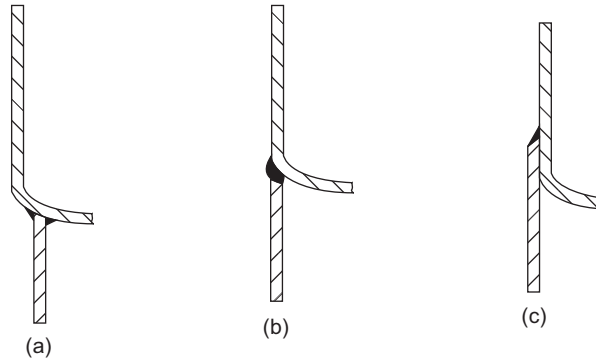


FIG. 14.18 Skirt-support welds.

σ_{ws} = the dead weight stress in the skirt,

$$= \frac{W_v}{\pi(D_s + t_{sk}) t_{sk}} \quad (14.44)$$

where

M_s = maximum bending moment, evaluated at the base of the skirt (due to wind, seismic, and eccentric loads; see Section 14.8),

W_v = total weight of the vessel and contents (see Section 14.8),

D_s = inside diameter of the skirt, at the base,

t_{sk} = skirt thickness.

The skirt thickness should be such that under the worst combination of wind and dead-weight loading, the following design criteria are not exceeded:

$$\sigma_s \text{ (tensile)} < S_s E \sin \theta_s \quad (14.45)$$

$$\sigma_s \text{ (compressive)} < 0.125 E_Y \left(\frac{t_{sk}}{D_s} \right) \sin \theta_s \quad (14.46)$$

where

S_s = maximum allowable design stress for the skirt material, normally taken at ambient temperature, 20 °C,

E = weld joint efficiency, if applicable,

θ_s = base angle of a conical skirt, normally 80 to 90 degrees.

The minimum thickness should be not less than 6 mm.

Where the vessel wall will be at a significantly higher temperature than the skirt, discontinuity stresses will be set up due to differences in thermal expansion. Methods for calculating the thermal stresses in skirt supports are given by [Weil and Murphy \(1960\)](#) and [Bergman \(1963\)](#).

Base ring and anchor bolt design

The loads carried by the skirt are transmitted to the foundation slab by the skirt base ring (bearing plate). The moment produced by wind and other lateral loads will tend to overturn the vessel; this will be opposed by the couple set up by the weight of the vessel and the tensile load in the anchor bolts. Various base ring designs are used with skirt supports. The simplest types, suitable for small vessels, are the rolled angle and plain flange rings shown in [Fig. 14.19](#) (a) and (b). For larger columns, a double ring stiffened by gussets ([Fig. 14.19c](#)) or chair supports are used. Design methods for base rings, and methods for sizing the anchor bolts, are given by [Brownell and Young \(1959\)](#). For preliminary design, the shortcut method and nomographs given by [Scheiman \(1963\)](#) can be used. Scheiman's method is based on a more detailed procedure for the design of base rings and foundations for columns and stacks given by [Marshall \(1958\)](#).

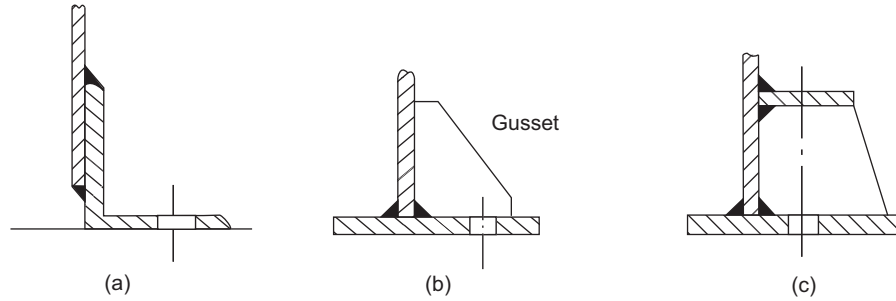


FIG. 14.19 Flange ring designs. (a) Rolled angle. (b) Single plate with gusset. (c) Double plate with gusset.

Example 14.3

Design a skirt support for the column specified in Example 14.2.

Solution

Try a straight cylindrical skirt ($\theta_s = 90^\circ$) of plain carbon steel, maximum allowable stress 89 N/mm^2 , and Young's modulus $200,000 \text{ N/mm}^2$ at ambient temperature.

The maximum dead weight load on the skirt will occur when the vessel is full of water.

$$\begin{aligned}
 \text{Approximate weight} &= \left(\frac{\pi}{4} \times 2^2 \times 50 \right) 1000 \times 9.81 \\
 &= 1,540,951 \text{ N} \\
 &= 1541 \text{ kN} \\
 \text{Weight of vessel from Example 14.2} &= 842 \text{ kN} \\
 \text{Total weight} &= 1541 + 842 = 2383 \text{ kN} \\
 \text{Wind loading, from Example 14.2} &= 2.79 \text{ kN/m} \\
 \text{Bending moment at base of skirt} &= 2.79 \times \frac{53^2}{2} \\
 &= 3919 \text{ kNm}
 \end{aligned} \tag{14.34}$$

As a first trial, take the skirt thickness as the same as that of the bottom section of the vessel, 18 mm.

$$\begin{aligned}
 \sigma_{bs} &= \frac{4 \times 3919 \times 10^3 \times 10^3}{\pi(2000 + 18) 2000 \times 18} \\
 &= 68.7 \text{ N/mm}^2
 \end{aligned} \tag{14.43}$$

$$\sigma_{ws} \text{ (test)} = \frac{2383 \times 10^3}{\pi(2000 + 18) 18} = 20.9 \text{ N/mm}^2 \tag{14.43}$$

$$\sigma_{ws} \text{ (operating)} = \frac{842 \times 10^3}{\pi(2000 + 18) 18} = 7.4 \text{ N/mm}^2 \tag{14.44}$$

Note: the "test" condition is with the vessel full of water for the hydraulic test. In estimating total weight, the weight of liquid on the plates has been counted twice. The weight has not been adjusted to allow for this, as the error is small and on the "safe side."

$$\text{Maximum } \hat{\sigma}_s \text{ (compressive)} = 68.7 + 20.9 = 89.6 \text{ N/mm}^2 \tag{14.42}$$

$$\text{Maximum } \hat{\sigma}_s \text{ (tensile)} = 68.7 - 7.4 = 61.3 \text{ N/mm}^2 \tag{14.41}$$

Take the joint efficiency E as 0.85.

Criteria for design:

$$\hat{\sigma}_s \text{ (tensile)} < S_s E \sin \theta \quad (14.45)$$

$$61.3 < 0.85 \times 89 \sin 90$$

$$61.3 < 75.6$$

$$\hat{\sigma}_s \text{ (compressive)} < 0.125 E_Y \left(\frac{t_{sk}}{D_s} \right) \sin \theta \quad (14.46)$$

$$89.6 < 0.125 \times 200,000 \left(\frac{18}{2000} \right) \sin 90$$

$$89.6 < 225$$

Both criteria are satisfied; add 2 mm for corrosion gives a design thickness of 20 mm.

14.9.3 Bracket supports

Brackets, or lugs, can be used to support vertical vessels. The bracket may rest on the building structural steel work, or the vessel may be supported on legs (see Fig. 14.16).

The main load carried by the brackets will be the weight of the vessel and contents; in addition, the bracket must be designed to resist the load due to any bending moment due to wind or other loads. If the bending moment is likely to be significant, skirt supports should be considered in preference to bracket supports.

As the reaction on the bracket is eccentric (Fig. 14.20), the bracket will impose a bending moment on the vessel wall. The point of support, at which the reaction acts, should be made as close to the vessel wall as possible, allowing for the thickness of any insulation. Methods for estimating the magnitude of the stresses induced in the vessel wall by bracket supports are given by Brownell and Young (1959) and by Wolosewick (1951). Backing plates or collars are often used to carry the bending loads.

The brackets and supporting steel work can be designed using the usual methods for structural steelwork. Suitable methods are given by Bednar (1986) and Moss and Basic (2013). A quick method for sizing vessel reinforcing rings (backing plates) for bracket supports is given by Mahajan (1977).

Typical bracket designs are shown in Fig. 14.21(a) and (b). The loads that steel brackets with these proportions will support are given by the following formula:

Single-gusset plate design (see Fig. 14.21a):

$$F_{bs} = 60 L_d t_c \quad (14.47)$$

Double-gusset plate design (see Fig. 14.21b):

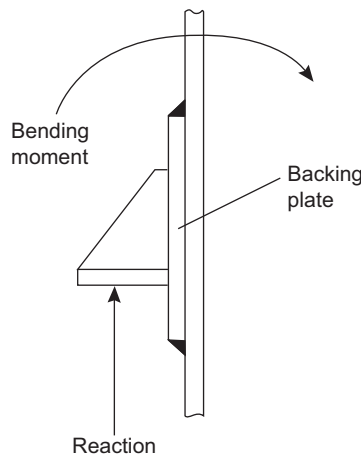


FIG. 14.20 Loads on a bracket support.

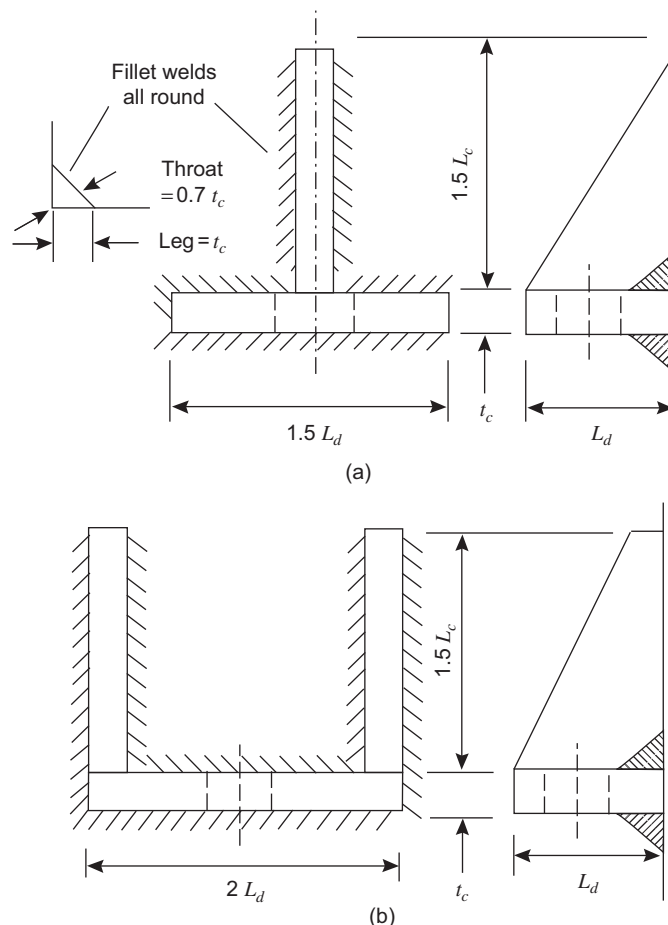


FIG. 14.21 Bracket designs. (a) Single gusset plate. (b) Double gusset plate.

$$F_{bs} = 120L_d t_c \quad (14.48)$$

where F_{bs} = maximum design load per bracket, N

L_d = the characteristic dimension of bracket (depth), mm

t_c = thickness of plate, mm

14.10 Bolted flanged joints

Flanged joints are used for connecting pipes and instruments to vessels, for manhole covers, and for removable vessel heads when ease of access is required. Flanges may also be used on the vessel body when it is necessary to divide the vessel into sections for transport or maintenance. Flanged joints are also used to connect pipes to other equipment, such as pumps and valves. Screwed joints are often used for small-diameter pipe connections, below 2 in (50 mm). Flanged joints are also used for connecting pipe sections where ease of assembly and dismantling is required for maintenance, but pipework will normally be welded to reduce costs.

Flanges range in size from a few millimeters in diameter for small pipes to several meters diameter for those used as body or head flanges on vessels.

14.10.1 Types of flanges and selection

Several different types of flanges are used for various applications. The principal types used in the process industries are:

1. Welding-neck flanges
2. Slip-on flanges, hub and plate types
3. Lap-joint flanges
4. Screwed flanges
5. Blank, or blind, flanges

Welding-neck flanges (Fig. 14.22a) have a long tapered hub between the flange ring and the welded joint. This gradual transition of the section reduces the discontinuity stresses between the flange and branch and increases the strength of the flange assembly. Welding-neck flanges are suitable for extreme service conditions, where the flange is likely to be subjected to temperature, shear, and vibration loads. They will normally be specified for the connections and nozzles on process vessels and process equipment.

Slip-on flanges (Fig. 14.22b) slip over the pipe or nozzle and are welded externally and usually also internally. The end of the pipe is set back from 0 to 2.0 mm. The strength of a slip-on flange is from one-third to two-thirds that of the corresponding standard welding-neck flange. Slip-on flanges are cheaper than welding-neck flanges and are easier to align but have poor resistance to shock and vibration loads. Slip-on flanges are generally used for pipe work. Fig. 14.22(b) shows a forged flange with a hub; for light duties, slip-on flanges can be cut from plate.

Lap-joint flanges (Fig. 14.22c) are used for pipework. They are economical when used with expensive alloy pipe, such as stainless steel, as the flange can be made from inexpensive carbon steel. Usually a short-lapped nozzle is welded to the pipe, but with some schedules of pipe, the lap can be formed on the pipe itself; this will give a cheap method of pipe assembly. Lap-joint flanges are sometimes known as "Van-stone flanges."

Screwed flanges (Fig. 14.22d) are used to connect screwed fittings to flanges. They are also sometimes used for alloy pipe that is difficult to weld satisfactorily.

Blind flanges (blank flanges) are flat plates used to blank off flange connections and as covers for manholes and inspection ports.

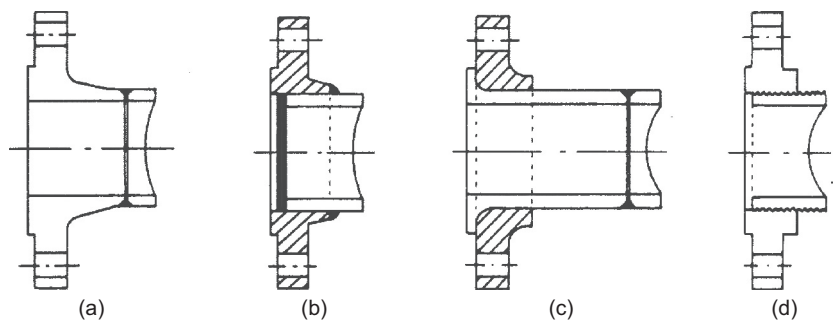


FIG. 14.22 Flange types. (a) Welding neck. (b) Slip-on. (c) Lap joint. (d) Screwed.

14.10.2 Gaskets

Gaskets are used to make a leak-tight joint between two surfaces. It is impractical to machine flanges to the degree of surface finish that would be required to make a satisfactory seal under pressure without a gasket. Gaskets are made from “semi-plastic” materials that will deform and flow under load to fill the surface irregularities between the flange faces yet retain sufficient elasticity to take up the changes in the flange alignment that occur under load.

A great variety of proprietary gasket materials is used, and reference should be made to the manufacturers’ catalogues and technical manuals when selecting gaskets for a particular application. Design data for some of the more commonly used gasket materials are given in [Table 14.4](#). Further data can be found in the ASME BPV Code Sec. VIII D.1 Mandatory Appendix 2, ASME B16.20 and in [Green and Southard \(2018\)](#). The minimum seating stress y is the force per unit area (pressure) on the gasket that is required to cause the material to flow and fill the surface irregularities in the gasket face.

The gasket factor m is the ratio of the gasket stress (pressure) under the operating conditions to the internal pressure in the vessel or pipe. The internal pressure will force the flanges’ faces apart, so the pressure on the gasket under operating conditions will be lower than the initial tightening-up pressure. The gasket factor gives the minimum pressure that must be maintained on the gasket to ensure a satisfactory seal.

The following factors must be considered when selecting a gasket material:

1. The process conditions: pressure, temperature, corrosive nature of the process fluid
2. Whether repeated assembly and disassembly of the joint are required
3. The type of flange and flange face (see Section 14.10.3)

Up to pressures of 20 bar, the operating temperature and corrosiveness of the process fluid will be the controlling factor in gasket selection. Vegetable fiber and synthetic rubber gaskets can be used at temperatures of up to 100 °C. Solid polyfluorocarbon (Teflon) and compressed asbestos gaskets can be used to a maximum temperature of about 260°C. Metal-reinforced gaskets can be used up to around 450 °C. Plain soft metal gaskets are normally used for higher temperatures.

14.10.3 Flange faces

Flanges are also classified according to the type of flange face used. There are two basic types:

1. Full-faced flanges ([Fig. 14.23a](#)), where the face contact area extends outside the circle of bolts over the full face of the flange.
2. Narrow-faced flanges ([Fig. 14.23b, c, and d](#)), where the face contact area is located within the circle of bolts.

Full face, wide-faced flanges are simple and inexpensive, but are only suitable for low pressures. The gasket area is large, and an excessively high bolt tension would be needed to achieve sufficient gasket pressure to maintain a good seal at high operating pressures.

The raised-face, narrow-faced flange shown in [Fig. 14.23\(b\)](#) is probably the most commonly used type of flange for process equipment.

Where the flange has a plain face, as in [Fig. 14.23\(b\)](#), the gasket is held in place by friction between the gasket and flange surface. In the spigot and socket and tongue and grooved faces (see [Fig. 14.23c](#)), the gasket is confined in a groove, which prevents failure by “blow-out.” Matched pairs of flanges are required, which increases the cost, but this type is suitable for high-pressure and high-vacuum service. Ring joint flanges ([Fig. 14.23d](#)) are used for high-temperature and high-pressure services.

14.10.4 Flange design

Standard flanges will be specified for most applications (see Section 14.10.5). Special designs would be used only if no suitable standard flange were available or for large flanges, such as the body flanges of vessels, where it may be cheaper to size a flange specifically for the duty required rather than to accept the nearest standard flange, which of necessity would be oversized.

[Fig. 14.24](#) shows the forces acting on a flanged joint. The bolts hold the faces together, resisting the forces due to the internal pressure and the gasket sealing pressure. As these forces are offset, the flange is subjected to a bending moment. It can be considered a cantilever beam with a concentrated load. A flange assembly must be sized so as to have sufficient strength and rigidity to resist this bending moment. A flange that lacks sufficient rigidity will rotate

TABLE 14.4 Gasket materials (based on Table 2-5.1 in ASME BPV Code Sec. VIII D.1 Mandatory Appendix 2 and a similar table in BS 5500—2003)

Gasket Material		Gasket Factor m	Min.Design Seating Stress y (N/mm ²)	Sketches	Minimum Gasket Width (mm)	
Rubber without fabric or a high percentage of asbestos fiber; hardness:						
below 75° IRH		0.50	0		10	
75° IRH or higher		1.00	14			
Asbestos with a suitable binder for the operating conditions	{ 3.2 mm thick	2.00	11.0			
		1.6 mm thick	2.75		25.5	
		0.8 mm thick	3.50	44.8		
Rubber with cotton fabric insertion		1.25	2.8		10	
Rubber with asbestos fabric insertion, with or without wire reinforcement	{ 3-ply	2.25	15.2		10	
		2-ply	2.50	20.0		
		1-ply	2.75	25.5		
Vegetable fiber		1.75	7.6		10	
Spiral-wound metal, asbestos filled	{ Carbon	2.50	20.0		10	
		Stainless steel	3.00			31.0
Corrugated metal, asbestos inserted or	Soft aluminum	2.50	20.0		10	
	Soft copper or brass	2.75	25.5			
	Iron or soft steel	3.00	31.0			
Corrugated metal, jacketed asbestos filled	Monel or 4 to 6% chrome	3.25	37.9			
	Stainless steels	3.50	44.8			
	Soft aluminum	2.75	25.5			
	Soft copper or brass	3.00	31.0			
	Iron or soft steel	3.25	37.9			
Corrugated metal	Monel or 4 to 6% Chrome	3.50	44.8		10	
	Stainless steels	3.75	52.4			
	Soft aluminum	3.25	37.9			
	Soft copper or brass	3.50	44.8			
	Iron or soft steel	3.75	52.4			
Flat metal jacketed asbestos filled	Monel	3.50	55.1		10	
	4 to 6% chrome	3.75	62.0			
	Stainless steels	3.75	62.0			
	Soft aluminum	3.25	37.9			
	Soft copper or brass	3.50	44.8			
Grooved metal	Iron or soft steel	3.75	52.4		10	
	Monel or 4 to 6% chrome	3.75	62.0			
	Stainless steels	4.25	69.5			
	Soft aluminum	4.00	60.6			
	Soft copper or brass	4.75	89.5			
Solid flat metal	Iron or soft steel	5.50	124		6	
	Monel or 4 to 6% chrome	6.00	150			
Ring joint	Stainless steels	6.50	179		6	
	Iron or soft steel	5.50	124			
	Monel or 4 to 6% chrome	6.00	150			
	Stainless steels	6.50	179			

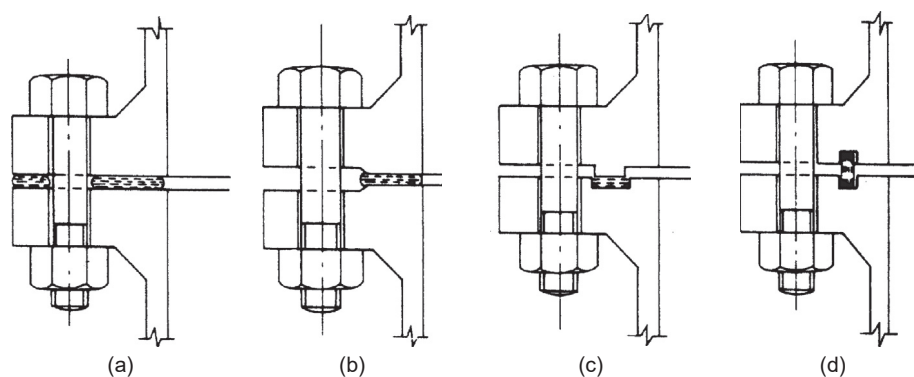


FIG. 14.23 Flange types and faces. (a) Full face. (b) Gasket within bolt circle. (c) Spigot and socket. (d) Ring-type joint.

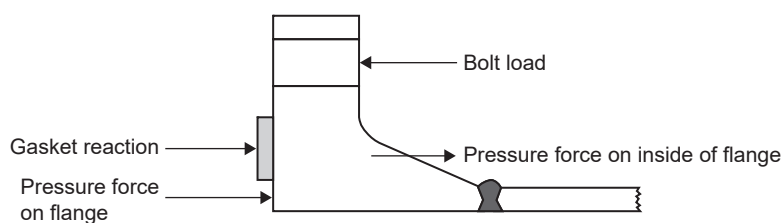


FIG. 14.24 Forces acting on an integral flange.

slightly, and the joint will leak (Fig. 14.25). The principles of flange design are discussed by Singh and Soler (1992) and Azbel and Cheremisinoff (1982). Singh and Soler give a computer program for flange design. Design procedures for pressure vessel flanges are given in ASME BPV Code Sec VIII D.1 Mandatory Appendix 2.

For design purposes, flanges are classified as integral or loose flanges. Integral flanges are those in which the construction is such that the flange obtains support from its hub and the connecting nozzle (or pipe). The flange assembly and nozzle neck form an “integral” structure. A welding-neck flange would be classified as an integral flange. Loose flanges are attached to the nozzle (or pipe) in such a way that they obtain no significant support from the nozzle neck and cannot be classified as an integral attachment. Screwed and lap-joint flanges are typical examples of loose flanges.

The number of bolts and the bolt size must be chosen such that the bolt load is less than the maximum allowable stress in the bolts. The bolt spacing must be selected to give a uniform compression of the gasket. It will not normally be less than 2.5 times the bolt diameter to give sufficient clearance for tightening with a wrench. The following formula can be used to determine the maximum bolt spacing:

$$p_b = 2d_b + \frac{6t_f}{(m + 0.5)} \quad (14.49)$$

where p_b = bolt pitch (spacing), mm,
 d_b = bolt diameter, mm,
 t_f = flange thickness, mm,
 m = gasket factor.

Bolting requirements are given in ASME B16.5. An overview of the B16.5 requirements is given by Stikvoort (2019).

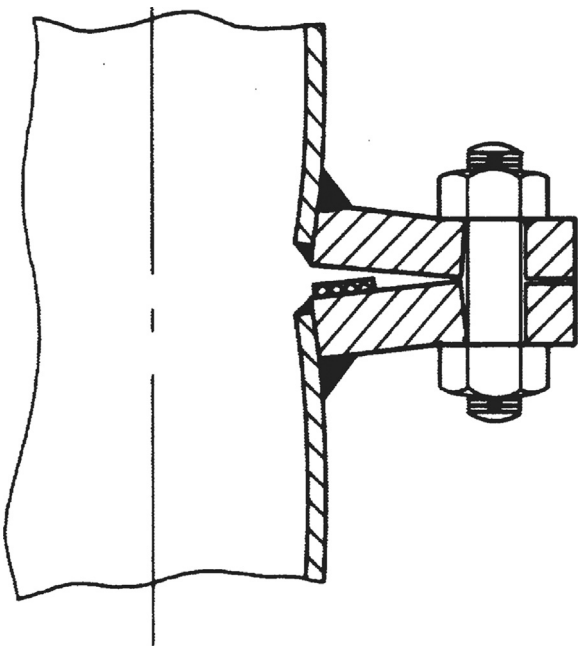
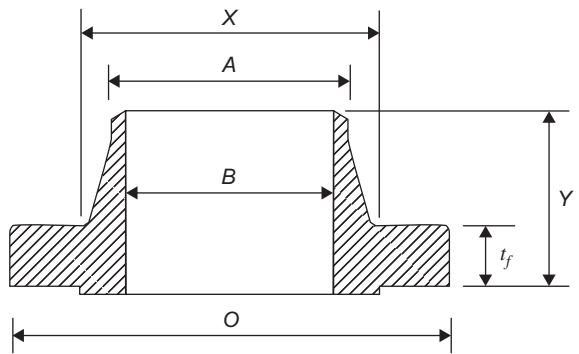


FIG. 14.25 Deflection of a weak flange (exaggerated).



Flange class	Nominal pipe size	Outside diameter of flange, O	Thickness of flange, t_f	Diameter of hub, X	Diameter beginning of chamfer, A	Length through hub, Y	Bore, B
150	1.00	4.25	0.50	1.94	1.32	2.12	1.05
	2.00	6.00	0.69	3.06	2.38	2.44	2.07
	4.00	9.00	0.88	5.31	4.50	2.94	4.03
	6.00	11.00	0.94	7.56	6.63	3.44	6.07
	8.00	13.50	1.06	9.69	8.63	3.94	7.98
	12.00	19.00	1.19	14.38	12.75	4.44	12.00
	24.00	32.00	1.81	26.12	24.00	5.94	TBS
300	1.00	4.88	0.62	2.12	1.32	2.38	1.05
	2.00	6.50	0.81	3.31	2.38	2.69	2.07
	4.00	10.00	1.19	5.75	4.50	3.32	4.03
	6.00	12.50	1.38	8.12	6.63	3.82	6.07
	8.00	15.00	1.56	10.25	8.63	4.32	7.98
	12.00	20.50	1.94	14.75	12.75	5.06	12.00
	24.00	36.00	2.69	27.62	24.00	6.56	TBS

Note: TBS = To be specified by purchaser.

FIG. 14.26 Standard flange dimensions for welding neck flanges based on ASME B16.5 Annex F.

14.10.5 Standard flanges

Standard flanges are available in a range of types, sizes, and materials and are used extensively for pipes, nozzles, and other attachments to pressure vessels.

Standards for flanges and pipe fittings are set by the ASME B16 committee. These include:

ASME B16.5	Pipe flanges and flanged fittings
ASME B16.9	Factory-made wrought butt welding fittings
ASME B16.11	Forged fittings, socket-welding and threaded
ASME B16.15	Cast bronze threaded fittings
ASME B16.24	Cast copper alloy pipe flanges and flanges fittings
ASME B16.42	Ductile iron pipe flanges and flanges fittings
ASME B16.47	Large diameter steel flanges

An abstract of the American standards is given by [Green and Southard \(2018\)](#).

A typical example of a standard flange design is shown in [Fig. 14.26](#). This was based on information in ASME B16.5 Annex F.

Standard flanges are designated by class numbers, or rating numbers, which roughly correspond to the primary service (pressure) rating of a steel flange of those dimensions at room temperature.

The flange class number required for a particular application will depend on the design pressure and temperature and the material of construction. The reduction in strength at elevated temperatures is allowed for by selecting a flange with a higher rating than the design pressure. For example, for a design pressure of 10 bar (150 psi), a class 150 flange would be selected for a service temperature below 300 °C, whereas for a service temperature of, say, 300 °C a 300-pound flange would be specified. A typical pressure–temperature relationship for carbon steel flanges is shown in [Table 14.5](#). Pressure–temperature ratings for a full range of materials can be obtained from the design codes.

Designs and dimensions of standard flanges over the full range of pipe sizes are given in ASME B16.5 Annex F. A summary of flange dimensions is given by [Green and Southard \(2018\)](#), which can be used for preliminary designs. The current standards and suppliers' catalogues should be consulted before firming up the design.

TABLE 14.5 Typical pressure–temperature ratings for carbon steel flanges, A350, A515, A516

Temperature	Working pressure by flange class (psig)						
	150	300	400	600	900	1500	2500
–20 to 100	285	740	985	1480	2220	3705	6170
200	260	680	905	1360	2035	3395	5655
300	230	655	870	1310	1965	3270	5450
400	200	635	845	1265	1900	3170	5280
500	170	605	805	1205	1810	3015	5025
600	140	570	755	1135	1705	2840	4730
700	110	530	710	1060	1590	2655	4425
800	80	410	550	825	1235	2055	3430

Adapted from ASME B16.5 Annex F Table F2-1.1.

14.11 Welded joint design

Process vessels are built up from preformed parts: cylinders, heads, and fittings, joined by fusion welding. Riveted construction was used extensively in the past (before the 1940s) but is now rarely seen except on very old plants.

Cylindrical sections are usually made up from plate sections rolled to the required curvature. The sections (strakes) are made as large as is practicable to reduce the number of welds required. The longitudinal welded seams are offset to avoid a conjunction of welds at the corners of the plates.

Many different forms of welded joints are needed in the construction of a pressure vessel. Some typical forms are shown in Figs. 14.27 to 14.29.

The design of a welded joint should satisfy the following basic requirements:

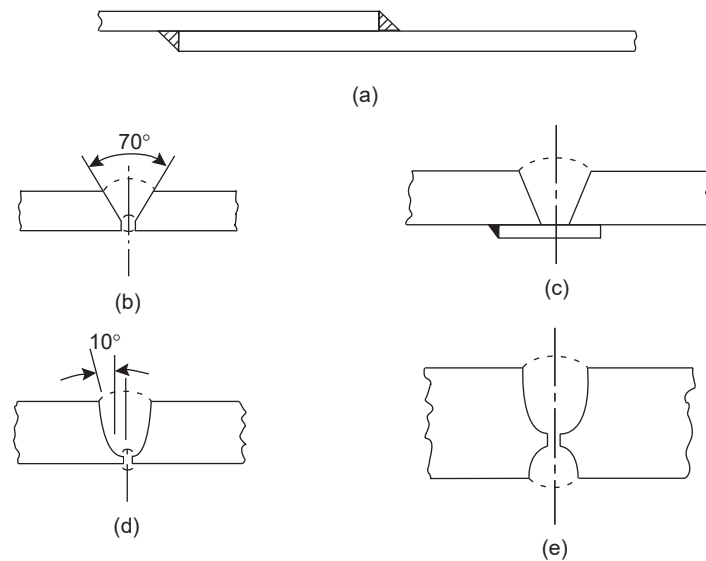


FIG. 14.27 Weld profiles (b to e) butt welds. (a) Lap joint. (b) Single "V." (c) Backing strip. (d) Single "U." (e) Double "U."

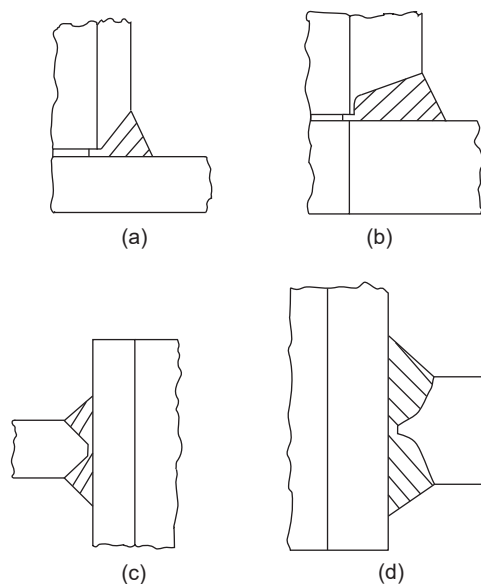


FIG. 14.28 Typical weld profiles. (a), (b) Branches. (c) Set-on branches. (d) Set-in branches.

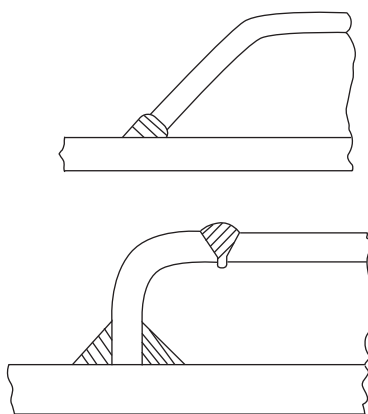


FIG. 14.29 Typical construction methods for welded jackets.

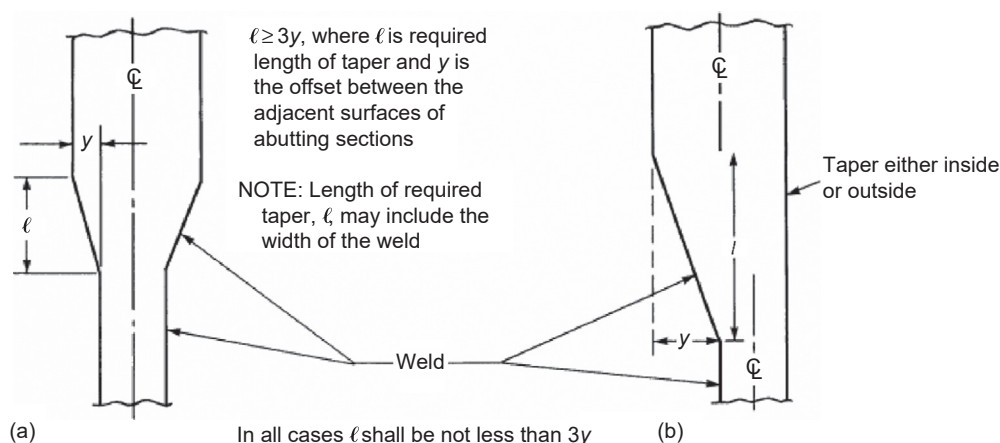


FIG. 14.30 Butt welding of plates of unequal thickness. Reprinted from ASME 2004 BPVC, Section VIII-Div.1 by permission of the American Society of Mechanical Engineers. All rights reserved.

1. Give good accessibility for welding and inspection.
2. Require the minimum amount of weld metal.
3. Give good penetration of the weld metal, from both sides of the joint, if practicable.
4. Incorporate sufficient flexibility to avoid cracking due to differential thermal expansion.

The preferred types of joints, and recommended designs and profiles, are given in the codes and standards. See, for example, ASME BPV Code Sec. VIII D.1, Part UW – Requirements for pressure vessels fabricated by welding.

The correct form to use for a given joint will depend on the material, the method of welding (machine or hand), the plate thickness, and the service conditions. Double-sided V- or U-sections are used for thick plates and single V- or U-profiles for thin plates. A backing strip is used where it is not possible to weld from both sides. Lap joints are seldom used for pressure vessel construction, but are used for atmospheric pressure storage tanks.

Where butt joints are made between plates of different thicknesses, the thicker plate is reduced in thickness with a slope of not greater than 1 in 3 (19 degrees) (ASME BPV Code Sec. VIII D.1 Part UW-9, shown in Fig. 14.30).

The local heating, and consequent expansion that occurs during welding can leave the joint in a state of stress. These stresses are relieved by postwelding heat treatment. Not all vessels will be stress relieved. Guidance on the need for postwelding heat treatment is given in ASME BPV Code Sec. VIII D.1 Part UW-40 and will depend on the service and conditions, materials of construction, and plate thickness.

To ensure that a satisfactory quality of welding is maintained, welding machine operators and welders working on the pressure parts of vessels are required to pass welder approval tests, which are designed to test their competence to make sound welds. Welding and brazing qualifications are discussed in Section IX of the ASME BPV Code.

14.12 Fatigue assessment of vessels

During operation, the shell or components of the vessel may be subjected to cyclic stresses. Stress cycling can arise from the following causes:

1. Periodic fluctuations in operating pressure
2. Temperature cycling
3. Vibration
4. "Water hammer"
5. Fluctuations in the flow of fluids or solids
6. Periodic fluctuation of external loads

A detailed fatigue analysis is required if any of these conditions is likely to occur to any significant extent. Fatigue failure will occur during the service life of the vessel if the endurance limit (number of cycles for failure) at the particular value of the cyclic stress is exceeded. The codes and standards should be consulted to determine when a detailed fatigue analysis must be undertaken.

14.13 Pressure tests

The pressure vessel codes and standards require that all pressure vessels be subjected to a pressure test to prove the integrity of the finished vessel (ASME BPV Code Sec. VIII D.1 Part UG-99). A hydraulic test is normally carried out, but a pneumatic test can be substituted under circumstances where the use of a liquid for testing is not practical. Hydraulic tests are safer because only a small amount of energy is stored in the compressed liquid. A standard pressure test is used when the required thickness of the vessel parts can be calculated in accordance with the particular code or standard. The vessel is tested at a pressure 30% above the design pressure. The test pressure is adjusted to allow for the difference in strength of the vessel material at the test temperature compared with the design temperature and for any corrosion allowance.

Formulae for determining the appropriate test pressure are given in the codes and standards; typically:

$$\text{Test pressure} = 1.30 \left[P_d \frac{S_a}{S_n} \times \frac{t}{(t-c)} \right] \quad (14.50)$$

where P_d = design pressure, N/mm²

S_a = maximum allowable stress at the test temperature, N/mm²

S_n = maximum allowable stress at the design temperature, N/mm²

c = corrosion allowance, mm

t = actual plate thickness, mm

When the required thickness of the vessel component parts cannot be determined by calculation in accordance with the methods given, the ASME BPV Code requires that a hydraulic proof test be carried out (Sec. VIII D.1 Part UG-101). In a proof test the stresses induced in the vessel during the test are monitored using strain gauges or similar techniques. In a proof test a duplicate of the vessel or part is tested until the part yields or bursts. The requirements for the proof testing of vessels are set out in ASME BPV Code Sec. VIII D.1 Part UG-101.

14.14 High-pressure vessels

High pressures are required for many commercial chemical processes. For example, the synthesis of ammonia is carried out at reactor pressures of up to 1000 bar, and high-density polyethylene processes operate up to 1500 bar.

Although there is no prescribed upper limit on pressure for vessels designed in accordance with ASME BPV Code Section VIII Division 1, the rules given in that section of the code usually cannot be economically satisfied for vessels designed to operate above 3000 psia (200 bar). For pressures greater than about 2000 psia, the alternative rules given in Section VIII Division 2 will usually lead to a more economical design. Division 2 restricts the materials that can be used, the allowable operating temperatures (not greater than 900 °F) and places stricter requirements on stress analysis and testing. The additional engineering and design costs are usually justified for high-pressure vessels because the Division 2 rules allow higher maximum allowable stresses and hence lead to thinner-walled vessels.

At the highest operating pressures, typically above 10,000 psia (680 bar), the alternative design rules given in ASME BPV Code Section VIII Division 3 can be followed.

A full discussion of the design and construction of high-pressure vessels and ancillary equipment (pumps, compressors, valves, and fittings) is given in the books by [Fryer and Harvey \(1997\)](#) and [Jawad and Farr \(2018\)](#). At high pressures it becomes increasingly difficult to fabricate single-walled vessels with sufficient strength because of the wall thickness and depth of welds required. Instead, compound vessels with several layers of vessel walls are often used. In a compound vessel, the outer layers can be used to place the inner layers in compression during manufacture, and hence offset the tensile forces that will act most strongly on the inner layers during operation.

14.14.1 Compound vessels

Shrink-fitted cylinders

Compound vessels can be made by shrinking one cylinder over another. The inside diameter of the outer cylinder is made slightly smaller than the outer diameter of the inner cylinder and is expanded by heating to fit over the inner. On cooling, the outer cylinder contracts and places the inner under compression. The stress distribution in a two-cylinder compound vessel is shown in [Fig. 14.31](#); more than two cylinders may be used.

Shrink-fitted compound cylinders are used for small-diameter vessels, such as compressor cylinder barrels. The design of shrink-fitted compound cylinders is discussed by [Manning \(1947\)](#) and [Jawad and Farr \(2018\)](#).

Multilayer vessels

Multilayer vessels are made by wrapping several layers of relatively thin plate round a central tube. The plates are heated, tightened, and welded, and this gives the desired stress distribution in the compound wall. The vessel is closed with forged heads. A typical design is shown in [Fig. 14.32](#). This construction technique is discussed by [Jasper and Scudder \(1941\)](#) and [Jawad and Farr \(2018\)](#).

Wound vessels

Cylindrical vessels can be reinforced by winding on wire or thin ribbons. Winding on the wire under tension places the cylinder under compression. For high-pressure vessels special interlocking strips are used, such as those shown in [Fig. 14.33](#). The interlocking gives strength in the longitudinal direction and a more uniform stress distribution. The strips may be wound on hot to increase the pre-stressing. This type of construction is described by [Birchall and Lake \(1947\)](#).

14.14.2 Autofrettage

Autofrettage is a technique used to prestress the inner part of the wall of a monobloc vessel to give a similar stress distribution to that obtained in a shrink-fitted compound cylinder. The finished vessel is deliberately overpressurized by hydraulic pressure. During this process, the inner part of the wall will be more highly stressed than the outer part and will undergo plastic strain. On release of the “autofrettage” pressure, the inner part, which is now oversized, will be placed under compression by the elastic contraction of the outer part, which gives a residual stress distribution similar to that obtained in a two-layer, shrink-fitted compound cylinder. After straining, the vessel is annealed at a relatively low temperature, approximately 300 °C. The straining also work-hardens the inner part of the wall. The vessel can be used at pressures up to the “autofrettage” pressure without further permanent distortion. The autofrettage technique is discussed by [Manning \(1950\)](#) and [Jawad and Farr \(2018\)](#).

Requirements for pressure vessels fabricated by layered construction are given in ASME BPV Code Sec. VIII D.1 Part ULW and Sec. VIII D.2 Articles D-11 and F-8.

14.15 Liquid storage tanks

Vertical cylindrical tanks, with flat bases and conical roofs, are universally used for the bulk storage of liquids at atmospheric pressure. Tank sizes vary from a few hundred gallons (tens of cubic meters) to several thousand gallons (several hundred cubic meters).

The main load to be considered in the design of these tanks is the hydrostatic pressure of the liquid, but the tanks must also be designed to withstand wind loading and, for some locations, the weight of snow on the tank roof.

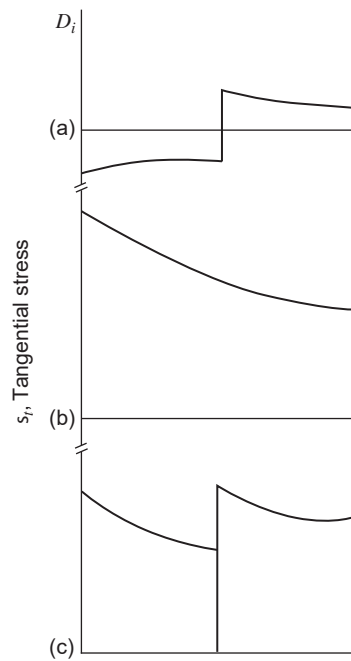


FIG. 14.31 Stress distribution in a shrink-fitted compound cylinder. (a) Due to shrinkage. (b) Due to pressure. (c) Combined (a + b).

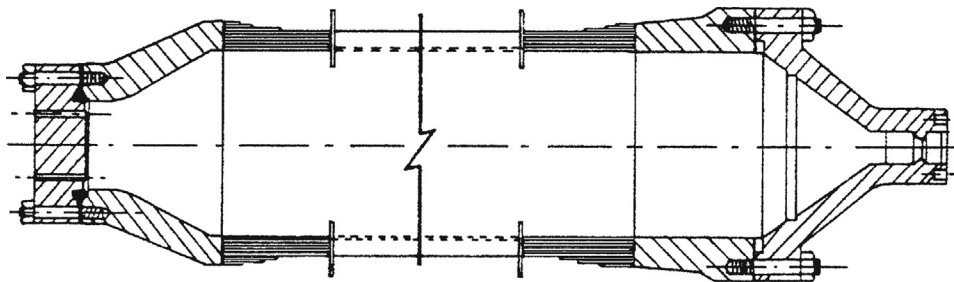


FIG. 14.32 Multilayer construction.

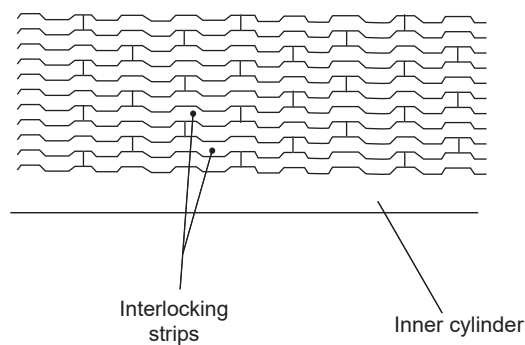


FIG. 14.33 Strip-wound vessel.

The minimum wall thickness required to resist the hydrostatic pressure can be calculated from the equations for the membrane stresses in thin cylinders (Section 14.3.4):

$$t_t = \frac{\rho_L H_L g}{2S_t E} \frac{D_t}{10^3} \quad (14.51)$$

where t_t = tank thickness required at depth H_L , mm

H_L = liquid depth, m

ρ_L = liquid density, kg/m³

E = joint efficiency (if applicable)

g = gravitational acceleration, 9.81 m/s²

S_t = maximum allowable stress for tank material, N/mm²

D_t = tank diameter, m

The liquid density should be taken as that of water (1000 kg/m³), unless the process liquid has a greater density.

For small tanks, a constant wall thickness would normally be used, calculated at the maximum liquid depth. With large tanks, it is economical to take account of the variation in hydrostatic pressure with depth by increasing the plate thickness progressively from the top to bottom of the tank. Plate widths of 2 m (6 ft) are typically used in tank construction.

The roofs of large tanks need to be supported by a steel framework supported on columns in very large-diameter tanks.

The design and construction of atmospheric storage tanks for the petroleum industry are covered by the American Petroleum Industry standards API 650 (2003) and 620 (2002). Other standards are also used internationally, for example, British Standard BS 2654 and BS EN 14015 for larger steel tanks and European Standard EN 13121 for glass-reinforced polymer tanks. The design of storage tanks is covered in the books by Myers (1997) and Jawad and Farr (2018). See also the papers by Debham et al. (1968) and Zick and McGrath (1968).

14.16 Capital cost of pressure vessels

The purchased equipment cost of a pressure vessel depends strongly on the complexity of the design. Preliminary costing methods will usually only generate very approximate answers, and consultation with a vessel manufacturer will be needed to firm up the estimate. Pressure vessels are often the most expensive pieces of equipment in a chemical plant. Vessels for high pressures (above 100 bar) will often be designed in accordance with the ASME BPV Code Section VIII Div.2 and may have long lead times because of the relative scarcity of fabricators that are qualified for Division 2 construction.

Most preliminary cost correlations for pressure vessels are based on the shell mass, so it is necessary to know the wall thickness as well as the general dimensions of the vessel. Wall thickness can be estimated using the BPV Code equations given in Section 14.5 and the combined loading analysis given in Section 14.8. It is very important to specify the design pressure and temperature correctly in these calculations. The choice of material of construction plays a key role in determining the wall thickness (through the maximum allowable stress and corrosion allowance), as well as affecting the cost through the materials cost factor. Preliminary cost correlations are discussed in Section 7.5.2, and correlations for pressure vessel costs are given in Table 7.2. The use of these correlations for costing horizontal and vertical pressure vessels is illustrated in Example 7.3.

Simple cost correlations cannot address any complexity in the design of a pressure vessel. If the vessel contains substantial internal features, has a large number of ports or manifolds, or requires stiffening rings to handle vacuum operation, then costs for these features must be added to the bare vessel cost.

More accurate cost estimates for pressure vessels are made by determining the parts and materials cost and fabrication work breakdown structure, and then assigning costs based on estimates of the labor and machine time required for vessel construction; see Section 7.5.3. This is the method used in the Aspen Capital Cost Estimator cost estimating software. Reasonably good estimates of pressure vessel costs can be obtained using commercial cost estimating programs if the design engineer is careful in specifying the vessel design conditions.

14.17 References

- Azbel, D. S., & Cheremisinoff, N. P. (1982). *Chemical and process equipment design: Vessel design and selection*. Ann Arbor Science.
- Bednar, H. H. (1990). *Pressure vessel design handbook* (2nd ed.). Krieger.
- Beer, F. P., Johnston, E. R., DeWolf, J. T., & Mazurek, D. F. (2014). *Mechanics of materials* (7th ed.). McGraw-Hill.
- Bhattacharyya, B. C. (2011). *Introduction to chemical equipment design, mechanical aspects*. CBS.
- Bergman, D. J. (1963). Temperature gradients for skirt supports of hot vessels. *Transactions of the American Society of Mechanical Engineers. J. Eng. for Ind.*, 85, 219.
- Birchall, H., & Lake, G. F. (1947). An alternative form of pressure vessel of novel construction. *Proceedings of the Institution of Mechanical Engineers*, 56, 349.

- Brownell, L. E. (1963). Mechanical design of tall towers. *Hydrological Processes, and Pet. Ref.* 42, 109. June.
- Brownell, L. E., & Young, E. H. (1959). *Process equipment design: Vessel design*. Wiley.
- Case, J., Chilver, A. H., & Ross, C. (1999). *Strength of materials and structures*. Butterworth-Heinemann.
- Debham, J. B., Russel, J., & Wiils, C. M. R. (1968). How to design a 600,000 b.b.l. tank. *Hydrological Processes*, 47 (May), 137.
- DeGhetto, K., & Long, W. (1966). Check towers for dynamic stability. *Hydrological Processes, and Pet. Ref.* 45(Feb.), 143.
- Ellenberger, J. P., Chuse, R., & Carson, B. E. (2004). *Pressure vessels: The ASME code simplified* (8th ed.). McGraw-Hill.
- Escoc, A. K. (1994). Mechanical design of process equipment, vol. 1 (2nd ed.). Piping and pressure vessels. Gulf.
- Farr, J. R., & Jawad, M. H. (2010). *Guidebook for the design of ASME Section VIII pressure vessels* (4th ed.). ASME.
- Freese, C. E. (1959). Vibrations of vertical pressure vessels. *Transactions of the American Society of Mechanical Engineers (J. Eng. Ind.)*, 81, 77.
- Fryer, D. M., & Harvey, J. F. (1997). *High pressure vessels*. Kluwer.
- Goodno, B. J., & Gere, J. M. (2017). *Mechanics of materials* (9th ed.). Cengage Learning.
- Green, D. W., & Southard, M. Z. (Eds.). (2018). *Perry's chemical engineers' handbook* (9th ed.). McGraw-Hill.
- Harvey, J. F. (1974). *Theory and design of modern pressure vessels* (2nd ed.). Van Nostrand-Reinhold.
- Henry, B. D. (1973). The design of vertical, free standing process vessels. *Australian Chemical Engineering*, 14(Mar.), 13.
- Hetenyi, M. (1958). *Beams on elastic foundations*. University of Michigan Press.
- Hibbeler, R. C. (2016). *Mechanics of materials* (10th ed.). Pearson.
- Jasper, McL, T., & Scudder, C. M. (1941). Multi-layer construction of thick wall pressure vessels. *Transactions of the American Institute of Chemical Engineers*, 37, 885.
- Jawad, M. H., & Farr, J. R. (2018). *Structural analysis and design of process equipment* (3rd ed.). Wiley.
- Mahajan, K. K. (1977). Size vessel stiffeners quickly. *Hydrological Processes*, 56(4), 207.
- Manning, W. R. D. (1947). The design of compound cylinders for high pressure service. *Engineering*, 163(May 2), 349.
- Manning, W. R. D. (1950). The design of cylinders by autofrettage (in three parts). *Engineering*, 169(April 28), 479 (May 5), 509; (May 15), 562.
- Marshall, V. O. (1958). In , 37. *Foundation design handbook for stacks and towers*. *Pet. Ref* (May) (supplement).
- Megyesy, E. F. (2008). Pressure vessel hand book. In *Pressure Vessel* (14th ed.). Hand Book Publishers.
- Moss, D. R., & Basic, M. M. (2013). *Pressure vessel design manual* (3rd ed.). Butterworth-Heinemann.
- Mott, R. L., & Untener, J. A. (2016). *Applied strength of materials* (6th ed.). CRC Press.
- Myers, P. E. (1997). *Above ground storage tanks*. McGraw-Hill.
- Nelson, J. G. (1963). Use calculation form for tower design. *Hydrological Processes, and Pet. Ref.* 42(June), 119.
- Scheiman, A. D. (1963). Short cuts to anchor bolting and base ring sizing. *Hydrological Processes, and Pet. Ref.* 42(June), 130.
- Seed, G. M. (2001). *Strength of materials: An undergraduate text*. Paul & Co. Publishing Consortium.
- Singh, K. P., & Soler, A. I. (1992). *Mechanical design of heat exchangers and pressure vessel components*. Springer-Verlag.
- Stikvoort, W. (2019). *ASME B16.5 Flanges: bolt tightening and target loads*. 126 p. 38). New York: Chemical Engineering, 12.
- Tang, S. S. (1968). Shortcut methods for calculating tower deflections. *Hydrological Processes*, 47(Nov), 230.
- Timoshenko, S., & Gere, J. M. (2009). *Theory of elastic stability* (2nd ed.). Dover.
- Weil, N. A., & Murphy, J. J. (1960). Design and analysis of welded pressure vessel skirt supports. *Transactions of the American Society of Mechanical Engineers (J. Eng. Ind.)*, 82(Jan.), 1.
- Wolosewick, F. E. (1951). Supports for vertical pressure vessels (in four parts). *Pet. Ref.* 30(July), 137. Aug.), 101; (Oct.), 143; (Dec.), 151.
- Yokell, S. (1986). *Understanding pressure vessel codes*, 93 p. 75). New York: Chemical Engineering.
- Zick, L. P. (1951). Stresses in large horizontal cylindrical pressure vessels on two saddle supports. *Weld. J. Res. Suppl.*, 30, 435.
- Zick, L. P., & McGrath, R. V. (1968). New design approach for large storage tanks. *Hydrological Processes*, 47(May), 143.

Bibliography

Useful references on pressure vessel design

- Annaratone, D. (2007). *Pressure vessel design*. Springer.
- Budynas, R. G., & Sadegh, A. M. (2020). *Roark's formulas for stress and strain* (9th ed.). McGraw-Hill.
- Gupta, J. P. (1986). *Fundamentals of heat exchanger and pressure vessel technology*. Hemisphere.

American standards

- API Standard 520 P1. (2014). Sizing, selection, and installation of pressure-relieving devices in refineries. *Part I: sizing and selection*. American Petroleum Institute.
- API Standard 520 P2. (2015). Sizing, selection, and installation of pressure-relieving devices in refineries. *Part II: installation*. American Petroleum Institute.
- API Standard 521. (2014). *Pressure-relieving and depressuring systems*. American Petroleum Institute.
- API Standard 526. (2017). *Flanged steel pressure relief valves* (7th ed.). American Petroleum Institute.
- API Standard 527. (2014). *Seal tightness of pressure relief valves* (4th ed.). American Petroleum Institute.
- API Standard 620. (2013). In *Design and construction of large, welded, low-pressure storage tanks* (12th ed.). American Petroleum Institute.
- API Standard 650. (2013). *Welded steel tanks for oil storage* (12th ed.). American Petroleum Institute.
- ASME Boiler and Pressure Vessel Code Section II. Materials. *ASME International*, (2019).
- Boiler, A. S. M. E., & Pressure Vessel Code Section VIII. (2019). Rules for the construction of pressure vessels. *ASME International*.
- Boiler, A. S. M. E., & Pressure Vessel Code Section, IX. (2019). Qualification standard for welding and brazing procedures, welders, brazers, and welding and brazing operators. *ASME International*.
- Boiler, A. S. M. E., & Pressure Vessel Code Section, X. (2019). Fiber-reinforced plastic vessels. *ASME International*.

- ASME B16.5-. (2017). Pipe flanges and flanged fittings NPS 1/2 through NPS 24 Metric/Inch Standard. *ASME International*.
- ASME B16.9-. (2018). Factory-made wrought buttwelding fittings. *ASME International*.
- ASME B16.11-2016. Forged fittings, socket-welding and threaded. *ASME International*.
- ASME B16.15-. (2018). Cast copper alloy threaded fittings classes 125 and 250. *ASME International*.
- ASME B16.20-. (2017). Metallic gaskets for pipe flanges. *ASME International*.
- ASME B16.24-. (2016). Cast copper alloy pipe flanges. *flanged fittings and valves, classes, 150, 300, 600, 900, 1500, and 2500*. *ASME International*.
- ASME B16.42-. (2016). Ductile iron pipe flanges and flanged fittings, class 150 and 300. *ASME International*.
- ASME B16.47-. (2017). Large diameter steel flanges, NPS 26 through NPS 60. *ASME International*.
- ASME PTC 25-. (2019). Pressure relief devices — performance test codes. *ASME International*.

Other standards

- BS 2654. (1989). *Specification for the manufacture of welded non-refrigerated storage tanks for the petroleum industry* (superseded by BS EN 14015).
- BS 4994. (1987). *Specification for vessels and tanks in reinforced plastics* (superseded by BS EN 13923).
- BS EN 13121. (2016). *GRP tanks and vessels for use above ground*.
- BS EN 13445-1. (2014). *Unfired pressure vessels — Part 1: General*.
- BS EN 13445-2. (2014). *Unfired pressure vessels — Part 2: Materials*.
- BS EN 13445-3. (2014). *Unfired pressure vessels — Part 3: Design*.
- BS EN 13445-4. (2014). *Unfired pressure vessels — Part 4: Fabrication*.
- BS EN 13445-5. (2014). *Unfired pressure vessels — Part 5: Inspection and testing*.
- BS EN 13923. (2006). *Filament-wound FRP pressure vessels. Materials, design, manufacturing and testing*.
- BS EN 14015. (2005). *Specification for the design and manufacture of site built, vertical, cylindrical, flat-bottomed, above ground, welded, steel tanks for the storage of liquids at ambient temperature and above*.
- BS PD 5500. (2018). *Specification for unfired fusion welded pressure vessels*.
- DIN 28020. (1998). *Horizontal pressure vessels 0.63 m³ up to 25 m³ capacity*.
- DIN 28022. (2006). *Vertical pressure vessels — vessels for process plants 0.063 m³ up to 25 m³ capacity for use in chemical process engineering*.

14.18 Nomenclature

Note that the same nomenclature has been used as in the ASME BPV Code and API recommended practices and standards as far as possible. This occasionally causes the same symbol to be used with different meanings in different contexts. Where the context is not clear and there is a possibility of confusion, a new symbol has been assigned.

		Dimensions in MLT
a_e	Acceleration due to an earthquake	LT^{-2}
C	Constant in Equation 14.16	—
C_d	Drag coefficient in Equation 14.36	—
C_e	Seismic constant	—
C_w	Weight factor in Equation 14.32	—
c	Corrosion allowance	L
D	Diameter	L
D_c	Diameter of cone at point of interest	L
D_e	Nominal diameter of flat end	L
D_{eff}	Effective diameter of column for wind loading	L
D_i	Internal diameter	L
D_m	Mean diameter	L
D_o	Outside diameter	L
D_s	Skirt internal diameter	L
D_t	Tank diameter	L
d_b	Bolt diameter	L
E	Joint efficiency, welded joint	—

		Dimensions in MLT
E_Y	Young's modulus	$ML^{-1}T^{-2}$
F_{bs}	Load supported by bracket	MLT^{-2}
F_L	Force in longitudinal (vertical) direction	MLT^{-2}
F_p	Local, concentrated, wind load	MLT^{-2}
F_s	Shear force due to an earthquake	MLT^{-2}
F_v	Force in horizontal direction	MLT^{-2}
g	Gravitational acceleration	LT^{-2}
H_L	Liquid depth	L
H_p	Height of local load above base	L
H_v	Height (length) of cylindrical section between tangent lines	L
I_p	Polar second moment of area	L^4
I_v	Second moment of area of vessel	L^4
L	Unsupported length of vessel	L
L_d	Bracket depth	L
L_o	Distance between center line of equipment and column	L
M	Bending moment	ML^2T^{-2}
M_e	Bending moment due to offset equipment	ML^2T^{-2}
M_p	Bending moment at base due to local load	ML^2T^{-2}
M_s	Bending moment at base of skirt	ML^2T^{-2}
M_x	Bending moment at point x from free end of column	ML^2T^{-2}
m	Gasket factor	—
P_d	Design pressure	$ML^{-1}T^{-2}$
P_i	Internal pressure	$ML^{-1}T^{-2}$
P_w	Wind pressure loading	$ML^{-1}T^{-2}$
P_b	Bolt pitch	L
R_c	Crown radius	L
R_k	Knuckle radius	L
R_p	Radius of curvature of plate	L
S	Maximum allowable stress (design stress)	$ML^{-1}T^{-2}$
S_a	Maximum allowable stress at test temperature	$ML^{-1}T^{-2}$
S_n	Maximum allowable stress at design temperature	$ML^{-1}T^{-2}$
S_s	Maximum allowable stress for skirt material	$ML^{-1}T^{-2}$
S_t	Maximum allowable stress for tank material	$ML^{-1}T^{-2}$
T	Torque	$ML^{-1}T^{-2}$
t	Thickness of plate or shell	L
t_c	Thickness of bracket plate	L
t_f	Thickness of flange	L
t_s	Thickness of shell	L
t_{sk}	Skirt thickness	L

		Dimensions in MLT
t_t	Tank wall thickness	L
u_w	Wind velocity	LT^{-1}
W	Wind load per unit length	MT^{-2}
W_e	Weight of ancillary equipment	MLT^{-2}
W_v	Weight of vessel and contents	MLT^{-2}
W_z	Weight of vessel and contents above a plane at elevation z	MLT^{-2}
x	Distance from free end of cantilever beam	L
y	Minimum seating pressure for gasket	$ML^{-1}T^{-2}$
α	Half cone apex angle	—
θ_s	Base angle of conical section	—
ν	Poisson's ratio	—
ρ_m	Density of vessel material	ML^{-3}
ρ_a	Density of air	ML^{-3}
ρ_L	Liquid density	ML^{-3}
σ	Normal stress	$ML^{-1}T^{-2}$
σ_b	Bending stress	$ML^{-1}T^{-2}$
σ_{bs}	Bending stress in skirt	$ML^{-1}T^{-2}$
σ_c	Critical buckling stress	$ML^{-1}T^{-2}$
σ_e	Stress at elastic limit of material	$ML^{-1}T^{-2}$
σ'_e	Elastic limit stress divided by factor of safety	$ML^{-1}T^{-2}$
σ_h	Circumferential (hoop) stress	$ML^{-1}T^{-2}$
σ_L	Longitudinal stress	$ML^{-1}T^{-2}$
σ_s	Stress in skirt support	$ML^{-1}T^{-2}$
σ_w	Stress due to weight of vessel	$ML^{-1}T^{-2}$
σ_{ws}	Stress in skirt due to weight of vessel	$ML^{-1}T^{-2}$
σ_x	Normal stress in x direction	$ML^{-1}T^{-2}$
σ_y	Normal stress in y direction	$ML^{-1}T^{-2}$
σ_z	Axial stresses in vessel	$ML^{-1}T^{-2}$
$\sigma_1, \sigma_2, \sigma_3$	Principal stresses	$ML^{-1}T^{-2}$
τ	Torsional shear stress	$ML^{-1}T^{-2}$
τ_e	Shear stress at elastic limit of material	$ML^{-1}T^{-2}$
τ_{xy}	Shear stress	$ML^{-1}T^{-2}$
τ_1, τ_2, τ_3	Shear stress maxima	$ML^{-1}T^{-2}$
<i>Superscript</i>		
$\wedge$	Maximum	

14.19 Problems

- 14.1.** A horizontal, cylindrical tank with hemispherical ends is used to store liquid chlorine at 10 bar. The vessel is 4 m internal diameter and 20 m long. Estimate the minimum wall thickness required to resist this pressure for the cylindrical section and the heads. Take the design pressure as 12 bar and the maximum allowable stress for the material as 110 MN/m^2 .
- 14.2.** The thermal design of a heat exchanger to recover heat from a kerosene stream by transfer to a crude oil stream was in [Chapter 19](#), Example 19.2. Make a preliminary mechanical design for this exchanger. Base your design on the specification obtained from the CAD design procedure used in the example. All material of construction to be carbon steel (semi-killed or silicon killed). Your design should cover:
- Choice of design pressure and temperature
 - Choice of the required corrosion allowances
 - Choice of the type of end covers
 - Determination of the minimum wall thickness for the shell, headers, and ends
 - A check on the pressure rating of the tubes
- 14.3.** Make a preliminary mechanical design for the vertical thermosyphon reboiler for which the thermal design was done as Example 19.9 in [Chapter 19](#). The inlet liquid nozzle and the steam connections will be 50 mm inside diameter. Flat plate end closures will be used on both headers. The reboiler will be hung from four bracket supports, positioned 0.5 m down from the top tube plate. The shell and tubes will be of semi-killed carbon steel. Your design should cover:
- Choice of design pressure and temperature
 - Choice of the required corrosion allowances
 - Selection of the header dimensions
 - Determination of the minimum wall thickness for the shell, headers, and ends
 - A check on the pressure rating of the tubes
- 14.4.** The specification for a sieve plate column is given here. Make a preliminary mechanical design for the column. Your design should include:
- Column wall thickness
 - Selection and sizing of vessel heads
 - The nozzles and flanges (use standard flanges)
 - Column supporting skirt

You need not design the plates or plate supports.

You should consider the following design loads:

- Internal pressure
- Wind loading
- Dead weight of vessel and contents (vessel full of water)

There will be no significant loading from piping and external equipment. Earthquake loading need not be considered. Column specification:

Length of cylindrical section, 37 m

Internal diameter, 1.5 m

Heads, standard ellipsoidal

50 sieve plates

Nozzles: feed, at mid-point, 50 mm inside diameter

vapor out, 0.7 m below top of cylindrical section, 250 mm inside diameter

bottom product, center of vessel head, 50 mm inside diameter

reflux return, 1.0 m below top of cylindrical section, 50 mm inside diameter

Two 0.6-m-diameter access ports (manholes) situated 1.0 m above the bottom and 1.5 m below the top of the column

Support skirt height, 2.5 m

Access ladder with platforms

Insulation, mineral wool, 50 mm thick

Materials of construction: vessel stainless steel, unstabilized (304)
 nozzles as vessel
 skirt carbon steel, silicon killed

Design pressure 1200 kN/m²

Design temperature 150 °C

Corrosion allowance, 2 mm

Make a dimensioned sketch of your design and fill out the column specification sheet given in Appendix G (available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793).

- 14.5.** A fixed-bed reactor is to be designed for a hydrocracking process. The reactor will treat 320,000 lb/h of vacuum gas oil (specific gravity 0.85) in the presence hydrogen at 650 °F, 2000 psig, 1.0 weight hourly space velocity (WHSV). The catalyst has bulk density of 50 lb/ft³ and void fraction 0.4. The catalyst is to be divided into four beds to allow a hydrogen quench to be brought in between the beds for temperature control. Make a preliminary mechanical design of the reactor(s). Your design should include:

- a. Selection of material of construction
- b. Sizing of the vessel(s), including allowance for any internals
- c. Determination of the required wall thickness
- d. Selection and sizing of vessel heads
- e. The nozzles and flanges (use standard flanges)
- f. A support skirt

You need not design the vessel internals.

You should consider the following design loads:

- a. Internal pressure
- b. Wind loading
- c. Dead weight of vessel and contents (vessel full of catalyst and gas oil)
- d. Hydraulic testing with no catalyst and vessel full of water

- 14.6.** A jacketed vessel is to be used as a reactor. The vessel has an internal diameter of 2 m and is fitted with a jacket over a straight section 1.5 m long. Both the vessel and jacket walls are 25 mm thick. The spacing between the vessel and jacket is 75 mm.

The vessel and jacket are made of carbon steel. The vessel will operate at atmospheric pressure, and the jacket will be supplied with steam at 20 bar. Check if the thickness of the vessel and jacket is adequate for this duty. Take the allowable design stress as 100 N/mm² and the value of Young's modulus at the operating temperature as 180,000 N/mm².

- 14.7.** A storage tank for concentrated nitric acid will be constructed from aluminum to resist corrosion. The tank is to have an inside diameter of 6 m and a height of 17 m. The maximum liquid level in the tank will be at 16 m. Estimate the plate thickness required at the base of the tank. Take the allowable design stress for aluminum as 90 N/mm².

Design of reactors and mixers

KEY LEARNING OBJECTIVES

- How to select and size a reactor
- Factors to consider in optimizing reaction conditions
- How to design mixers
- How to design reactors for heating, cooling, and catalysis
- How to design reactors for multiphase reactions
- How to design reactors for biological processes
- Factors to consider in scaling up reactions from the laboratory to plant scale

15.1 Introduction

The reactor is the heart of a chemical process, where chemical reactions are carried out to transform feeds into products. Reactors are usually designed as special items for a given project. Many reactor designs are unique and proprietary, particularly when the reaction system involves catalysis or multiphase flow. Even conventional stirred tanks are often customized for the application by optimization of internal mixing pattern, feed addition points, heat transfer surface, or control instrumentation.

Numerous textbooks have been published on the subjects of chemical kinetics and reaction engineering, and a selection is given in the bibliography at the end of this chapter. These subjects will not be covered here. Instead, the focus will be on translating the reactor design parameters into process equipment specifications that can be used to lay out a preliminary reactor design and develop drawings and specifications for use by detailed equipment design groups.

Reactor design is a vital step in the overall design of the process, and it is important to ensure that the equipment specified will be capable of achieving the desired yields and selectivity when operated at full scale. Few industrial reactors are designed solely based on detailed modeling of the kinetics and hydrodynamics; instead, reactors are usually scaled up from pilot plant reactors or previous designs, making suitable allowance for heat transfer, mass transfer, residence time, or whatever other parameters have been determined to govern the size. The final volume of real reactors is often determined more by the needs for mixing, segregation, or heat transfer than by the residence time required for the reaction.

15.2 Reactor design: General procedure

This section provides a generalized overall procedure for reactor design that applies to most reacting systems. Further information on the steps in this procedure will be given in subsequent sections.

15.2.1 General procedure for reactor design

The design of a chemical reactor should not be carried out in isolation from overall process design. It will usually be necessary to determine optimal conditions of reactor conversion, selectivity, and yield of by-products, as described in [Chapter 2](#). Optimization of the reactor performance thus requires a design of the process separations and heat recovery, as well as costing and economic analysis. Although the reactor itself is usually a relatively small fraction of the process fixed capital cost, the reactor performance can have a very significant impact on capital and operating costs elsewhere in the process.

The design of a reactor should therefore begin with an overall process design to set rough targets for yield and selectivity, as discussed in Sections 2.3.3 and 2.6.1. At this stage, there may be very little reaction kinetics information available; indeed, it is very unlikely that there will be sufficient information to develop a model that accurately predicts the effect of feed contaminants and the formation of by-products. All that the designer needs at this point is a rough idea of what yields are feasible in a given residence time or space velocity; a few good data points from the laboratory are usually sufficient to get started.

The overall procedure for reactor design is given in [Fig. 15.1](#). The steps in this procedure are described next. During several of the steps it may be necessary for the design engineer to interact with researchers to collect additional data, as indicated in the figure.

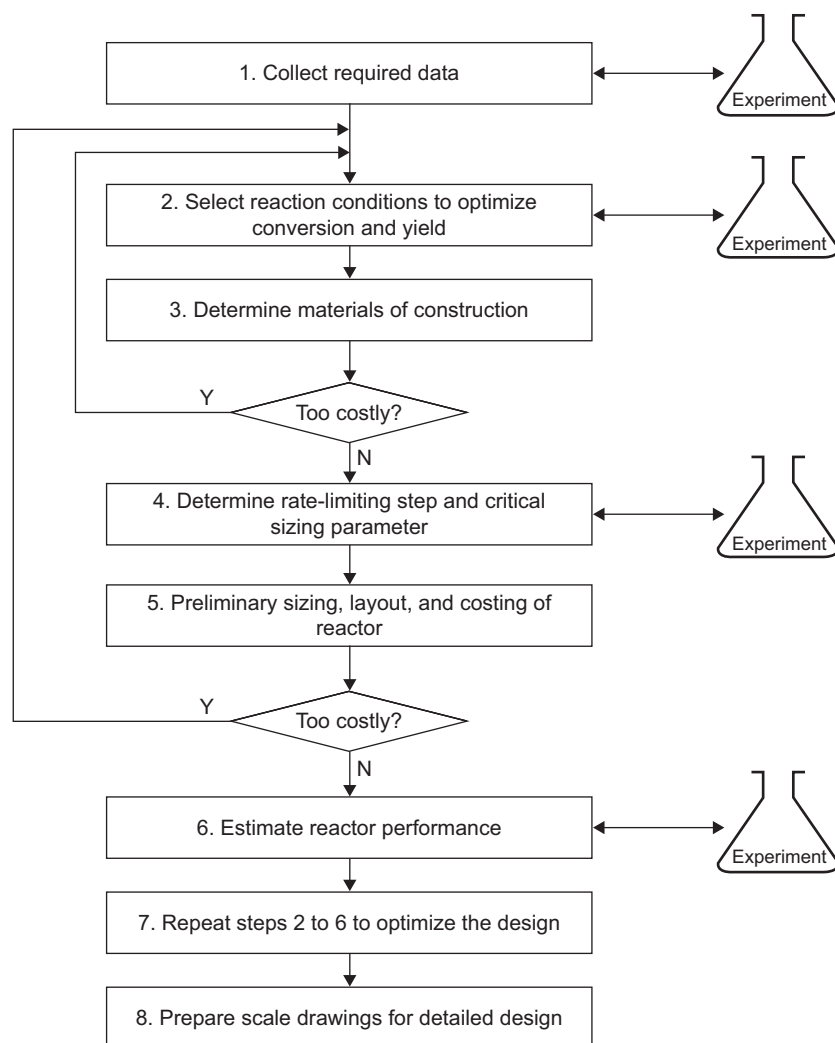


FIG. 15.1 General procedure for reactor design.

Step 1: Collect required data

Reactor design always requires more experimental input than the design of other process equipment. Some required parameters, such as heats of reaction and phase equilibrium constants, can be estimated using commercial process simulation models, while some parameters, such as diffusion coefficients and heat and mass transfer coefficients, can be predicted using correlations from the literature; however, almost all reaction rate constants must be measured experimentally.

Sources of reaction engineering data and methods for predicting heats of reaction and transport properties are given in Section 15.3.

The collection of reaction engineering data is often an iterative process. As the designer and researchers gain more insight into reactor performance, they may decide to operate the reactor under different conditions or may become aware of different by-products or feed contaminants that affect reaction rates. It is unlikely that the initial experiments or literature search will provide comprehensive information that covers all possible reaction conditions. It will thus be necessary to return to the laboratory and collect more data in the new region of interest at subsequent stages in the reactor design.

Step 2: Select reaction conditions

The reaction conditions should be chosen to optimize the reactor conversion, selectivity, and yield, within a reasonable reactor cost and allowing for a safe and controllable reactor design. The choice of reactor conditions is probably the most critical step in reactor design. Some of the conditions, such as the presence of contaminants in recycle streams, may be determined by other process considerations. The selection of reaction conditions is discussed in more detail in Section 15.4. The selection of reaction conditions will often govern the selection of reactor type. For example, if the reaction conditions cause all the reagents and products to be in the vapor phase, a stirred tank reactor will not be applicable.

Step 3: Determine materials of construction

When the reaction conditions have been specified, the designer can make a preliminary analysis of the materials of construction for the reactor. If the temperature, the pressure, or the presence of a particular component indicates that the reactor will need to be made from a costly alloy, the designer may be tempted to go back to the research team and try to identify a new set of reactor conditions. The selection of materials of construction is discussed in [Chapter 6](#).

Step 4: Determine the rate limiting step and critical sizing parameters of the reactor

In this step the skills and knowledge of reaction engineering are deployed in elucidating the key parameters that determine the extent of reaction under the conditions previously specified. An experimental plan must be carried out under a broad enough range of conditions to determine what physical process limits the extent of reaction. In general, the reaction rate is usually limited by one of the following fundamental processes:

- **Intrinsic kinetics:** The rate of the reaction itself. In multistep reactions there will usually be one slowest step that governs the overall rate.
- **Mass transfer rate:** Mass transfer can be particularly important in multiphase reactions, where reagents must transfer between phases, and in reactions that use porous heterogeneous catalysts, where reagents and products must diffuse in and out of catalyst pores. Often, the measured rate of reaction will be confounded with the rate of mass transfer, and careful experimentation will be needed to separate the two effects and determine which is rate limiting for scale-up.
- **Heat transfer rate:** If a reaction is endothermic, then the temperature of the reaction mixture will decrease unless heat is added. The rate of heat addition can become the governing parameter in reactor sizing, and the reaction may even need to be carried out in a heat transfer device such as a heat exchanger or fired heater. Heat transfer to and from reactors is discussed in more detail in Section 15.6.
- **Feed addition rate:** If the reaction is exothermic, is very rapid, or requires a very low concentration of one of the components for optimal yield, the designer may choose to “starve” the reaction by limiting the feed rate of one of the reagents. Starving a reaction can be a good reactor control strategy for smaller reactors, but can be difficult to accomplish in larger reactors where uniform mixing is more difficult.
- **Mixing rate:** For very fast reactions, the time taken to mix the reagents can be the limiting step. If the reaction yield is sensitive to feed concentrations, the rate of mixing must be fast enough to ensure that the desired concentration

is achieved before reaction occurs (i.e., the reactor must be designed so that mixing is no longer rate limiting). Mixing is discussed in more detail in Section 15.5.

The rate-limiting step can be determined experimentally by collecting rate data and fitting a suitable model of reaction kinetics; see any of the references in the bibliography at the end of this chapter. Once the rate-limiting step has been determined, the designer will be able to specify a critical sizing parameter for the reactor. This will usually be one of the parameters given in Table 15.1. These parameters allow the reactor volume, volume or mass of catalyst, or contacting area for mass transfer to be scaled up relative to the mass or volumetric flow of reactor feed. The most common case that does not have a simple scaling parameter is when an endothermic reaction is limited by the rate of heat addition, which is discussed in more detail in Section 15.6.

The rate-limiting step can depend on the type of reactor that is selected for the experimental program. Care must be used when scaling up data to ensure that the same fundamental process will be rate limiting in a larger vessel.

TABLE 15.1 Reactor sizing parameters

Sizing parameter	Definition	Units	Comments
Residence time	$= \frac{\text{reactor volume}}{\text{volumetric flow rate}}$	Time	Most widely used reactor sizing parameter. Mainly used for homogeneously catalyzed reactions. Note that for liquid-phase reactions reactor volume refers to volume occupied by liquid, not total reactor volume. Volumetric flow rate should be calculated under average reactor conditions, which can be difficult for compressible gases if there is a significant change in temperature along the reactor.
Space velocity (also GHSV = gas hourly space velocity, LHSV = liquid hourly space velocity)	$= \frac{\text{volumetric flow rate}}{\text{reactor (or more often catalyst) volume}}$	Time ⁻¹ (usually by convention hr ⁻¹)	Usually used for reactors that employ solid catalysts. Volume usually refers to volume occupied by packed bed of catalyst, regardless of whether catalyst bed is expanded under process conditions. Volumetric flow rate should be calculated under average reactor conditions as noted above.
Weight or mass space velocity (WHSV = weight hourly space velocity)	$= \frac{\text{mass flow rate}}{\text{mass of catalyst}}$	Time ⁻¹ (usually by convention hr ⁻¹)	Usually used for reactors that employ solid catalysts. Use of WHSV avoids complications of allowing for changes in volumetric flow rate along reactor and variations in catalyst bed loading density between small laboratory reactors and full-scale reactors.
Number of transfer units	See Section 17.14.2. The definition is always a path integral of the inverse of concentration or partial pressure driving force. Different definitions can be used for vapor or liquid phase.	Dimensionless	Used for reactors where mass transfer between vapor and liquid phases is the rate-controlling step. See Section 17.14.2 for design of mass transfer contacting devices and Section 15.8 for more on multiphase reactors. Must be used with the appropriate definition of height of a transfer unit, which always includes the molar flow rate of one of the phases.

Step 5: Preliminary sizing, layout, and costing of reactor

Once the sizing parameter of the reactor has been obtained, the designer can estimate the reactor volume, catalyst volume, etc., that will be required at the design rate. The volume estimated is only the active reacting volume, and the reactor layout must also consider the following factors that may add to the volume required for the reactor vessel:

- Additional space needed for any internal heat transfer devices such as coils, stab-in bundles, quench zones, etc. (see Section 15.6).
- Additional space needed for vapor–liquid distribution, spargers, vapor–liquid segregation or redistribution (see Section 15.7).
- For liquid-phase stirred tank reactors, some inert vapor space should always be allowed so that the vessel is not filled with liquid. This makes pressure control much easier and reduces the possibility of hydraulic overpressure due to minor changes in flow during operation. A stirred tank reactor should not be designed to operate more

than 90% filled, and 65% to 75% is a better design guideline. If foaming can occur, then the level should be at the lower end of the range to allow space for a foam breaker.

- For packed or moving beds, space needed for inert catalyst support materials or support grids.
- For fluidized beds, space needed for fluid distribution grids, fluid–solid segregation, cyclones, transfer lines, allowance for bubble volume, etc. (see Section 15.7.3).

Other factors that can affect the overall reactor volume are discussed in Sections 15.6 to 15.10. Specific sizing guidelines are given for different reactor layouts in these sections.

The geometry of the reactor will depend on the desired flow pattern and mixing requirements. For simple reactors, the designer will select a vessel design that approximates one of the ideal mixing patterns described in Section 15.2.2. For more complex designs that involve several reaction zones, the designer will seek a geometry that arranges the different zones in the correct sequence, preferably (but not necessarily) within the same vessel. See Section 15.11.2 for more discussion of complex reactor networks. In some cases, the requirements of the reaction will dictate that a particular piece of equipment should be used. For example, if the reaction leads to precipitation of a product, it might be logical to use a crystallizer as the reactor, or if the reaction is endothermic and must be carried out at high temperature, then it may make sense to use a fired heater as the reactor.

Most reactors fall under the definition of pressure vessels, so once the length and diameter of the reactor have been selected, the wall thickness can be found using the methods described in [Chapter 14](#). The cost of the reactor can then be built up from the cost of the pressure vessel plus the cost of the internals listed earlier. Methods for estimating capital costs for equipment components are given in [Section 7.5](#). The sizing and costing of reactors are illustrated in Examples 15.4, 15.5, and 15.6.

If the preliminary reactor design is too costly, the designer should revisit Step 2 and see if other conditions can be found that will lead to an improved design.

Step 6: Estimate reactor performance

It is important to confirm that the reactor design will actually achieve the target conversion and selectivities for main products and by-products. Unfortunately, it is often impossible to be fully satisfied without building and testing a full-scale reactor, which is a very expensive proposition.

Historically, chemical companies would go through multiple steps of pilot plant scale-up to validate their reactor designs. Successive pilot plants would be built with the scale of production increasing by one or two orders of magnitude each time. This gradual approach to scale-up was expensive and delayed the market launch of new products and process technologies.

Nowadays, a more common approach is to use a combination of experimental methods and computer modeling to attempt to predict the full-scale performance. This allows the design team to eliminate some of the intermediate steps in scale-up and go from well-designed small pilot plants to a demonstration unit that may be approaching full scale in a single step. Computer simulation methods for reactor analysis are discussed in Section 15.11, and methods for determining actual reactor performance are described in Section 15.12.

Step 7: Optimize the design

If necessary, Steps 2 to 6 can be repeated to optimize the design. Optimization to reduce the reactor cost is usually a waste of time because the cost of the reactor is typically a relatively small fraction of the total capital cost; however, if the reactor validation experiments showed the presence of unexpected components or showed different selectivities than were found at smaller scale, it will be necessary to reevaluate the overall process optimization and confirm that the target conversion, yields, and selectivities still apply.

Step 8: Prepare scale drawings for detailed design

The final step for the chemical engineer is to make an effective hand-off of the design to the specialized mechanical engineering designer or design team. Because most reactors require substantial internals and ancillary equipment such as agitators, heat transfer coils, etc., the mechanical design of the vessel is rarely trivial. The mechanical engineer needs a complete specification of the vessel and internals, so as to be able to make detailed designs of nozzles, reinforcing rings, internal and external support brackets, and other features. The best means for transmitting all of the required specifications is usually for the reactor designer and the mechanical designer to collaborate on producing scale drawings that show all of the necessary features. Often the mechanical engineer will be involved throughout the design process and will already have a good understanding of the reactor requirements.

15.2.2 Ideal and real reactors

It is convenient to describe reactors in terms of how closely they approximate the behavior of the two extreme types of ideal reactor: the plug flow reactor (PFR) and the well-mixed reactor (WMR). In fact, it can be shown that real reactors can be modeled as networks or combinations of multiple plug flow and well-mixed reactors; see Section 15.11.2.

Plug-flow reactor (PFR)

In perfect plug flow, each batch of material that enters the reactor moves through the reactor together, mixing and reacting only with material that entered at the same time, and all leave the reactor together. In a PFR with tubular geometry, there is perfect radial mixing but no axial mixing. All material that flows through the reactor has exactly the same residence time, τ , and all material experiences the same profile of temperature and concentration as it moves along the reactor. A perfectly mixed batch reactor behaves the same as a PFR, with the material moving through time instead of space.

A PFR can be sized by considering an incremental slice of the reactor, as shown in Fig. 15.2. The change in molar flow rate of a reacting component (usually a feed component) due to the reaction is equal to the rate of reaction in the increment:

$$dM = \mathcal{R}dV \quad (15.1)$$

where

M = molar flow rate = volume flow rate $\times$ molar concentration

dV = incremental volume

$\mathcal{R}$ is an expression for the rate of reaction per unit volume

Because both the rate of reaction and the molar flow rate can be written in terms of the concentration or partial pressure of the reacting components, Equation 15.1 can be integrated along the reactor length to give a relationship between reactor residence time and concentration or conversion. Different relationships are obtained depending on the form of expression for the rate of reaction; see any of the books cited in the bibliography at the end of this chapter for multiple solved examples.

Well-mixed reactor (WMR)

In a perfect WMR, the entire contents of the vessel are at the same temperature and concentration and there is no spatial variation in any mixture property. The fluid leaving the WMR is therefore at the same concentration and temperature as the vessel contents. Material entering the WMR is immediately mixed with the contents and comes to the same conditions. Some of the material that enters can leave immediately, whereas some leaves much later, so there is a broad distribution in residence time, as shown in Fig. 15.3.

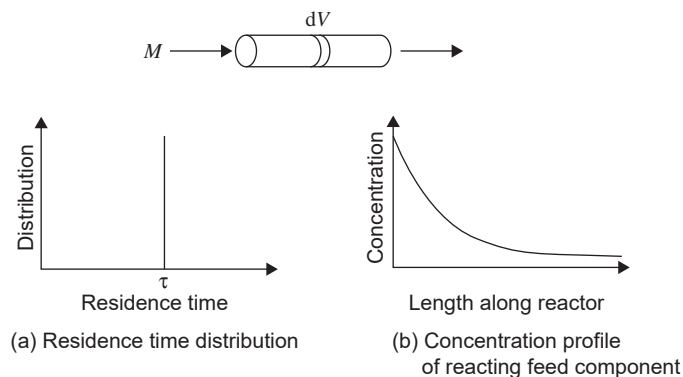


FIG. 15.2 Plug-flow reactor.

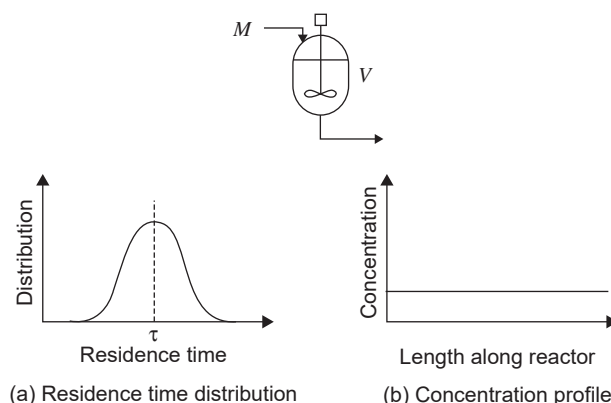


FIG. 15.3 Well-mixed reactor.

A WMR can be sized by solving a material balance across the reactor:

$$M_{in} - M_{out} = \mathcal{R}V \quad (15.2)$$

where

M_{in} = molar flow rate of any component into the reactor

M_{out} = molar flow rate of any component out of the reactor

V = reactor volume

As with the PFR, the molar flow rate can be expressed in terms of volume flow and concentration, and the rate of reaction can also be written in terms of concentrations, allowing Equation 15.2 to be restated as a relationship between residence time and concentration or conversion. The books cited in the bibliography provide examples of different expressions obtained with different reaction rate equations.

Real reactors

Few real reactors achieve ideal reactor performance, although many reactors are close enough that ideal reactor performance can be assumed for modeling purposes; see Section 15.11. Some real reactor types that have performance close to that of ideal reactors are shown in Fig. 15.4.

Near plug flow conditions can be obtained in tubular or serpentine reactors, but allowance must be made for some degree of axial dispersion. Simple tubular (pipeline) reactors are normally used for homogeneous gas-phase reactions, for example, in the thermal cracking of petroleum crude oil fractions to ethylene and the thermal decomposition of dichloroethane to vinyl chloride. Tubular reactors can also be used for homogeneous liquid-phase reactions. Single-phase flow through packed beds or heat exchanger or furnace tubes is a good approximation of plug flow, as is the flow in a transport-phase fluidized bed. It is not necessary for a reactor to be long and thin to be in plug flow. A thin bed of catalyst is often loaded into a radial-flow reactor, in which the fluid flows radially through a bed of catalyst retained between perforated screens (see Section 15.7.3 and Fig. 15.29). Although the path length through the bed is short, there is little opportunity for back-mixing, and plug flow is a good approximation.

Continuous stirred tank reactors often provide a reasonable approximation of WMR performance, although mixing is never perfect. The effect of poor mixing is usually to skew the residence time distribution, giving a longer tail to the distribution. Sparged tanks and lift-tube reactors are well-mixed in the liquid phase, although plug flow with respect to the vapor phase. A fluidized bed operated in bubbling-bed mode is well-mixed with respect to the solids and can be either well-mixed or plug flow with respect to the liquid or vapor that is causing fluidization; see Section 15.7.3.

Examples of real reactors will be introduced in the following sections, particularly in Sections 15.5 to 15.10. The use of networks of ideal reactors to model real reactor performance is discussed in Section 15.11.2. The volumes by Rase (1977, 1990) cover the practical aspects of reactor design and include case studies of industrial reactors. Henkel (2005) catalogs different types of industrial reactors and gives examples of the types of reaction carried out in each.

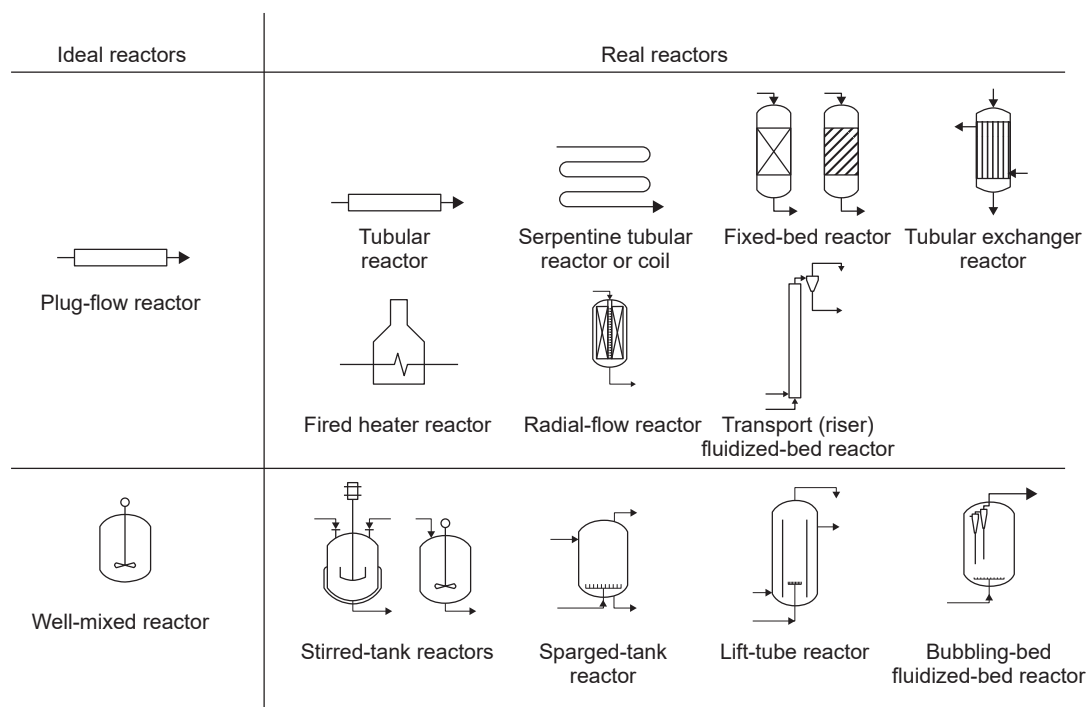


FIG. 15.4 Ideal reactors and some real reactors that approximate the same flow pattern.

15.3 Sources of reaction engineering data

This section provides a brief overview of sources of reaction engineering data such as enthalpy and Gibbs free energy of reaction, equilibrium constants, reaction rate constants, activation energies, and mass transfer properties.

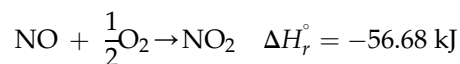
15.3.1 Enthalpy of reaction

The amount of heat given out in a chemical reaction depends on the conditions under which the reaction is carried out. The standard heat of reaction is the heat released when the reaction is carried out under standard conditions: pure components, pressure 1 atm (1.01325 bar), and temperature usually, but not necessarily, 25 °C. Care must be taken to correct the heat of reaction to the temperature and pressure of the process.

Values for the standard heats of reaction for the reactions used in the production of commercial chemicals can usually be found in the literature or can be calculated from heats of formation or combustion. A useful source of data on heats of formation and combustion is the critical review of the literature by Domalski (1972).

Benson developed a detailed group contribution method for the estimation of heats of formation; see Benson (1976) and Benson et al. (1969). He estimated the accuracy of the method to be from ± 2.0 kJ/mol for simple compounds to about ± 12 kJ/mol for highly substituted compounds. Benson's method and other group contribution methods for the estimation of heats of formation are described by Reid et al. (1987). Group contribution approaches are used by commercial process simulation programs to estimate enthalpies of formation for user-specified compounds if the user does not manually enter a heat of formation. Group contribution approaches are also sometimes used for library compounds.

When quoting heats of reaction, the basis should be clearly stated, either by giving the chemical equation, for example:

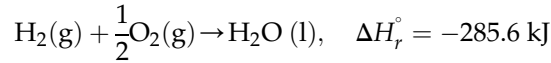
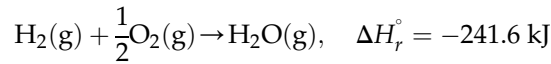


(the equation implies that the quantity of reactants and products are moles), or by stating to which quantity the quoted value applies:

$$\Delta H_r^\circ = -56.68 \text{ kJ per mol NO}_2$$

The reaction is exothermic, and the enthalpy change ΔH_r° is therefore *negative*. The heat of reaction $-\Delta H_r^\circ$ is *positive*. The superscript $^\circ$ denotes a value at *standard* conditions, and the subscript *r* implies that a chemical reaction is involved.

The state of the reactants and products (gas, liquid, or solid) should also be given if the reaction conditions are such that they may exist in more than one state; for example:



The difference between the two heats of reaction is the latent heat of the water formed.

In process design calculations, it is usually more convenient to express the heat of reaction in terms of the moles of product produced for the conditions under which the reaction is carried out, kJ/mol product.

Effect of temperature on heat of reaction

The heat of reaction must be estimated at process conditions for use in reactor design. Failure to correct the heat of reaction can lead to gross overestimates of the required heating or cooling duty.

Standard heats of reaction can be converted to other reaction temperatures by making a heat balance over a hypothetical process, in which the reactants are brought to the standard temperature, the reaction carried out, and the products then brought to the required reaction temperature, as illustrated in Fig. 15.5.

$$\Delta H_{r,T} = \Delta H_r^\circ + \Delta H_{\text{prod}} + \Delta H_{\text{react}} \quad (15.3)$$

where :

$-\Delta H_{r,T}$ = heat of reaction at temperature T

$\Delta H_{\text{react.}}$ = enthalpy change to bring reactants to standard temperature (which will be negative, as the reagents are being cooled)

$\Delta H_{\text{prod.}}$ = enthalpy change to bring products to reaction temperature, T

Effect of pressure on heat of reaction

Equation 15.3 can be written in a more general form:

$$\begin{aligned} \Delta H_{r,P,T} = \Delta H_r^\circ &+ \int_1^P \left[\left(\frac{\partial H_{\text{prod.}}}{\partial P} \right)_T - \left(\frac{\partial H_{\text{react.}}}{\partial P} \right)_T \right] dP \\ &+ \int_{298}^T \left[\left(\frac{\partial H_{\text{prod.}}}{\partial T} \right)_P - \left(\frac{\partial H_{\text{react.}}}{\partial T} \right)_P \right] dT \end{aligned} \quad (15.4)$$

Where: P = pressure

If the effect of pressure is likely to be significant, the change in enthalpy of the products and reactants, from the standard conditions, can be evaluated to include both the effects of temperature and pressure (for example, by using tabulated values of enthalpy) and the correction made in a similar way to that for temperature only.

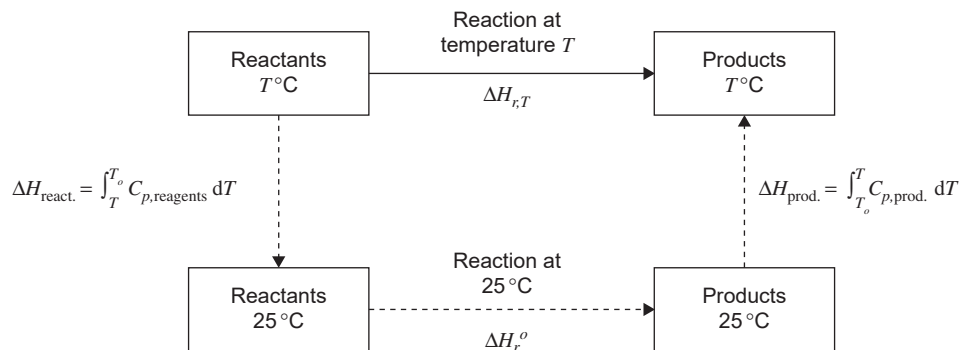


FIG. 15.5 ΔH_r at temperature T .

Estimation of heat of reaction using process simulation programs

Commercial process simulation programs can easily be manipulated to give quick estimates of reactor heating or cooling duty at process temperature and pressure. For most library compounds, the heat of reaction will be calculated using tabulated heats of formation and heat capacities that have been measured experimentally, but for user-specified compounds and possibly some library compounds, group contribution methods will be used to estimate the heat of formation and heat capacity, and so there will be more uncertainty in the answer.

When literature data on standard heat of reaction are available, a quick check of the process simulation software can be made by simulating an isothermal reactor at 25 °C. The heating or cooling duty calculated should match the standard heat of reaction.

Simulation programs are particularly helpful when several reactions are occurring at the same time, in which case calculating the overall reactor duty manually becomes tedious.

When using simulation programs to estimate reactor heating or cooling requirements, the following points should be kept in mind:

- Make sure the feeds to the reactor in the simulation model are at the temperature and pressure that the real feeds will be at. If a feed is hotter or colder than the reaction temperature, the change in sensible heat can have a significant effect on the reactor heat balance.
- Be sure to include all reactions that have a significant extent of reaction or conversion. It is best to use a conversion or yield reactor when predicting reactor heating or cooling duty unless the process gives products at equilibrium. See Section 4.5.1 for a discussion of different standard reactor models.
- To estimate the heating or cooling requirement, provide a heat stream into the reactor. Do not specify the duty of this heat stream. Instead, specify the desired temperature of the reactor products; the simulator will then calculate the required heating or cooling duty.

The heating or cooling duty predicted from a simulation model should be benchmarked against experimental data whenever possible. It may be difficult to accurately measure heat of reaction in a small pilot plant, and allowance for additional heating or cooling above that predicted by the simulation should be made during scale-up.

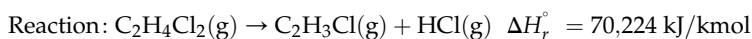
Example 15.1

This example illustrates the manual calculation of a reactor heat balance.

Vinyl chloride (VC) is manufactured by the pyrolysis of 1,2-dichloroethane (DCE). The reaction is endothermic. The flow rates to produce 5000 kg/h at 55% conversion are shown in Fig. 15.6.

The reactor is a pipe reactor heated with fuel gas, gross calorific value 33.5 MJ/m³. Estimate the quantity of fuel gas required.

Solution



The small quantity of impurities, less than 1%, that would be present in the feed has been neglected for the purposes of this example. Also, the selectivity for VC has been taken as 100%. It would be in the region of 99% at 55% conversion.

Heat capacity data, for vapor phase:

$$C_p^\circ = a + bT + cT^2 + dT^3 \text{ kJ/kmolK}$$

	<i>a</i>	<i>b</i> × 10 ²	<i>c</i> × 10 ⁵	<i>d</i> × 10 ⁹
VC	5.94	20.16	−15.34	47.65
HCl	30.28	−0.761	1.325	−4.305
DCE	20.45	23.07	−14.36	33.83

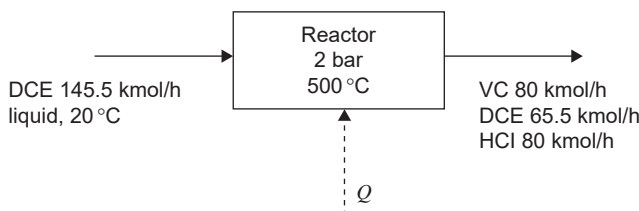


FIG. 15.6 Vinyl chloride reactor example.

for liquid phase: DCE at 20 °C, $C_p = 116 \text{ kJ/kmol K}$, taken as constant over a temperature rise from 20 to 25 °C.

Latent heat of vaporization of DCE at 25 °C = 34.3 MJ/kmol.

At 2 bar pressure, the change in C_p with pressure will be small and will be neglected. Take base temperature as 25 °C (298 K), the standard state for ΔH_r° .

$$\text{Enthalpy of feed} = 145.5 \times 116(293 - 298) = -84,390 \text{ J/h} = -84.4 \text{ MJ/h}$$

$$\text{Enthalpy of product stream} = \int_{298}^{773} \sum (n_i C_p) dT$$

Component	$n_i \text{ (mol/h)}$	$n_i a$	$n_i b \times 10^2$	$n_i c \times 10^5$	$n_i d \times 10^9$
VC	80	475.2	1612.8	-1227.2	3812.0
HCl	80	2422.4	-60.88	106.0	-344.4
DCE	65.5	1339.5	1511.0	-940.6	2215.9
$\sum n_i C_p$		4237.1	3063.0	-2061.8	5683.5

$$\int_{298}^{773} n_i C_p dT = \int_{298}^{773} (4237.1 + 3063.0 \times 10^{-2} T - 2061.8 \times 10^{-5} T^2 + 5683.5 \times 10^{-9} T^3) dT$$

$$= 7307.3 \text{ MJ/h}$$

Heat consumed in system by the endothermic reaction = $\Delta H_r^\circ \times \text{moles produced}$

$$= 70,224 \times 80 = 5,617,920 \text{ kJ/h} = 5617.9 \text{ MJ/h}$$

Heat to vaporize feed (gas phase reaction)

$$= 34.3 \times 145.5 = 4990.7 \text{ MJ/h}$$

Heat balance:

$$\text{Output} = \text{Input} + \text{Consumed} + Q$$

$$Q = H_{\text{product}} - H_{\text{feed}} + \text{Consumed}$$

$$= 7307.3 - (-84.4) + (5617.9 + 4990.7) = 18,002.3 \text{ MJ/h}$$

Taking the overall efficiency of the furnace as 70%, the gas rate required:

$$= \frac{\text{Heat input}}{(\text{calorific value} \times \text{efficiency})}$$

$$= \frac{18,002.3}{33.5 \times 0.7} = 768 \text{ m}^3/\text{h}$$

15.3.2 Equilibrium constant and Gibbs free energy

The reaction equilibrium constant is related to the change in Gibbs free energy of reaction by the equation:

$$\Delta G = -RT \ln K \quad (15.5)$$

where:

$-\Delta G$ = change in Gibbs free energy as a result of reaction at temperature T

R = ideal gas constant

K = reaction equilibrium constant, defined as:

$$K = \prod_{i=1}^n a_i^{\alpha_i} \quad (15.6)$$

where:

a_i = activity of component i

α_i = stoichiometric coefficient of component i , defined such that stoichiometric coefficients are positive for products and negative for reagents (so reagents end up on the denominator)

n = total number of components

Equilibrium constants for many commercial process reactions can be found in the literature using academic search tools such as the ACS chemical abstracts tool *SciFinder*.

The equilibrium constant is useful for understanding the equilibrium of the main process reaction and for assuring thermodynamic consistency between the rates of forward and reverse reactions; but when there are several simultaneous reactions, the equilibrium concentration is more easily evaluated by minimizing Gibbs free energy. This is the method used in the Gibbs reactor model that is found in all commercial process simulation programs. The use of Gibbs reactors and equilibrium reactors in process simulation is discussed in Section 4.5.1.

Many of the comments made about the enthalpy of reaction also apply to the Gibbs energy of reaction and by extension to the estimation of equilibrium constants. Care must be taken to correct the Gibbs energy to the process conditions, and the equations for variation of Gibbs energy and equilibrium constant with temperature and pressure can be found in most reaction engineering or thermodynamics textbooks. When using a process simulation program to estimate equilibrium constants (or equilibrium composition), the designer should be aware that the program may be using group contribution methods to estimate Gibbs energies and heat capacities, thereby introducing a higher margin of error into the result.

The measurement of actual chemical equilibrium is often more complex than might be expected. In principle, a reaction system that is held at the same conditions for a long enough time will reach equilibrium, allowing the equilibrium composition to be measured. In practice, this can be difficult because of the following factors:

- In real reacting systems, often many components are present, and it may be difficult to determine which species are able to react with each other and contribute to the overall equilibrium. This is particularly problematic for processes that involve electrolyte solutions (including most biological processes) and processes that deal with high-molecular-weight hydrocarbon compounds. It may not be practical to carry out enough experiments to determine all of the possible equilibrium constants.
- If the composition is not measured *in situ* in the reaction mixture, reactions may occur during sample withdrawal, preparation, and analysis, leading to a false estimate of composition. This is a significant problem for high-temperature reactions, as the sample will typically become cooled as it is withdrawn for analysis, and so the composition that is measured may have moved away from the high-temperature equilibrium composition. Rapid sample quench methods can improve accuracy, but for some very fast reactions, there will still be an offset.
- Slow reactions, such as thermal degradation, may affect the ultimate equilibrium, but may or may not be important for process purposes.

When using equilibrium constants from the literature, the design engineer should pay careful attention to the experimental design and test methodology to be satisfied that the values reported are consistent with the conditions of the test reactor.

15.3.3 Reaction mechanisms, rate equations, and rate constants

One of the most difficult concepts in reaction engineering for novice engineers to grasp is that you don't need to know the reaction rate to make a preliminary design of a reactor. If the required residence time or space velocity has been determined experimentally, the reactor design can be completed and scaled up without requiring any knowledge of the kinetics. In the context of industrial process development, it is usually necessary to complete a preliminary reactor design and process economic evaluation long before sufficient data have been collected to fit a predictive model of all the reactions involved.

The reaction rate equation expresses the rate of removal of a reagent or formation of a product, usually as a function of the concentrations of components present in the mixture, temperature, pressure, and other properties such as adsorption equilibrium and mass transfer properties. Few reactions of industrial significance have simple first- or second-order rate equations, for the following reasons:

- Most industrial processes use heterogeneous catalysis or enzymes, leading to Langmuir–Hinshelwood–Hougen–Watson or Michaelis–Menten kinetics.
- Many industrial reactions require mass transfer between vapor and liquid or between two liquid phases, leading to overall rate expressions that are modified to include mass transfer effects.

- Many industrial reactions have multistep mechanisms, leading to rate expressions that do not necessarily follow the overall reaction stoichiometry.
- In most industrial processes there are several competing side reactions in addition to the main reaction. The rate of feed consumption or product removal is then the sum of contributions from each of these reactions.

Having said this, in many cases, the main process reaction can be approximated as first or second order over a narrow range of temperature, pressure, and concentration for purposes of estimating the required residence time to achieve a given conversion.

Reaction rate equations and rate constants cannot reliably be predicted from first principles and must be determined by fitting to experimental data. The books listed in the bibliography at the end of this chapter give details of how to measure rate constants experimentally; see also [Green and Southard \(2018\)](#). [Stewart and Caracotsios \(2008\)](#) provide an excellent introduction to techniques for model discrimination and parameter estimation.

The chemical engineering literature contains an abundance of papers that report reaction mechanisms and rate equations. These can be located through the ACS Chemical Abstract Service using *SciFinder*. Before using kinetics data from the literature, the designer should make a careful search for other papers that confirm the results, and if possible, should validate the rate model against commercial or pilot plant data.

Reaction mechanisms can be very sensitive to process or experimental conditions, and the rate-limiting step may change as a result of relatively small changes in temperature or concentrations, particularly for solid-catalyzed reactions. Rate equations are always a fit of experimental or plant data and should only be used for interpolation within the range of conditions spanned by the data. If extrapolation is necessary, more data must be collected to confirm that the rate model still applies. This is particularly important for exothermic reactions, which have the potential to suffer runaway. The mechanism and kinetics of exothermic reactions should always be studied over a wide enough range of temperature to allow safe design of the reactor system and collect necessary data for venting and relief load calculations; see Section 15.13.3.

15.3.4 Transport properties

Heat transfer

Heat transfer coefficients are needed for the design of internal heating or cooling devices and for reactions that are carried out in fired heaters or heat exchangers. Heat transfer coefficients can be predicted using the methods given in [Chapter 19](#) for tubular exchangers, fired heaters, internal coils, jacketed vessels, and stirred tanks.

When catalyst is packed inside the tubes of a heat exchanger, the internal heat transfer coefficient is increased. The tube-side heat transfer coefficient for a tube packed with pellets can be estimated from the correlations of [Leva \(1949\)](#).

$$\text{For heating: } \frac{h_i d_t}{\lambda_f} = 0.813 \left(\frac{\rho_f u d_p}{\mu} \right)^{0.9} e^{-6d_p/d_t} \quad (15.7)$$

$$\text{and for cooling: } \frac{h_i d_t}{\lambda_f} = 3.50 \left(\frac{\rho_f u d_p}{\mu} \right)^{0.7} e^{-4.6d_p/d_t} \quad (15.8)$$

Where:

- h_i = tube-side heat transfer coefficient for a packed tube
- d_t = tube diameter
- λ_f = fluid thermal conductivity
- ρ_f = fluid density
- u = superficial velocity
- d_p = effective particle diameter
- μ = fluid viscosity

Diffusivities

Diffusion coefficients are needed in the design of catalytic reactions where mass transfer can limit the rate of reaction, as well as in mass transfer processes, such as gas absorption, distillation, and liquid–liquid extraction. Experimental values for some common systems can be found in the literature, but for most design work, the values must be estimated.

For gases, the equation developed by Fuller et al. (1966) is easy to apply and gives reliable estimates:

$$D_v = \frac{1.013 \times 10^{-7} T^{1.75} \left(\frac{1}{M_a} + \frac{1}{M_b} \right)^{1/2}}{P \left[\left(\sum_a v_i \right)^{1/3} + \left(\sum_b v_i \right)^{1/3} \right]^2} \quad (15.9)$$

where:

D_v = diffusivity, m^2/s

T = temperature, K

M_a, M_b = molecular masses of components a and b

P = total pressure, bar

$\sum_a v_i, \sum_b v_i$ = the summation of the special diffusion volume coefficients for components a and b , given in Table 15.2

Fuller's method is illustrated in Example 15.2.

The diffusivity of components in the liquid phase can be predicted using the equation developed by Wilke and Chang (1955), given here.

$$D_L = \frac{1.173 \times 10^{-13} (\phi M_w)^{0.5} T}{\mu V_m^{0.6}} \quad (15.10)$$

where D_L = liquid diffusivity, m^2/s .

ϕ = an association factor for the solvent.

= 2.6 for water (some workers recommend 2.26).

= 1.9 for methanol.

= 1.5 for ethanol.

= 1.0 for unassociated solvents.

M_w = molecular mass of solvent.

μ = viscosity of solvent, $\text{mN s}/\text{m}^2$.

T = temperature, K.

V_m = molar volume of the solute at its boiling point, m^3/kmol . This can be estimated from the group contributions given in Table 15.3.

TABLE 15.2 Special atomic diffusion volumes

Atomic and structural diffusion volume increments			
C	16.5	Cl	19.5*
H	1.98	S	17.0*
O	5.48	Aromatic or hetrocyclic rings	-20.0
N	5.69*		
Diffusion volumes of simple molecules			
H ₂	7.07	CO	18.9
D ₂	6.70	CO ₂	26.9
He	2.88	N ₂ O	35.9
N ₂	17.9	NH ₃	14.9
O ₂	16.6	H ₂ O	12.7
Air	20.1	CCL ₂ F ₂	114.8*
Ne	5.59	SF ₆	69.7*
Ar	16.1	Cl ₂	37.7*
Kr	22.8	Br ₂	67.2*
Xe	37.9*	SO ₂	41.1*

*Value based on only a few data points.

From Fuller et al. (1966).

TABLE 15.3 Structural contributions to molar volumes, m^3/kmol

Molecular volumes							
Air	0.0299	CO ₂	0.0340	H ₂ S	0.0329	NO	0.0236
Br ₂	0.0532	COS	0.0515	I ₂	0.0715	N ₂ O	0.0364
Cl ₂	0.0484	H ₂	0.0143	N ₂	0.0312	O ₂	0.0256
CO	0.0307	H ₂ O	0.0189	NH ₃	0.0258	SO ₂	0.0448
Atomic volumes							
As	0.0305	F	0.0087	P	0.0270	Sn	0.0423
Bi	0.0480	Ge	0.0345	Pb	0.0480	Ti	0.0357
Br	0.0270	H	0.0037	S	0.0256	V	0.0320
C	0.0148	Hg	0.0190	Sb	0.0342	Zn	0.0204
Cr	0.0274	I	0.037	Si	0.0320		
Cl, terminal, as in RCl			0.0216	in higher esters, ethers		0.0110	
medial, as in R—CHCl—R			0.0246	in acids		0.0120	
Nitrogen, double-bonded			0.0156	in union with S, P, N		0.0083	
triply bonded, as in nitriles			0.0162	three-membered ring		−0.0060	
in primary amines, RNH ₂			0.0105	four-membered ring		−0.0085	
in secondary amines, R ₂ NH			0.012	five-membered ring		−0.0115	
in tertiary amines, R ₃ N			0.0108	six-membered ring as in benzene, cyclohexane, pyridine		−0.0150	
Oxygen, except as noted below			0.0074				
in methyl esters			0.0091	Naphthalene ring		−0.0300	
in methyl ethers			0.0099	Anthracene ring		−0.0475	

From Gambill (1958).

The Wilke–Chang method is illustrated in Example 15.3.

The Wilke–Chang correlation is shown graphically in Fig. 15.7. This figure can be used to determine the association constant for a solvent from experimental values for D_L in the solvent.

The Wilke–Chang equation gives satisfactory predictions for the diffusivity of organic compounds in water, but not for water in organic solvents.

Example 15.2

Estimate the diffusivity of methanol in air at atmospheric pressure and 25 °C.

Solution

Diffusion volumes from Table 15.2; methanol:

Element	v_i		No. of	
C	16.50	×	1 =	16.50
H	1.98	×	4 =	7.92
O	5.48	×	1 =	5.48
			$\sum_a v_i$	29.90

Diffusion volume for air = 20.1

1 standard atmosphere = 1.013 bar

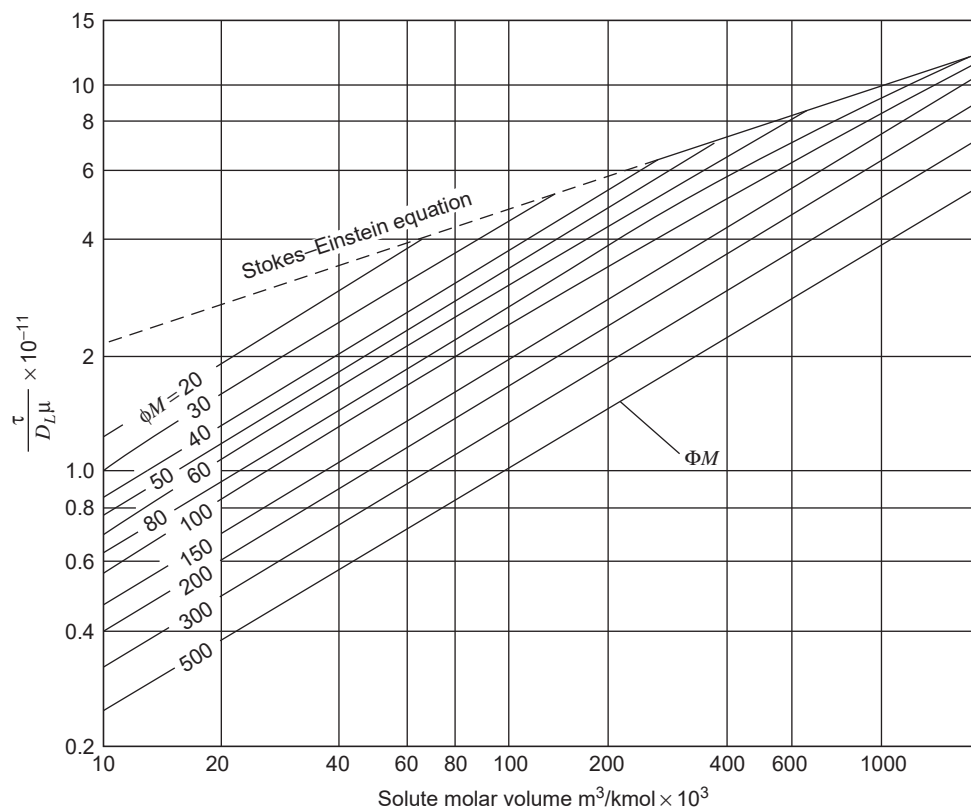


FIG. 15.7 The Wilke–Chang correlation.

molecular mass $\text{CH}_3\text{OH} = 32$, air = 29

$$D_v = \frac{1.013 \times 10^{-7} \times 298^{1.75} (1/32 + 1/29)^{1/2}}{1.013 \left[(29.90)^{1/3} + (20.1)^{1/3} \right]^2} \quad (15.9)$$

$$= \underline{\underline{16.2 \times 10^{-6} \text{ m}^2/\text{s}}}$$

Experimental value, $15.9 \times 10^{-6} \text{ m}^2/\text{s}$

Example 15.3

Estimate the diffusivity of phenol in ethanol at 20 °C (293 K).

Solution

Viscosity of ethanol at 20 °C, 1.2 mNs/m²

Molecular mass, 46

Molar volume of phenol, $\text{C}_6\text{H}_5\text{OH}$, from Table 15.3:

Atom	Vol.	No. of	
C	0.0148 ×	6 =	0.0888
H	0.0037 ×	6 =	0.0222
O	0.0074 ×	1 =	0.0074
ring	−0.015 ×	1 =	<u>−0.015</u>
			0.1034 m ³ /k mol

$$D_L = \frac{1.173 \times 10^{-13} (1.5 \times 46)^{0.5} 293}{1.2 \times 0.1034^{0.6}} = \underline{\underline{9.28 \times 10^{-10} \text{ m}^2/\text{s}}} \quad (15.10)$$

Experimental value, $8 \times 10^{-10} \text{ m}^2/\text{s}$, error 16%.

Mass transfer coefficients

Mass transfer coefficients are used in multiphase reactors, for example, in calculating the flux of material to and from the surface of solid catalysts and in calculating the mass flux across a vapor–liquid interface. It is very unlikely that a suitable mass transfer coefficient for a given reactor design can be found in the literature, so the mass transfer coefficient must usually be estimated using correlations.

For a single pellet or droplet suspended in a fluid, the mass transfer coefficient can be predicted using the Frössling equation (Frössling, 1938):

$$\text{Sh} = 2.0 + 0.552 \text{Re}^{0.5} \text{Sc}^{0.33} \quad (15.11)$$

where: Sh = Sherwood number = $k d_p / D$

k = mass transfer coefficient, m/s

d_p = particle diameter, m

D = diffusivity, m^2/s

Re = Reynolds number = $\rho_f u_p d_p / \mu_f$

u_p = particle velocity relative to the fluid, m/s

Sc = Schmidt number = $\mu_f / \rho_f D$

For mass transfer to a packed bed of particles, the equation of Gupta and Thodos (1963) can be used:

$$\text{Sh} = 2.06 \frac{1}{\varepsilon} \text{Re}^{0.425} \text{Sc}^{0.33} \quad (15.12)$$

where: ε = bed void fraction

Re is calculated using the superficial velocity of the fluid through the bed

Mass transfer between vapor and liquid in packed beds is discussed in Section 17.14. Mass transfer from a falling drop to a vapor can be estimated using Frössling's equation, Equation 15.11. Mass transfer between vapor and liquid in an agitated vessel can be predicted using the equations given by Van't Riet (1979):

$$\text{For air – water : } k_L a = 0.026 \left(\frac{P_a}{V_{liq}} \right)^{0.4} Q^{0.5} \quad (15.13)$$

$$\text{and for air – water – electrolyte: } k_L a = 0.002 \left(\frac{P_a}{V_{liq}} \right)^{0.7} Q^{0.2} \quad (15.14)$$

where:

k_L = mass transfer coefficient, m/s

a = interfacial area per unit volume, m^2/m^3

Q = gas volumetric flow rate, m^3/s

V_{liq} = liquid volume, m^3

P_a = agitator power input, W

The mass transfer coefficient for other low-viscosity systems can be calculated from the air–water coefficient using the method suggested by Fair (1967):

$$\frac{(k_L a)_{\text{system}}}{(k_L a)_{\text{air–water}}} = \left(\frac{D_{L,\text{system}}}{D_{L,\text{air–water}}} \right)^{0.5} \quad (15.15)$$

where D_L = liquid phase diffusivity, m^2/s

Green and Southard (2018) recommend caution when using mass transfer correlations for vapor–liquid systems if surfactants may be present. Whenever possible, the results predicted from correlations should be benchmarked against pilot plant measurements.

15.4 Choice of reaction conditions

The choice of operating conditions for carrying out a reaction plays a large role in determining what type of reactor will be used. Although the best conditions will typically be determined experimentally by the research team, the reactor designer can guide the chemists to conditions that will lead to a more optimal overall design. Chemists are generally trained to optimize syntheses by maximizing the reactor yield of desired product; however, the conditions that maximize reactor yield will usually not be optimal from a process standpoint, as operation at lower feed conversion usually improves selectivity and hence overall process yield. By performing a preliminary economic analysis, as described in Section 2.6, the design engineer can set yield and selectivity targets that will focus the experimentation on conditions that are more likely to lead to an optimum overall process.

When the final reaction conditions have been selected, it is important to confirm experimentally that the target process yields and selectivities are obtainable in practice. The validation of reactor designs is discussed in Section 15.12.

15.4.1 Chemical or biochemical reaction

Biochemical reactions involve the use of microorganisms, cells, or enzymes to carry out the process chemistry. Many compounds can be made by both chemical and biochemical routes, but some highly valuable products can only be synthesized using biochemical techniques.

Biochemical reactions must be carried out under conditions that preserve the effectiveness of the biological agent: the microorganism, cells, or enzymes that carry out the reaction. Although life has evolved to exploit almost every condition encountered on our planet, most living systems are not robust to a wide variation in environmental conditions. Even isolated enzymes cannot be used over a wide range of temperature, as proteins thermally denature at relatively low temperatures. The conditions that support life therefore dictate the reaction conditions in biochemical reactors, and a particular biochemical reaction will usually be constrained to relatively narrow ranges of temperature, shear rate, concentration of oxygen, and other solutes and pH. The optimal range of these conditions depends on the organism or enzyme chosen. Almost all biological processes are carried out in an aqueous solvent, and most are operated at temperatures slightly above ambient. The design of bioreactors is discussed in more detail in Section 15.9.

15.4.2 Catalyst

A catalyst increases the rate of a chemical reaction without itself becoming permanently changed by the reaction. By increasing the reaction rate, catalysts allow reactions to be run in smaller reactors, or in some cases allow a reaction to be operated at lower temperature. Most important of all, catalysts can be found that are more selective for a desired reaction, and hence the desired reaction can be promoted relative to other reactions and the process selectivity can be improved.

Catalytic reactors are discussed in more detail in Section 15.8. If a catalyst that is selective for the desired reaction can be found, it will almost always lead to a more economically attractive process than a noncatalytic route. Much of the research and development effort of the chemical industry therefore goes into searching for better catalysts.

The use of a catalyst usually introduces additional constraints into the selection of reactor operating conditions. Regardless of whether the catalyst is homogeneous (same phase as the reagents) or heterogeneous (different phase from the reagents—usually a solid), the reactor conditions must maintain catalyst activity for a reasonable length of time between catalyst regenerations. Catalysts are in general less stable and more prone to deactivation at high temperatures. Catalysts can also be sensitive to feed contaminants and may impose constraints on the concentration of certain components that could be present in the feed or might accumulate in recycle streams.

15.4.3 Temperature

Higher temperatures always increase the rate of reaction, diffusivities, and mass transfer rates. Higher temperatures increase the equilibrium constant of endothermic reactions and decrease the equilibrium constant of exothermic reactions (Fig. 15.8), but it should be noted that many exothermic reactions have very high equilibrium constants even at high temperature.

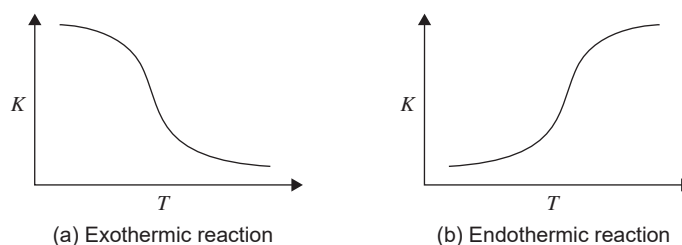


FIG. 15.8 Effect of temperature on equilibrium constant.

In general, increasing reaction temperature will lead to a less costly reactor design until one of the following constraints or trade-offs is encountered:

- Biological agents such as microorganisms, cells, and enzymes are killed or denatured at relatively low temperatures. Most biological processes are operated in the range 20 °C to 50 °C, and very few biological processes can be operated above 100 °C.
- Most organic compounds undergo thermal degradation reactions at high temperature. The presence of certain substituents, such as oxygen and nitrogen, can accelerate thermal degradation, and more complex molecular structures are typically more sensitive to pyrolysis. As temperature is raised, a point will be reached where thermal degradation will begin to have a detrimental effect on reactor selectivity. Some organic compounds can be sensitive to thermal decomposition at temperatures as low as 100 °C.
- At higher temperatures it can be harder to suppress side reactions such as polymerization or auto-oxidation. These reactions can pose a safety hazard, as well as reducing selectivity.
- The selectivity of oxidation reactions (and exothermic reactions in general) usually decreases as temperature is increased, as it becomes harder to maintain locally optimal concentrations of feed, oxidant, and product, and thus the tendency to oxidize the product increases. Selective oxidation reactions are therefore usually operated at as low a temperature as is practical and often use solvents or diluents to reduce the impact of the reaction exotherm.
- It is more difficult to maintain temperature control of exothermic reactions at high temperature, and there is a greater potential for the reaction to run away. Safety considerations in the design of reactors for exothermic processes are discussed in Section 15.13.
- At very high temperatures the cost of constructing a reactor becomes prohibitive. Maximum allowable stresses for steels start to decrease significantly with temperature above 300 °C. The pressure vessel design codes limit the maximum temperature at which alloys can be used; see Section 6.3.7. For reactions carried out under pressures greater than about 70 bar, an important threshold is reached at 482 °C (900 °F), as this is the maximum temperature limit for using the ASME Boiler and Pressure Vessel Code Section VIII Division 2 design rules. Any vessel operating with a wall temperature above this limit would need to be designed under the more conservative (hence higher cost) Division 1 rules. Higher temperatures can be accommodated by using *cold-wall* reactors, in which the reactor is lined with refractory material to reduce the temperature at the pressure-retaining wall, but this approach leads to larger reactors, and cold-wall reactors must be inspected regularly to ensure integrity of the lining.

15.4.4 Pressure

Several factors must be considered in selecting the reactor pressure. The primary consideration is usually to ensure the reactor contents are in the desired phase under the temperature conditions selected.

In some cases, the pressure is chosen so as to allow or prevent vaporization of a component. For example, it may be advantageous to allow a product to evaporate during the reaction so that it can be recovered from the vapor phase. Alternatively, in some cases a stoichiometric by-product can be allowed to evaporate, causing the equilibrium of the main reaction to be shifted. Feed, product, or solvent are sometimes allowed to evaporate as a means of removing heat from the reactor.

For gas-phase reactions, increasing the reactor pressure has the effect of increasing reactant activity, and hence increases the reaction rate. For reactions that are approaching equilibrium, the reactor yields change in accordance with Le Chatelier's principle. If the reaction leads to an increase in total number of moles, then a higher equilibrium conversion will be obtained at low pressure. Conversely, if the reaction leads to a reduction in total number of moles, then the equilibrium conversion will be higher at high pressure.

For gas-liquid reactions, increasing pressure will increase the solubility of the gas-phase components in the liquid, increasing the reaction rate. The effect of pressure on mass transfer rate is usually small. The savings in reactor cost resulting from higher gas solubility will, however, often be negligible compared with the increased

cost of compressing the gas. An additional important benefit of high-pressure operation is that less liquid will be lost by evaporation into any gas that is vented from the process. The savings in vent gas recovery equipment from operating at higher pressure are often greater than the savings in reactor cost.

15.4.5 Reaction phase

Reactions are usually not carried out in the solid phase unless the reagent is an insoluble solid such as coal, wood pulp, mineral ore, recycled polymer, etc. Fluids are easier to handle, compress, meter (flow control), heat and cool, and transport between process operations. Solid-phase reagents or products such as cells, polymer beads, and minerals are usually suspended in a gas or liquid for reaction.

The reaction temperature often dictates the reaction phase. In general, operation in the liquid phase gives the highest concentrations and most compact reactor, but at temperatures above the critical temperature, it is not possible to form a liquid phase. If all the reacting components cannot be maintained in the liquid phase at a reasonable pressure, a multiphase reactor will be needed, which will introduce mass transfer resistances and require a reactor with high vapor–liquid interfacial area; see Section 15.8. In some cases it will then make sense to decrease the pressure to the point where the reaction temperature is above the dew point and the reaction can be carried out in the vapor phase.

Some reaction systems are designed to operate with two liquid phases. Liquid–liquid reactors can have several advantages over single liquid-phase reactors and are discussed in more detail in Section 15.7.2.

15.4.6 Solvent

Many liquid-phase reactions are carried out with the reacting components dissolved in an inert solvent. A solvent can serve several purposes:

- Dilution of feed components to reduce the rates of side reactions and improve selectivity
- Increasing the solubility of gas-phase reactants
- Allowing components that would otherwise be solids to be dissolved in the reacting phase
- Increasing thermal mass and reducing the temperature change due to reaction
- Bringing components that would be mutually insoluble together so that they can react

A good solvent should be inert to the main reaction and to reactions with by-products and feed contaminants. A good solvent should also have low cost, should be easily separated from the reaction products, and should not introduce serious safety or environmental concerns. Properties of some of the solvents that are most widely used in the process industries are given in [Table 15.4](#).

TABLE 15.4 Commonly used process solvents

Solvent	Advantages	Disadvantages
Water	• Low cost, readily available • Easy and inexpensive to dispose of waste • Nontoxic, not flammable • Natural environment for most biological agents • Strong polar solvent (dielectric constant = 80.1) with ability to hydrogen bond, hence high solubility for many organic and inorganic compounds • High density allows easy separation from insoluble organic liquids, enabling liquid–liquid separations • High specific heat capacity gives high thermal mass and moderates temperature change due to heat of reaction, making temperature control easier • Remains liquid over a broad range of temperature at moderate pressures	• Presence of ions can make water corrosive • Low solubility for many nonpolar organic compounds • Presence of salts or ions can cause unwanted formation of second liquid phase • Heterogeneous catalysis can be more difficult in aqueous phase (due to leaching, galvanic effects, etc.) • Forms azeotropes with many compounds, making downstream separations more difficult • Reactive with many compounds • High latent heat can make recovery by distillation energy intensive • Contaminated water must be carefully handled to prevent mixing with groundwater

TABLE 15.4 Commonly used process solvents—cont'd

Solvent	Advantages	Disadvantages
Light alcohols (methanol, ethanol, n-propanol, isopropanol)	• Low-cost commodity chemicals • Moderate polar solvents (dielectric constants MeOH = 33, EtOH = 25, n-PrOH = 20, i-PrOH = 18) with ability to hydrogen bond • Liquid over broad range of temperature at moderate pressures, but easily recovered by fractionation as long as azeotropes are not present	• Toxic • Flammable • Tendency to form azeotropes with water and oxygenated compounds (except methanol), requiring more complex distillation for solvent recovery • Easily oxidized to aldehydes and acids (except isopropanol)
Acetic acid	• High-dipole moment makes acetic acid a good polar solvent, despite low dielectric constant (6.2). • Able to hydrogen bond • Resistant to oxidation, so widely used as solvent in oxidation reactions • Can be recovered by fractionation at moderate temperatures	• Toxic • Corrosive in presence of water • Flammable • Tendency to form azeotropes with other oxygenated compounds
Acetone	• Good polar aprotic solvent with high solubility for many organic compounds, including polymers • Miscible with water • Low chronic and acute toxicity in normal use	• Flammable with low flash point (−20 °C) and broad range of flammability in air, so easily ignited
Acetonitrile	• Medium polarity solvent (dielectric constant 37.5) with ability to dissolve many electrolytes as well as organic compounds • Low viscosity; hence widely used in chromatography	• Flammable • Moderate toxicity because acetonitrile is metabolized to hydrogen cyanide
Diethyl ether	• Good nonpolar solvent with high solubility for many organic compounds • Low solubility in water allows liquid–liquid extraction of products • Low toxicity in normal use	• Low autoignition temperature (160 °C) and flash point (−45 °C) make ether highly flammable • Prone to formation of explosive peroxides
Halogenated solvents (e.g., chloroform, dichloromethane, fluorinated solvents)	• Variation of the type and extent of halogenation allows a wide range of compounds to be developed with the ability to tailor polarity and volatility to specific needs • Reactivity with other compounds is usually low • Generally can be designed to be nonflammable and nontoxic • High density, allowing easier separation from insoluble liquids	• Some halogenated solvents (e.g., chloroform) are carcinogenic • Disposal costs can be high due to concerns about dioxin formation if incinerated, global warming potential
Benzene	• Good nonpolar solvent with low reactivity and low solubility in water, but highly carcinogenic and so usually substituted with toluene, cyclohexane, or other solvents whenever possible	• Flammable • Toxic, carcinogenic, and highly regulated
Toluene	• Low polarity • Low miscibility with water • More reactive than benzene, but much less toxic, so preferred over benzene as aromatic solvent	• Toxic • Flammable • Possible teratogen
Cyclohexane	• Low polarity • Low miscibility with water • Low reactivity with many compounds	• Extremely flammable (flash point −20 °C, autoignition temperature 245 °C) • Toxic (causes skin and lung irritation)
Paraffins (pentane, hexane, decane, etc.)	• Low polarity, only dissolve nonpolar compounds • Low miscibility with water • Inexpensive • Low reactivity with many compounds • Low acute toxicity	• Flammable • Volatility varies with carbon number, but light paraffins are easily vaporized and ignited

Note: Consult individual MSDS forms for additional safety information, including chemical incompatibility.

In some cases, a solvent can be found that will allow reaction and separation to be carried out simultaneously. For example, if the reaction feeds are soluble in the solvent but the product is not, product can be precipitated and removed continuously, shifting the equilibrium of the reaction to full conversion.

When selecting a solvent, care must be taken to ensure that the solvent will be compatible with all the components present in the reacting mixture. Some common solvents can be highly reactive with components such as oxidizing agents.

15.4.7 Concentrations

The concentration of components in the liquid phase can be increased up to the limit of solubility in the solvent, although operation at the limit of solubility is usually avoided to prevent problems due to precipitation or formation of a second liquid phase. The concentration of components in the gas phase (partial pressure) can be increased by increasing system pressure.

Feeds

Higher concentrations of feed compounds usually lead to a higher rate of reaction and a smaller reactor. In some situations, such as highly exothermic reactions, high concentrations should be avoided, in which case either a large excess of one feed is used or an inert material is added, as described next.

Feed components do not need to be supplied in stoichiometric ratio, and in fact very few processes use a stoichiometric ratio of feeds. Using a higher concentration of one feed can improve the selectivity of a reaction or the conversion of another compound. For example, the alkylation of benzene with propylene to form cumene is usually carried out with excess benzene. This increases the reaction selectivity by reducing the formation of diisopropyl- and triisopropyl-benzene, while also allowing full conversion of propylene and obviating propylene recovery for recycle.

By-products and contaminants

The concentrations of all possible feed contaminants and by-products are important variables that must be understood by the reactor designer and the research team. The effect of these components on catalysts and reaction rates can play a critical role in determining reactor performance and overall selectivities. It is very important to ensure that reactor performance has been tested under conditions that are representative of the expected levels of these components, particularly when recycle streams are present.

It may be advantageous to allow some by-product species to be recycled to the reactor. If the by-products are formed through reversible reactions that are not part of the desired reaction stoichiometry, they can be recycled to extinction, leading to greater overall selectivity to the desired product. Allowing some buildup of by-products in recycles can also sometimes suppress the rate of by-product formation. Allowing some buildup of by-products in a feed recycle can significantly reduce the cost of downstream separation processes.

Feed contaminants are generally more problematic than by-products, particularly for biological and catalytic processes. The feeds to fermentation reactors must be sterilized to prevent contamination by bacteria, viruses, fungal spores, etc.; see Section 15.9.3. Catalysts can be sensitive to poisons such as metals, compounds containing sulfur, oxygen or nitrogen, carbon monoxide (which adsorbs strongly on noble metals), and even water, which can strongly adsorb on solid acid catalysts. Compounds that have a tendency to form carbonaceous deposits on a catalyst (coke precursors) will accelerate the rate of catalyst deactivation by coking if they are allowed to accumulate in recycle streams.

If the effect of a feed contaminant is known to be particularly severe, the design should be modified to remove the contaminant upstream of the reactor. This may be as simple as adding a guard bed of a suitable adsorbent, as described in Section 16.2.1, or may require a more complex treatment process.

Inerts

The addition of inert compounds to a reacting mixture will usually increase the reactor cost, as well as adding downstream separation costs; however, there are some situations where it may still be worthwhile:

- The addition of an inert compound to a gas-phase reaction reduces the partial pressure of the reagents. This can increase the equilibrium conversion in reactions that lead to an increase in number of moles.
- Diluents can reduce the rate of reaction of a feed compound with itself, or with products, and thereby increase selectivity.

- Addition of an inert diluent can allow operation outside the flammability envelope. Oxygenation reactions are often carried out using depleted air (enriched in nitrogen) to reduce the likelihood of forming a flammable mixture.
- The presence of a diluent increases the heat capacity flow rate ($m.C_p$) of the mixture, and hence reduces the change in temperature caused by the heat of reaction. Moderation of the change in temperature has safety benefits for exothermic reactions and reduces the number of reheats required in endothermic reactions. It can also help prevent the formation of hot or cold spots in the reactor.
- Inert compounds are sometimes added to solutions to buffer the solution and allow more stable control of pH or ionic strength.

One of the best-known examples of use of a diluent is the use of steam in ethylene cracking furnaces, where the steam reduces the feed partial pressure, suppresses side reactions, and also suppresses coking.

15.5 Mixing

In addition to playing a critical role in reactor performance, mixing processes are important in many other processing stages, from the preparation of reagents through to the final blending of products. The equipment used depends on the nature of the materials and the degree of mixing required.

In many processes, some or all of the reactor feeds are premixed before being pumped to reactor pressure and heated to reaction temperature. Premixing is commonly used when solid feeds must be dissolved or fed as a slurry, and liquid and solids mixing operations are often carried out as batch processes.

This section addresses the mixing of fluids and mixing of solids into fluids. The blending of mixtures of dry solids is addressed in Section 18.4.11. An overview of scale-up rules for mixing is given by [Post \(2010\)](#).

15.5.1 Gas mixing

Specialized equipment is rarely needed for mixing gases, which because of their low viscosities mix easily. The mixing given by turbulent flow in a length of pipe is usually sufficient for most purposes. Turbulence promoters, such as orifices, vanes, or baffles, can be used to increase the rate of mixing. The piping arrangements used for inline mixing are discussed in the section on liquid mixing.

15.5.2 Liquid mixing

The following factors must be taken into account when choosing equipment for mixing liquids:

1. Batch or continuous operation
2. Nature of the process: miscible liquids, preparation of solutions, or dispersion of immiscible liquids
3. Degree of mixing required
4. Physical properties of the liquids, particularly the viscosity
5. Whether the mixing is associated with other operations such as reaction or heat transfer

Inline mixers can be used for the continuous mixing of low-viscosity fluids. For other mixing operations, stirred vessels or proprietary mixing equipment will be required.

Inline mixing

Static devices that promote turbulent mixing in pipelines provide an inexpensive way of continuously mixing fluids. Some typical designs are shown in [Fig. 15.9](#). A simple mixing tee (see [Fig. 15.9a](#)), followed by a length of pipe equal to 10 to 20 pipe diameters, is suitable for mixing low-viscosity fluids (≤ 50 mNs/m²), providing the flow is turbulent and the densities and flow rates of the fluids are similar.

With injection mixers (see [Figs. 15.9b and c](#)), in which the one fluid is introduced into the flowing stream of the other through a concentric pipe or an annular array of jets, mixing will take place by entrainment and turbulent diffusion. Such devices should be used where one flow is much lower than the other and will give a satisfactory blend in about 80 pipe diameters. The inclusion of baffles or other flow restrictions will reduce the mixing length required.

The static inline mixer shown in [Fig. 15.10](#) is effective in both laminar and turbulent flow and can be used to mix viscous mixtures. The division and rotation of the fluid at each element cause rapid radial mixing; see [Rosennzweig](#)

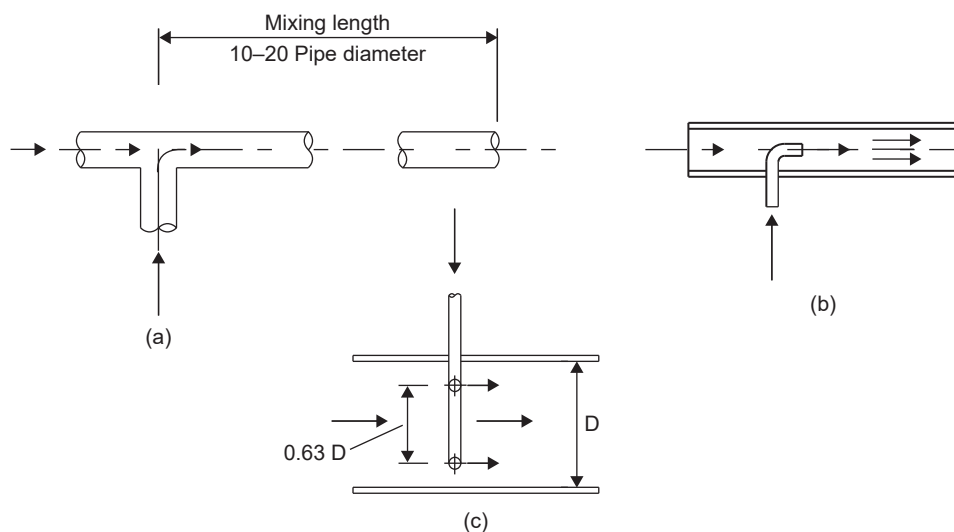


FIG. 15.9 Inline mixers. (a) Tee. (b) Injection. (c) Annular.



FIG. 15.10 Static mixer. Courtesy Kenics Corporation.

(1977) and Baker (1991). There is a great variety of different proprietary designs for static mixers, and they are easily found by searching on the Internet. The dispersion and mixing of liquids in pipes is discussed by Zughi et al. (2003) and Lee and Brodkey (1964).

Centrifugal pumps are effective inline mixers for blending and dispersing liquids. Various proprietary motor-driven inline mixers are also used for special applications; see Green and Southard (2018).

Stirred tanks

Mixing vessels fitted with some form of agitator are the most commonly used type of equipment for blending viscous liquids and preparing solutions of dissolved solids.

The stirred tank reactor can also be considered to be the basic chemical reactor, modeling on a large scale the conventional laboratory flask. Tank sizes range from a few liters to several thousand liters. They are used for homogeneous and heterogeneous liquid–liquid and liquid–gas reactions and for reactions that involve finely suspended solids, which can be held in suspension by the agitation. As the degree of agitation is under the designer's control, stirred tank reactors are particularly suitable for reactions where good mass transfer or heat transfer is required.

Most stirred tank reactors are not designed to operate completely filled with liquid. A vapor space is left at the top of the vessel to make pressure control easier. Stirred tanks should not be designed to operate more than 90% full, and 60% to 70% full is most typical. Lower liquid levels are used in reactors that have gas feeds (to allow for droplet disengagement) or are prone to foaming.

Liquid mixing in stirred tanks is covered in several textbooks: Uhl and Gray (1967), Harnby et al. (1997), Tatterson (1991, 1993), McCabe et al. (2004), Chhabra and Shankar (2017), and Paul et al. (2003).

A typical arrangement of the agitator and baffles in a stirred tank, and the flow pattern generated, is shown in Fig. 15.11. Mixing occurs through the bulk flow of the liquid and, on a microscopic scale, by the motion of the turbulent eddies created by the agitator. Bulk flow is the predominant mixing mechanism required for the blending of miscible liquids and for solids suspension. Turbulent mixing is important in operations involving mass and heat transfer, which can be considered as shear-controlled processes.

The most suitable agitator for a particular application will depend on the type of mixing required, the capacity of the vessel, and the fluid properties, mainly the viscosity.

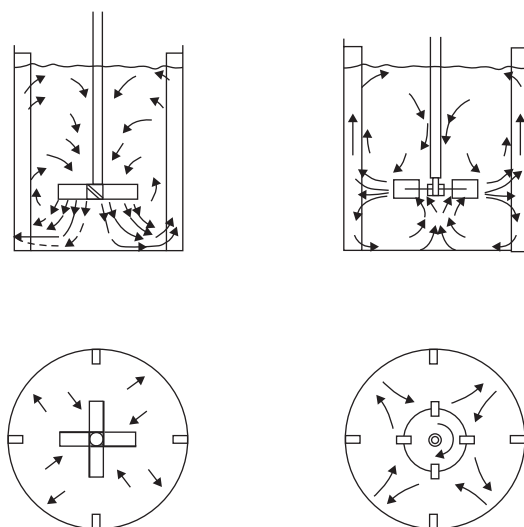


FIG. 15.11 Agitator arrangements and flow patterns.

The three basic types of impeller that are used at high Reynolds numbers (low viscosity) are shown in Fig. 15.12. They can be classified according to the predominant direction of flow leaving the impeller. The flat-bladed (Rushton) turbines are essentially radial-flow devices, suitable for processes controlled by turbulent mixing (shear-controlled processes). The propeller and pitched-bladed turbines are essentially axial-flow devices, suitable for bulk fluid mixing.

Paddle, anchor, and helical ribbon agitators (Fig. 15.13) and other special shapes are used for more viscous fluids.

The selection chart given in Fig. 15.14, which has been adapted from a similar chart given by Penny (1970), can be used to make a preliminary selection of the agitator type, based on the liquid viscosity and tank volume. Impeller selection is also discussed in the articles by Fasano (2015) and Grenville et al. (2017). Specially designed impellers are used in situations when high shear is needed, such as when dispersing fine powders into a liquid. See Dickey (2015) for examples.

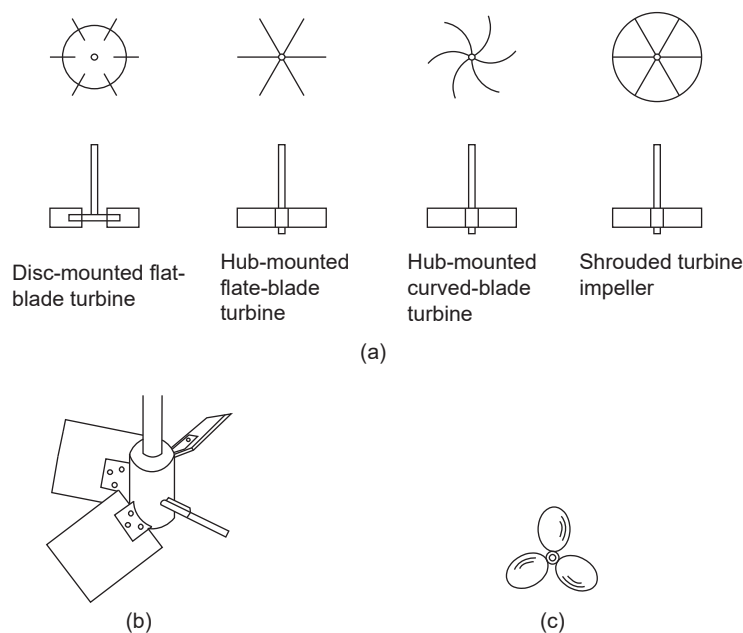


FIG. 15.12 Basic impeller types. (a) Turbine impeller. (b) Pitched bladed turbine. (c) Marine propeller.

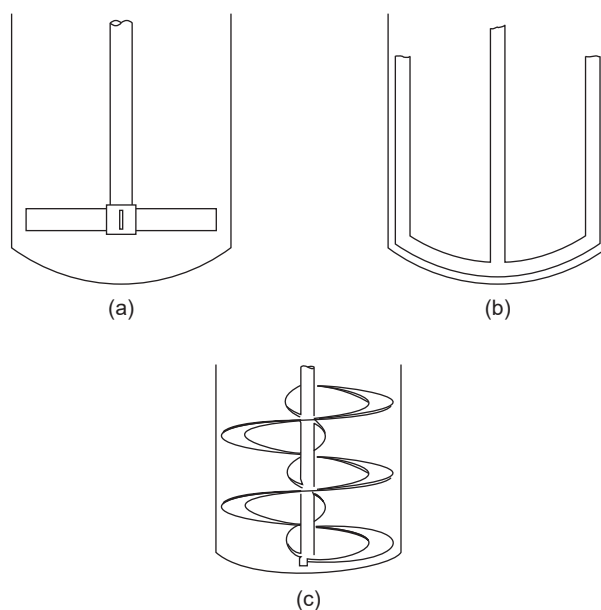


FIG. 15.13 Low-speed agitators. (a) Paddle. (b) Anchor. (c) Helical ribbon.

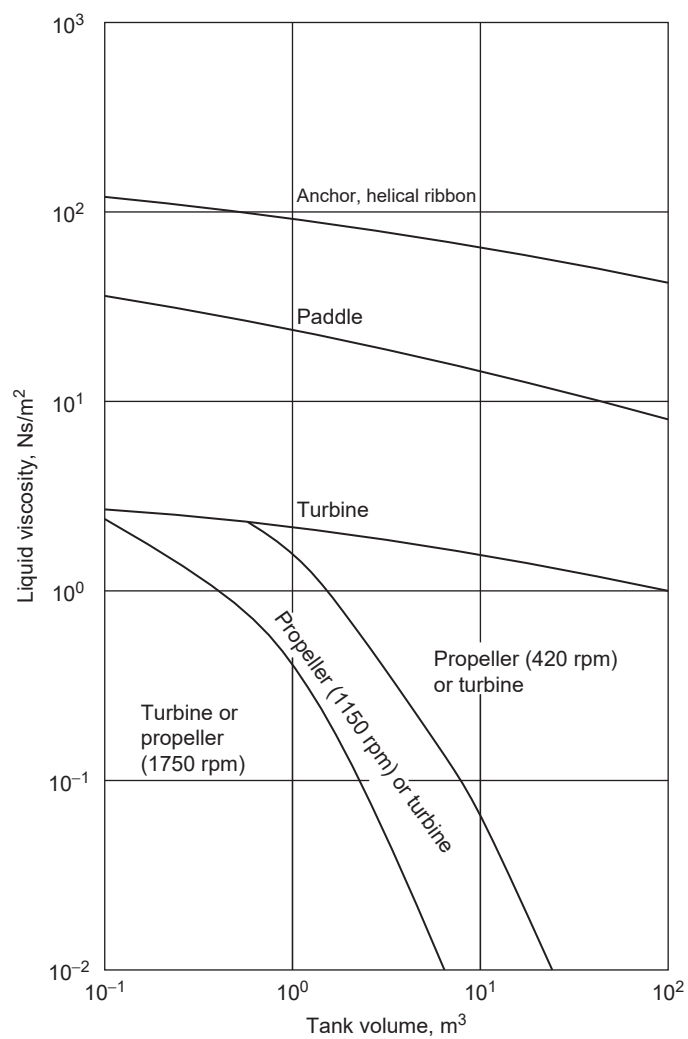


FIG. 15.14 Agitator selection guide.

For turbine agitators, impeller-to-tank diameter ratios of up to about 0.6 are used, with the depth of liquid equal to the tank diameter. Baffles are normally used to improve the mixing and reduce problems from vortex formation. Anchor agitators are used with close clearance between the blades and vessel wall, with anchor-to-tank diameter ratios of 0.95 or higher. The selection of agitators for dispersing gases in liquids is discussed by Hicks (1976).

Agitator power consumption

The power requirements for agitation will depend on the degree of agitation required and will range from about 0.2 kW/m³ for moderate mixing to 2 kW/m³ for intense mixing.

The shaft power required to drive an agitator can be estimated using the following generalized dimensionless equation, the derivation of which is given in Chhabra and Shankar (2017).

$$N_p = K Re^b Fr^c \quad (15.16)$$

where:

$$N_p = \text{power number} = \frac{P_a}{d_a^5 N^3 \rho}$$

$$Re = \text{Reynolds number} = \frac{d_a^2 N \rho}{\mu}$$

$$Fr = \text{Froude number} = \frac{d_a N^2}{g}$$

P_a = shaft power, W

K = a constant, dependent on the agitator type, size, and the agitator tank geometry

ρ = fluid density, kg/m³

μ = fluid viscosity, Ns/m²

N = agitator speed, s⁻¹ (revolutions per second, rps)

d_a = agitator diameter, m

g = gravitational acceleration, 9.81 m/s²

Values for the constant K and the indices b and c for various types of agitator, tank agitator geometries, and dimensions can be found in the literature; see Rushton et al. (1950). A useful review of the published correlations for agitator power consumption and heat transfer in agitated vessels is given by Wilkinson and Edwards (1972); they include correlations for non-Newtonian fluids. Typical power curves for propeller and turbine agitators are given in Figs. 15.15 and 15.16. In the laminar flow region, the index " b " = 1, and at high Reynolds number the power number is independent of the Froude number: index " c " = 0.

An estimate of the power requirements for various applications can be obtained from Table 15.5.

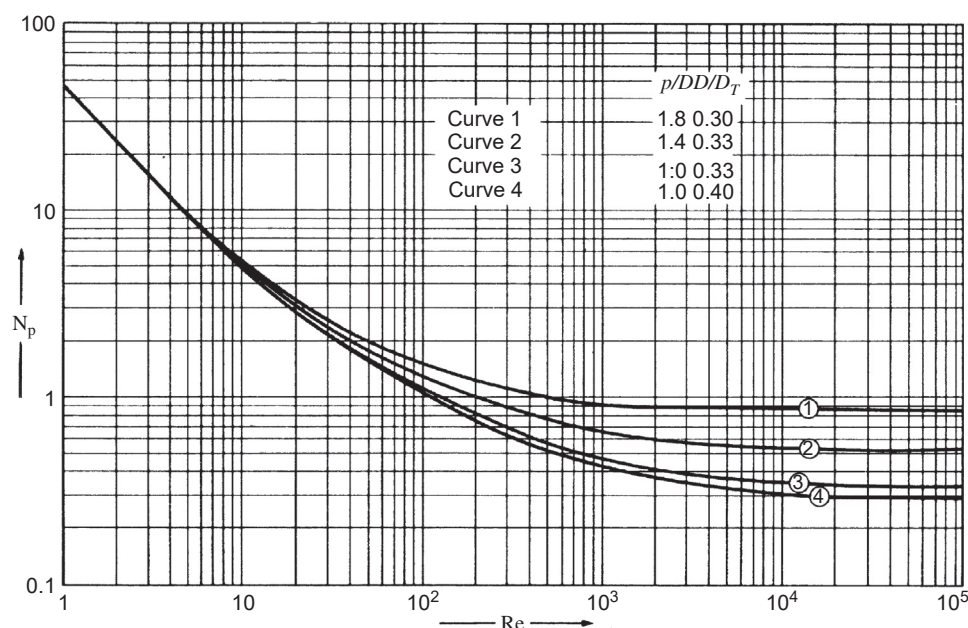


FIG. 15.15 Power correlation for single three-bladed propellers baffled. p = blade pitch, D = impeller diameter, DT = tank diameter. From Uhl & Gray, 1967, with permission.

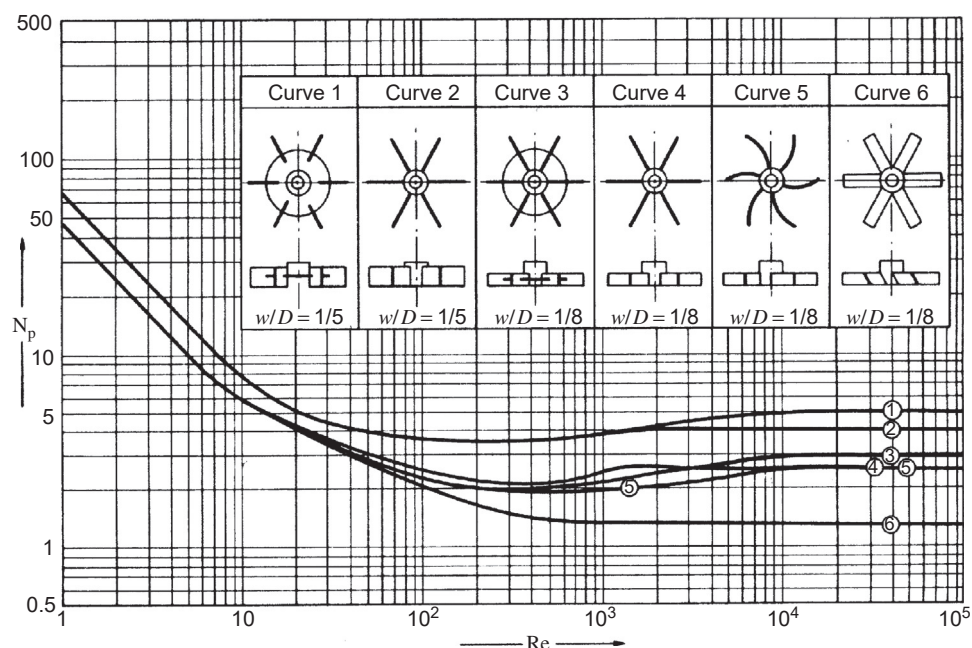


FIG. 15.16 Power correlations for baffled turbine impellers for tank with four baffles. w = impeller width, D = impeller diameter. From Uhl & Gray, 1967, with permission.

TABLE 15.5 Power requirements: Baffled agitated tanks

Agitation	Applications	Power, kW/m ³
Mild	Blending, mixing	0.04–0.10
	Homogeneous reactions	0.01–0.03
Medium	Heat transfer	0.03–1.0
	Liquid–liquid mixing	1.0–1.5
Severe	Slurry suspension	1.5–2.0
	Gas absorption	1.5–2.0
	Emulsions	1.5–2.0
Violent	Fine slurry suspension	>2.0

Side-entering agitators

Side-entering agitators are used for blending low-viscosity liquids in large tanks, where it is impractical to use conventional agitators supported from the top of the tank; see Oldshue et al. (1956).

Where they are used with flammable liquids, particular care must be taken in the design and maintenance of the shaft seals, as any leakage may cause a fire.

For blending flammable liquids, the use of liquid jets should be considered as an “intrinsically” safer option; see Fossett and Prosser (1949).

15.5.3 Gas–liquid mixing

Gases can be mixed into liquids using either inline mixing, stirred vessels, or the vapor–liquid contacting devices described in Chapter 17.

When a small amount of gas is fed or the gas dissolves completely, inline mixing can be used. The most common arrangement is an injection mixer (see Fig. 15.9b) followed by a static mixer. In some cases, a long injection tube with multiple holes drilled in it is used. This is known as a *sparger* (Fig. 15.17).

If a gas is injected into a stirred tank, the location of the gas injection must be chosen based on the mixing pattern obtained with the impeller that has been selected. The gas injection device is usually an annular ring with multiple small openings, and the openings are oriented to promote the desired circulation of gas bubbles. Methods such as computational fluid dynamics (CFD) are used to analyze the gas bubble flow pattern and ensure that the gas hold-up and interfacial area are adequate; see Section 15.11.3. The rate of mass transfer from gas to liquid can be predicted using Equations 15.13 to 15.15. It should always be assumed that any excess vapor leaving a stirred tank will be saturated with all the components of the liquid, and it is reasonable to assume at least 1 to 2 volume percent entrainment of liquid in the vapor if the rate of agitation is high.

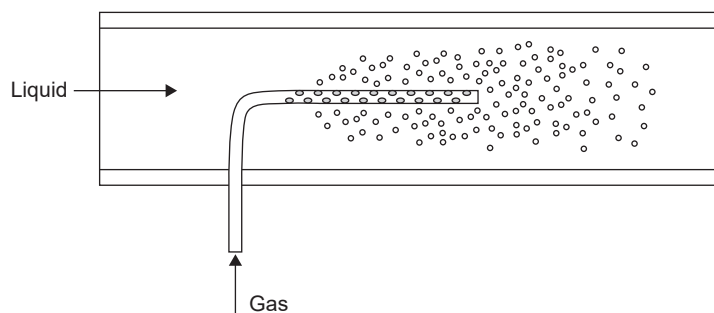


FIG. 15.17 Gas sparger.

The extent of agitation caused by injecting a gas into a liquid can be seen from Table 15.6, which is based on information from Green and Southard (2018).

A small flow of liquid can be dispersed into a gas stream using a spray nozzle (Fig. 15.18). Many different proprietary spray nozzles are available, and the nozzle is usually selected in consultation with a vendor.

When large flow rates of vapor and liquid are to be contacted to carry out reaction, mass transfer, or direct heat transfer, plate or packed columns are usually used; these are discussed in detail in Chapter 17.

TABLE 15.6 Air rate ($\text{ft}^3/\text{ft}^2 \cdot \text{min}$) required to cause agitation of water at 1 atm

Degree of agitation	Liquid depth 9 ft	Liquid depth 3 ft
Moderate	0.65	1.3
Complete	1.3	2.6
Violent	3.1	6.2

From Green and Southard (2018).

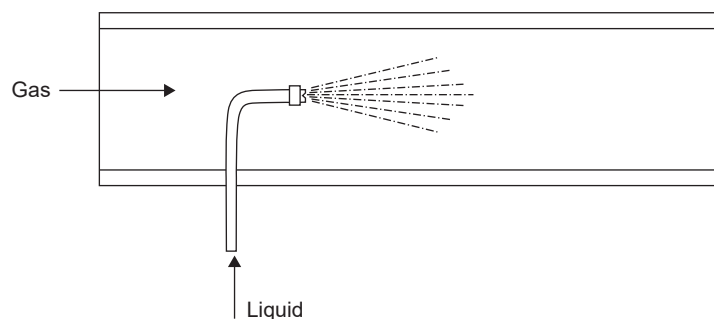


FIG. 15.18 Liquid injection into gas.

15.5.4 Solid–liquid mixing

Solids are mixed into liquids to dissolve solid feeds and to form slurries for transport and reaction of insoluble solids. Solids are usually added to a liquid in a stirred tank, although liquids are sometimes introduced into screw conveyors, which then act as mixing troughs. The design of stirred tanks is covered in Section 15.5.2.

Liquids and solids are usually mixed at atmospheric pressure to make handling and flow of the solid easier. The resulting mixture or slurry can then be pumped to the process pressure.

Mixing of liquids and solids is often carried out as a batch operation to allow more accurate control of dissolved solid concentration. A modest inventory of solution or slurry can then be used to maintain a constant feed to continuous downstream operations. If an inventory of slurry is maintained, it must be continuously agitated to prevent settling.

The formation of slurry feeds for hydraulic conveying is discussed in Section 18.3.5. Many of the same considerations apply when feeding a solid to a liquid-phase reactor.

15.6 Heating and cooling of reacting systems

Heat addition and removal are important for maintaining temperature control of exothermic and endothermic reactions. For endothermic reactions, the rate of heat input can sometimes be the rate-limiting step.

15.6.1 Heating and cooling reactors: Basic principles

Adding heating or cooling to a reactor design always increases the reactor cost. The designer should first consider the following:

1. Can the reaction be carried out adiabatically? If the heat of reaction is small, a small change in temperature across the reactor may be acceptable and heating or cooling can be avoided. If a diluent is added to the system, the thermal mass of the reagents will increase and the temperature change across the reactor will be reduced, so adding a diluent can enable adiabatic operation.
2. Can the feeds provide the required heating or cooling? Staged addition of hot feed to a reactor can provide additional heat to an endothermic reaction. Similarly, staged addition of cold feed can provide quench to an exothermic reaction. A commercial example of this is the use of hydrogen quenches to control the temperature increase during hydrocracking reactions (Fig. 15.19). In a stirred tank or other reactor that approximates well-mixed reactor behavior, the heat of reaction can sometimes be balanced by a sensible heat change between a hot or cold feed temperature and the reactor temperature.
3. Would it be more cost-effective to carry out the heat exchange outside the reactor? If the required heat transfer surface is large, it may be difficult to fit into a simple reactor design, in which case the designer should consider using an external heat exchange system. Examples are given in the following sections.
4. Would it be more effective to carry out the reaction inside a heat transfer device? If the required residence time is small, or if the volume of catalyst is small, the reaction can sometimes be carried out inside the tubes of a heat exchanger or fired heater. The use of heat-exchange devices as reactors is discussed in Section 15.6.4.
5. Does the proposed design allow the process to be started up and shut down smoothly? The heating or cooling system must be flexible to respond to nonsteady operating conditions of the plant, as well as steady-state operation.
6. Are there safety concerns with heating or cooling the reactor? For exothermic reactions, the designer should consider what would happen if there was a failure of the cooling system. How quickly can the control system respond and shut the reaction down before the exothermic reaction runs away? Exothermic reactors should not require a heater, and feedback between feed heaters, feed-effluent heat exchangers and the reactor must be considered. Are there concerns about the heat transfer medium leaking into the reactor or reactor contents leaking into the heat transfer medium?

The heating or cooling duty of the reactor is usually most easily estimated using commercial process simulation software. The designer must take care to ensure that the enthalpy of reaction and the heat capacities used by the simulator are sufficiently accurate for design purposes; see Section 15.3.1.

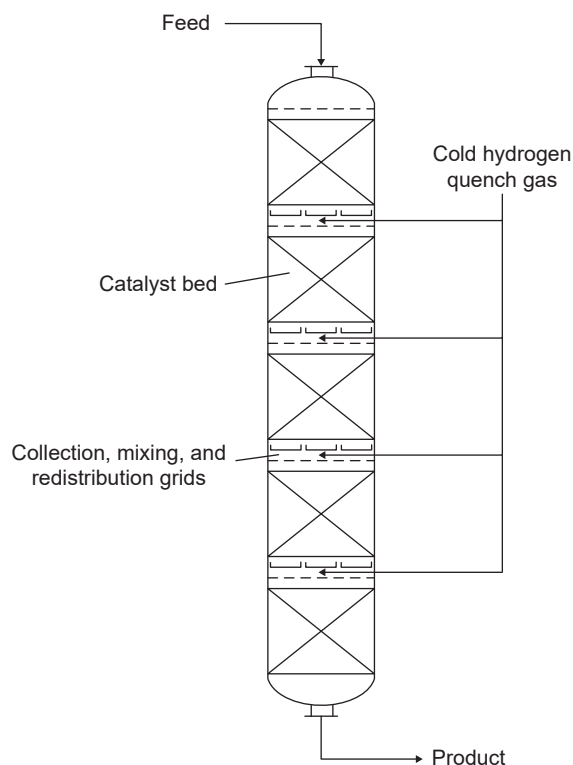


FIG. 15.19 Use of hydrogen quench in hydrocracking reactor.

Novice designers often make the mistake of forgetting to pre-heat (or pre-cool) the reactor feeds, leading to unreasonably large reactor heating or cooling duties. The reactor feed should be at the desired reaction temperature unless hot or cold feed is being used as a temperature control strategy.

15.6.2 Heating and cooling stirred tank reactors

A stirred tank reactor can be operated in batch or continuous mode. In either case, the objective in heating or cooling the contents is to achieve a uniform temperature throughout the vessel and eliminate hot or cold spots that could cause fouling, nonselective reaction, or lead to hazards such as reaction runaway. The heating or cooling should be accomplished without disrupting the mixing in the vessel and creating undermixed zones or dead spots.

Indirect heat transfer

Heat transfer to stirred vessels is described in detail in Section 19.18, and correlations for heat transfer coefficients are given in Section 19.18.3.

Fig. 15.20 shows the most common approaches to providing heating or cooling to stirred vessels. Low heating duties can be accommodated with a jacketed vessel (see Fig. 15.20a). The mechanical and thermal design of jacketed vessels is discussed in Section 19.18.1. Different utility flow patterns can be accommodated in a jacketed vessel. For

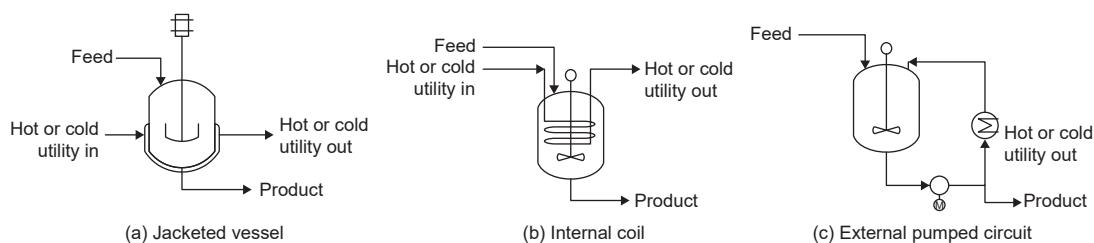


FIG. 15.20 Heating or cooling stirred-tank reactors.

example, when steam is used for heating a reactor, the steam usually flows in at the top of the jacket and condensate is drained near the bottom, whereas cooling water is usually fed from the bottom up.

If the area of a jacket is insufficient, an internal coil can be used (see Fig. 15.20b). Heat transfer to internal coils and coil pressure drop are covered in Section 19.18.2. Although coils can provide much more area, the volume of the reactor must be increased to accommodate the coil, and coils can be difficult to clean and sterilize.

For high rates of heat addition or removal, an external pumped circuit should be considered, as shown in Fig. 15.20(c). The use of an external heat exchanger removes any volume limit on accommodating the heat transfer surface and allows the heat exchange to be optimized independently of reactor design considerations. The residence time of the fluid in the pump, piping, and heat exchanger should be subtracted from the required residence time of the reactor. If a shell and tube heat exchanger is used, the reacting fluid will usually (but not always) be placed on the tube side to minimize the likelihood of forming dead spots or undermixed zones. Plate heat exchangers are often used in reactor pump-around circuits because of their low inventory of fluid, short residence time, and ease of cleaning.

In any reactor that uses indirect heat transfer, the designer must check carefully to ensure that local hot spots or cold spots on the heat transfer surface will not lead to fouling, excessive corrosion, or poor reaction selectivity. A calculation of the hottest and coldest metal surface temperatures should be made. These will normally be in regions close to where the utility enters and exits. The calculation can be based on a sophisticated model of reactor hydrodynamics, as described in Section 15.11.3, or can be as simple as an estimate based on internal and external heat transfer coefficients with suitably conservative assumptions. If there is a concern that either the hottest or coldest surface temperature could have a serious impact on reactor performance, the designer should reconsider the choice of hot or cold utility to attempt to reduce the temperature difference between the utility and reactor contents. This will generally lead to an increase in required area and may cause the selection of a different reactor design.

Direct heat transfer: Heating using live steam

If the reacting mixture is compatible with addition of water and the temperature and pressure are suitable, direct heat transfer to a vessel can be accomplished by injecting live steam into a reactor (Fig. 15.21). Steam is usually injected through a distribution ring or sparger, but if the vessel is well agitated, a simple dip tube may suffice.

The use of live steam eliminates the need for heat transfer surface and saves capital cost in the reactor. The trade-off is that the steam is lost to the process fluid and condensate cannot be recovered, so the cost of live steam must include the cost of replacing the boiler feed water. Process wastewater costs are also increased.

When live steam is used, it is unlikely that all the steam will condense, so provision must be made for a net vapor flow from the reactor. Any vapor leaving the reactor will be saturated with all the components of the reacting mixture, so the off-gas will normally be cooled and condensed to recover material that was vaporized. The cost impact of adding a vapor recovery system must also be considered when evaluating use of live steam if such a system was not already in place.

Direct heat transfer: Evaporative cooling

If the process feed, solvent, or product can be made to evaporate at the reaction temperature, evaporative cooling can be used to cool the reactor.

In an evaporative cooling system, the vapor from the reactor is usually cooled externally and condensed to recover process fluids. In some cases, two stages of cooling are used, with an air or water cooler followed by a refrigerated chiller. The condensed fluid is usually refluxed to the reactor to provide additional cooling (Fig. 15.22).

If the overhead system has adequate capacity, evaporative cooling can provide highly sensitive temperature control. A small decrease in pressure will quickly cause more evaporation and remove heat from the reactor, so the reactor temperature control can be cascaded onto the pressure control.

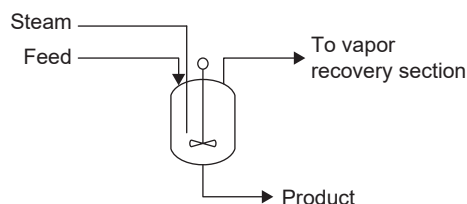


FIG. 15.21 Live steam injection.

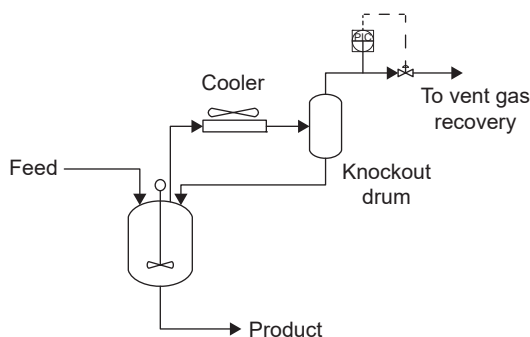


FIG. 15.22 Evaporative cooling.

15.6.3 Heating and cooling catalytic reactors

Slurry reactors

If a solid catalyst is mixed into a reacting liquid to form a slurry, heat transfer can be accomplished using the methods described in Section 15.6.2. Because slurries can cause erosion, the use of internal coils is not preferred. If an external pumped circuit is used, the pump design must be suitable for slurry service, and the fluid should be placed on the tube side of the exchanger to prevent filtering out of catalyst particles in dead spots. Plate heat exchangers can be used for slurries, but the designer should consult with a plate exchanger vendor to ensure that the gap spacing is adequate to prevent plugging.

Fixed-bed reactors

It is usually difficult to heat or cool a fixed-bed reactor by indirect heat transfer, as it is hard to maintain a uniform temperature profile across a radial section of the bed. Radial variation in temperature can cause faster deactivation of catalyst in the hotter zones and shorten the time between plant shutdowns for catalyst change-out.

The most common approach for heating or cooling a fixed-bed reactor is to divide the reactor into a series of smaller adiabatic beds and use interstage heating or cooling, as shown in Fig. 15.23. Each bed is sized to have an acceptable adiabatic change in temperature. In the case of an endothermic reaction, the inlet temperature must not be so hot that selectivity is compromised, whereas the outlet temperature must be hot enough to maintain

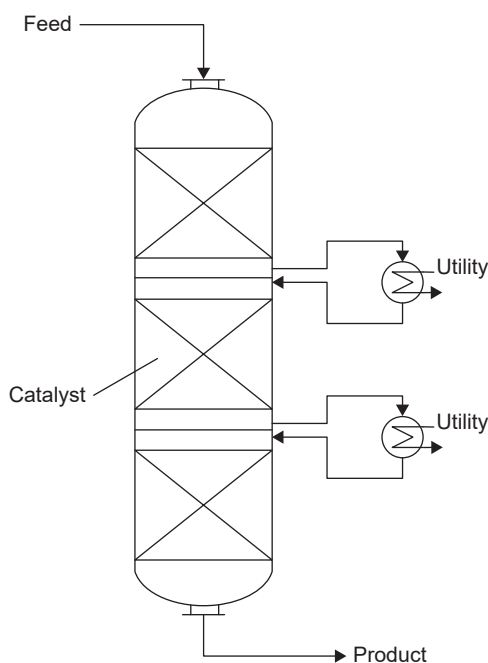


FIG. 15.23 Fixed-bed reactor with interstage heating or cooling.

reasonable catalyst activity. For an exothermic reaction, the situation is reversed, and care must be taken to ensure that the reaction will not run away at the outlet temperature. The bed sizes are not necessarily equal, as the reaction rate may vary from bed to bed.

In scaling up nonisothermal packed-bed reactors, the designer must ensure that the temperature change across each bed remains about the same as the capacity is increased. Because the temperature across any given bed is not uniform, both the bed inlet temperature and the average bed temperature should be kept the same during scale-up.

Occasionally, it will be possible to pack the catalyst into tubes of a heat exchanger or fired heater. The use of heat-exchange devices as reactors is discussed in Section 15.6.4.

Fluidized-bed reactors

Fluidized beds have high heat transfer coefficients; see [Zenz and Othmer \(1960\)](#). Indirect heat transfer to fluidized-bed reactors using coils or stab-in heat exchange bundles is therefore often effective.

The heat capacity of the solids in a fluidized bed allows the solid particles themselves to be used as a heat transfer medium. In processes such as fluidized catalytic cracking (FCC), the catalyst is heated to a temperature above the reaction temperature in a regeneration reactor. Hot catalyst is then circulated back to the reactor, where it provides the heat of reaction and heat needed to vaporize the feed. Deactivated catalyst is returned to the regeneration reactor. This process is discussed in more detail in Section 15.8.4 and illustrated in [Fig. 15.42](#).

15.6.4 Heat-exchange devices as reactors

When it is necessary to operate a reactor isothermally and there is a large heat of reaction, the heat transfer requirements dominate the design, and the reactor must be designed as a heat transfer device. Some common situations where this occurs include:

- High-temperature endothermic reactions that quickly quench without continuous heat input.
- Low-temperature exothermic reactions that must be kept at constant temperature to maintain selectivity or for safety reasons. Many selective oxidation reactions fall into this category.

Reactions can be carried out inside any heat transfer device, but the most commonly used arrangements are shell and tube heat exchangers or fired heaters. Plate heat exchanger reactors have been the subject of much research, but are not yet in widespread commercial use.

Homogeneous reaction

If the reaction mixture is a single phase and there is no need for a catalyst, then the mechanical design and layout of a heat exchanger reactor is the same as a conventional heat transfer device. The thermal design of a heat exchanger reactor, however, is much more complex than a conventional heat exchanger.

The heat exchanger design equations that are normally used for heat transfer will not apply to the design of a heat exchanger reactor due to the nonlinear behavior of the reaction rate with temperature and hence the variation in heat release (or uptake) along the exchanger tubes. For example, in the case of an exothermic reaction, the rate of reaction might be expected to be highest at the inlet, where the concentrations of feeds are highest, but as the reaction proceeds and temperature increases, the rate of reaction will increase, so the hot spot in the tube may actually be at some distance in from the inlet, as illustrated schematically in [Fig. 15.24](#). The location of the hot spot can be further complicated by the flow pattern on the shell side of the heat exchanger; hence determination of the effective temperature difference for the exchanger is not trivial. Many shell and tube heat exchanger reactors are operated using a constant temperature utility such as boiling water or a boiling refrigerant as coolant to simplify the thermal design and ensure good control of utility-side temperature.

It is usually not wise to make a conservative assumption of the temperature difference and proceed with a standard heat exchanger design. A design with excessive area will not achieve the desired reaction temperature and, if placed in operation, may require changes in utility temperature that could compromise process safety margins. Although such an approach can be useful as a means of generating an approximate capital cost for the reactor in the early stages of design, great care must be taken to ensure that the design is revised before detailed design begins.

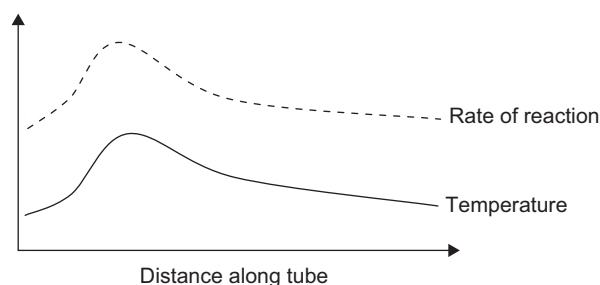


FIG. 15.24 Schematic profiles of temperature and reaction rate for an exothermic reaction in a heat exchange tube.

The approach that is typically followed in industrial practice is to construct a detailed model of the reaction kinetics and hydraulics, as described in Section 15.11.3. The model can then be fitted to experimental data to give confidence in its validity for scale-up of the design.

Heterogeneous reaction

When a catalyst is used inside a heat-exchange device, the designer faces the same issues as for homogeneous reactions, with the added complication of modifying the exchanger mechanical design to contain the catalyst.

Catalyst can be loaded into the tubes of a shell and tube heat exchanger if the exchanger is mounted vertically and a suitable retaining screen is used at the end of each tube or immediately below the bottom tubesheet. This arrangement is used in the oxidation of orthoxylene to phthallic anhydride, where the reaction is cooled using a circulating molten salt, as illustrated in Fig. 15.25. Heat from the molten salt is used to raise steam.

For high-temperature endothermic reactions, heat input from steam or molten salt is not practical, and the reactor must be designed for heat input from a fired heater. The rate of heat input in the convective section of a heater is usually too low to be useful because of the low heat transfer coefficient of the flue gas, so the reactor tubes must normally be placed in the radiant section of the furnace. It is difficult to maintain a uniform loading of catalyst in a horizontal tube, so the reactor tubes are typically hung vertically. The design must allow for thermal expansion of the tubes, which makes the use of U-tubes difficult, as the cold spot may be in one side of the U, causing the tube to bend. Several ingenious proprietary designs have been developed to overcome this problem. For example, ICI developed a tube-in-tube design for use in the steam reforming of methane to hydrogen, which is illustrated in Fig. 15.26. The chemistry of methane steam reforming is discussed in more detail in Example 4.5. In the ICI design

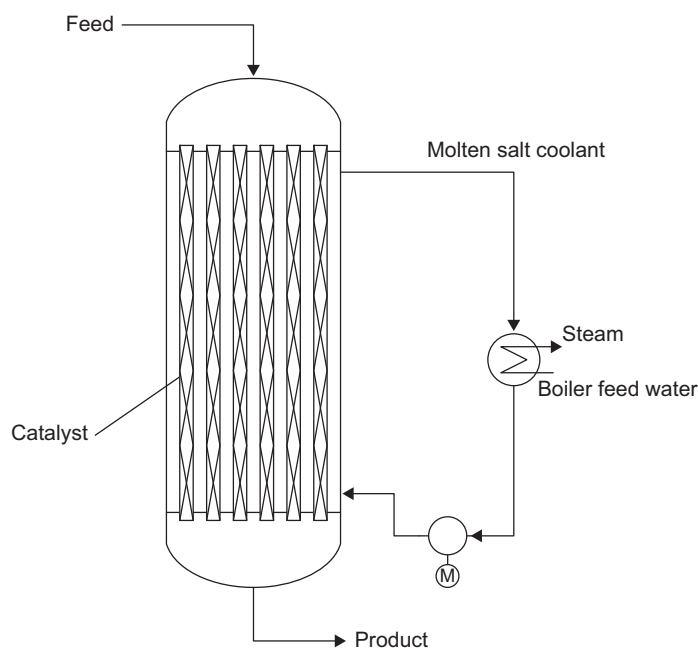


FIG. 15.25 Phthalic anhydride reactor.

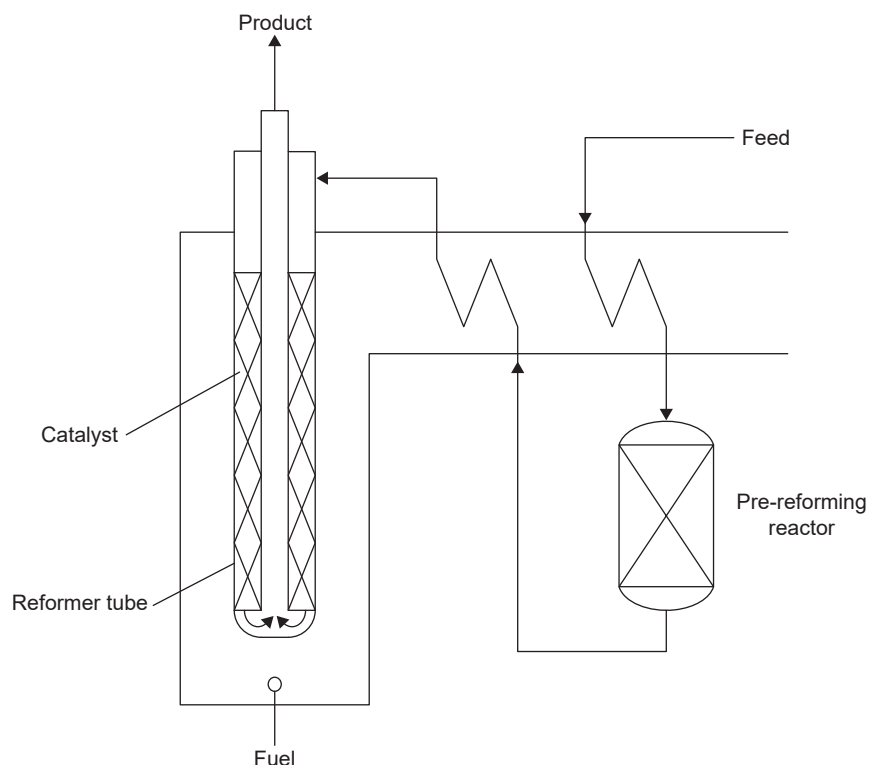


FIG. 15.26 ICI tube-in-tube steam reforming reactor. From US 4,985,231.

the feed is pre-heated in the convection section of the furnace, passed through a pre-reforming reactor, and reheated in the convection section before entering the reforming tubes. The reforming tubes have a tube-in-tube arrangement, with catalyst loaded in the annular space. The process gas passes down through the catalyst and then exits through the inner tube. It can be seen that if the inner and outer tubes experience different thermal expansion, there will be no loss of containment of the catalyst or bending of the tubes.

Heat transfer to packed beds of catalyst can be modeled using the equations developed by [Leva \(1949\)](#); see Section 15.3.4. The same cautions apply as for design of homogeneous heat exchanger reactors, and detailed models of the kinetics and hydrodynamics must be used to scale up a design.

15.7 Multiphase reactors

In multiphase reactors, the design is complicated by the need to promote mass transfer between the phases, which is usually accomplished by providing a high interfacial area. If more than one phase is flowing, the design may also need to allow for phase separation.

15.7.1 Vapor–liquid reactors

Vapor–liquid reactions are important in many chemical processes. Many oxygenation and hydrogenation reactions of organic compounds are carried out with the organic component in the liquid phase.

Vapor–liquid contacting columns are preferred for vapor–liquid reactors if the residence time requirements are short enough because of the high area for mass transfer. Packed columns are usually used; see Section 17.14. When long residence time is needed for the liquid phase, stirred tanks or tubular reactors are used. Methods of mixing a gas into a liquid are described in Section 15.5.3, and the mass transfer rates obtained when gas is sparged into a stirred tank can be predicted using [Equations 15.13 to 15.15](#); see Section 15.3.4.

The kinetics of gas–liquid reactions depend on the relative rates of mass transfer and chemical reaction. Consider the case where a gas species A reacts with a liquid species B. If the reaction is second order, the rate of reaction per unit volume can be written as:

$$\mathcal{R} = k_2 C_A C_B \quad (15.17)$$

where:

C_A = concentration of A in the liquid phase
 C_B = concentration of B in the liquid phase
 k_2 = second-order rate constant

If component B is present in excess in the liquid, we can assume that its composition is constant, in which case we can rewrite Equation 15.17 as:

$$\mathcal{R} \approx k_1 C_A \quad (15.18)$$

where:

$$k_1 = k_2 C_B$$

Now consider mass transfer of component A through the boundary layer at the vapor–liquid interface, shown schematically in Fig. 15.27.

The rate of mass transfer per unit volume of liquid can be written as:

$$\text{Rate of mass transfer} = k_L a (C_{A,i} - C_{A,bulk}) \quad (15.19)$$

where:

k_L = mass transfer coefficient for component A in the liquid phase
 a = interfacial area per unit volume of liquid
 $C_{A,i}$ = concentration of A in the liquid at the interface (as predicted by phase equilibrium)
 $C_{A,bulk}$ = concentration of A in the bulk liquid

If most of the reaction occurs in the bulk liquid, then the rate of reaction in the liquid must be equal to the rate of mass transfer through the film, and we can equate Equations 15.18 and 15.19:

$$k_1 C_{A,bulk} = k_L a (C_{A,i} - C_{A,bulk}) \quad (15.20)$$

hence:

$$C_{A,bulk} = \frac{k_L a}{(k_1 + k_L a)} C_{A,i} \quad (15.21)$$

and we can rewrite the rate of reaction as:

$$\mathcal{R} = k_1 C_{A,bulk} = k_L a C_{A,i} \frac{k_1}{(k_1 + k_L a)} \quad (15.22)$$

Two cases are apparent from Equation 15.22. If $k_L a \gg k_1$:

$$\mathcal{R} \approx k_1 C_{A,i} \quad (15.23)$$

This is known as the slow kinetics–controlled regime, and the reaction occurs at the rate that would be predicted with the liquid-phase concentration predicted by phase equilibrium. In this regime the rate is not sensitive to the interfacial area and increasing the agitation rate or packing area per unit volume will not affect the conversion.

Alternatively, if $k_1 \gg k_L a$:

$$\mathcal{R} \approx k_L a C_{A,i} \quad (15.24)$$

This is known as the slow mass-transfer–controlled regime, in which the reaction rate is equal to the rate of mass transfer that would be obtained if the concentration of A in the bulk were zero. The rate of reaction is directly proportional to the interfacial area.

If the thickness of the boundary layer is δ , and the condition for either of these regimes is that the reaction rate in the bulk is greater than the reaction rate in the film, we can write the condition as:

$$a\delta k_1 C_{A,i} \ll a k_L (C_{A,i} - C_{A,bulk}) \quad (15.25)$$

The definition of the mass transfer coefficient is $k_L = D_A / \delta$, and if $C_{A,bulk} \approx 0$, we can derive:

$$\frac{D_A k_1}{k_L^2} \ll 1 \quad (15.26)$$

or $Ha^2 \ll 1$, where Ha is the Hatta number:

$$Ha = \frac{\sqrt{D_A k_1}}{k_L} \quad (15.27)$$

and D_A = diffusion coefficient of A in the liquid phase.

If Ha is roughly equal to or greater than 1.0, then the reaction occurs substantially in the boundary layer and the analysis becomes more complicated. Details of the equations obtained in the resulting regimes can be found in reaction engineering textbooks such as [Froment et al. \(2010\)](#) and [Levenspiel \(1998\)](#).

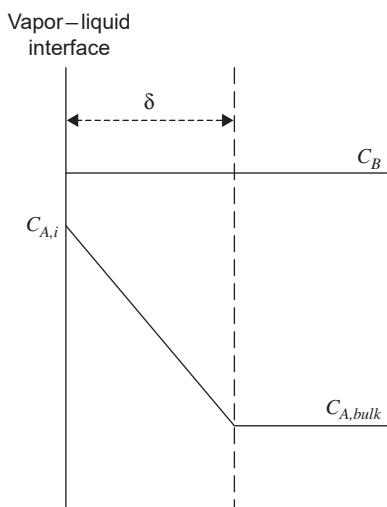


FIG. 15.27 Mass transfer from vapor to liquid.

Many vapor–liquid reactions of commercial significance occur in the slow regimes because oxidation and hydrogenation reactions are exothermic and are usually carried out at low temperature to improve selectivity. The use of falling-film wetted column reactors of known area can allow measurement of k_L and k_1 for use in reactor scale-up. Most gas–liquid reactions are operated under conditions where the gas component is not fully converted so as to ensure that the reaction does not become starved of reagent. Unreacted gas can be recycled (with a purge to prevent buildup of impurities), but inexpensive gases such as air are usually used once through.

15.7.2 Liquid–liquid reactors

Liquid–liquid reactions occur between immiscible liquid phases, usually an organic phase and an aqueous phase. Examples include reactions such as the nitration of toluene or benzene with mixed acids, emulsion polymerizations, and many reactions that are catalyzed by liquid acids, such as motor fuel alkylation.

Liquid–liquid reactions are usually carried out in stirred tank reactors so that agitation can be used to generate a high liquid–liquid area for mass transfer. The reactor is typically followed by a settling tank, similar to the mixer–settler arrangement used for single-stage extraction, so that the phases can be separated for further processing, although settling is usually carried out in the reactor itself in a batch process. The decision of which phase will be continuous and which phase will be dispersed is usually made based on the required hold-up of each phase, ease of dispersing one phase in the other, or safety considerations. In most cases, reaction only occurs in one phase or at the interface between the phases.

The equations used to describe mass transfer with reaction in a liquid–liquid system are analogous to those for vapor–liquid reactions, and the overall rate is often determined by mass transfer rather than the intrinsic kinetics. Mass transfer control of a liquid–liquid reaction can be detected experimentally by varying the agitation rate with other conditions kept constant. When increasing the agitation rate no longer affects the conversion, the reaction is no longer mass transfer controlled and the intrinsic kinetics can be found.

Prediction of liquid–liquid mass transfer rates is difficult due to surfactant effects on droplet coalescence and breakup, and the equations used for vapor–liquid mass transfer coefficient cannot be extrapolated to liquid–liquid systems. Scale-up is usually carried out by finding a condition at which mass transfer is not controlling and then assuming the same agitator power input per unit volume of reactor.

Some examples of liquid–liquid reactor designs are given in the discussion on liquid–liquid heterogeneous catalysis in Section 15.8.2.

The design of settling tanks is relatively straightforward and is discussed in Section 16.4.1.

15.7.3 Vapor–solid reactors

Gas-phase reactions that use solid catalysts can be carried out in fixed, moving or fluidized bed reactors. When a gas reacts with a solid reagent, a fluidized-bed reactor is usually used.

Fixed-bed reactors

In a fixed-bed reactor, the gas flows over a stationary packed bed of catalyst. Fixed-bed catalytic reactors are probably the most common type of reactor for high-temperature catalytic processes and for gas conversion. Examples of processes that use fixed beds include ammonia synthesis, oxidation of sulfur dioxide, isomerization of n-alkanes (paraffins), and the reaction of benzene with ethylene or propylene to form ethylbenzene or cumene.

Fixed-bed reactors can range in size from a few centimeters diameter inside a heat exchanger tube up to several meters diameter for large-scale processes in oil refineries. Reactors greater than 14 ft (4.27 m) are generally avoided because of the need for site fabrication, but if the reactor is to be designed for high-pressure operation, then one larger, site-fabricated vessel may still be cheaper than using two parallel reactors.

In most gas-phase fixed-bed reactors, the gas enters at the top of the reactor and flows down through the bed of catalyst. Up-flow can be used, but the designer must take care to avoid conditions that would cause fluidization; see Section 18.2.2. Distribution of the gas across the bed is easy, as the pressure drop along the bed is much greater than the pressure drop in the head space of the vessel. In some cases, the head space is filled with inert ceramic balls so that catalyst is only loaded to the tangent line of the vessel. A layer of inert ceramic balls is also usually used at the bottom of the reactor to help contain the catalyst (Fig. 15.28).

The outlet of a fixed-bed reactor must be designed with a suitable collector to retain the catalyst and prevent it from being carried out into downstream operations. The outlet collector is usually a can with perforated walls that is welded inside the outlet nozzle.

Fixed-bed reactors can be sized as follows:

1. From the space velocity determine the bed volume of catalyst required.
2. Assume the bed volume is contained in the tangent-to-tangent section of a pressure vessel (not the head space).
The vessel will be a cylinder, usually with aspect ratio between 1:1 and 4:1.
3. Check the pressure drop. If pressure drop is too high, then use a lower aspect ratio.
4. If multiple reheats or quenches are needed, make allowance for internals between the beds.
5. Design the pressure vessel using the methods given in [Chapter 14](#).

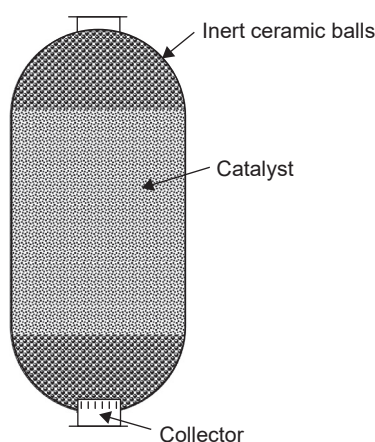


FIG. 15.28 Fixed-bed reactor loading.

The pressure drop in a fixed-bed reactor can be calculated using the Ergun equation (Ergun, 1952):

$$\frac{\Delta P}{L_b} = 150 \frac{(1-\varepsilon)^2}{\varepsilon^3} \frac{\mu u}{d_p^2} + 1.75 \frac{(1-\varepsilon)}{\varepsilon^3} \frac{\rho_f u^2}{d_p} \quad (15.28)$$

where:

- ΔP = pressure drop, N/m²
- L_b = length of the bed of solids, m
- ε = void fraction
- μ = fluid viscosity, Ns/m²
- u = fluid superficial velocity, m/s
- d_p = particle effective diameter, m
- ρ_f = fluid density, kg/m³

If there is a constraint on the pressure drop, this will set a limit on the length of the bed and cause the designer to select a vessel with lower aspect ratio. The design of a fixed-bed reactor is illustrated in Example 15.4.

If a very low-pressure drop is required, then instead of designing a “pancake” reactor with very low aspect ratio, a radial-flow packed bed can be used. In a radial-flow reactor the catalyst is contained in an annulus between vertical perforated or slotted screens. The fluid flows radially through the bed, and the direction of flow can be either inwards or outwards. A schematic radial-flow reactor is shown in Fig. 15.29. The use of radial flow allows for a much smaller bed length while maintaining a compact vertical cylindrical pressure vessel.

Packed-bed reactors are normally operated adiabatically, although packed beds can be accommodated inside heat exchanger or furnace tubes. See Section 15.6.3 for a discussion of heat addition and removal from fixed-bed reactors. When a fixed bed of catalyst is divided into smaller beds to allow reheats or quenches, some volume allowance must be made for collection and redistribution of the gas. The vessel design must also allow for the additional nozzles (branches) needed for the flows to and from the heaters or coolers.

Many fixed-bed reactors operate at temperatures substantially above ambient. The design must make allowance for bed settling due to thermal expansion of the vessel. Frequent heating and cooling of fixed-bed reactors should be avoided, as contraction of the vessel during cooling will compress the catalyst and can crush the catalyst, leading to fines formation and pressure-drop problems.

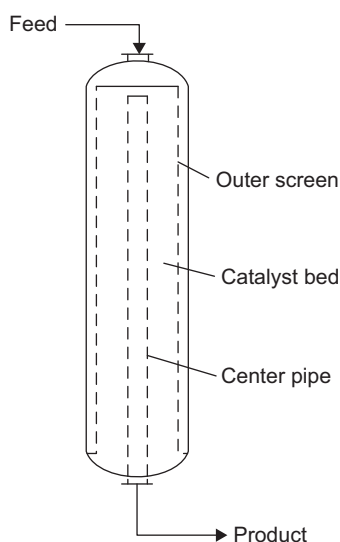


FIG. 15.29 Radial-flow reactor.

Moving-bed reactors

A moving-bed reactor is essentially a fixed-bed reactor that accommodates slow movement of the solids. Moving-bed reactors are used when the solids require gradual change-out—for example, when the catalyst deactivates at a rate that is too rapid for periodic regeneration *in situ* but not rapid enough to warrant the extra cost of using a fluidized-bed system.

Most industrial moving-bed reactors are radial-flow reactors in which the catalyst flows downwards between the retention screens, while the gas flows radially across the moving bed. An example is the UOP CCR Platforming process for catalytic reforming of naphtha, which uses a moving bed to enable catalyst regeneration. Three or four beds in series are used, with reheats between the beds, and the beds are stacked vertically to allow catalyst to flow through all the beds before being sent to a separate regeneration reactor. The design is illustrated in Fig. 15.30, reproduced with permission of UOP, LLC.

Many other proprietary moving-bed reactor designs have been developed, and these can easily be found by searching in patent databases such as www.uspto.gov or www.delphion.com.

The design of a moving-bed reactor must allow for the space needed for catalyst transfer lines, as well as the bed volume and other reactor internals. A detailed mechanical layout of the reactor should be made before sizing the pressure vessel. When evaluating a new design, large-scale, cold-flow models are often built to ensure that the solids flow properly.

Fluidized-bed reactors

In a fluidized-bed reactor the solid particles are fluidized by the gas flow. Several different types of fluidized-bed reactors are possible, corresponding to the different regimes of fluidization. Fluidized-bed reactors are widely used for conversion of solid reagents, for example, in coal combustion, coal gasification, and biomass pyrolysis; for processes that form a solid at high temperature, such as fluidized coking or polymerization; and for catalytic processes where the rate of catalyst deactivation is high and frequent regeneration is needed, as in the catalytic cracking of heavy oils in oil refining.

The physics of fluidization and the properties of fluidized beds are discussed in Section 18.2.2. It can be seen from Fig. 18.8 that fluidization spans a broad range of behavior, from expanded beds through bubbling to spouting and ultimately pneumatic transport. A reaction can be carried out under any of these conditions, leading to a combination of possible mixing patterns, as illustrated in Fig. 15.31.

Near to incipient fluidization the gas flows through the bed with little back-mixing. Although the solid particles are moving, there is no long-range mixing in the solid bed (see Fig. 15.31a). When the gas flow increases, bubbles begin to form. The bubbles agitate the fluidized bed, causing the solids to approach a well-mixed state, while the gas phase remains in plug flow, but with some degree of bypass due to the bubbles (see Fig. 15.31b). In turbulent

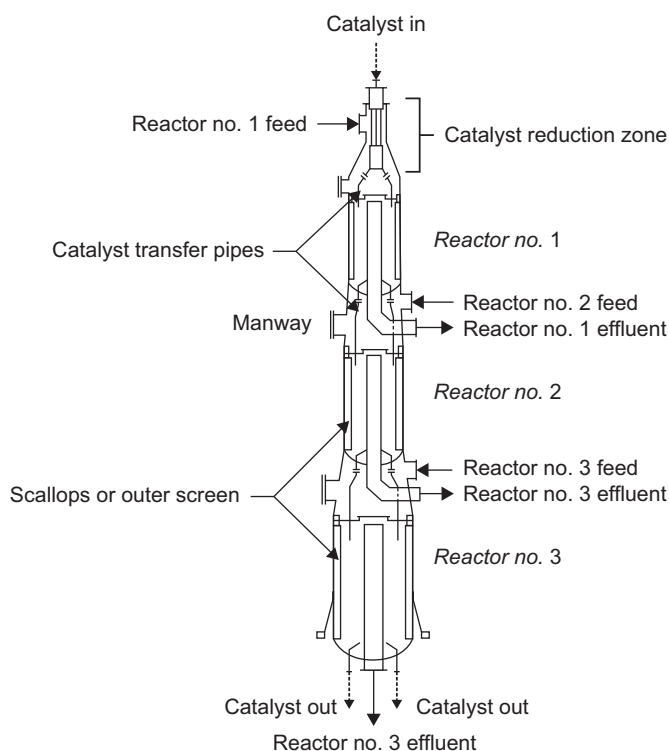


FIG. 15.30 UOP catalytic reforming reactor.

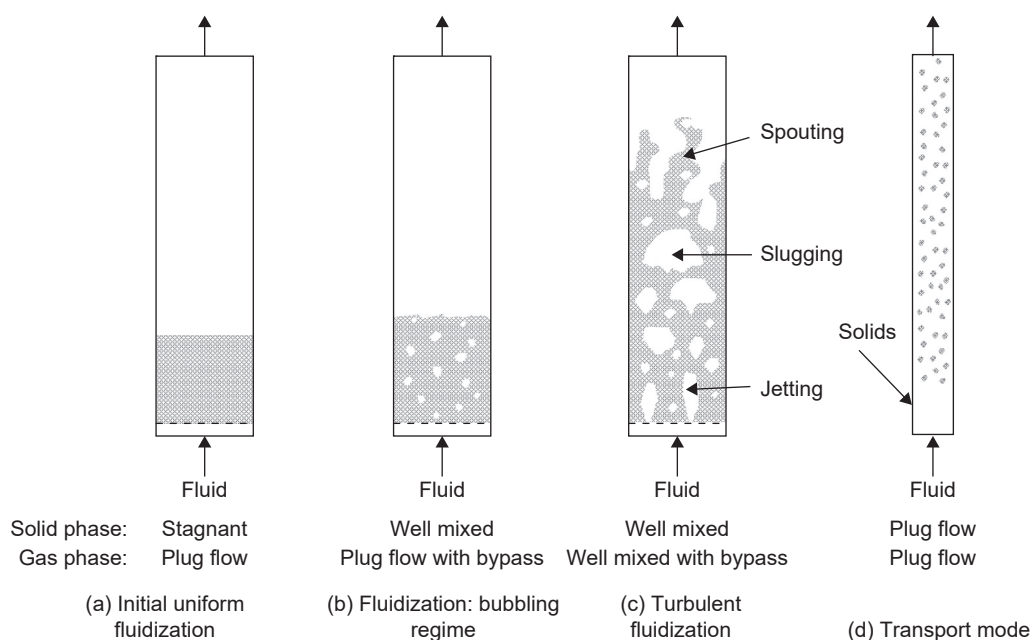


FIG. 15.31 Fluidized-bed reactor modes.

fluidization, the solids are well mixed, and the violent agitation of the bed causes back-mixing of the gas phase, causing it to also approach well-mixed behavior (see Fig. 15.31c). In this regime there can be substantial bypass of the bed by the gas phase due to the presence of jets, spouts, and large bubbles. Finally, when the gas velocity is high enough to entrain the solids, pneumatic conveying occurs and the solids flow with the gas in transport mode, with both phases approximating plug flow (see Fig. 15.31d). Note that in a transport fluidized bed the velocity of the solid particles is not necessarily the same as the velocity of the gas, depending on the flow direction. The slip velocity between the gas and solid particles can be found from a force balance between buoyancy and drag (essentially the same balance used to calculate terminal velocity).

The rates of heat and mass transfer in fluidized beds are much higher than in fixed beds, and external mass transfer is rarely the rate-controlling process. Heat transfer coefficients on the order of $200 \text{ W/m}^2 \text{ } ^\circ\text{C}$ to jackets and internal coils are typically obtained.

Fluidization can only be used with relatively small sized particles: $<300 \text{ }\mu\text{m}$ with gases. The solid material must be strong enough to withstand attrition in the fluidized bed and cheap enough to allow for make-up to replace attrition losses. The solids may be a catalyst, a reactant in fluidized combustion processes, or an inert powder added to promote heat transfer.

Modeling the kinetics of reactions in fluidized beds can be challenging because of the complex mixing that occurs, the possibility of bypass due to bubbles, and the difficulty of predicting the residence time distributions of the solids and the fluid phase. An introduction to modeling fluid bed reactions is given in Froment et al. (2010); see also Rase (1977), Grace et al. (1996), and Basu (2006). The use of combined models of particle dynamics and kinetics is described in Jung et al. (2009).

In addition to accommodating the expanded fluidized bed of solids, a fluidized-bed reactor must make allowance for separating the gas-phase product from entrained solids so that solids are not carried out of the reactor. This is usually accomplished using one or two stages of cyclones at the reactor exit. The cyclones can be mounted externally to the reactor, or internally, with dip legs that ensure return of solids into the bed, as illustrated in Fig. 15.32. The design of cyclones is described in Section 18.5.3. After bulk solids recovery, methods such as fabric filters or electrostatic precipitation are used for control of fine particulates. These and other processes for recovering residual particles from the reactor product are described in Section 18.5. Any equipment downstream of the particulate control equipment must still be designed for the presence of dust.

Depending on the hardness of the particles and the operating temperature of the reactor, it may be necessary to provide a fluidized-bed reactor with a refractory lining to prevent erosion by the particles.

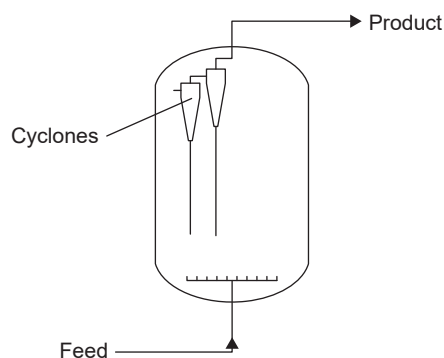


FIG. 15.32 Fluidized-bed reactor with internal cyclones.

15.7.4 Liquid–solid reactors

The reactors used for vapor–solid reactions can all be used for liquid–solid reactions if the liquid is flowing continuously. More care is needed when specifying up-flow of a liquid through a packed bed, as the smaller difference in density between a liquid and a solid increases the buoyancy of the solid and makes fluidization more likely. Down-flow is consequently preferred for packed-bed liquid–solid reactors.

Liquid–solid reactions can also be carried out in a slurry phase, in which the solid is mixed up in the liquid. Slurry reactors are prone to attrition of the solids, caused by pumping or agitation of the liquid. In some processes this attrition is desirable, for example, in reactions that convert solid minerals. In processes where a solid phase is being formed, as in slurry polymerization or reactive crystallization, the shearing or attrition of the solids can be traded off against the particle growth rate to obtain a desired particle size distribution. The formation of slurries is discussed in Section 18.3.5.

Slurry-phase operation is usually not preferred for processes that use heterogeneous catalysts because the catalyst tends to become eroded and can be difficult to recover from the liquid. Packed beds are usually used instead. If well-mixed reactor performance is desired, a packed bed with an effluent recycle can be used, as shown in Fig. 15.33.

15.7.5 Vapor–liquid–solid reactors

A three-phase reactor is required when a gas and liquid react over a solid catalyst, as in amination of alcohols or hydrogenation reactions, such as the saturation of fats or the desulfurization of petroleum fractions. All fermentation processes are three-phase due to the presence of cells as the solid phase and the need to provide oxygen and remove carbon dioxide from the fermentation broth.

Most vapor–liquid–solid reactions are carried out in either a slurry reactor or a trickle-bed reactor. Packed bubble column reactors are more rarely encountered.

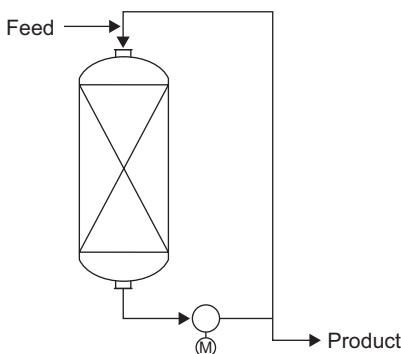


FIG. 15.33 Packed bed with effluent recycle.

Slurry reactors

The use of slurry reactors for liquid–solid reactions was discussed in Section 15.7.4. It is relatively straightforward to bubble or sparge a gas or vapor into a slurry reactor, and the mass transfer rate from gas to liquid can be predicted using the same correlations used for stirred tanks.

At high gas rates, the presence of gas bubbles can help mix the liquid and reduce the need for external agitation. This is exploited in loop reactors and draft-tube reactors. In a loop reactor, the gas is fed at the bottom of one side of a U-shaped reactor loop (the riser) and bubbles up through the liquid. At the top of the riser the gas and liquid disengage. The difference in density between the riser and the other side of the loop creates a hydraulic gradient that causes circulation of the liquid (Fig. 15.34). A draft-tube reactor uses the same concept with the gas fed to a central, cylindrical riser and down-flow occurring in the outer annulus between the draft tube and vessel wall (Fig. 15.35).

When a gas is fed to a slurry reactor, provision must be made for separation of the vapor and liquid in the head space and for recovery of entrained liquid from the exiting gas. If the gas flow is low, droplets will easily settle in the space above the liquid surface and no additional equipment will be needed. If the gas velocity is high, a horizontal settler can be used downstream of the reactor to allow liquid drops to settle and be returned to the reactor. A vent condenser may also be needed if volatile components from the liquid must be recovered. If foaming occurs in the reactor, then the rate of liquid carry-over can be excessive, and an antifoaming agent would typically be added to the reactor.

Trickle-bed reactors

In a trickle-bed reactor the liquid flows down over the surface of a stationary bed of solids. The gas phase usually also flows downwards with the liquid, but countercurrent flow is feasible as long as flooding conditions are avoided. The surface area provided by the solid helps promote vapor–liquid mass transfer, and both the liquid and vapor phase have little back-mixing and approximate plug flow behavior. Examples of processes that use trickle-bed reactors include the hydrocracking of heavy oils and reactions over immobilized organisms in gel columns.

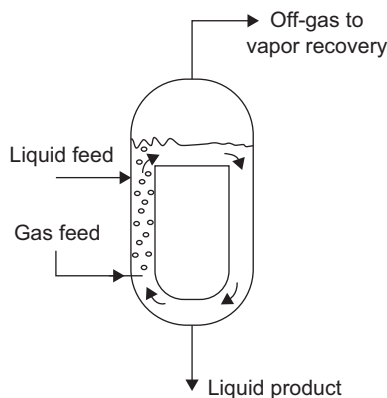


FIG. 15.34 Loop reactor.

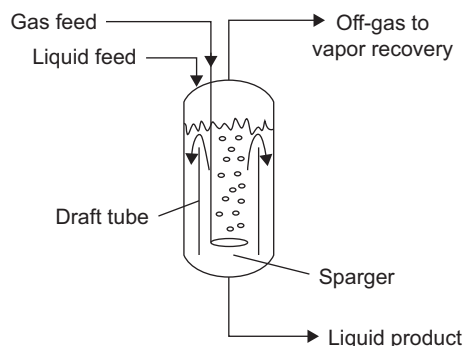


FIG. 15.35 Draft-tube reactor.

One of the most important design aspects of a trickle bed is to ensure proper wetting of the solids. If channeling of the liquid occurs, regions of the bed can be flooded and experience poor vapor–liquid contacting, whereas other regions may have no liquid present at all. Maldistribution of liquid can also cause local hot or cold spots in the reactor, which can affect selectivity and be a safety concern if the reaction is exothermic. Good wetting is particularly difficult to ensure if the heat of reaction causes a significant fraction of the liquid to vaporize as it passes down the catalyst bed or if the reaction forms compounds that vaporize under reaction conditions. The distributors that are used to ensure good liquid coverage of the solids are similar to those used in packed columns; see Section 17.14.5. The reactor designer must make allowance for the space needed for these internals.

Most trickle-bed reactors operate adiabatically. If reheats or quenches are needed between stages, the vapor and liquid must be collected and redistributed onto the stage below.

The pressure drop in a trickle bed is higher than predicted by the Ergun equation; see [Al-Dahhan and Dudukovic \(1994\)](#).

Modeling the kinetics and hydraulics of trickle flow is not trivial. Trickle-bed reactors can be sized following the same method as packed-bed reactors if the space velocity is known from pilot plant data, but prediction of performance from kinetics measured in laboratory-scale slurry reactors is challenging. More information on modeling the performance of trickle-bed reactors is given in the books by [Harriott \(2002\)](#) and [Ranade et al. \(2011\)](#).

15.8 Reactor design for catalytic processes

Most nonbiological industrial reactions are carried out in the presence of a catalyst. If a catalyst can be found that selectively promotes the desired reaction, it will have the following benefits:

- Improved selectivity for the desired product relative to by-products
- Operation at a lower temperature, giving improved selectivity relative to thermal reactions
- Faster rate of reaction, leading to a smaller, less costly reactor

Catalysts are classified as homogeneous if they are in the same phase as the reagents and heterogeneous if they are in a different phase. Most industrial catalytic processes use solid catalysts, as they are easier to contain, handle, and recover, but solid catalysts are more susceptible to poisoning by feed contaminants and to deactivation by accumulation of material inside the pores of the catalyst.

A wide range of materials can serve as catalysts. Some catalysts, such as liquid sulfuric acid, are inexpensive and can be treated as a consumable, with little attempt made to recover the catalyst when it is spent. At the other end of the cost spectrum, many reactions use catalysts that contain noble metals such as platinum and palladium. The value of a reactor load of noble metal catalyst can be several million dollars, and the design must guarantee that catalyst is properly contained and not lost from the system.

When a catalyst is specified, the reactor design must ensure that catalyst activity is maintained and that the catalyst is recovered for reuse, regeneration, or safe disposal. Other specific requirements for different types of catalyst are discussed in the following sections.

15.8.1 Design for homogeneous catalysis

A homogeneous catalyst is a catalyst that is in the same phase as the reagents. Homogeneous catalysts are relatively common in the process industries, but are not preferred because of the difficulty and expense of recovering the catalyst from the reaction mixture. Homogeneous catalysts are most often used in the liquid phase. Some examples include water-soluble enzymes, the use of organometallic compounds of cobalt in oxidation processes, and the use of sodium hydroxide as catalyst in the transesterification of triglycerides with methanol to produce fatty acid methyl esters (FAME, commonly known as biodiesel).

A homogeneous catalyst must either be cheap enough to be used once through in the process and not recovered or else chemically stable enough to withstand the conditions of a downstream recovery process. Salts or metal–ligand complexes can often be recovered from reaction products relatively easily by extraction or by causing the reaction products to separate into two liquid phases. If the product is volatile relative to the catalyst and solvent and the catalyst is thermally stable, it may be possible to flash or distill off the product, leaving a catalyst-rich residue that can be recycled.

Some homogeneous catalysts can be converted into heterogeneous catalysts by immobilizing the catalytic species on the surface or in the pores of a solid. For example, a wide variety of methods have been developed for immobilization of enzymes; see [Storey and Schafhauser-Smith \(1994\)](#). This approach is not always feasible, particularly when the active catalytic species is soluble in the reaction mixture and will leach from the support.

From a safety perspective, an important class of homogeneously catalyzed reactions is autocatalytic reactions. In an autocatalytic reaction, a product or reaction intermediate catalyzes the reaction and increases the reaction rate. Examples include many nitration reactions, reactions that involve peroxides that form hydroperoxide radicals, and other free radical reactions where free radical branching mechanisms can occur. Autocatalytic reactions have the potential to run away and are therefore a serious safety concern. Great care is needed in design and scale-up to ensure that the reaction is controllable, particularly when the reaction is exothermic or is not easily quenched by the addition of a free radical inhibitor. If an alternative, safer route cannot be found, the designer should apply inherently safer design principles to minimize the hazard; see Section 15.13. The design must consider reaction runaway as a relief scenario and as a scenario during hazard and operability (HAZOP) studies.

15.8.2 Design for heterogeneous catalysis

Heterogeneous catalysis is usually preferred over homogeneous catalysis, as it is much easier to recover and reuse the catalyst if it is in a different phase from the reagents. Examples of heterogeneous catalysis can be found in all possible phase combinations, but in the process industries the most common situations are liquid–liquid catalysis and fluid–solid catalysis.

Liquid–liquid catalysis

Reactions that are catalyzed by a different liquid phase include reactions where the catalyst is dissolved or suspended in water and the reagents are in the organic phase, as in the Asahi process for adipic acid by hydration of cyclohexene, and reactions that use liquid acids as catalysts, such as the reactions of alkenes with benzene and the alkylation of isobutane with alkenes to form alkylate for use as a gasoline blend stock. An emerging area of liquid–liquid catalysis is the use of ionic liquids as catalysts. Ionic liquids are salts with very low melting points that are liquid at ambient conditions. Ionic liquids have very low volatility, and variation of the anion and cation can allow optimization of their properties as solvents. Ionic liquids have been proposed as a means of immobilizing transition metal salts as catalysts and, in some cases, as catalysts in their own right; see [Stark and Seddon \(2007\)](#) for multiple examples.

In most liquid–liquid catalytic reactions the reaction occurs at the interface or in one phase only. Surfactant effects can be important, as surfactant molecules can stabilize micelles (or reverse micelles) and have a significant effect on interfacial area. It is not necessary for any of the reagents to be soluble in the catalyst phase to any great extent.

The design of reactors for liquid–liquid catalysis is the same as the design of liquid–liquid reactors; see Section 15.7.2. Mixer–settler reactors are most common, although extraction column reactors can also be used. In a mixer–settler arrangement, the stirred tank is usually operated at a high rate of agitation to maximize the interfacial area.

Several proprietary mixer–settler designs have been developed for liquid–liquid catalyzed processes. For example, isobutane can be alkylated with 2-butene to form 2,2,4-trimethylpentane (iso-octane), as shown in [Fig. 15.36](#). This reaction is important in oil refining, as it allows C₄ compounds that would be too volatile for use in fuels to be converted into a high-octane gasoline blendstock. The reaction is exothermic and is catalyzed by strong acids such as sulfuric acid and hydrofluoric acid.

In the reactor design developed by Stratco (currently licensed by DuPont) and shown in [Fig. 15.37](#), the reaction takes place in an agitated horizontal contacting vessel. The agitator mixes the sulfuric acid and organic phases and causes a circulation over heat-exchange tubes that contain refrigerant. A second vessel is mounted above the reactor and serves as a separator, allowing the organic product to be drawn off and acid to be returned to the reactor.

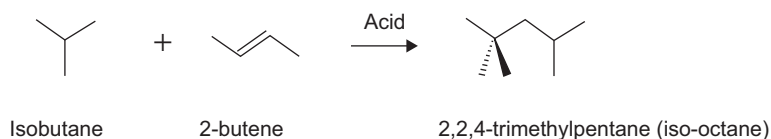


FIG. 15.36 Alkylation of isobutane with 2-butene.

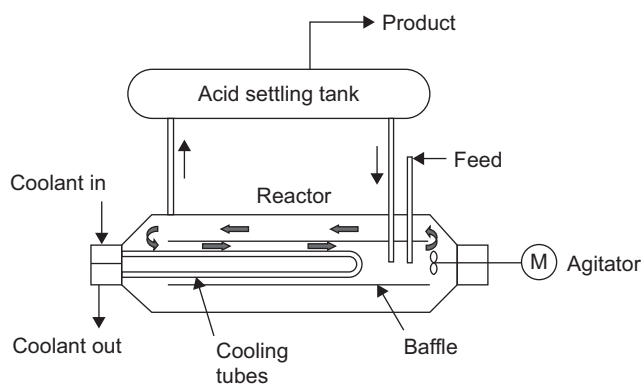


FIG. 15.37 Stratco sulfuric acid alkylation reactor.

An alternative sulfuric acid alkylation reactor design, developed by ExxonMobil, is illustrated in Fig. 15.38. The ExxonMobil design uses auto-refrigeration instead of indirect heat transfer to remove the heat of reaction. Part of the isobutane is boiled off from the reactor, compressed, condensed, and returned to the reactor to provide the necessary cooling. The ExxonMobil reactor has a series of agitated cells with staged feed addition and approximates plug flow with staged feed. The acid and hydrocarbon phases are separated in a separate vessel, and acid is returned to the first reactor cell.

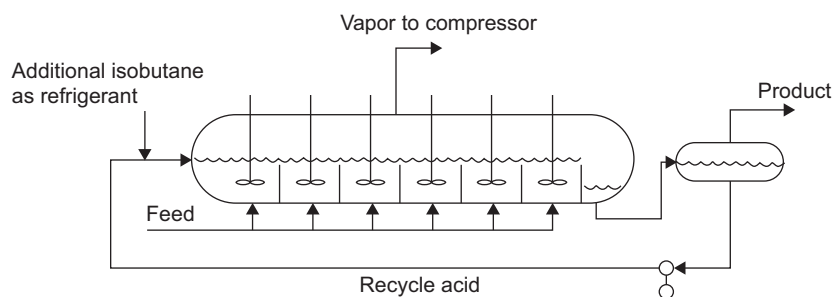


FIG. 15.38 ExxonMobil sulfuric acid alkylation reactor.

Fluid–solid catalysis

In most catalytic processes the catalyst is in a solid form and the reagents are a gas, liquid, or vapor–liquid mixture. The use of solid-phase catalysts is preferred because the separation of a solid from a fluid allows relatively easy recovery and reuse of the catalyst, but solid catalysts have many other advantages that come from the chemical and physical structure of the solid; these are discussed in Section 15.8.3.

The kinetics of a reaction in a porous solid catalyst can be complex, as there are multiple steps to the reaction:

1. External mass transfer of the reagents from the bulk fluid to the catalyst surface.
2. Internal mass transfer of reagents through macropores within the catalyst. Most catalysts consist of smaller particles or crystallites stuck together with a binder (Fig. 15.39). The macroporosity comes from the spaces between crystallites.
3. Internal mass transfer of reagents within micropores or within the crystals themselves.
4. Adsorption of reagents onto the active site on the catalyst surface.
5. Reaction.
6. Desorption of products from the catalyst surface.
7. Internal mass transfer of products within micropores.
8. Internal mass transfer of products through macropores.
9. External mass transfer of products through the boundary layer to the bulk fluid.

Any of these steps can be rate limiting. Multiple mechanisms can be written for adsorption, as well as for the chemical reaction step, so a large number of possible rate equations can be developed for solid catalyzed reactions.

The development of these equations is described in all of the reaction engineering textbooks listed in the bibliography at the end of this chapter. It is often difficult to discriminate between alternative mechanisms, and a given mechanism may be valid over a very narrow range of process conditions. Care must be taken to ensure that rate expressions are developed under conditions that match the conditions expected in a plant reactor; see Section 15.3.3.

Solid catalysts can be used in any of the reactor designs introduced in Section 15.7. Packed-bed reactors are most common, as they minimize the reactor volume required to contain the catalyst, but slurry reactors, moving-bed reactors, trickle-bed reactors, and fluidized-bed reactors are all widely used for catalytic processes.

Solid catalysts can become deactivated by a variety of mechanisms. Some catalysts deactivate slowly over many years, whereas other catalysts can lose their activity in a few seconds. The mechanisms of deactivation and design approaches to catalyst deactivation and regeneration are discussed in Section 15.8.4.

15.8.3 Design and selection of solid catalysts

The process design engineer will very rarely need to design the catalyst for use in the process; catalyst design will usually be carried out by research chemists in a catalyst manufacturing company. It is nonetheless important to understand how catalysts are formulated and manufactured to be able to select the best catalyst for a process application.

Structure and formulation of catalysts

Most solid catalysts are composite materials. In a typical catalyst the active material is either dispersed on the surface of a highly porous support or intermingled with a porous material so as to create a high internal surface area for adsorption and reaction and high porosity for good mass transfer. Some materials such as natural and synthetic zeolites have an open crystalline lattice with channels that are large enough for organic molecules to diffuse inside the crystal. The shape and size of these pores can be exploited to make highly selective catalysts. Many aluminosilicate and aluminophosphate materials have zeolite structures. Varying the ratio of aluminum to silicon or substituting metals into the lattice of these crystals allows adjustment of the Brønsted or Lewis acidity and further increases the options for catalysis. Many examples of zeolite catalysts are given in the books by [Breck \(1974\)](#) and [Kulprathipanja \(2010\)](#).

The manufacture of catalysts usually involves the following steps:

1. Synthesis of small crystals of the active crystalline component.
2. Ion exchange to adjust the acidity.
3. Blending of the active component with support material and binder to form a paste or sol. The function of the binder is to hold the crystallites together (see [Fig. 15.39](#)). Clays are usually used as binders, but any material can be used that will form a bond between the crystals without reducing the surface area or adversely affecting chemical or physical properties of the finished catalyst.
4. Forming of the mixture into particles of the desired shape and size. The most common method of forming is to extrude a paste through a die and form a shaped extrudate catalyst. A range of shapes can be formed, as illustrated in [Fig. 15.40](#), but cylinders are most common. Small spherical catalysts are made by spray drying, whereas larger spheres can be made by oil dropping or granulation methods; see Section 18.8.
5. Drying and calcining of the particles. Drying is carried out to remove any residual solvent used in the mixing process. Calcining the catalyst at higher temperature causes the particles to harden and ensures that the formed shape is retained.
6. Impregnation with metals. In many catalysts the active sites are small crystallites or even individual atoms of metal dispersed on the catalyst surface. Metals are usually loaded by wet impregnation methods, in which the catalyst is soaked in a solution of a thermally unstable metal salt and the metal salt is subsequently decomposed at high temperature to leave the metal on the catalyst surface in a reduced form.

The sequence of steps can vary, and some catalysts may not require all of these steps. The resulting catalyst pellets should have an open internal structure that offers a high active surface area for reaction.

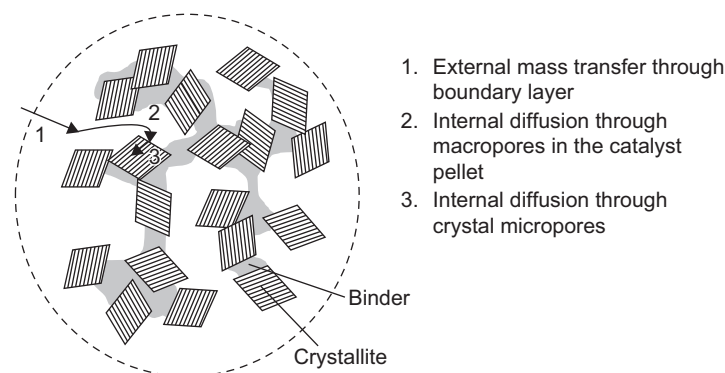


FIG. 15.39 Diffusion steps in a microcrystalline porous solid catalyst.

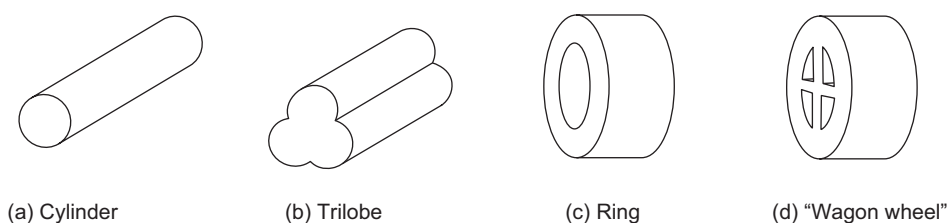


FIG. 15.40 Extruded catalyst shapes.

Physical properties of catalysts

The reactor design engineer must pay attention to some of the physical properties of the catalyst, as well as its chemical effectiveness. The following properties can affect reactor design and performance:

- **Shape:** Extruded catalyst shapes, such as those shown in Fig. 15.40, have higher external surface area per unit volume, which can reduce external mass transfer and heat transfer limitations. Spherical catalysts are preferred in applications where the catalyst is moving, such as slurry, fluidized-bed, or moving-bed reactors. If catalysts with sharp edges are used in these reactors, they will become rounded due to erosion. Part of the catalyst will be lost as fines, and there may be problems with dust carrying into downstream equipment.
- **Size:** Smaller catalyst particles have lower external mass transfer resistance but higher pressure drop when used in a packed bed. Smaller particles are harder to recover from slurries. Fluidized-bed reactors require particles to be in a certain size range to give the desired fluidization behavior; see Section 18.2.2.
- **Composition:** Aside from the active component of the catalyst, the composition of other materials used in the formulation can affect the catalyst properties. If a binder is used, it must be inert under all the expected process conditions, including catalyst regeneration conditions.
- **Porosity:** The external macroporosity of a catalyst formulation can be important in some processes. If the active catalyst has low porosity, then it is usually dispersed onto the surface of a more porous support material to increase the available surface area while retaining a large enough particle for convenient solids handling.
- **Strength:** The catalyst must be strong enough to withstand impacts and attrition that occur during loading and reactor operation. The strength of particulate materials is discussed in Section 18.2. Catalysts are usually composite materials, and the strength can be strongly affected by the choice and amount of binder used.
- **Thermal stability:** Many catalyst supports and catalytically active materials are sensitive to loss of internal surface area by sintering at high temperature. Reaction with steam at high temperature can similarly cause hydrothermal damage. Some catalysts, such as the ultra-stable Y zeolite used in catalytic cracking reactors, retain adequate porosity and activity to still be used after multiple exposures to high temperature. When testing such catalysts, it is important to age the catalyst properly so that the performance reflects the performance the catalyst will have after it has experienced several regenerations.

The characterization and properties of granular solids are discussed in more detail in Section 18.2.

Catalyst testing and selection

Usually several catalyst vendors can provide suitable catalyst for use in a given process application. Catalysts are normally selected based on performance rather than price, as the cost of the catalyst is typically a small fraction of

process operating costs, and a catalyst that has better selectivity will almost always quickly pay back any price premium expected by the manufacturer.

Because the rate-limiting step of the reaction is sensitive to several aspects of the catalyst formulation, it is important to test the catalyst under conditions that are representative of process conditions. Pilot plants can be used if they are benchmarked against full-scale plants and if they properly simulate the effect of recycles, feed contaminants, etc. See Section 15.12.1 for further discussion of experimental validation of reactor performance.

15.8.4 Design for catalyst deactivation and regeneration

Catalysts in process service can lose their activity over time. In some cases, deactivation can be reversed and the catalyst performance can be restored, whereas other deactivation mechanisms are permanent and require the operator to replace the catalyst load. If the design engineer expects that deactivation cannot be avoided, the reactor design must account for catalyst regeneration or replacement.

Catalyst deactivation mechanisms

Catalyst deactivation may be reversible or irreversible. Some of the more common methods of catalyst deactivation and the design methods that are used to prevent them from occurring are described in [Table 15.7](#).

Reactor design for catalyst deactivation

Some of the strategies that are used to prevent catalyst deactivation are listed in [Table 15.7](#). If the rate of deactivation of the catalyst is slow, it is also possible to design the reactor to tolerate a certain extent of deactivation. The two strategies that are most often used are temperature ramping and use of excess catalyst.

In a temperature-ramping strategy, the fresh catalyst is operated at a temperature that is below the maximum temperature recommended for the reaction. As the catalyst activity declines, the operators slowly raise the reactor temperature to maintain the same conversion across the reactor. The operators continue increasing temperature until a temperature is reached at which the reaction selectivity is compromised or the rate of catalyst deactivation is accelerated, at which point it becomes necessary to shut down and regenerate or replace the catalyst. The amount of temperature ramp that can be tolerated is usually recommended by the catalyst vendor. Typical values are 20 to 40 °C over a period that can range from days to several years. When a temperature-ramping strategy is planned, it is important to understand the reactor performance and yields under both start-of-run and end-of-run conditions, as changes in selectivity can occur, creating a need for the designer to examine the impact on downstream equipment.

If excess catalyst is loaded in the reactor, then some of the catalyst can become deactivated without overall performance declining. The use of excess catalyst is limited to reactions where the selectivity at the reactor outlet is insensitive to the amount of catalyst present, for example, processes that approach equilibrium or processes that can afford to overshoot the required conversion. An example of such a process is the hydrodesulfurization of petroleum fractions, where excess sulfur removal exceeds process requirements but does not usually cause significant financial loss.

Reactor design for catalyst regeneration

Catalysts that have become deactivated by coke formation can usually be reactivated by burning off the coke under controlled conditions. The frequency of regeneration required depends on the rate of coke formation. Some catalysts are regenerated after a few years or months of operation, whereas the catalysts used in fluid catalytic cracking of gas oils are regenerated after a few seconds on stream and experience multiple regenerations per hour.

When catalyst regeneration is infrequent, there is usually no need to make special modifications to the reactor design. If the process can afford to be shut down for a period for catalyst regeneration, the regeneration can sometimes be carried out *in situ* in the reactor. In this case, the ancillary equipment needed for regeneration must be added to the process flowsheet. Ancillary equipment may include such items as an air blower, air pre-heater, dust control equipment, etc. More often, the catalyst is unloaded and sent off-site for regeneration. In the meantime, a different batch of catalyst is loaded, allowing the plant to come back on stream faster.

If an interruption in process operation is not acceptable, a cyclic or swing-bed regeneration approach can be used. In cyclic designs, the plant has several reactors and is designed so that one reactor can be taken off-stream for regeneration without affecting the operation of the other reactors. Cyclic regeneration can occur as often as several times an hour, but longer cycles are preferred because they cause fewer thermal cycles and reduce the likelihood of fatigue damage. Cyclic regeneration designs need particular attention to the selection of the switching valves that are used

TABLE 15.7 Catalyst deactivation mechanisms

Mechanism	Causes	Design approaches to mitigate deactivation
Reversible poisoning	Feed components or contaminants that adsorb reversibly on the catalyst active site. Common reversible poisons include ammonia, arsine, phosphine, water, hydrogen sulfide, oxygen, and carbon monoxide.	If the feed concentration of a reversible poison is higher than the design basis value, the effect can often be reversed by returning to the design level. Upstream separation processes can be used to remove contaminants. Guard beds of adsorbent can also be used.
Irreversible poisoning	Contaminants that are strongly adsorbed on the catalyst or react irreversibly with the active site. Examples include sulfur dioxide and halogens, although water and ammonia can irreversibly damage strong acid catalysts.	If the presence of irreversible poisons is expected, then a guard bed of adsorbent should be placed upstream of the reactor. See Section 16.2.1 for design of adsorption processes. Irreversible adsorbents are usually used to ensure a strong interaction that will tie up the contaminant. A sacrificial bed of low-activity catalyst at the inlet of the reactor can also be used as a mitigation technique.
Amalgamation	If mercury is present in a feed, it can amalgamate with metals on the catalyst and destroy the metal function.	Guard beds containing copper- or silver-based adsorbents are used to protect catalysts from mercury.
Coking	Accumulation of carbon-rich deposits on the catalyst. Coke is usually formed by condensation reactions between unsaturated organic compounds. Coke can cover the active sites or block the pores that lead to them.	The effect of coking can usually be reversed by burning off the coke. Care must be taken to ensure that the conditions during coke burn do not lead to deactivation by other mechanisms.
Sintering and hydrothermal damage	Sintering is caused by exposure to high temperatures; hydrothermal damage occurs when high temperatures are combined with concentrations of steam. The effect of both is a reduction in internal surface area, and in some cases loss of crystallinity. Thermal damage of enzyme-based catalysts can occur at relatively low temperatures.	Sintering and hydrothermal damage are prevented by avoiding the conditions under which they can occur. Particular care is needed during catalyst regeneration. Regeneration is often carried out using depleted air that has been enriched in nitrogen or using a recycle of flue gas, so as to limit the temperature rise and the concentration of water that occurs. Thermal damage of enzyme-based catalysts is avoided by using cooling and temperature control to keep the temperature in the desired range.
Agglomeration	Metals on the catalyst can become agglomerated instead of dispersed over the surface, reducing the number of available active sites. Agglomeration can be accelerated by tramp metals that find their way onto the catalyst from the feed or from corrosion products.	In some cases it may be possible to redisperse the metals during a regeneration process, as occurs in catalytic reforming, where some agglomeration of the noble metals on the catalyst occurs during regeneration and a chlorination step is used to redisperse the metal; however, agglomeration is often irreversible.
Leaching	Metals on the catalyst or immobilized catalysts such as enzymes become dissolved in the process fluid and are elutriated from the reactor.	Leaching is irreversible, and the design should try to avoid conditions that can cause it to occur.
Fouling and plugging	Materials such as dust, catalyst fines, corrosion products, gums, dirt, etc., can be filtered by a fixed bed, causing the bed or reactor internals to become plugged and leading to an unacceptable increase in pressure drop.	Filters can be used upstream of the reactor to remove material that can cause plugging. In packed beds, a layer of highly porous inert ceramic material is sometimes placed on top of the catalyst to trap out plugging agents. A periodic back-flush can sometimes reverse the effects of plugging.

to isolate the process conditions from the regeneration conditions, and these valves can be a substantial component of the reactor section cost.

Several commercially important processes make use of continuous catalyst regeneration. If the rate of catalyst deactivation is fast, continuous regeneration offers several advantages relative to cyclic regeneration:

- The process can be operated under conditions where deactivation is more rapid, as regeneration can be more frequent.
- The catalyst maintains optimum activity and selectivity in the reactor, leading to a less costly reactor and better process performance.
- Maintenance and reliability problems due to the switching valves are eliminated.
- Thermal cycling of vessels is eliminated, reducing stresses on vessels and supports.
- Catalyst damage due to stresses on the catalyst bed from thermal cycling is eliminated.
- Regeneration equipment is smaller and operates more efficiently in continuous mode.

Continuous regeneration designs use at least two reactors: one for the process reaction and one for regeneration. Catalyst circulates between the process reactor and the regeneration reactor. Many variations are possible, and some designs have multiple reactors, either in parallel or in series, connected to the same regenerator. Examples of different proprietary reactor–regenerator designs can be found in the patent literature at www.uspto.gov.

Moderate rates of coke formation can be addressed by use of a moving-bed continuous catalyst regeneration design. In a moving-bed reactor the catalyst is slowly circulated between the reactor and regenerator, and the residence time can vary from a few hours to a few days in each zone. A simplified schematic diagram of a moving-bed continuous regeneration design is shown in Fig. 15.41. The proprietary designs used commercially are substantially more complex and can involve several regeneration zones and multiple reactors, with reheats of the process fluid between reactors.

Care must be taken to ensure separation between the hydrocarbon-rich atmosphere of the reactor and the oxygen-rich atmosphere of the regenerator. In moving-bed regeneration systems, inert gas purges or lock hoppers can be integrated into the catalyst lift and transfer system to ensure that a flammable mixture is not formed.

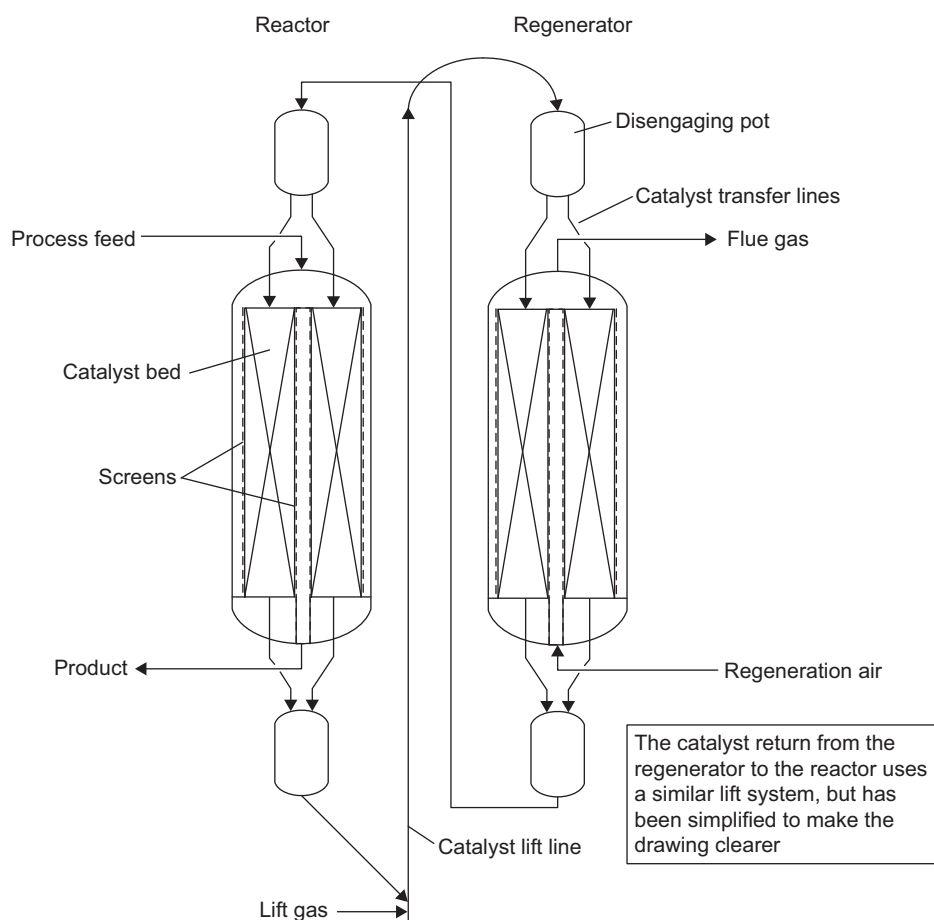


FIG. 15.41 Simplified schematic of a moving-bed reactor system for continuous catalyst regeneration.

In a moving-bed reactor, the catalyst flows as a packed bed moving under gravity, and the catalyst is only fluidized during pneumatic transport in the lift lines between the vessels (see Fig. 15.41). Radial-flow reactors are often used, as they give a more uniform catalyst residence time than moving packed beds. The hydraulic design of moving-bed reactors can be complex, and the designer must ensure that the design allows for controlled flow of the solids as well as process fluids.

Moving-bed reactors are most commonly applied to gas–solid reactions, although they can also be used for liquid–solid reactions. Moving-bed reactors allow flow of solids without the high rates of attrition that are encountered in slurries and fluidized beds, and are therefore attractive when the catalyst is expensive or prone to attrition. Moving-bed reactors are used in the catalytic reforming of naphtha and the catalytic dehydrogenation of propane to propylene. For more details on these processes, see Meyers (2016).

At high rates of coke formation, slow movement of the catalyst is no longer feasible. Instead, very high rates of catalyst circulation can be achieved by using fluidized-bed reactors for both reaction and regeneration and transferring fluidized catalyst between the reactor and regenerator. This approach is used in the FCC of gas oils to lighter hydrocarbons in oil refining.

A catalytic cracking reactor–regenerator system is illustrated in Fig. 15.42, which is reproduced with permission of UOP, LLC. In catalytic cracking, hot regenerated catalyst enters the bottom of a transport riser reactor. The catalyst is fluidized using steam as lift medium and flows up the riser. Gas oil feed is sprayed onto the catalyst, where a variety of cracking reactions occur, forming lighter hydrocarbon products that can be used to make petrochemicals and transportation fuels such as gasoline and diesel. At the top of the riser the solids are separated from the reaction mixture using cyclones, and the reactor product is sent to a fractionation section for product recovery. The catalyst from the top of the riser is stripped with steam to recover hydrocarbons and then passes through a stand pipe to the regenerator. In the regenerator, coke that formed in the reactor is burned off the catalyst, restoring the catalyst activity so that the catalyst can be returned to the riser reactor. The heat from burning the coke is used to vaporize the gas oil feed and to provide the heat needed to give the desired riser reactor outlet temperature.

Over 350 FCC units are in operation worldwide, and almost a quarter of the world's oil production is processed through catalytic cracking. Many different proprietary designs of FCC reactors and regenerators have been developed and can be found in the patent literature; see www.uspto.gov. Catalyst residence times in the riser are typically a few seconds, but can be as short as fractions of a second. The rate of catalyst circulation is typically between five and ten times the feed rate on a mass basis, and large catalytic cracking reactors can have catalyst circulation rates as high as 1.6 metric tons per second. More information on FCC is given in Meyers (2016).

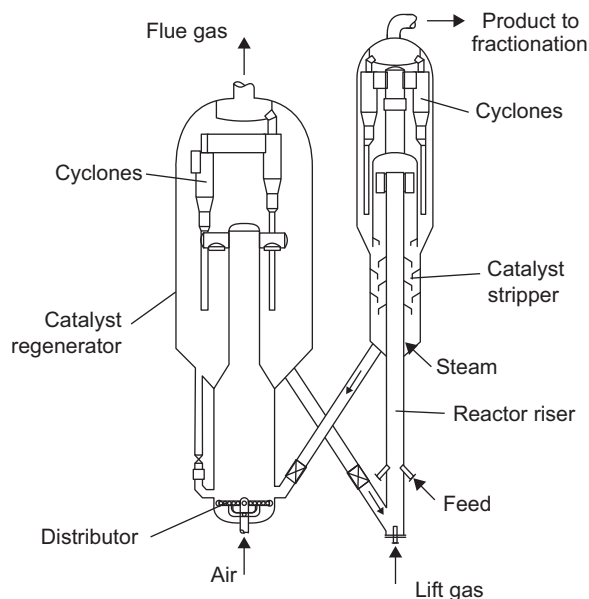


FIG. 15.42 UOP fluidized catalytic cracking reactor and regenerator.

Example 15.4

A new process is being developed for converting benzene to cyclohexane by hydrogenation over a noble metal catalyst. The reaction is carried out at 50 bara, at a feed temperature of 220 °C, and the reactor contents should not exceed 300 °C to minimize formation of methylcyclopentane (MCP) by-product. The reaction is essentially irreversible under these conditions, and the temperature can be controlled by staged addition of cold hydrogen. Six beds are to be used, each converting the same amount of benzene, with overall conversion of 100%. The average weight hourly space velocity based on benzene is 10 h^{-1} for each adiabatic bed, and a pressure drop of 0.5 bar is acceptable across each bed. The catalyst is in the form of 1/16 in (1.588 mm) beads with average bulk density of 700 kg/m^3 . Design, size, and cost a reactor for making 200,000 metric tons per year of cyclohexane using this process.

Cautionary note: This example is hypothetical, and all numbers in the problem statement are fictitious. Any resemblance to any commercial cyclohexane technology is purely coincidental.

Solution

The first step is to determine the flow rates around the reactor. Assuming 8000 hours operation per year, the product rate is $200,000/8000 = 25 \text{ t/h}$. To make 1 kg of cyclohexane (C_6H_{12}) requires $78/84 = 0.929 \text{ kg}$ of benzene (C_6H_6), so the plant feed rate is $0.929 \times 25 = 23.2 \text{ t/h}$.

The hydrogen quench rates must be calculated from a heat balance at process conditions. These calculations are most easily performed using a commercial process simulation program.

Fig. 15.43 shows a screen shot of a reactor model built using UniSim Design R390. Six conversion reactors were used, with the specified conversion of benzene adjusted to give the same amount of benzene converted in each (i.e., 16.7% conversion in

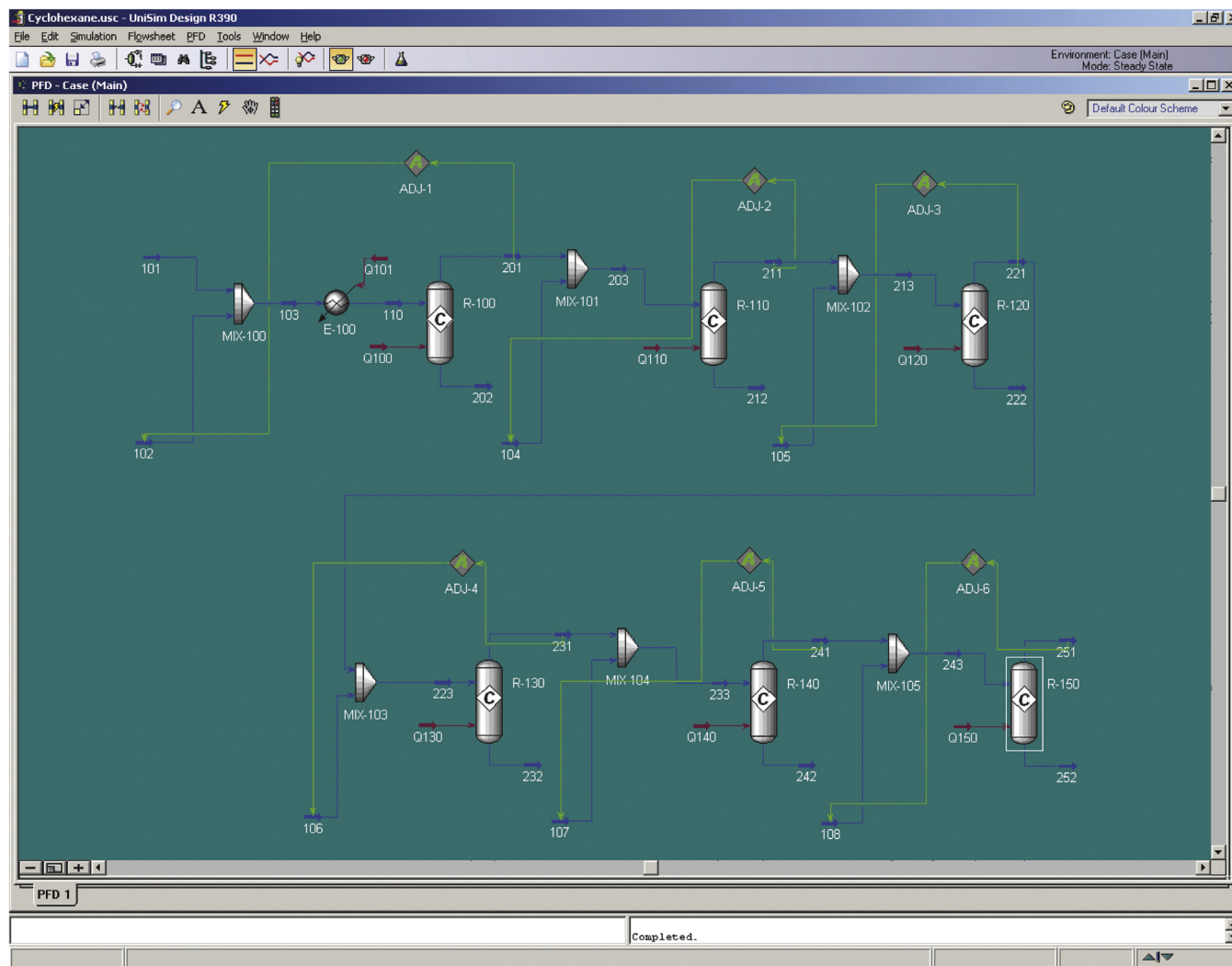


FIG. 15.43 UniSim Design model of cyclohexane reactor for Example 15.4.

TABLE 15.8 Stream flows and properties for Example 15.4

Bed	R100	R110	R120	R130	R140	R150
Inlet temperature (°C)	220	239	251	259	264	269
Outlet temperature (°C)	300	300	300	300	300	300
Feed flow (kmol/h)	3445	4781	6065	7403	8744	10030
Feed flow (kg/h)	29546	32539	35427	38424	41426	44320
Feed flow (act. m ³ /h)	2860	4172	5465	6843	8254	9642
Product flow (act. m ³ /h)	3214	4566	5892	7299	8738	10151
Feed density (kg/m ³)	10.33	7.80	6.48	5.61	5.02	4.60
Product density (kg/m ³)	9.19	7.13	6.01	5.26	4.74	4.37
Feed viscosity (Ns/m ²)	1.62E-05	1.59E-05	1.57E-05	1.57E-05	1.56E-05	1.56E-05
Product viscosity (Ns/m ²)	1.87E-05	1.77E-05	1.72E-05	1.69E-05	1.66E-05	1.65E-05

R100, 20% in R110, 25% in R120, 33.3% in R130, 50% in R140, and 100% in R150). Adjust controllers were used to set the interbed quench hydrogen flow rate so that the reactor outlet temperature was 300 °C on each reactor. The heat duty to each reactor was specified as zero. A pressure drop of 0.5 bar was allowed on each reactor. From the simulation, we can download the necessary stream flows and properties, which are given in Table 15.8.

Next we must calculate the bed volume to get an idea of what bed diameters will be suitable. With a feed rate of 23,200 kg/h of benzene and WHSV of 10 h⁻¹, the mass of catalyst in each bed is 23,200/10 = 2320 kg. Because the catalyst has average bed density 700 kg/m³, catalyst bed volume = 2320/700 = 3.314 m³.

For a given column diameter we can now calculate the bed cross-sectional area and length of bed. In addition to the height required for the catalyst, we must leave some allowance for the hydrogen inlet piping and quench zone between beds. Examining the hydrogen quench flow rates, this will need at least 8-in piping, so a 3-ft (0.914-m) height allowance should be made between beds. For example:

If bed diameter = 6ft = 6 × 0.3048 = 1.829 m

Bed cross-sectional area = $\pi D_b^2/4 = 3.142 \times (1.829)^2/4 = 2.627 \text{ m}^2$

Bed length for 3.314 m³ bed volume = 3.314/2.627 = 1.262 m

Length of catalyst bed plus quench zone = 1.262 + 0.914 = 2.176 m

Total reactor length for six beds, noting that we only need five quench zones = 6 × 2.176 – 0.914 = 12.142 m. We also need some space for a bottom support grid, say 0.30 5m, so total length = 12.447 m.

Vessel aspect ratio = length/diameter = 12.447/1.829 = 6.8

This calculation is easily coded in a spreadsheet and repeated for other bed diameters, giving the results in Table 15.9.

Using the actual volume flow and cross-sectional area, we can now estimate the superficial velocity, and hence use the Ergun equation (Equation 15.28) to estimate the pressure drop. It can be seen from Table 15.8 that the increase in temperature across the reactor has a stronger effect on density than the decrease in molar flow rate due to reaction, so the outlet conditions govern the pressure drop. We could attempt to break each bed into sections to get a more accurate estimate of pressure drop,

TABLE 15.9 Bed and reactor dimensions for Example 15.4

	6	7	8
Diameter (ft)			
Diameter (m)	1.83	2.13	2.44
Area (m ²)	2.63	3.58	4.67
Bed length (m)	1.262	0.93	0.71
Quench zone length (m)	0.914	0.91	0.91
Tangent length (m)	12.446	10.438	9.135
Aspect ratio (L/D)	6.8	4.9	3.7

but a conservative simplifying assumption is to just use the outlet condition. For example, for R150 (the final bed), with bed diameter 1.829 m:

$$\begin{aligned}\text{Reactor outlet actual volume flow} &= 10,151 \text{ m}^3/\text{h} \\ \text{Superficial velocity} &= 10151/(2.627 \times 3600) = 1.073 \text{ m/s}\end{aligned}$$

From Table 15.8, at the bed outlet, the density is 4.37 kg/m^3 and the viscosity is $1.65 \times 10^{-5} \text{ Ns/m}^2$. The bed void fraction is not given, but 0.4 is a reasonable value for beads, so substituting in Equation 15.28:

$$\begin{aligned}\frac{\Delta P}{L_b} &= 150 \frac{(1-\epsilon)^2}{\epsilon^3} \frac{\mu u}{d_p^2} + 1.75 \frac{(1-\epsilon)}{\epsilon^3} \frac{\rho_f u^2}{d_p} \\ \frac{\Delta P}{1.262} &= 150 \frac{(1-0.4)^2}{0.4^3} \frac{1.65 \times 10^{-5} \times 1.073}{0.001588^2} + 1.75 \frac{(1-0.4)}{0.4^3} \frac{4.37 \times 1.073^2}{0.001588} \\ \text{hence: } \Delta P &= 73027 \text{ N/m}^2 \approx 0.73 \text{ bar}\end{aligned}\quad (15.28)$$

In this case the pressure drop is higher than the allowable 0.5 bar, and a larger bed diameter should be considered. This calculation is easily coded into a spreadsheet and repeated for different diameters, giving the results in Table 15.10.

From Table 15.10, either a 7-ft or 8-ft reactor would have an acceptable pressure drop across all the beds. We have not made any allowance for pressure drop in the quench zone, so if we assume this will need ~ 0.25 bar, then the 8-ft vessel should be selected.

We can now make an estimate of the required wall thickness.

Design pressure, take as 10% above operating gauge pressure:

$$\begin{aligned}&= (50 - 1) \times 1.1 = 53.9 \text{ bar} \\ &= 5.39 \text{ N/mm}^2\end{aligned}$$

Maximum operating temperature is 300°C (572°F). With a 50°F design margin, design temperature = 622°F .

From Table 14.2, if we select a $1\frac{1}{4}$ Cr steel such as A387, the maximum allowable stress = $16.6 \times 10^3 \text{ psi} = 114.5 \text{ N/mm}^2$. Assuming full radiographic examination, the welded joint efficiency is 1.0, so we can substitute in Equation 14.13:

$$t = \frac{P_i D_i}{2SE - 1.2 P_i} \quad (14.13)$$

$$t = \frac{5.39 \times 2.44 \times 10^3}{(2 \times 114.5 \times 1) - (1.2 \times 5.39)} = 59.1 \text{ mm} \quad (14.13)$$

$$\text{add corrosion allowance } 59.1 + 3 = 62.1 \text{ mm}$$

So the wall thickness is 61 mm, or roughly a 2.5-in plate.

Under these conditions of temperature and pressure, it is probably worth specifying a hemispherical head, so the head thickness can be found using Equation 14.17:

$$t = \frac{P_i D_i}{4SE - 0.4 P_i} \quad (14.17)$$

$$t = \frac{5.39 \times 2.44 \times 10^3}{(4 \times 114.5 \times 1) - (0.4 \times 5.39)} = 28.9 \text{ mm} \quad (14.17)$$

So with a corrosion allowance, the head thickness needed is 32 mm, or $1\frac{1}{4}$ in.

TABLE 15.10 Bed pressure drop as a function of reactor diameter for Example 15.4

Bed	R100	R110	R120	R130	R140	R150
Pressure drop (bar): 6 ft	0.165	0.253	0.349	0.464	0.592	0.730
Pressure drop (bar): 7 ft	0.069	0.105	0.145	0.192	0.244	0.300
Pressure drop (bar): 8 ft	0.033	0.050	0.068	0.090	0.114	0.140

Knowing the vessel wall thickness and dimensions, we could now calculate the shell weight and use Equation 7.9, together with the correlation in Table 7.2, to obtain an estimate of the reactor cost; however, a more accurate approach is to use a commercial cost estimating program, such as Aspen Capital Cost Estimator. Fig. 15.44 shows the data entered into ACCE. This gives a total vessel cost of \$697,600 on a January 2006 basis.

The final step in completing the design would be to work with a mechanical engineer to prepare a scale design drawing for detailed mechanical design of the vessel and internals. A preliminary sketch for such a design is shown in Fig. 15.45, but there are many details still missing, such as the location of thermowells for temperature control, support brackets for internals and vessel skirt, design of the quench zone, etc.

Example 15.5

An alternative process for converting benzene to cyclohexane also uses a supported noble metal catalyst, but carries out the reaction in the liquid phase, at a temperature of 160 °C and pressure of 100 bara, in a slurry reactor. The catalyst has piece density 1100 kg/m³. Laboratory-scale experiments in an isothermal, continuous well-mixed reactor, cooled in a bath of coolant, have determined that with a high rate of hydrogen agitation and 10wt% loading of catalyst, a conversion of 95% can be attained with 40 minutes' residence time.

To scale up this reaction, it is proposed to operate a slurry reactor, agitated by hydrogen, at 95% conversion and to recycle unconverted benzene. Design and size a reactor for making 200,000 metric tons per year of cyclohexane using this process.

Cautionary note: This example is hypothetical and all numbers in the problem statement are fictitious. Any resemblance to any commercial cyclohexane technology is purely coincidental.

Solution

The first step is to estimate stream flow rates and properties and determine the rate of heat removal. These calculations can easily be carried out using a commercial process simulation program.

Fig. 15.46 shows a simulation model of the slurry reactor. The conversion is known, so a conversion reactor can be used. The hydrogen feed can be set in ratio to the benzene feed using a "Set" controller. The hydrogen feed must be in excess of the stoichiometric requirement to provide agitation, so it is reasonable to assume that we will need to recover evaporated liquid from the hydrogen off-gas. Returning this cold liquid as a recycle to the reactor feed reduces the reactor heat duty. We can then use an "Adjust" controller to manipulate the benzene feed rate to give the desired cyclohexane product rate.

Because the reactor temperature was specified, in the simulation model the reactor outlet stream temperatures are specified, and the heat duties will then be calculated by the simulation.

A quick examination of the stream data for the flow scheme in Fig. 15.46 reveals two things:

- The liquid recycle in stream 207 contains 90.9 mol% cyclohexane, whereas the liquid product in stream 202 is only 87 mol% cyclohexane. As might be expected, the condensate is enriched in the lighter product.
- The duty of the condenser, Q102, is 688 kW, whereas the duty of the reactor, Q100, is 14,480 kW.

It is therefore immediately apparent that a recycle of condensate makes very little sense. We would be recycling a product-rich stream, thereby potentially losing selectivity, and the sensible heat required to heat the recycle back to reaction temperature would be only a fraction of the heat of condensation, which itself is less than 5% of the total cooling duty. We can therefore simplify the flow scheme by eliminating the condensate recycle.

Fig. 15.47 shows a revised simulation model without recycle of condensate. The condensate stream is now recovered into the product. The hydrogen flow rate was arbitrarily set at 110% of stoichiometric to ensure that sufficient hydrogen would be available for agitation. The results of this simulation are given in Table 15.11 and can be used as stream data for sizing the reactor.

From Table 15.11, reactor liquid feed (stream 101) volume flow rate = 28.4 m³/h

Reactor residence time = 40 minutes; therefore liquid volume in reactor = $28.4 \times 40/60 = 18.93 \text{ m}^3$

We also need to add the volume for the catalyst in the slurry. The liquid density in the reactor is the same as that of the reactor liquid product (stream 202), that is, 624.5 kg/m³, so:

Mass of liquid in reactor = $624.5 \times 18.93 = 11824 \text{ kg}$

Mass of catalyst in reactor = 10% of total = $11824/9 = 1314 \text{ kg}$

Volume occupied by catalyst = $1314/1100 = 1.19 \text{ m}^3$

Total reactor volume occupied by slurry = $1.19 + 18.93 = \underline{20.13 \text{ m}^3}$

CEP Equip list 2010 (BaseCase) - Aspen Icarus Process Evaluator 2006 - aspenONE - [R100]

File Run View Tools Window Help

OK Cancel Apply Options Size Evaluate P&ID Add Delete

All components

H:\Classes\Process Design\Pr
CED Equip list
CED Equip list 2010
CED MEK
C:\Program Files\AspenTech\Additional Project Compo
Sample Projects
C:\Program Files\AspenTech\Additional Project Compo
Sample Projects

Projects Libraries Compo...

R100 - Vertical process vessel

Name	Units	Item 1
Item Reference Number		1,384
Remarks 1		
Remarks 2		
Item description		R100
User tag number		
Structure tag		
Component WBS		
Quoted cost per item	USD	
Currency unit for mat cost		
Number of identical items		1
Application		CONT
Shell material		A387B
Liquid volume	GALLONS	
Vessel diameter	FEET	8
Vessel tangent to tangent height	FEET	30
Design gauge pressure	PSIG	800
Vacuum design gauge pressure	PSIG	-13
Design temperature	DEG F	625
Operating temperature	DEG F	575
Skirt height	FEET	6
Vessel leg height	FEET	
Wind or seismic design		WWS
Fluid volume	PERCENT	20
Manhole diameter	INCHES	18
Number of manholes		2
Allowance for internals	PERCENT	10
Demister thickness	INCHES	
Demister area	SF	
Base material thickness	INCHES	
Corrosion allowance	INCHES	0.125
Number of body flange sets	PAIR	
Weld efficiency	PERCENT	100
Stress relief		
Cladding material		
Cladding thickness	INCHES	
Stiffening ring spacing	INCHES	
Head type		HEMI
Head thickness Top	INCHES	
Head thickness Bottom	INCHES	
Diameter option		ID

R100 - Vertical process vessel

Name	Design gauge pressure
Data Type	Real
Min	
Max	
Default	15
Description	If pressure and vacuum entered, design is for worst case. Default pressure design.

Ready

FIG. 15.44 Aspen ICARUS data entry for Example 15.4.

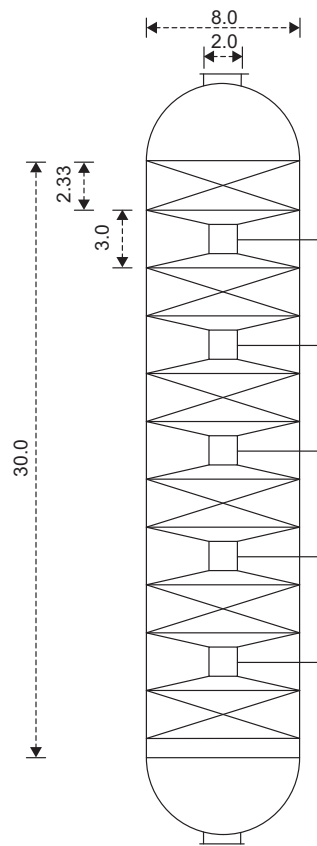


FIG. 15.45 Preliminary reactor design for Example 15.4.

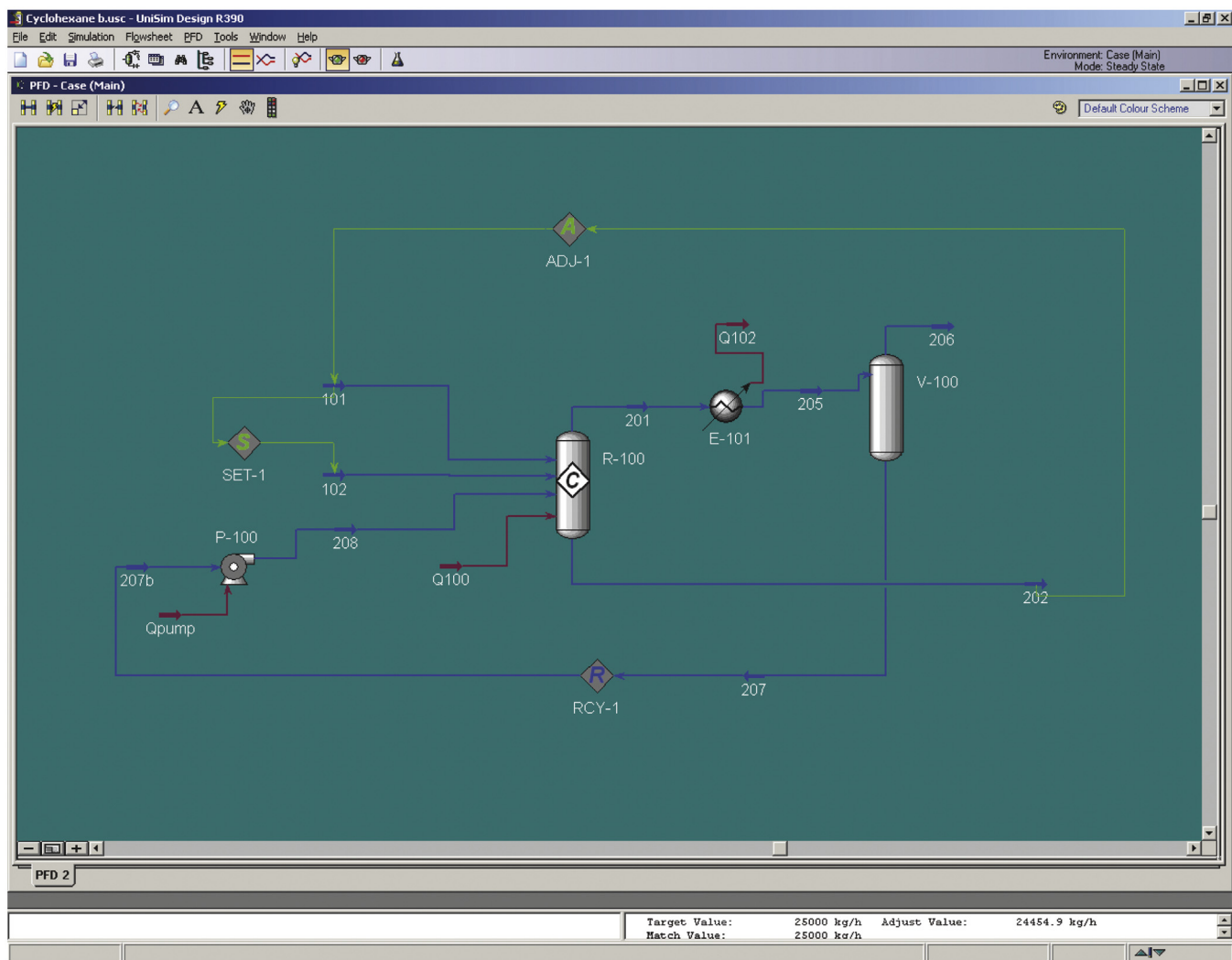


FIG. 15.46 UniSim Design model of cyclohexane slurry reactor with condensate recycle.

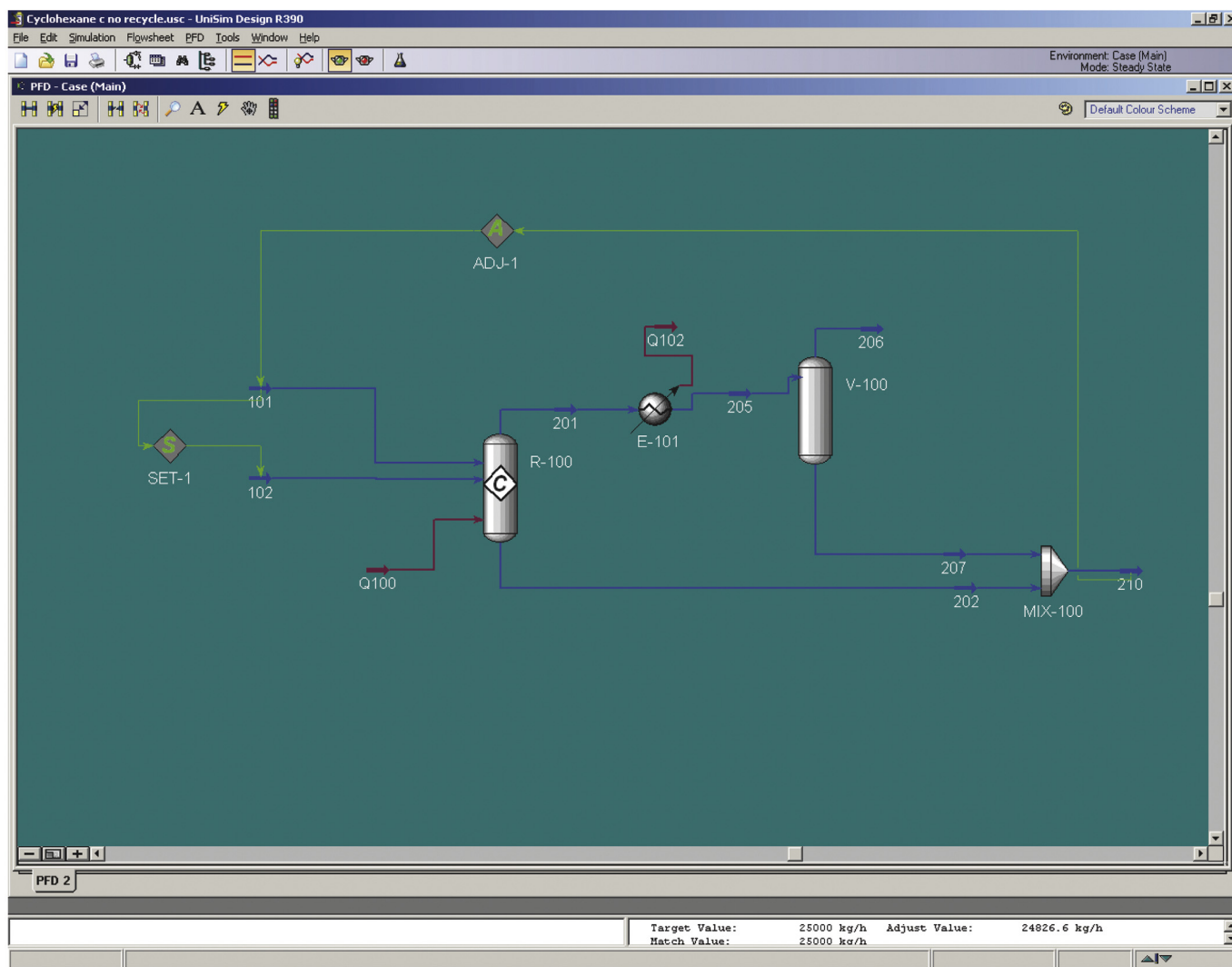


FIG. 15.47 UniSim Design model of cyclohexane slurry reactor.

TABLE 15.11 Simulation results for Example 15.5

Stream	101 Liq feed	102 Gas feed	201 Gas Product	202 Liq Product	205 Cooled Gas	206 Net gas	207 Condensate	210 Product
Vapor fraction	0	1	1	0	0.9069	1	0	0.0003
Temperature (C)	40	40	160	160	50	50	50	156.6
Pressure (kPa)	10020	10020	10000	10000	9980	9980	9980	9980
<i>Molar flow rates (kmol/h)</i>								
Benzene	313.31	0.00	0.63	15.04	0.63	0.03	0.59	15.63
Hydrogen	0.00	1033.93	114.30	26.69	114.30	113.78	0.53	27.21
Cyclohexane	0.00	0.00	11.22	286.43	11.22	0.59	10.62	297.05
Total	313.31	1033.93	126.15	328.16	126.15	114.41	11.74	339.90
<i>Mass flow rates (kg/h)</i>								
Benzene	24472.8	0.0	49.0	1174.6	49.0	2.7	46.4	1221.0
Hydrogen	0.0	2084.4	230.4	53.8	230.4	229.4	1.1	54.9
Cyclohexane	0.0	0.0	943.9	24106.0	943.9	49.9	894.0	25000.0
Total	24472.8	2084.4	1223.4	25334.5	1223.4	282.0	941.4	26275.8

TABLE 15.11 Simulation results for Example 15.5—cont'd

Stream	101 Liq feed	102 Gas feed	201 Gas Product	202 Liq Product	205 Cooled Gas	206 Net gas	207 Condensate	210 Product
<i>Properties</i>								
Mass density (kg/m ³)	862.1	7.5	26.3	624.5	36.8	8.8	751.1	628.3
Act. volume flow (m ³ /h)	28.4	278.9	46.5	40.6	33.2	32.0	1.3	41.8
Mass heat capacity (kJ/kg-C)	1.57	14.40	4.37	2.51	4.17	11.99	1.83	2.49
Thermal conductivity (W/m-K)	0.127	0.188	0.181	0.082		0.189	0.114	
Viscosity (cP)	0.498	9.40E-03	1.59E-02	0.1523		9.65E-03	0.5616	

TABLE 15.12 Gas superficial velocity as a function of diameter for Example 15.5

Diameter (ft)	Area (ft ²)	Gas flow top (ft/min)	Gas flow bottom (ft/min)
4	12.57	2.18	13.06
5	19.64	1.39	8.36
6	28.28	0.97	5.81
7	38.49	0.71	4.27
8	50.27	0.54	3.27
9	63.63	0.43	2.58
10	78.55	0.35	2.09

We can now try different reactor geometries that could accommodate this volume. Because the reactor is agitated using hydrogen, it is important to understand the extent of agitation, and hence the gas superficial velocity. The gas actual volume flow at the bottom and top of the reactor can be read from Table 15.11 as the actual flow rates of streams 102 and 201. These can be used to calculate the superficial velocity at the top and bottom of the reactor assuming a cylindrical geometry. The results are shown in Table 15.12, in units of ft/min, so as to be consistent with the velocities specified in Table 15.6.

From Table 15.6, we would ideally prefer the gas velocity to be in the range of 1 to 3 ft/min. The change in gas flow rate due to the reaction is large enough that no diameter gives an ideal velocity at both ends. The diameters above 7 ft appear to give velocities that are a bit low at the outlet, whereas 4 ft and 5 ft may be a bit high at the bottom, so it looks like 6 ft is about right.

For a 6-ft (1.829-m)-diameter cylindrical vessel, cross-sectional area = $\pi (1.829)^2/4 = 2.627 \text{ m}^2$

Height for 20.13 m³ volume = $20.13/2.627 = 11.00 \text{ m}$ or 36 ft

It is not clear that a vessel 36 ft tall by 6 ft diameter would be well mixed by bubbling gas through it. A better design might be to use a draft tube reactor, with the hydrogen flow driving liquid circulation, as discussed in Section 15.7.5 and illustrated in Fig. 15.35. The slurry volume is then set by the outer diameter. If we choose an outer diameter of 10 ft (3.048 m), then:

For a 10-ft-diameter cylindrical vessel, cross-sectional area = $\pi (3.048)^2/4 = 7.298 \text{ m}^2$

Height for 20.13 m³ volume = $20.13/7.298 = 2.76 \text{ m}$ or 9 ft

Note that this is the height of the section filled with liquid, and we will require some additional space above the liquid. We also need to make some allowance for the gas hold-up (i.e., the void space filled by bubbles).

The next thing to consider is the required heat removal. From the simulation model, the reactor cooling duty is 14.94 MW, at isothermal temperature of 160 °C. If boiling water is used as coolant (for constant temperature and high heat transfer coefficient), then low-pressure steam could be raised at 2 bar, 120 °C, giving a 40 °C temperature difference. A quick estimate of the overall heat transfer coefficient can be made from Fig. 19.1. If the process side is a light organic and utility side is boiling water, then an overall heat transfer coefficient of about 650 W/m²K is reasonable. The area can then be estimated using Equation 19.1:

$$Q = UA\Delta T_m \quad (19.1)$$

where : Q = heat transferred per unit time, W

U = the overall heat transfer coefficient, W/m²°C

A = heat transfer area, m²

ΔT_m = the mean temperature difference, the temperature driving force, °C

$$\text{Substituting : } 14.94 \times 10^6 = 650 \times A \times 40$$

$$\text{Required area} = 14.94 \times 10^6 / (650 \times 40) = 574.6 \text{ m}^2$$

This is quite a large area and obviously too large to use a cooling jacket or coil. We can investigate how many tubes would be needed if we used a stab-in heat exchange bundle:

$$\text{Area of 1-inch diameter tube, 9 ft long} = \pi \times 0.0254 \times 2.76 = 0.220 \text{ m}^2$$

$$\text{Number of tubes required for heat transfer (if bundle is inside reactor)} = 574.6/0.22 = 2612$$

Such a large number of tubes would clearly not be easy to accommodate inside the draft tube without a significant effect on hydraulics. Instead, we must consider an external exchanger. If we choose to pump liquid from the reactor, cool it with cooling water to 60 °C (allowing a 20 °C approach so that a cross-flow exchanger with a X-shell can be used to minimize fouling or blocking by catalyst fines), and then return it to the reactor, then a heat balance gives the required flow rate:

$$\text{Specific heat capacity} = \text{heat capacity of stream 202} = 2.52 \text{ kJ/kg}^\circ\text{C}$$

$$\text{Mass flow rate} = 14.94 \times 10^6 / (2.52 \times 10^3 \times 100) = 54.05 \text{ kg/s} = 194.6 \text{ t/h, or roughly 7.6 times the product rate.}$$

This is not unacceptably high, given that we have 11824 kg of liquid in the reactor, but the pump-around loop must circulate the entire reactor contents every $11824 / (54 \times 60) = 3.6$ minutes. Finding a way to withdraw this amount of liquid without carrying catalyst with it could be difficult (a large screen area would be required), so a better approach might be to use flash vaporization for cooling:

$$\text{From the simulation, the heat of vaporization} = 377 \text{ kJ/kg}$$

$$\text{Evaporation needed} = 14.94 \times 10^6 / (377 \times 10^3) = 39.6 \text{ kg/s}$$

This probably actually increases the required circulation rate, as it would likely be difficult to evaporate much more than about half of the liquid in a flash cooler.

So we must design the reactor circuit to include a heat exchanger that can handle a slurry flow of catalyst. A flow diagram for the proposed reactor design is given in Fig. 15.48. A condenser is used to recover liquid product from the net gas. Some space should also be left at the top of the reactor to allow for vapor–liquid disengagement and for the void space occupied by bubbles, say, an additional 3.66 m, giving an overall tangent length of 6.42 m or 21 ft. The reactor and separation drum can then be designed and costed as pressure vessels; see Chapter 14 and Example 15.4. The heat exchanger and the condenser can be designed using the methods given in Chapter 19. The overall cost will most likely be dominated by the heat exchanger cost.

This design has a number of potential flaws that require further evaluation. The hydraulic performance and mass transfer rate have not been validated, and it would be prudent to construct a pilot plant–scale reactor to confirm the performance. The gas hold-up and bubble volume fraction should be estimated more accurately or determined experimentally in a pilot plant

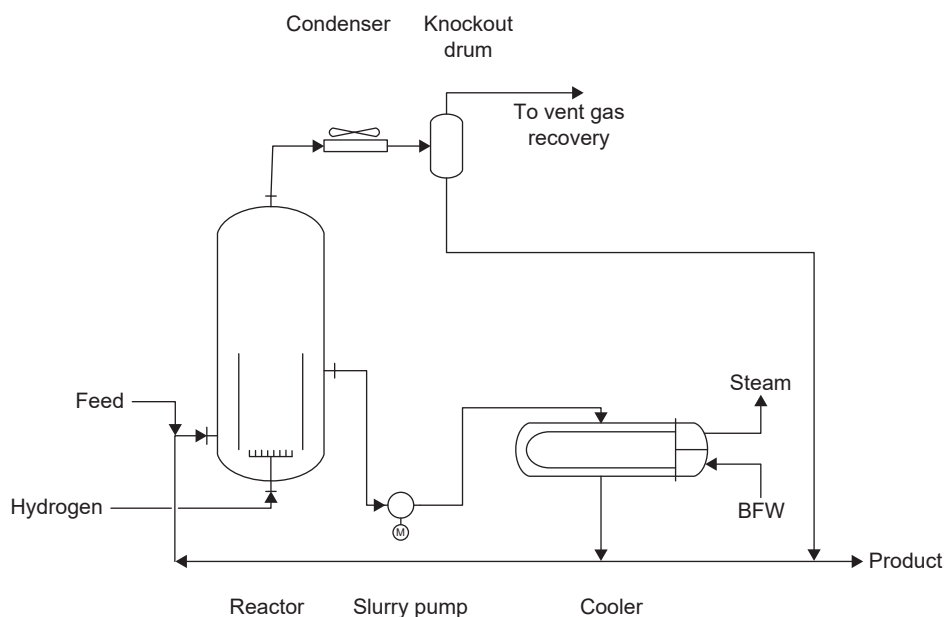


FIG. 15.48 Slurry reactor design for Example 15.5.

reactor to ensure that we have left sufficient space for expansion of the liquid due to bubbles. The use of a cross-flow exchanger will probably give a low F-factor and high area requirement; a different exchanger arrangement might be better. There is also the possibility that catalyst could accumulate in dead spots in the exchanger shell, so flowing the slurry on the tube side might be better.

Several alternative designs could be contemplated. For example, the reaction could be carried out inside a heat exchanger with a suitable design that allowed for gas injection, a loop reactor could be used with the heat exchange tubes in one side of the loop, or the reaction could be combined with product fractionation as a reactive distillation process.

15.9 Design of bioreactors

Biological processing is of increasing importance in the chemical, food, and pharmaceutical industries. Many valuable chemicals and active pharmaceutical ingredients have complex molecular structures that can only be synthesized by biological routes. Some molecules can be made synthetically, but the biological route gives greater selectivity for the desired product or enantiomer. Biological processing can also allow simple chemicals to be produced from biomass feedstocks, as in the fermentation of sugars to ethanol for use as a gasoline additive. Examples of biological processes can be found in almost every sector of the chemical industry; see the design project problems in Appendix E (available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793) for multiple examples.

Biological processing introduces additional constraints on process and reactor design. This section addresses the major concerns that face the design engineer in selecting and sizing a biological reactor and specifying the ancillary equipment required for feed preparation and aseptic operation. There is not sufficient space here to cover the fundamentals of biological processes or the kinetics of biological reactions. These subjects are addressed in biochemical engineering textbooks; see Bailey and Ollis (1986), Blanch and Clark (1996), Shuler et al. (2017), and Krahe (2005).

15.9.1 Enzyme catalysis

If an enzyme can be isolated or expressed from host cells and maintains its activity, it can be used as a catalyst. Enzymes are proteins, usually with between 100 and 2000 amino acid residues, and the active catalytic site is formed by folding of the protein. Enzyme catalysts can be homogeneous in the liquid phase or can be made heterogeneous by immobilization onto a solid support.

Enzyme reactions usually have low reaction rates compared with catalytic reactions of simple molecules at high temperatures, but enzymes can be very selective, particularly when stereoselectivity is required for enantiomeric products.

Enzymes can be permanently deactivated (*denatured*) by high temperatures and can be inhibited by molecules that are able to bind to or block the active site of the enzyme. Enzymes can also lose their activity if the shape of the enzyme molecule is altered, for example, by changes in pH, solution ionic strength or as a result of immobilization. Most enzymes are used in water and lose their activity in organic solvents.

The feedstock for an enzyme reaction is known as a substrate. Substrates can be dissolved in the liquid or can be suspended solids such as starch and fibers in biomass processing. High concentrations of substrate often inhibit the reaction, leading to the selection of a well-mixed reactor or fed-batch reactor. Likewise, enzymes are often inhibited by high concentrations of product.

Some enzymes require the presence of a *cofactor* or *coenzyme* to maintain their function. Cofactors can be as simple as metal ions, ammonia, oxygen, small organic compounds, or vitamins. Cells naturally regulate the levels of cofactors to maintain optimal enzyme performance. In the design of enzyme-catalyzed bioreactors, some experimentation is needed to determine whether make-up of cofactors is needed, as some cofactors are strongly bound, whereas others can be eluted and must be continuously fed.

The kinetics of enzyme-catalyzed reactions are described in all the biochemical engineering books cited in the introduction to this section, as well as in all the reaction engineering books cited in the bibliography at the end of this chapter. Rate equations usually have the form of the Michaelis–Menten equation, which is similar to the Langmuir–Hinshelwood–Hougen–Watson (LHHW) equation for gas–solid catalysis. As is often the case,

biochemical engineering led the way: the Michaelis–Menten equation was developed in 1913, whereas the LHHW equation did not follow until 1947.

The design of a reactor for enzyme catalysis is substantially easier than design for cell cultivation. One of the advantages of enzyme catalysis relative to cell culture is that enzymes do not respire. Consequently, a gas–liquid reactor is not needed unless a gas-phase cofactor is required, which is unusual. Enzyme-catalyzed reactions are therefore usually carried out in the liquid phase or slurry phase if the substrate is a solid. The choice of reactor depends largely on the method of enzyme confinement, as discussed next.

Enzyme confinement and immobilization

Enzymes that can be produced inexpensively can be used once through in a bioreactor; examples include amylase for starch digestion in bread manufacture and the production of high-fructose corn syrup; glucose isomerase for converting glucose to fructose in the manufacture of high-fructose corn syrup and ligninases in paper-making. Enzymes that act on solid substrates must usually be used once through unless the enzyme converts the substrate into soluble components. Enzymes that are more expensive must be recovered for recycle or confined to the reactor system.

Most enzymes have large enough molecular size that they can be retained during nanofiltration or ultrafiltration using a membrane filter. If the product molecule is small enough to pass through the filter while the enzyme is retained, the reactor effluent can be sent to a cross-flow filter and the retentate returned to the reactor (Fig. 15.49). This allows the enzyme to be used in solution and confined within the reactor-membrane circuit. The design of membrane filtration processes is discussed in Section 16.5.4.

Many methods have been developed for immobilizing enzymes. The most common approaches are adsorption onto activated carbon, ion exchange resin, functionalized beads or fibers, or encapsulation in a gel. Care must be taken to ensure that the immobilization method does not alter the protein structure of the enzyme and affect its activity. Methods for enzyme immobilization are reviewed by Storey and Schafhauser-Smith (1994). Immobilized enzymes on solid supports can be used in packed-bed reactors if the substrate is dissolved in the liquid phase. A packed bed with recycle is used to achieve a well-mixed performance if the substrate inhibits the reaction (see Fig. 15.33).

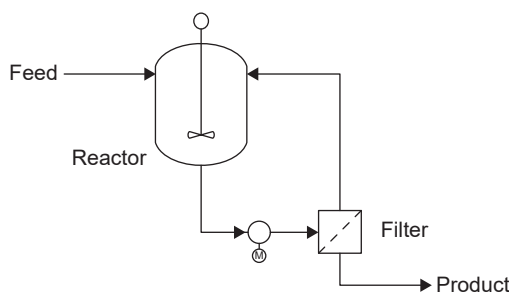


FIG. 15.49 Reactor-membrane circuit for enzyme catalysis.

15.9.2 Cell cultivation

Many biological products are formed through complex reaction paths that are not easily accomplished outside of living cells. In some cases, a species can be found (or bred or otherwise genetically modified) that expresses the desired product into solution. Examples include algae and cyanobacteria that express lipids and yeasts (such as *Saccharomyces cerevisiae*) that express ethanol. For more complex molecules, for example, monoclonal antibodies, it is necessary to break the cell wall (a process known as *lysis*) to harvest the product from inside the cell. The method of product recovery has a large impact on reactor design and even strain selection. For example, if a product is recovered after lysis, the cells should be able to tolerate the product at high concentrations, and a batch or plug flow system is preferred to maximize cell productivity. Conversely, if the cells express product continuously, a fed-batch or continuous well-mixed reactor will often have higher productivity.

Cell cultivation and growth cycle

A wide range of microorganisms can be grown in bioreactors. Most biological processes use simple microorganisms such as bacteria, molds, and yeast. Bacteria and yeast are single-celled organisms, with sizes typically in the range of 0.5 to 2 μm for bacteria and 5 to 10 μm for yeast. Molds are multicellular and have sizes from 5 μm to much larger. Animal and plant cells can also be grown in bioreactors, but the formation of multicellular structures (tissue culture) is much more difficult and is discussed later.

In all cases, the productivity of a bioreactor depends on the concentration of cells in the reactor. The cell concentration will vary with time in a batch or plug flow process and will reach a steady state in a fully continuous process. The reactor concentration of an *intracellular* product (a product inside the cells) will not necessarily be maximized when the concentration of cells is maximized, as long as the product remains stable after cell death. This is illustrated in Fig. 15.50, which illustrates the concentration of live cells and intracellular product during batch fermentation. The fermentation proceeds through the following steps, labeled in the figure:

- I. After initial inoculation with cells, there is a short lag phase while the cells adapt to the new environment. During this phase, some of the cells in the inoculum die and balance the rate of cell growth so that overall cell concentration increases only slowly.
- II. Once the cells have adjusted to the new environment, their numbers begin to increase rapidly. Because the rate of cell formation is proportional to the number of cells, this is termed the exponential growth phase.
- III. Eventually the rate of growth begins to slow, either as a result of nutrient depletion, accumulation of toxic products or by-products, difficulty of obtaining oxygen, or overheating. With careful control of these parameters in a fed-batch or continuous system, a linear growth phase can be maintained for some time. This is important in continuous fermentation for intracellular products, as a fraction of the cells must be harvested continuously to obtain the product.
- IV. At some point the rates of cell growth and cell death are in balance and a stationary phase is reached. The goal of continuous fermentation for *extracellular* products (products that are expressed outside the cell) is to sustain this condition by control of nutrient addition, product removal, oxygen supply, pH, and heat removal. In batch fermentation, the stationary phase is often very short.
- V. In the final phase, cells die or sporulate and the live cell concentration decreases exponentially. If the desired product is not degraded during or after cell death, then the product concentration in the reactor can continue to increase, even as the live cell population decreases. It may therefore be optimal to continue batch fermentation beyond the end of the stationary phase.

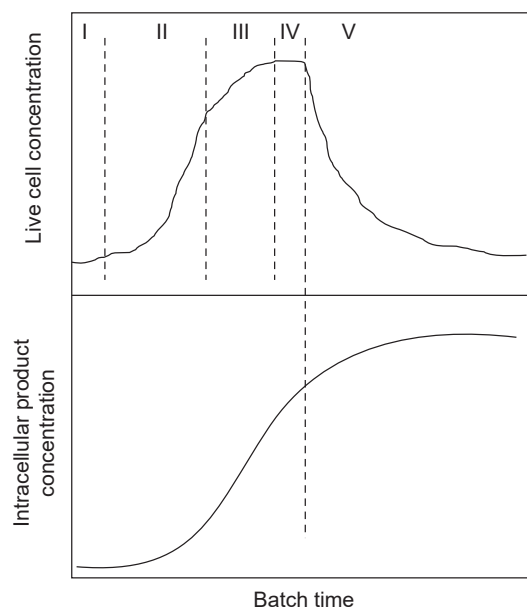


FIG. 15.50 Cell growth and product formation in batch fermentation.

The kinetics of each of these steps can be modeled to develop an overall reaction engineering model of the cell growth cycle and the product formation cycle; see [Bailey and Ollis \(1986\)](#) or [Blanch and Clark \(1996\)](#). In industrial practice, scale-up of fermentation processes is usually accomplished by taking multiple product assays during a small-scale fermentation to determine the optimal cycle experimentally.

Cell immobilization

Cells can be immobilized using many of the same methods used for enzymes, but cell immobilization is less common. Cells already have a large enough size that they are easily retained by filters or even hydrocyclones or centrifuges, and many species naturally flocculate to larger clusters that are even easier to retain. Immobilizing cells onto a support can introduce mass-transfer limitations that reduce cell growth rates and productivity. Consequently, the disadvantages of cell immobilization outweigh the advantages for most processes. The most widely used application of cell immobilization is in biofilters for wastewater treatment, where a film containing many different naturally occurring species forms on a packing and wastewater is passed over the packing. See [Bailey and Ollis \(1986\)](#) for more information on the design of wastewater treatment plants.

Tissue culture

Tissue culture is used to describe the growth of multicellular structures separate from an organism in an artificial environment. The technique has been practiced by biologists on a small laboratory scale for over a hundred years and is used as a method of propagating plants, but is not yet in widespread use as a processing technique. It is, however, of considerable interest to the biomedical industry because of the possibility of growing tissue grafts or even organs for therapeutic uses; see [Lavik and Langer \(2004\)](#) and [Xu et al. \(2008\)](#).

Tissue culture is very difficult to scale up because of the competing needs for high mass transfer and low shear. High mass transfer rates are needed to supply nutrients and oxygen to the cells, whereas low shear rates are necessary to prevent damage to cells and allow multicellular structures to form. [Martin and Vermette \(2005\)](#) and [Curtis et al. \(2001\)](#) review the challenges of tissue culture on a process scale and describe different reactor design approaches, for example, immobilizing the cells on a scaffold or support, such as a fiber bed. This area continues to be the subject of a great deal of research.

15.9.3 Prevention of contamination in biological systems

Biological reactors and their feed systems must be designed to prevent ingress of unwanted species into the reaction medium. The design must consider both chemical and biological contamination, as well as cleaning to prevent batch-to-batch contamination.

Chemical contamination

Cell growth and enzyme activity can be strongly affected by relatively low concentrations of poisons in the growth medium. The reactor and feed system must be designed to ensure that contamination by these chemicals does not occur.

Most biological reactors and feed systems are constructed from low-carbon austenitic stainless steels because of their low rate of corrosion and ease of electropolishing ([Krahe, 2005](#)). Although expensive, 316L stainless steel is preferred, but the less expensive 304 and 304L are sometimes used in food processing. Copper and its alloys were used historically in brewing, but have been shown to have a strong inhibitory effect on many fermentation processes, and so copper, bronze, and brasses are usually eliminated from vessels, instruments, valve trims, etc.

Polymers that are used in O-rings, gaskets, valve packing, and diaphragms must withstand the process conditions and be rated as safe for use in food or food packaging if the product is for human or animal consumption. Plasticizers or other additives in the polymer must not leach into solution. Viton, Teflon, and ethylene-propylene-diene rubber (EPDM) are most often used.

Process feed quality must be carefully controlled for the presence of potential contaminants or poisons. Processes that make products for human or animal consumption do not necessarily require USP-grade feeds, but the feed quality or processing must be such that USP-grade product quality is attained.

Additional requirements on feed purity and quality control may be set by Good Manufacturing Practice (GMP) regulations; see Section 15.9.8.

Biological contamination and design for sterile operation

If a different species enters a fermenter, it can compete with the design species for substrate, cause infection and cell loss, or contaminate the products with unwanted and potentially toxic compounds. Biological processes are therefore almost always designed for aseptic operation. The maintenance of aseptic conditions requires careful cleaning between batches or continuous runs, sterilization of the plant before operation, and sterility of every feed other than the inoculum that is introduced during normal operation.

Sterilization is itself a reaction process, and the rate of death of biological contaminants can usually be described with first-order kinetics; however, because biological contaminants must be essentially completely removed, the process is often treated in probabilistic terms.

Production equipment is too large to be disassembled and sterilized in autoclaves, so sterilization in place (SIP) is used for the plant. Sterilization of the plant is carried out in batch mode, typically by using steam to heat the plant to a high enough temperature (usually 120 °C or higher) followed by holding at temperature for a predetermined time and then cooling back to reaction conditions. During cooling, steam in the plant will condense, and air must be admitted to prevent underpressuring, so it is important that the air source also be sterile.

Feed media pose several sterilization challenges. Simple compounds such as sugars and salts can be dissolved in water and sterilized by heating to high temperature, but some nutrients such as vitamins are thermally sensitive and cannot be heat treated without losing some material to degradation; see [Leskova et al. \(2006\)](#). In batch fermentation, those feed components that are not thermally sensitive can be charged to the reactor and sterilized along with the plant. In fed-batch and continuous processes, the feed must be continuously sterilized. Because of the importance of ensuring complete conversion (death of biological contaminants), plug flow devices must be used.

The most common flow schemes for continuous sterilization are shown in [Figs. 15.51 and 15.52](#). [Fig. 15.51](#) shows a steam injection sterilization process, in which steam is injected into the liquid feed to cause heating. The liquid is then passed through a serpentine tubular holding section to achieve the desired residence time for sterilization and cooled to the process temperature by flash cooling. Flash cooling reduces the possibility of contamination from cooling water in a cooler. [Fig. 15.52](#) shows a heat exchange sterilization process using plate heat exchangers. Heat from the product is recovered to pre-heat the feed before it enters a steam heater. The feed is then held at temperature in a tubular holding section before being cooled against incoming feed and cooling water. Plate exchangers are typically used because of their ease of cleaning and inspection. Heat-exchange sterilization has lower energy costs but has a higher possibility of feed-to-product contamination or contamination from cooling water.

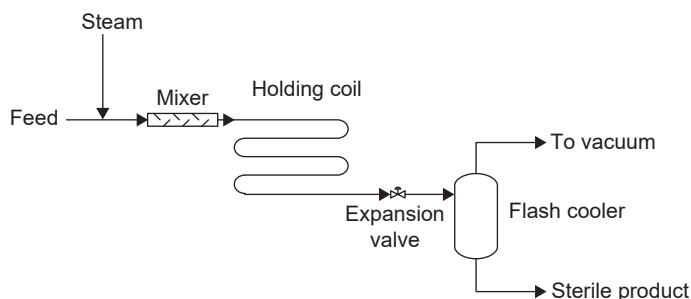


FIG. 15.51 Steam injection sterilization process.

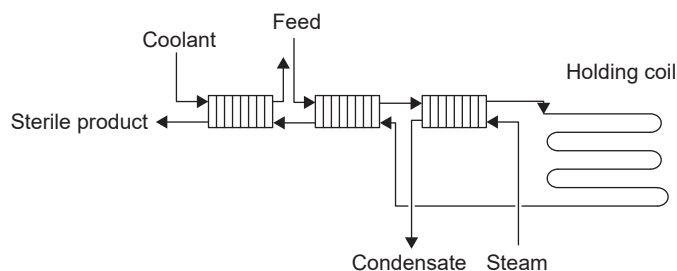


FIG. 15.52 Heat exchange sterilization process.

The holding time required for sterilization depends on the temperature used, the degree of sterility required, and the types of organism that might be present in the feed. Fifteen minutes at 120 °C or three minutes at 135 °C are typical treatments that are adequate for bacteria, molds, fungi, and bacterial spores, but more extreme treatments are used in some cases.

If thermally sensitive compounds such as vitamins are subjected to heat treatment, then high temperatures must be used with short residence times to minimize thermal degradation of the desired component (*high-temperature short-time*, or HTST, sterilization). It may be necessary to prepare a feed that contains a higher concentration of the desired component to allow for some thermal loss during sterilization. Continuous sterilization processes are more suited to HTST conditions than batch sterilization in the reactor, as cooling occurs more quickly in continuous flow. In extreme cases, a sterile feed may need to be purchased or synthesized.

Some components that are introduced into a fermenter may not require sterilization. For example, if acids or bases are used to control pH, their concentration may be high enough to guarantee sterility without sterilization.

Air that is brought in as a source of oxygen is usually sterilized by membrane filtration to remove bacteria and particulates. Air leakage into the plant from the atmosphere is normally prevented by operating the process under pressure. The exception is for species that are strong pathogens, where the hazard posed to workers from a potential leak is too great and the plant is run under slight vacuum.

Cleaning

Cleaning of the plant is required at the end of a production run to remove residual substrates that can lead to contamination and to prevent batch-to-batch contamination. Bioreactors and their feed systems are usually designed for cleaning-in-place (CIP), but disassembly and manual cleaning are used for equipment, such as heat exchangers, that is prone to fouling.

Bioreactor systems should be designed to be free-draining and to avoid corners, crevices, and dead-ended pipe legs to facilitate cleaning. Plate heat exchangers are preferred over shell and tube heat exchangers for ease of cleaning, as they have no internal dead spots where material can accumulate and are easily taken apart and inspected; see Section 19.12. Diaphragm valves are also preferred because of their ease of cleaning.

Cleaning cycles typically consist of the following steps:

1. Wash with high-pressure water jets
2. Drain
3. Wash with alkaline cleaning solution (typically 1M NaOH)
4. Drain
5. Rinse with tap water
6. Drain
7. Wash with acidic cleaning solution (typically 1M phosphoric or nitric acid)
8. Drain
9. Rinse with tap water
10. Drain
11. Rinse with deionized water
12. Drain

The acid and alkaline washes may be reversed, depending on the system.

Cleaning cycles typically require filling and emptying the reactor repeatedly and can contribute significantly to the time between batches in batch fermentation. The process design must include the necessary equipment for preparing the cleaning fluids (typically by dilution of stronger acid or base), as well as neutralizing and disposing of the effluents from cleaning. Equipment cleaning has a substantial impact on wastewater production in biochemical processing, and hence a significant environmental impact. [Previs and Roche \(2018\)](#) discuss methods to reduce the environmental impact of cleaning processes.

15.9.4 Feed preparation and consumption

Even single-celled organisms require a balanced diet to grow happily and reproduce. The feed medium must contain sources of carbon, nitrogen, phosphorus, sulfur, and required metal ions. In some cases, more complex components such as vitamins or enzymes may be added.

TABLE 15.13 Typical feed medium composition

Component	Composition (g)
Water	1000
Glucose	5
NH ₄ Cl	1
K ₂ HPO ₄	1
MgSO ₄ ·7H ₂ O	0.2
FeSO ₄ ·7H ₂ O	0.01
CaCl ₂	0.01
Mn, Mo, Cu, Co, Zn (as salts)	2×10^{-5} to 5×10^{-4} of each

From Stanier et al. (1986).

The optimum feed medium depends on the organism (or enzyme), desired product, method of production (extra-cellular or intracellular), and choice of fermentation reactor (batch or continuous). The recipe is usually determined experimentally. A typical composition, taken from Stanier et al. (1986) is given in Table 15.13.

Cell metabolism and productivity can be manipulated by changes in feed composition. It may be desirable to change the feed during different phases of the growth cycle shown in Fig. 15.50. For example, more minerals may be needed during the exponential growth phase, but a change in composition may prolong the stationary phase or lead to a higher titer of product. Mead and Van Urk (2000) describe a process for controlling medium concentration in response to a by-product concentration, and Shibuya et al. (2010) describe a method for controlling feed composition by mixing several media to optimize overall productivity.

It is important to note that in most biological processes the rate of feed consumption is determined primarily by cell metabolism and growth of new cells and has no strong correlation to the rate of product formation. A stoichiometric relationship between the feed and product (or even feed and cell mass) may have no value for design purposes. Cells require nutrients both to maintain life and to grow new cells. If the rate of growth of new cells, μ_g , is defined by:

$$\frac{d x}{d t} = \mu_g x \quad (15.29)$$

where:

x = concentration of cells, g/l

t = time, s

μ_g = growth rate, s⁻¹

then the rate of consumption of any substrate can be written as:

$$\frac{d s_i}{d t} = \left(m_i + \frac{\mu_g}{Y_i} \right) x \quad (15.30)$$

where:

s_i = concentration of substrate i , g/l

m_i = rate of consumption of substrate i to maintain cell life, g of substrate/g cells.s

Y_i = yield of new cells on substrate i , g of cells/g substrate

Some example values of m and Y for glucose and oxygen for different species are given in Table 15.14, based on data from Solomon and Erickson (1981).

TABLE 15.14 Substrate consumption for cell maintenance and growth

Microorganism	m_{glucose} (g/g.hr)	Y_{glucose} (g/g)	m_{oxygen} (g/g.hr)	Y_{oxygen} (g/g)
<i>Escherichia coli</i>	0.072	0.35	0.6	24.7
<i>Escherichia coli</i>	0.090	0.53	3.0	42
<i>Saccharomyces cerevisiae</i>	0.018	0.51	0.6	34.5
<i>Aerobacter aerogenes</i>	0.054	0.38	1.4	44

From Solomon and Erickson (1981).

Rates of cell growth vary widely by species. [Green and Southard \(2018\)](#) state typical rates of biomass growth are in the range 2 to 5 g/liter.h, and typical rates of oxygen consumption are in the range 1.5 to 4 g/liter.h.

15.9.5 Batch fermentation

Most biological processes are carried out using batch fermentation reactors. Batch operation is easier to scale up from laboratory-scale fermentation, and batch integrity is an important method of quality control.

In strict batch cultivation, the substrate is charged to the reactor, sterilized and cooled, and then the inoculum of live cells is added. No additional substrate is added, although air is usually added continuously to maintain the oxygen concentration above a critical level. The critical oxygen level varies by species, but is typically above 1 ppm. After a set period of time, the fermentation is stopped and the reactor contents sent to downstream processes for harvesting.

Strict batch processing is rather inefficient, as the rate of cell growth and the product titer quickly become limited by consumption of substrate. A more common approach is fed-batch cultivation, in which additional feed medium is added as the cell concentration increases, allowing the process to proceed further into the growth cycle and achieve higher product titers. In a fed-batch process, the reactor may start out only 20% to 40% filled to allow space for adding more growth medium, and the design must ensure that mixing and aeration will be sufficient at all fill levels. Different fed-batch processing strategies are described in [Krahe \(2005\)](#).

Fermenter design

Batch fermenters are usually stirred tanks with jackets and/or coils for heating and cooling and spargers or other means for introducing air. A typical design is shown in [Fig. 15.53](#).

Good mixing is important in fermentation to ensure that all the microorganisms in the fermenter have access to the desired concentrations of substrates and oxygen and to maintain isothermal conditions. Baffles are usually used to improve the mixing pattern in the vessel and prevent swirl (see Section 15.5.2), but baffles can make cleaning and sterilization more difficult. Other mixing criteria for stirred tanks are discussed in Section 15.5.2.

The rate of agitation in a fermenter is usually determined primarily by the rate of oxygen mass transfer that is needed, and the agitation rate can be adjusted to give the desired mass transfer parameter, $k_L a$; see [Equations 15.13 to 15.15](#) in Section 15.3.4. The cells have a density that is very close to the density of water, so they are easily suspended in solution, and biomass suspension rarely limits the agitation rate. Very high agitation rates are avoided, as high shear can cause breakage of the cell walls, causing death.

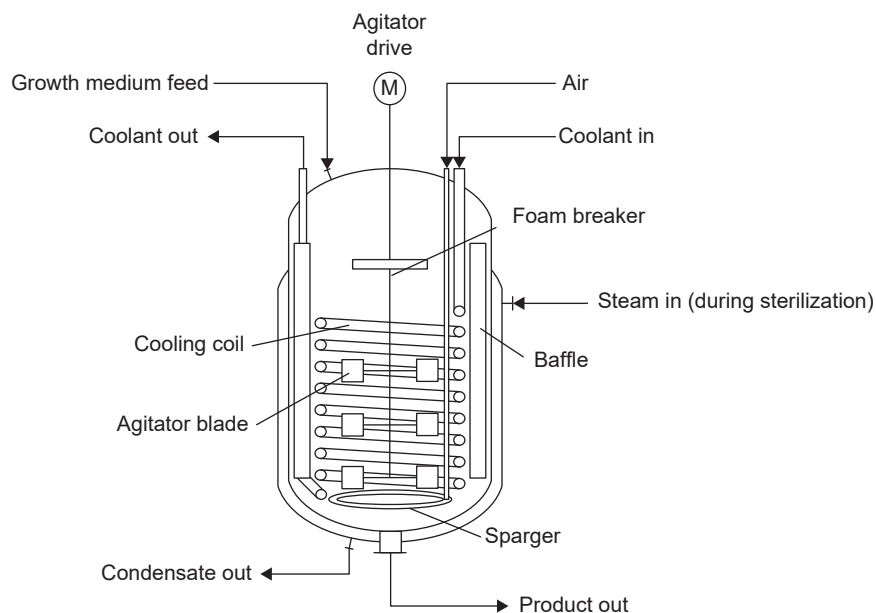


FIG. 15.53 Fermentation reactor.

TABLE 15.15 Standard fermenter sizes

Vessel size (m ³)	0.5	1.0	1.5	3	5	7.5	15	25	30
Vessel size (gal)	150	300	400	800	1500	2000	4000	7000	8000

Foaming can be a serious problem in fermentation. Surfactants may be present in the growth media or formed during fermentation. The bubbling of air naturally causes froth to form at the vapor–liquid interface. If foaming is excessive, cells and product can be lost to the vapor recovery system and reactor productivity impaired. Mechanical foam breakers mounted on the agitator shaft can be used to break up large bubbles. Antifoaming agents (antifoam) can be added if a suitable compound can be identified that does not interfere with the cells or impede oxygen transfer. The reactor is usually designed to operate less than 75% filled to allow space for foam breaking and vapor–liquid segregation.

The rate of heat release in fermentation processes is usually relatively low, and adequate cooling can be provided by an external jacket or internal coils for smaller fermenters. As was the case for baffles, the presence of coils can make cleaning and sterilization more difficult. The rate of heat transfer can be found using the correlations given in Section 19.18.

The size of a batch fermenter is determined by the species productivity, required residence time and desired plant attainment rate. Larger-size fermenters are custom-built, but standard vessel sizes are used at small and intermediate scales. Standard sizes are usually stated in liters or m³, and the more common sizes are given in Table 15.15, with approximate equivalents in U.S. gallons. The vessel aspect ratio is usually between 2 and 4. Krahe (2005) and Benz (2019) discuss factors to consider in choosing between a taller or shorter vessel.

When the batch residence time is long, it is common to use several reactors in parallel so as to maximize the productivity of downstream separation equipment. Very large fermenters are only used for inexpensive products so that contamination of a batch does not cause excessive financial loss. For example, Lee (2010) stated that a typical new plant for monoclonal antibody production would have 4 to 12 fermenters, each of 10 to 25 m³. After 10 to 14 days of culture time, each fermenter would attain a titer of about 4 g/l of product. The total plant cost, including feed preparation and product recovery, would be between \$400 million and \$1 billion, and the plant would take about 6 years to build from groundbreaking to Food and Drug Administration (FDA) approval.

Most production-scale batch fermenters are made from austenitic stainless steel, typically 316L, to avoid contamination of the growth medium with corrosion products. Stainless steel fermenters are designed as pressure vessels so as to withstand sterilization conditions; see Chapter 14. At smaller production volumes, there is a growing market for single-use disposable plastic reactors, which now account for about 75% of clinical-scale production of FDA GMP–certified materials and about 5% of commercial-scale production (Mullin, 2016). Disposable reactors are available in sizes up to 2 m³ at the time of writing. Thayer (2010) gives details of suppliers and applications of disposable vessels.

Smaller batch fermenters are often emptied by pressuring the vessel with air to force the liquid out through the bottom drain line. This obviates using a bottoms pump and removes a potential source of contamination at the pump shaft seal.

Scale-up considerations

Batch fermentation is used industrially at very large scales. Beer and wine brewing can be carried out using either continuous or batch fermentation, with fermenter volumes up to 200 m³ quite common. Even very expensive products such as monoclonal antibodies are routinely produced in 25-m³ batches (Lee, 2010).

Scale-up of batch fermenters is simple in the sense that small-scale fermentations are also batch processes, so scaling of kinetics data is relatively straightforward as long as the same temperature and concentrations of substrates and oxygen can be achieved in the larger vessel. Scale-up is not trivial, however, as rates of heat and mass transfer become increasingly important as fermenter size is increased.

The most important factor in scale-up of a batch fermentation process is usually ensuring that the same oxygen concentration can be maintained. The designer should attempt to ensure that the new design will obtain the same mass transfer parameter, $k_L a$, which may require increasing the agitation rate or air flow rate compared with a smaller vessel. The rate at which air can be supplied is often the limit on the maximum concentration of live organisms that can be attained.

The rate of heat transfer can also be an important consideration in scaling up a batch fermentation process. Most fermentation processes are mildly exothermic. If heat is not removed, the temperature will slowly increase until cell

productivity decreases or cells begin to die. A larger fermenter has a lower external area per unit volume, and so the rate of heat loss to the environment is lower in proportion to the rate of heat generation. Additional cooling means, such as cooling coils or an external pumped heat exchange circuit, may need to be considered.

The scale-up of batch fermentation processes is discussed in the context of pharmaceuticals manufacture in the book by [Levin \(2011\)](#).

15.9.6 Continuous fermentation

In continuous fermentation, the plant is operated such that the rate of live cell loss (either by death or elutriation from the fermenter) matches the growth rate of new cells. A stable population balance is thereby achieved, and with careful control, this steady state can be maintained for days, weeks, or even months. Steady continuous operation maximizes the volumetric productivity of the fermenter, as the fraction of time spent in draining, cleaning, filling, and sterilizing operations is dramatically reduced.

The design of a continuous fermentation reactor is strongly dependent on whether the product is extracellular or intracellular.

Extracellular products can be recovered from the fermentation broth without requiring removal of the cells. The cells can therefore be contained in the reactor loop, either by immobilization or by using a reactor-membrane circuit, as shown in [Fig. 15.49](#). Reactor productivity will usually be optimized if the cells are in the stationary phase of the growth cycle, shown as phase IV in [Fig. 15.50](#), with the highest stable concentration of live cells. The rates of substrate addition, dilution (by water coming in with the substrate), oxygen addition, carbon dioxide removal, and heat removal must all be controlled to maintain the optimal conditions for sustaining this steady state. The most common industrial example of continuous fermentation for an extracellular product is the use of *S. cerevisiae* to ferment sugars into alcohol in production of ethanol for use in gasoline and in large-scale brewing of wine and beer.

When the product is the cells themselves or the product must be harvested from inside the cell by lysis, a continuous fermentation process must maximize the production of live cells. Cells are usually removed by elutriation, either from the reactor itself or by using a settling vessel, ultrafiltration module, or hydrocyclone to form a product stream that has a higher cell concentration than the fermentation broth. The reactor is usually a continuous stirred tank fermenter (CSTF), often referred to as a *chemostat*. Reactor productivity is maximized by operating at a point on the growth cycle with a high cell growth rate, typically in the linear phase of the growth cycle, shown as phase III in [Fig. 15.50](#), and reactor conditions are chosen to stimulate cell growth. Most nutrients are supplied in excess so that one nutrient becomes the limiting feed and can be controlled to control the growth rate. Continuous fermenters with cell harvest are used in the production of bakers' yeast and in the manufacture of certain bulk enzymes.

Continuous fermenter design and scale-up

All of the issues discussed under batch fermentation in Section 15.9.5 also apply to CSTFs, with the additional requirement that the reactor must be designed for cell retention or elutriation, depending on the method of product recovery. Continuous stirred tank fermenters have the same appearance as batch fermenters (see [Fig. 15.53](#)).

The maximum concentration of cells that can be achieved in the fermentation broth is typically only a few weight percent, so it is advantageous to concentrate the cells into a thicker slurry product when the product is intracellular or is the cells themselves. Concentrating the cells reduces the volume flow of water through the plant, reduces the size of product recovery operations, allows use of more concentrated feeds, and reduces the sterilization requirements. If the cells are not too sensitive to shear, hydrocyclones can be used. Alternatively, ultrafiltration can be used to reject water, or a settling tank can be used. The design of equipment for separating solids from liquids is discussed in Section 18.6.

Sterility of all feeds to the process is critical to the operation of a continuous fermenter; see Section 15.9.3. Sterile conditions must also be maintained in immediate downstream operations, as bacteria are able to travel upstream against the product flow. Product quality must be monitored regularly to ensure that contamination has not occurred. Continuous processes cannot use batch integrity as a means of quality control. If contamination occurs, the run must be stopped and the vessel emptied, cleaned, and sterilized before being placed back in service.

As with batch fermentation processes, the primary concerns in scale-up are ensuring adequate rates of mass transfer for oxygen and carbon dioxide and ensuring adequate heat removal for good control of temperature. Heat removal is somewhat easier in a continuous fermentation than in batch fermentation, as the continuous flow of feed and removal of product allows use of a cold feed, and hence reduces the cooling requirement.

Reactors for immobilized cells can have a variety of geometries. Immobilized cells still require oxygen for respiration, so slurry reactors, trickle beds, or fluidized reactors must be used. The design of these reactors is similar to reactors for solid-catalyzed vapor–liquid reactions and is discussed in Section 15.7.5. Immobilization of cells that can be grown in suspension is rarely practiced industrially, as it is much harder to provide good heat and mass transfer to immobilized cells. Cell immobilization is discussed in more detail in Section 15.9.2.

15.9.7 Bioreactor instrumentation and control

The conditions that must be controlled in a bioreactor typically include the following:

- **Temperature:** is usually measured by one or more thermocouples and controlled by manipulating the rate of flow of steam or coolant to the jacket, coils, or external heat exchanger.
- **Pressure:** is usually measured on the vent gas line and controlled by a vent gas pressure control valve. Pressure is usually maintained above ambient to prevent contamination from air leaks into the plant.
- **Liquid level:** is important in control of fed-batch and continuous fermenters. Liquid level can be difficult to measure due to gas bubbles and foam in the reactor. Load cells are sometimes used to infer the total vessel contents for smaller reactors. Alternatively, the pressure difference between the top and bottom of the reactor can be measured and hence the liquid volume inferred from the static head. This method determines the liquid content but does not locate the vapor–liquid interface.
- **Feed rate:** in fed-batch and continuous fermentation processes, the rate of substrate addition must be maintained in the desired range. Feed rate is often varied during a batch or during a continuous production cycle, and the rates of individual feed components may also be varied to maximize productivity. It is common to use a more dilute growth medium at the start of the fermentation and then add more concentrated feeds later in the run to limit dilution and cell washout.
- **Agitation rate:** stirrer speed is usually continuously controlled to maintain the desired level of agitation.
- **pH:** *in situ* pH sensors are used to measure pH. The pH is controlled by discrete (on/off) addition of small amounts of a concentrated acid or base such as sulfuric acid, sodium hydroxide, or ammonia solution.
- **Dissolved oxygen:** the oxygen concentration is usually measured by an *in situ* oxygen probe. The oxygen detector can control the rate of air addition or can be cascaded onto the pressure controller (to increase oxygen partial pressure by increasing overall pressure) or the agitation rate controller (to increase mass transfer by higher agitation).
- **Foaming:** many fermenters are fitted with a foam detector to control the rate of antifoam addition. This is a simple conductivity probe placed at a suitable height above the liquid level and shielded from splashing.

Gas outlet composition is sometimes monitored as a means of measuring the oxygen availability or carbon dioxide production rate. Liquid concentration is usually determined using off-line chromatography methods. Methods for measuring cell concentrations are described in [Bailey and Ollis \(1986\)](#).

Batch fermenters use programmable logic controllers to control the sequence of events during a batch run and to ensure that all steps are carried out in the correct order. Programmable logic controllers are also used for discrete (on/off) control of some operations in CSTFs and for control of cleaning and sterilization operations.

More information on the control of biological reactors is given by [Alford \(2009\)](#).

15.9.8 Safety and quality control of bioreactors

Biological processes are typically carried out in aqueous solution under conditions that sustain life, and consequently are substantially less hazardous than nonbiological processes that handle flammable mixtures at high temperatures and pressures. Nonetheless, microbes and biologically active products can be a hazard to workers and the environment and must be contained in the plant. In addition to complying with safety regulations, plants that make products that are intended for human or animal consumption are required by law to maintain rigorous quality control standards and must undergo periodic inspection by regulatory agencies.

Batch processing is an important quality control method in biological processing. The use of batch processing allows operators to maintain batch integrity throughout processing, that is, to keep the same batch of material together and not mix it with other batches as it moves through the different stages of the process. The plant is cleaned and sterilized between batches so that batch-to-batch contamination cannot occur. If a batch is found to be contaminated or fails to meet quality control requirements, the entire batch can be rejected and sent to waste disposal without affecting any other batch, because no batch should ever come into contact with any other batch.

Good Manufacturing Practices

The widespread use of biological processing in the manufacture of food additives, personal care products, cosmetics, and drugs brings many products under the jurisdiction of the FDA in the United States, and therefore subject to production using CGMP (GMP for short). Other countries have their own GMP requirements, but all of them are similar to those of the World Health Organization (WHO) for pharmaceuticals manufacture. The WHO guidelines are set out in WHO Technical Report Series 908, which is available for free download from www.who.int/medicines/areas/. Information on U.S. GMP requirements is given in 21 CFR Part 210 and 21 CFR Part 211 and can be obtained from the FDA at www.fda.gov; see also Willig and Stoker (1996), Nally (2006), and Bunn (2019).

Food ingredients and drugs manufactured in the United States must be made to the standards specified by the United States Pharmacopeia (USP). Food ingredients are covered in the USP Food Chemicals Codex (FCC) set of standards. USP and FCC standards are recognized internationally and have been adopted as legal standards in many other countries.

Similar requirements are in force in Canada, and details of Canadian GMP guidelines can be obtained from Health Canada at www.hc-sc.gc.ca.

The GMP guidelines spell out the requirements for quality assurance, including maintenance of a quality control system, plant and facility design and operation, sanitation and hygiene, training, record keeping, inspection, subcontractor management, and materials testing.

Containment

Containment of biological material inside the plant is important in the following situations:

- When the microorganisms in the plant are pathogenic to humans, wildlife, or plants.
- When the microorganisms produce compounds that are toxic through their metabolism.
- When exposure of workers to the microorganism can lead to allergic reactions.
- When the microorganisms have been genetically modified to have desirable properties and there is a concern that the organism could interbreed with wild species if released into the environment.

For example, at the time of writing there is a great deal of research interest in genetically modified algae that have rapid growth rates and use sunlight to convert carbon dioxide into oils with high efficiency. These species might be very attractive as a means of mitigating carbon dioxide emissions, but could be difficult to control if they were to escape into the environment and hybridize with natural algae.

The primary means of containment of the liquid phase is the plant itself. Open-air transfers are avoided for sterility and quality-control reasons. Leakage through valve-stem or pump-shaft seals is also usually prevented by the selection of leakproof equipment as a means of preventing contamination from entering the plant.

Containment of biological hazards in the reactor off-gas is more difficult. The agitated, bubbling conditions in the fermenter are likely to lead to some entrainment of spray in the off-gas. If the spray contains microorganisms or their spores, then it must be treated to prevent leakage to the environment. High-efficiency particulate air (HEPA) filters are usually used on the off-gas, with multiple stages for more hazardous organisms. In extreme cases, incineration of the off-gas may be required. Krahe (2005) provides a guide to the selection of off-gas treatment technology.

Example 15.6

Riboflavin (vitamin B₂) is a nutrient that is used as a food additive (mainly in animal feed, but also in breakfast cereal). It can be produced by fermentation of glucose using strains of the yeast species *Candida famata*, as described in US 5,164,303 (Heefner et al., 1992). Example IV in the patent gives the following yields in a 450-liter fed-batch fermentation tank operated at 30 °C and 40% of oxygen saturation. The glucose concentration was maintained above a critical level of 30 g/l by feeding additional 600 g/l glucose solution during the fermentation.

Time	Optical density	Riboflavin	Glucose
(h)	(620 nm)	(g/l)	(g/l)
0	0.06	0.0	68.8
40	25.0	0.0	47.8
60	50.0	2.5	32.2
80	150	6.0	43.0
100	175	9.0	47.9
140	225	15.0	44.5
200	210	21.0	47.2

Design a reactor system to produce 40 metric tons per year of riboflavin using the strain given in the patent.

Solution

Estimate reactor volume

A quick calculation of the overall productivity will give an initial estimate of the required reactor volume.

$$\text{Riboflavin production} = 21 \text{ g/l in } 200 \text{ h} = 21/200 = 0.105 \text{ g/l.h or } 0.105 \text{ kg/m}^3\text{.h}$$

$$\text{Assuming } 8000 \text{ operating hours per year, productivity} = 8000 \times 0.105 = 840 \text{ kg/m}^3\text{.y}$$

$$\text{So to produce } 40 \text{ t/y, at } 100\% \text{ fermenter utilization would require } 40/0.84 = 47.6 \text{ m}^3$$

We do not expect that batch fermentation could run at 100% utilization, as we need to allow time for CIP and SIP between batches. If we make an initial assumption that CIP and SIP account for half the total time, then the utilization would be 50% and the required volume $47.6 \times 2 = 95.2 \text{ m}^3$. Assuming that the reactors are 75% full at the end of a batch, the total volume needed is $95.2/0.75 = 127 \text{ m}^3$. If we add another reactor to allow for one to be under extended maintenance at any time, then six 25-m^3 standard fermentation reactors looks like a good initial design.

Determine cleaning time

We now need to determine the cleaning and heat transfer requirements to confirm that the design basis CIP and SIP times can be accomplished.

$$\text{Liquid content of } 75\% \text{ full } 25\text{-m}^3 \text{ fermenter} = 0.75 \times 25 = 18.75 \text{ m}^3$$

$$\text{Time to drain vessel through a 6-inch pipe at } 1 \text{ m/s} = \frac{18.75}{1 \times \pi \times (6 \times 0.0254)^2 / 4} = 1028 \text{ s}$$

This is about 17 minutes, which is not unreasonable. Using an 8-inch pipe would give a drain time of 9.6 minutes, which might be preferred, but there are probably fewer pipe fittings and valves suitable for sterile service at the larger size, so let us select 6-inch for now and assume a drain time of 20 minutes to allow some time for pressuring the vessel during draining.

The washing step depends on the number of nozzles and the flow through each nozzle. BETE Fog Nozzle, Inc., is a nozzle manufacturer that sells a wide range of nozzles for cleaning applications. Their SC24 deluge washing nozzle can deliver 272.5 l/min at 4.5 bar. If we assume that we will fill the vessel to 80% full during cleaning (20 m^3), then at this flow rate, using six nozzles, the time required to fill the reactor to 20 m^3 would be $20 \times 10^3 / (6 \times 272.5) = 12.2$ minutes. The time to drain from 80% full is $1028 \times 80/75 = 1097 \text{ s}$, or about 18.3 minutes, so still under the allowed 20 minutes.

Following the guidance in Section 15.9.3, we can now propose the following CIP cycle:

1. Wash with high-pressure water jets (12 min spray + 8 min agitation)
2. Drain (20 min)
3. Wash with 1M NaOH (12 min spray + 8 min agitation)
4. Drain (20 min)
5. Rinse with tap water (12 min spray + 8 min agitation)
6. Drain (20 min)
7. Wash with 1M phosphoric acid (12 min spray + 8 min agitation)
8. Drain (20 min)
9. Rinse with tap water (12 min spray + 8 min agitation)
10. Drain (20 min)
11. Rinse with deionized water (12 min spray + 8 min agitation)
12. Drain (20 min)

$$\text{Total CIP time} = 12 \times 20 = 240 \text{ minutes} = 4 \text{ hours}$$

Determine SIP time

The sterilization and cool-down is a little more difficult to evaluate. The reactor is only partially filled at the start of the run, as this is a fed-batch process. It is clear from the glucose concentration data that additional glucose is being fed after 60 hours, so we need to make a rough estimate of the initial volume in the reactor. This information is not available in the patent, so some creative interpretation is required.

Fig. 15.54 shows the optical density (a measure of total cell mass, including live and dead cells) and riboflavin concentration plotted against time. Note that the figure is similar to Fig. 15.50, with the maximum product concentration attained after the peak in cell mass. It is impossible to tell from this figure where the maximum optical density was obtained, but it looks like

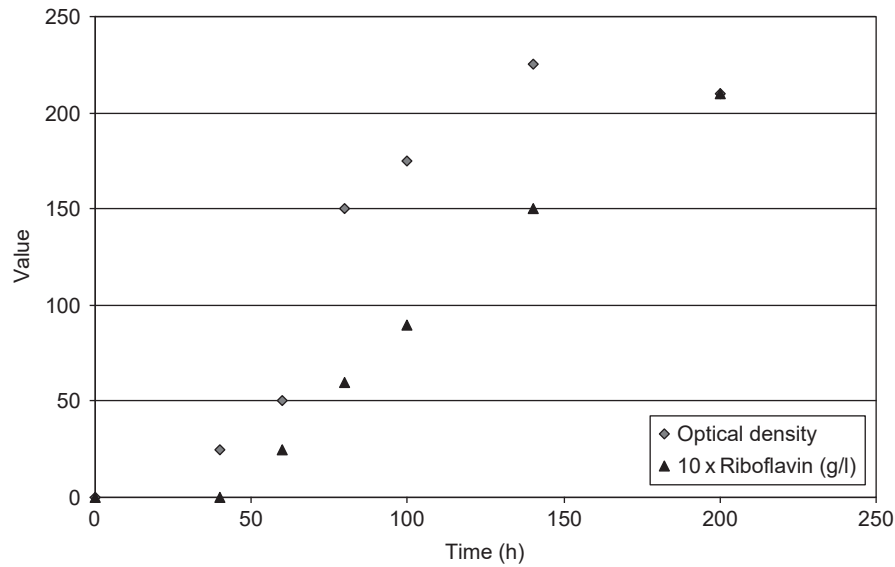


FIG. 15.54 Cell and product yields for Example 15.6.

it was some time between 140 and 200 hours. Because this is a batch process, the only way that the cell mass can decrease is by dilution. We can hence calculate how much liquid was added in the last 60 hours:

$$\text{If the cell mass is } x, \text{ then the density at the end of the run is : } 210 = \frac{x}{18.75}$$

$$\text{If the liquid volume added is } y, \text{ then the density at 140 hours is } 225 = \frac{x}{18.75 - y}$$

$$\text{Hence: } \frac{18.75 - y}{18.75} = \frac{210}{225}, y = 1.25 \text{ m}^3$$

The rate of dilution with fresh growth medium is highest at the end of the run when the concentration of cells is highest. Earlier in the run, beginning some time around 60 minutes, the required rate of glucose addition would be lower. The total time over which material is fed to the reactor is $200 - 60 = 140$ minutes, and 1.25 m^3 is added in the last 60 minutes, so a reasonable assumption of the total volume added is somewhere between 2 and 2.5 m^3 . For the purposes of designing the sterilization, we should take a conservative estimate and use 2 m^3 , as this gives the largest estimate for the initial volume in the reactor. The initial volume in the fermenter is then $18.75 - 2 = 16.75 \text{ m}^3$ of growth medium. The growth medium is largely water, and we can assume the properties of water for the heat transfer calculations.

To determine the heating and cooling times, we need to make an unsteady-state heat balance on the fermenter. If the vessel is heated using a steam jacket, with constant wall temperature, the rate of change in temperature of the contents is equal to the rate of heat transfer across the wall:

$$m_L C_p \frac{dT}{dt} = U A (T_w - T) \quad (15.31)$$

$$\text{Hence : } T = T_w - (T_w - T_o) \exp \left\{ - \left(\frac{U A}{m_L C_p} \right) t \right\} \quad (15.32)$$

where:

m_L = mass of liquid in the fermenter = $16.75 \times 10^3 \text{ kg}$

C_p = specific heat capacity of contents = $4.2 \text{ kJ/kg}^\circ\text{C}$

U = the overall heat transfer coefficient, $\text{kW/m}^2\text{C}$

A = heat transfer area, m^2

T = temperature, $^\circ\text{C}$

T_w = wall temperature, $^\circ\text{C}$

T_o = initial temperature, $^\circ\text{C}$

t = time, s

The jacket area can be estimated from the reactor dimensions. A 25-m³ vessel with 2:1 aspect ratio has diameter $(25 \times 2/\pi)^{1/3} = 2.515$ m. If this vessel is filled with 16.75 m³ of liquid, then it is filled to a depth of $2.515 \times 2 \times 16.75/25 = 3.37$ m, and the available area for heat transfer is $\pi \times 2.515 \times 3.37 = 26.6$ m².

The overall heat transfer coefficient can be estimated from Table 19.1 to be about 700 W/m²K. To confirm this value, we can use Equation 19.70c:

$$\text{Nu} = 0.74 \text{Re}^{0.67} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70c)$$

where μ_w is the viscosity at the wall. The Prandtl number for water over the temperature range of interest (20 °C to 120 °C) varies from 7 to about 1.4, so as a conservative estimate use 1.5.

The Reynolds number for an agitated vessel was given in Equation 15.16 as:

$$\text{Re} = \frac{d_a^2 N \rho}{\mu}$$

Where:

ρ = fluid density ≈ 1000 kg/m³

μ = fluid viscosity $\approx 0.4 \times 10^{-3}$ Ns/m² over the temperature range of interest

N = agitator speed, s⁻¹ (revolutions per second) (rps)

d_a = agitator diameter = $0.6 \times$ vessel diameter = 1.51 m

The power number, N_p , was also given in Equation 15.16:

$$N_p = \text{power number} = \frac{P_a}{d_a^5 N^3 \rho}$$

Where:

P_a is the power consumption in W.

From Table 15.5, for mild agitation we need a power input of ~ 0.05 kW/m³, so:

$$P_a = 0.05 \times 1000 \times 16.75 = 837.5 \text{ W}$$

From Fig. 15.16, the power number is about 4 at high values of Re (using curve 1), so:

$$N_p = 4 \times \frac{837.5}{1.51^5 \times N^3 \times 1000}$$

$N = 0.299$ revolutions per second, or about 18 rpm.

$$\text{Hence Re} = \frac{1.51^2 \times 0.299 \times 1000}{0.4 \times 10^{-3}} = 1.30 \times 10^6$$

$$\text{Neglecting the wall viscosity correction, } \text{Nu} = 0.74 \times (1.3 \times 10^6)^{0.67} \times (1.5)^{0.33} = 10,550$$

The thermal conductivity of water is about 0.66 W/m.K over this temperature range, so the inside heat transfer coefficient = $10550 \times 0.66/1.51 = 4611$ W/m²K. This seems a bit high, so we might be outside the size range for which the correlation was developed. When combined with a condensing steam coefficient of ~ 4000 W/m²K, this will give an overall coefficient of ~ 2140 W/m²K, which is substantially higher than the estimate of 700 W/m²K from Table 19.1. As a compromise, we can use an intermediate value, say 1000 W/m²K, or 1 kW/m²K.

If the initial temperature of the vessel contents is 20 °C and we use medium-pressure steam at 180 °C as heat source, we can substitute into Equation 15.32 to find the time required to reach 120 °C:

$$120 = 180 - (180 - 20) \exp \left\{ - \left(\frac{1 \times 26.6}{16.75 \times 10^3 \times 4.2} \right) t \right\} \quad (15.32)$$

Hence, $t = 2594$ s, or 43 minutes.

According to the patent, an additional 30 minutes is required at temperature for sterilization.

Determine cool-down time

We now need to estimate the rate of cooling back to the fermentation temperature of 30 °C. Table 19.1 gives an estimate of the cooling heat transfer coefficient using an internal cooling coil to be in the range 400 to 700 W/m²K. We can confirm this by making a layout of the cooling coil design.

The agitator diameter is 1.51 m, and we need space for baffles, so select coil diameter of 1.83 m (6 ft). If the coil is made from 1-inch tubing on a 2-inch pitch, then in 3.37 meters of filled depth, the number of coils is $3.37/(3 \times 0.0254) = 44$ coils. Each coil

has area $\pi \times 1.83 \times 0.0254 = 0.146 \text{ m}^2$, so the total coil area is $44 \times 0.146 = 6.42 \text{ m}^2$. This is not particularly large, and we could consider using a double coil if we need more area.

For heat transfer to the coil, we can use Equation 19.70f:

$$\text{Nu} = 0.87 \text{Re}^{0.62} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70f)$$

Neglecting the viscosity correction, $\text{Nu} = 0.87 \times (1.3 \times 10^6)^{0.62} \times (1.5)^{0.33} = 6142$

So external heat transfer coefficient = $6142 \times 0.66/1.51 = 2684 \text{ W/m}^2\text{K}$

Typical inside heat transfer coefficient for water in a 1-inch pipe $\sim 1000 \text{ W/m}^2\text{K}$

So overall heat transfer coefficient = $((1000^{-1}) + (2684^{-1}))^{-1} = 730 \text{ W/m}^2\text{K}$,

which is not too far from the value found in Table 19.1.

To solve for the cool-down time, we should make an unsteady-state energy balance on the vessel during cooling. This is more complex than Equation 15.32, as the cooling water is not at constant temperature, and the outlet temperature of the cooling water will vary as the fermenter cools down.

If the cooling water has a maximum flow rate, then we can write:

$$m_L C_p \frac{dT}{dt} = U A \Delta T_{eff} \quad (15.33)$$

where ΔT_{eff} = effective temperature difference = $\frac{(T - T_1) - (T - T_2)}{\ln \left(\frac{T - T_1}{T - T_2} \right)}$, and

$$m_w C_{pw} (T_2 - T_1) = U A \Delta T_{eff} \quad (15.34)$$

where T_1 = cooling water inlet temperature, °C, which is constant

T_2 = cooling water outlet temperature, °C, which is a function of time

m_w = mass flow rate of cooling water, kg/s

C_{pw} = specific heat capacity of cooling water = $4.2 \text{ kJ/kg}^\circ\text{C}$

Equations 15.33 and 15.34 are coupled and must be solved simultaneously. This can easily be done using programs such as Mathcad or by discretizing in the space of vessel temperature, solving for T_2 and hence finding the heat transfer rate and the time to reach the next temperature. Assuming we use chilled water with a feed temperature of 10°C and maximum velocity of 4 m/s , the cooling time required turns out to be 660 minutes or 11 hours. This is a lot less than the allowed time, so we do not need to add another coil (which would make cleaning more difficult).

Determine overall batch sequence and productivity

We now have the following overall batch sequence:

1. Fill with batch of feed medium (30 minutes, assuming same time as drain time)
2. Sterilization heat-up (45 minutes)
3. Sterilization hold (30 minutes)
4. Sterilization cool-down (11 hours)
5. Production cycle (200 hours)
6. Product drain (30 minutes)
7. Cleaning in place (4 hours)

The total nonproduction time is $0.5 + 0.75 + 0.5 + 11 + 0.5 + 4 = 17.25$ hours, which is much less than the 200 hours we allowed when we assumed 50% utilization. It might be prudent to allow some time at the end of the cool-down to confirm that the growth medium is sterile. If we allow 48 hours of hold time followed by sampling to ensure sterility, then we reduce the possibility of a contamination event and increase the nonproduction time to 66 hours. With $200/266 = 75.2\%$ utilization, we could in principle reduce the total required reactor volume to $47.6/(0.75 \times 0.75) = 84.6 \text{ m}^3$. This would allow us to reduce the number of 25-m^3 reactors to five, while still allowing for one spare fermenter in case one requires extended maintenance or manual cleaning.

Further calculations are needed to confirm that the rates of aeration and heat removal are adequate during the production cycle. The patent does not give sufficient detail to make this evaluation, so additional experimental data would be required.

15.10 Multifunctional batch reactors

In a batch process, some of the reagents are added at the start; the reaction proceeds and the compositions change with time. Additional reagents may be added as the reaction proceeds, and changes in temperature may also be made. At the end of the recipe the reaction is stopped when the required conversion has been reached and the products are withdrawn.

Batch processes are suitable for small-scale production and for processes where several different products, or grades, are to be produced in the same equipment; for instance, pigments, dyestuffs and polymers. Batch reactors are widely used in the manufacture of specialty chemicals, pharmaceuticals, and food additives, where the production volumes are relatively small, the chemistry is relatively complex, and batch integrity provides an important means of quality control. Batch processing is discussed in more detail in Section 2.3.2.

Most biochemical reactions are carried out in batch reactors, but because of the many additional constraints in biochemical reaction engineering, these are discussed separately in Section 15.9. An example of batch bioreactor design is given in Example 15.6.

Batch reactors are mainly used for liquid-phase reactions, although some reactions involving the conversion or formation of solids are carried out in batch mode. Gases may be added or removed during a batch reaction, but gas-phase reactions are almost never carried out in batch mode.

Processes that do not fit the definition of batch or continuous are often referred to as semi-continuous or semi-batch. In a semi-batch reactor, some of the products may be withdrawn as the reaction proceeds. A semi-continuous process can be one that is interrupted periodically for some purpose, for instance, for the regeneration of catalyst.

15.10.1 Design of batch reactors

Most batch reactors are stirred tanks. Mixing in stirred tanks is discussed in Section 15.5, and heat transfer to stirred vessels is addressed in Sections 15.6.2 and 19.18.

Although a batch reactor is a stirred tank and the contents are spatially well-mixed at any given time, the reactor approximates the same performance as a PFR, as back-mixing in the time dimension cannot occur.

It is not necessary to add all the reagents at the start of a batch reaction, and many batch reactions are actually operated semi-continuously, with one feed added at the start and others added during the reaction. This is almost always the case when one of the reagents is in the gas phase, in which case the gas is sparged into the reactor to maintain a constant concentration of the gas in the liquid phase.

Several reactions can be carried out in series in the same batch reactor: This is one of the advantages of batch processing. Separation steps can also be carried out between reaction steps, as discussed in the following section.

The design of a batch reactor must consider not only the rate of reaction but also the rates of all the other process steps that will be carried out in the vessel. At a minimum, these will include:

- Charging the reactor with feed, solvent, catalyst, etc.
- Heating the reactor contents to reaction temperature.
- Holding the reagents at reaction temperature for the desired residence time. In some cases, a temperature ramp profile may be applied and additional feeds may be added during the reaction.
- Cooling the reaction products to suppress reaction.
- Pumping out the reaction products to storage or downstream process operations.

The design engineer must calculate the time required for each step. The sequence of steps is often presented as a Gantt chart of the overall reaction process.

If the downstream process is operated continuously, it may make sense to operate several batch reactors, each at different points in the sequence of operations, so that at any given time one reactor is always at the pumping-out step and can feed the downstream process. This is the situation in batch fermentation of sugars to ethanol, where 6 to 12 fermenters are used to feed the continuous distillation section. Batch scheduling is discussed in Section 3.6.2 in the context of energy management: the same considerations can apply when scheduling batch reactors.

An important feature of batch plants is that they can often be used to make multiple different products in the same equipment. Multiproduct plants may use variations on the same recipe or may be scheduled to make a range of products by developing recipes that fit the plant. When an existing batch plant is evaluated (rated) for producing

a different product, the same calculations are carried out as in the design of the plant, but instead the engineer determines what production rate will be feasible within the constraints of the existing equipment.

15.10.2 Multifunctional batch reactors

It is quite common to use a batch reactor to carry out other steps in the batch processing sequence:

- The reactor can be used to heat the feeds or cool the reaction products (reactor as heater or cooler).
- At the end of a liquid–liquid reaction the products can be allowed to settle so that the liquid phases can be removed separately (reactor as decanter).
- A second solvent can be added to extract a product from the reaction mixture (reactor as extractive mixer-settler).
- The reaction mixture can be cooled or evaporated to cause crystallization of a product (reactor as crystallizer).
- A component can be added to cause a product to flocculate or precipitate from the reaction mixture (reactor as precipitator).
- The reactor can be heated to cause evaporation of a product (reactor as batch distillation still).

If other processing steps are to be carried out in the reaction vessel, the design engineer must rate the vessel for each of these steps and determine what time will be needed to achieve the desired process performance in each step. Allowance must be made for any time between steps and for time to change temperature between different operations. The total time required for all steps will determine the productivity (product capacity) of the plant. If productivity is not acceptable, the designer should consider adding another reactor or carrying out some steps in other equipment.

The design and sequencing of batch reactors and other batch operations are discussed in the books by [Sharatt \(1997\)](#) and [Korovessi and Linninger \(2005\)](#).

15.11 Computer simulation of reactors

It is not necessary to build a computer model to complete the design of a reactor. Very few industrial reactors were initially designed using computer models, although many designs have been substantially improved over the years by applying insights learned from computer simulation.

Reactor modeling is a complex, multiscale problem, spanning molecular-scale interactions, surface and diffusion effects in catalysts, mass transfer between phases, mixing and heat transfer in bulk fluids, and hydraulics of multi-phase and particulate flows. These processes occur over dimensions from the order of an angstrom to the order of tens of meters. It is not usually feasible to construct a single model that treats all of these processes rigorously. Instead, reactor designers use different models to address microscopic and macroscopic phenomena.

In the early stages of design, relatively simple models can be used to estimate reactor size for the purpose of cost estimating. If the project goes ahead, it is more important to ensure that the reactor will attain the desired performance, and it may be worthwhile to build more detailed models for use in scale-up and validation of the design. It is very important to ensure that these models are fitted to realistic plant data, either from an existing full-scale plant or from a pilot plant that is able to mimic the conditions of a full-scale reactor.

Creating a model of the main process reaction is usually relatively straightforward; however, capturing the effects of side reactions, contaminants, inhibition, cell life cycle, or catalyst deactivation can be much more difficult and may require a substantial and expensive experimental program. Failure to properly include these effects may render the model inaccurate or even misleading and can lead to inferior and possibly unsafe designs. Most industrial processes use catalytic or biological routes that are sensitive to many input parameters, so it is important that these parameters are properly studied in the experimental program. The development of reactor models is discussed in all the reaction engineering textbooks cited in the bibliography. [Stewart and Caracotsios \(2008\)](#) give an excellent introduction to reactor modeling.

15.11.1 Commercial process simulation models

Modeling of reactors using commercial process simulation programs was discussed in Section 4.5.1, where the standard library reactor models are described. These models are adequate for determining the overall process mass and energy balance and can be very useful for calculating heats of reaction, as discussed in Section 15.3.1.

The library models in commercial process simulation programs can be used for reactor sizing if the rate of the main reaction is known or can be approximated as a simple expression under the conditions where the reactor will be operated. Instructions on how to size reactors using commercial simulation programs are given in the user manuals or online help links.

Commercial simulation programs can also be used to model more complex reactors by forming networks of simple reactors to account for mixing effects, as discussed in the next section, or by building sub-flowsheets that incorporate simple reactors and other unit operations to simulate a more complex reactor design. For example, the reactor designed in Example 15.4 and shown in Fig. 15.45 would be a single pressure vessel, but was modeled with the set of simple reactor models and other operations shown in Fig. 15.43.

15.11.2 Network models

Combinations of simple reactor models can be used to model the mixing and reaction behavior of more complex reactors. Fig. 15.55 shows some examples of simple reactor networks that can be used to model nonideal reactor behavior.

Networks of simple reactor models can also be used to determine optimum mixing patterns and heat addition profiles in real reactors, as long as a sufficiently accurate model of the reaction kinetics is available. The concept of using mixed-integer nonlinear programming (MINLP) to optimize a superstructure of simple reactor models was introduced by Kokossis and Floudas (1990) and is discussed in Section 12.11.2 as an example of the application of MINLP approaches in process synthesis. Kokossis and coworkers have extended this approach to nonisothermal, catalytic and multiphase reactors, and to reaction-separation processes. The optimal networks found from superstructure optimization can be used to infer what mixing pattern should be used in a real reactor and can lead to nonintuitive designs when the reaction mechanism is complex.

15.11.3 Hydrodynamic models

Realistic models of mixing, multiphase flow, and solids flow cannot be built using process simulation programs. To determine the effect of reactor geometry and hydraulics on the spatial variation of temperature and concentration and hence on reaction rates, it is necessary to combine a model of the reaction kinetics with a hydrodynamic model.

Hydrodynamic simulation models of any reactor geometry can be built using CFD programs. CFD programs allow the user to define a two- or three-dimensional geometry that is then discretized into a mesh of points. Boundary conditions are set for the edges of the mesh, and initial conditions are defined. The program then solves the Navier–Stokes equations for all points in the mesh in either steady-state or transient mode, and hence determines the hydraulic and mixing behavior. CFD programs were initially developed for aerospace applications, and the need for detailed models of jet engines led to the incorporation of chemical reactions into the simulation. Computational fluid dynamics programs are now routinely used for modeling reactors, as well as other process equipment; see Aglave and Eppinger (2018). Details of CFD programs, their capabilities, and their application in reaction engineering

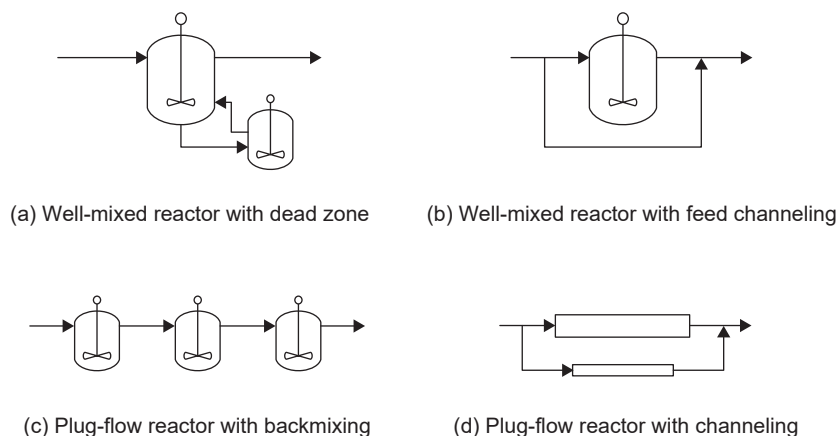


FIG. 15.55 Network models of nonideal reactors.

can be found on the websites of CFD software vendors; see, for example, <http://ansys.com/> or www.cpf-d-software.com/.

The development and testing of a CFD model can be expensive and time-consuming. It is essential to have good experimental or plant data to use in validating the model, so large-scale cold-flow experiments using air, water, and inert solids such as glass beads or sand are often carried out as a means of testing the hydraulic performance of the model before reaction effects are introduced.

Reaction systems that involve flowing solids present an even greater modeling challenge. Flowing solids can sometimes be treated as a pseudo-continuous fluid phase, but this approximation ignores many of the realities of particulate flow. Discrete element modeling (DEM) treats each particle separately and simulates the movement of particles and fluid through a meshed geometry. DEM methods are advancing rapidly, but at the time of writing are not yet able to solve problems with a realistic number of particles to model industrial heterogeneous catalysis. Jung et al. (2009) discuss the challenges of simulating fluidized-bed reactions using computational hydrodynamics models.

15.12 Determining actual reactor performance

The performance of full-scale commercial reactors can be difficult to predict from laboratory experiments and computer models. Many factors can affect reactor performance, particularly for catalytic or biological processes. If the experimental system does not carefully control these factors and provide a realistic model of the mixing and heat transfer that will be achieved in the real reactor, different performance may be obtained when the reaction is carried out at full scale.

Experience with similar reaction systems is usually the best way to ensure successful scale-up. The design of experimental reactors and scale-up are discussed by Rase (1977) and Bisio and Kabel (1985).

In most cases, any new reactor design will require testing and validation at the pilot-plant scale before a full-scale unit is built. This section discusses some of the important issues to address and techniques that can be deployed when determining whether a proposed reactor design will work as intended at full scale.

15.12.1 Measuring experimental reactor output

For a laboratory or pilot-plant reactor to successfully model the behavior of a full-scale reactor, the following factors must be considered:

- **Mixing pattern:** The mixing pattern in the experimental reactor should model that of the real reactor as closely as possible. This does not mean that the experimental reactor must be a smaller version of the commercial-scale reactor. The reaction engineer must give some thought as to whether to mix the feeds before heating, mix them inside the reactor itself, etc. At the small scale, mixing will generally be quicker than at the large scale, but there is a greater chance of channeling, bypassing, wall effects, and laminar flow effects.
- **Material balance:** It is very important to properly close a material balance across the experimental reactor for every data point. This is not as easy as it sounds, particularly when some products are in the vapor phase. Vaporized products must be collected using vent condensers or gas-phase sampling, and the composition of the vapor phase must be tied to the composition of the liquid phase at the time the sample was taken. The research engineer should be able to complete a material balance for every element (C, H, O, etc.), as well as an overall material balance. If an element cannot be materially balanced, that can be an indication of poor test methodology or of accumulation of species in the reactor or on the catalyst. In some cases, as in the growth of microorganisms, the accumulation is desired, and it is important to know uptake rates of feed species such as minerals to determine the required rates for processing.
- **Energy balance and temperature profile:** Small-scale reactors are difficult to operate adiabatically, as the surface area-to-volume ratio is large and so heat is easily lost to (or gained from) the environment. When a reaction is run at or near ambient conditions, the temperature must be carefully controlled to eliminate diurnal or seasonal variations in temperature. Reactors should be well insulated and should be instrumented to allow measurement of the temperature in multiple locations so that the temperature profile is properly understood. If possible, the temperature profile in the experimental reactor should match that expected in the commercial reactor, and multiple heating zones or reaction zones may need to be used to correctly model the commercial reactor temperature pattern.
- **Mass transfer:** Mass transfer rates between phases or inside solid catalysts can govern the overall rate of reaction. If mass transfer is likely to be limiting at the commercial scale, the experimental reactor should allow

measurement of the mass transfer rate or allow reaction kinetics and mass transfer to be de-confounded so that models can be developed that predict performance at full scale.

- **Feed composition:** Whenever possible, the experimental feed should be the same as the feed that will be used in the commercial plant. Laboratory-grade chemicals will have higher purity than commercial grades and are less likely to contain contaminants that may play an important role in reactor performance. Experiments using pure model compounds may be useful for determining reaction mechanisms, but can overlook significant effects such as catalyst inhibition. It may not always be possible to use the commercial feed, for example, if the plant is to be fed from an upstream process that has not yet been built; however, when the real feed is not available, the test program should err on the side of caution by dosing potential contaminants into the model feed to understand their impact.
- **Recycle structure:** If the process uses a recycle, then the pilot plant or laboratory reactor should also be designed to incorporate a recycle. Trace components that are formed in the reaction can accumulate in recycle streams and affect reactor performance. It is important to identify these effects experimentally before scale-up so that the correct reactor performance is predicted. In some cases, the process may even need to be modified by adding a separation process to control the recycle composition.
- **Steady state:** The experimental design must allow sufficient time for the reactor to reach steady-state performance if a continuous system is used. This requires achieving thermal steady state, stable conversion, and stable concentration of components in recycle streams. If process conditions are modified, the system must again return to steady state before data can be collected.
- **Catalyst or cell age:** For catalytic and biological processes, the behavior of the catalyst or cell population can vary with time. Catalysts can lose their activity due to inhibition, coking, or damage suffered during regeneration cycles. Microorganisms may pass through many generations during a processing run (even in a batch process) and consequently can experience evolutionary effects that change productivity. The experimental design must allow for these effects and ensure that the experimental run length is adequate to predict reactor performance over the anticipated process time between catalyst or cell culture replacement.

15.12.2 Measuring commercial reactor behavior

It is much more difficult to carry out experiments using large-scale commercial reactors, as they are not as highly instrumented as pilot plants, and the primary imperative for operators is to maintain safe and steady operation at optimal conditions to generate revenue. Nonetheless, data taken from operating units can play an important role in reactor design and development. The techniques used to collect data from commercial plant must cause no interruption to normal operation, require no shutdown or plant modification, and cause no hazard to workers or the plant.

Tracer studies

Experiments using tracer molecules can be used to determine the residence time distribution in a real reactor. The most common noninvasive approach is to use a radioactive tracer species that has a half-life that is long enough for detection, but short enough that there will be no residual radiation in the plant or products. The species must also be chosen so that the radioactive decay products are not toxic, reactive, or otherwise problematic for the reactor or downstream process operations.

In a tracer experiment, detectors are positioned around the reactor or in downstream piping. The tracer molecule is injected upstream of the reactor, usually into one of the feed lines, and the signal at each detector is then logged as a function of time. The signal strength gives a measure of the concentration of tracer passing that point, and hence allows the residence time distribution to be constructed.

Tracer studies provide insight into the distribution of residence times and can be very useful in identifying deviations from the expected mixing pattern. Tracer experiments can identify channeling, bypassing, unwanted backmixing, and the presence of dead zones. The use of tracer studies on an existing reactor can help suggest modifications to reactor geometry and internals that would improve the performance of the existing reactor or of future reactors in a new design. Examples of commercial applications of tracer studies can be found from vendor websites; see, for example, www.tracerco.com.

Residence time distributions measured from real plant can be used to construct models of commercial reactors that predict plant performance better than idealized reactor models. The use of networks of simple models to fit real reactor residence time distributions is discussed in Section 15.11.2.

Reactor tomography

Tomography is imaging of the internal structure of objects by carrying out multiple sectional scans using penetrating radiation. Tomography can be applied to reactors using x-rays or gamma rays, depending on the reactor wall thickness, and can reveal regions of local high or low density inside the reactor. Tomography studies can be carried out while the reactor is in operation at design conditions, as the technique is noninvasive.

Tomography studies are very useful for understanding particulate and multiphase flows. Tomography results can be used to validate CFD models, which can then be used to propose improvements to reactor internals and other features of the design. Tomographic studies of existing commercial-scale reactors can be quite valuable as a means of validating models that can then be used to design novel reactors. An example of the application of tomography to FCC is given by [Wolschlag and Couch \(2010\)](#).

15.13 Safety considerations in reactor design

The reactor is often the highest temperature point in the process. It is also where reactive species are brought together and where heat of reaction may be released. The residence time required for reactors can be long compared with processing times in other operations, so the inventory of chemicals in the reactor can be relatively high. All of these factors make reactors hazardous and require the design engineer to pay particular attention to safety when developing a reactor design.

15.13.1 Inherently safer design principles applied to reactors

The concept of inherently safer design was introduced in Section 10.1.3. The basic principle of inherently safer design is to remove or reduce process hazards and thereby reduce the impact of undesired or unforeseen events. Inherently safer design methods are complementary to adding passive, active, and procedural layers of protection to the design and do not guarantee a safe process or remove the need for extrinsic safety systems. Inherently safer design methods should be applied at an early stage in the design as part of good engineering practice and is not something that can be added later by a specialist in process safety.

Guidelines for inherently safer design have been articulated by Kletz and others and are set out in the books by [Kletz and Amyotte \(2010\)](#), [CCPS \(2008\)](#), [Cameron and Raman \(2005\)](#), [Mannan \(2012\)](#), and [Crowl and Louvar \(2011\)](#). The core principles are to eliminate hazardous conditions (*eliminate*), reduce inventories (*minimize*), use less hazardous materials (*substitute*), operate under less hazardous conditions (*moderate*), and eliminate complexity so that errors are less likely to occur (*simplify*). It should be noted that often one of these principles will have to be traded off against another. For example, moderating reaction conditions by operating at lower temperature will lead to a lower reaction rate and hence require a longer residence time and higher inventory.

[Table 15.16](#) gives some examples of application of inherently safer design principles in reactor design and points out some of the possible trade-offs that may be introduced. It is important to consider potential trade-offs so that the hazard is truly reduced and not just shifted elsewhere in the process, or even made worse. The list in [Table 15.16](#) is not exhaustive, and the reader should refer to the books cited above for additional examples.

15.13.2 Designing for exothermic reactions

Exothermic reactions are particularly hazardous because heat released by the reaction can increase the temperature of the reaction mixture, thereby increasing the reaction rate, releasing more heat, and ultimately allowing the reaction to run away. When an exothermic reaction is unavoidable, the designer must ensure that the temperature change of the reaction mixture can be carefully controlled and that the reaction can quickly be shut down if temperature control is lost.

The use of solvent or inert diluent allows moderation of the temperature change by increasing the heat capacity flow rate (thermal mass) of the mixture relative to the rate of heat release per unit mass of mixture. Provision can be made to flood the reactor with cold diluent or solvent as an emergency response if the temperature exceeds a critical limit.

If a cooling system is used, it should be designed so that it has the capacity to handle the rate of heat release at the maximum temperature conditions expected in operation. The cooling system should then be able to return the process to the desired temperature range if the maximum temperature is reached and the cooling system is operational.

TABLE 15.16 Some applications of inherently safer design approaches in reactor design

Inherently safer design principle	Reactor application	Rationale	Possible trade-offs
<i>Minimize</i>	Use continuous processing instead of batch processing	Pipe reactor has less inventory than a tank and is easier to close off.	Fed-batch system may be safer if reaction rate can be limited by carefully controlling addition of one component.
<i>Minimize</i>	Use large number of small reactors (process miniaturization)	Impact from a hazardous event in one reactor is reduced.	Higher probability of a hazardous event occurring. More complex system. Higher production costs.
<i>Minimize</i>	Use more active catalyst	More active catalyst allows smaller reactor volume.	Higher rate of heat release if reaction is exothermic.
<i>Minimize</i>	Combine reaction and separation, as in reactive distillation	Fewer process vessels.	Total liquid inventory may increase. Conditions that are effective for separation may lead to longer residence times for reaction.
<i>Minimize</i>	Operate at higher conversion	Reduces recycle of material, hence inventory of unconverted feed in separation section.	Usually leads to poor selectivity, which may actually increase complexity of the separation section. Inventories of by-products may increase.
<i>Minimize</i>	Improve mixing	Reduces reaction time, hence inventory.	Higher rate of heat release if reaction is exothermic.
<i>Substitute</i>	Use nonflammable solvent instead of flammable	Nonflammable solvent cannot cause a fire.	Many nonflammable solvents are halogenated compounds that have high global warming potential.
<i>Substitute</i>	Use biological processing route	Biological routes are carried out in aqueous media at ambient conditions, therefore less hazardous.	Much larger reactors. Increase in wastewater production.
<i>Substitute</i>	Use different chemical route	An alternative chemical route may be available that avoids the use of explosive or highly toxic compounds.	
<i>Substitute</i>	Use steam as heat source instead of fired heat or electric heaters	Steam heating is intrinsically limited to the temperature of the steam; therefore is safer than electric heaters or fired heaters.	Harder to reach high temperatures; therefore may have the effect of reducing reaction rate. May require high-pressure steam or introduce possibility of water leaking into process.
<i>Moderate</i>	Operate at lower temperature	Leaking material will be further from flash point, autoignition temperature. Liquids below atmospheric boiling point are less likely to suffer boiling liquid expanding vapor explosion (BLEVE).	Reaction rate is reduced, increasing residence time and inventory. Reactor is larger and more expensive.
<i>Moderate</i>	Operate at lower pressure	Reduced consequences of overpressure, reduced inventory.	Lower reaction rates, so longer residence times. Much larger vessels are required.
<i>Moderate</i>	Operate at higher dilution	Reduced reaction intensity, less change in temperature due to heat of reaction.	Additional complexity due to additional species (solvent, diluent) introduced into the process, possibly requiring additional separations or other processing.
<i>Simplify</i>	Eliminate discontinuous operations	Failure of a switching valve could cause a hazardous situation.	
<i>Simplify</i>	Carry out reaction inside heat exchanger	Fewer vessels, lower inventory.	Possibility of leaks from process to utility or vice-versa.

Shutting off the feed to an exothermic reaction is not always a safe strategy for preventing reaction runaway. If feed-effluent heat exchange was used, then shutting off the feed reduces cooling of the reactor product and allows overheated product to enter downstream operations. If the reactor relied upon cold feed to provide some of the cooling requirement, then shutting off the feed can reduce the rate of cooling and worsen the runaway condition. It may be more effective to shut off only one of the reagents or to substitute a reagent with a cold inert material that has the same cooling effect but no reaction.

The consequences of loss of cooling, loss of diluent, and reactor runaway should always be considered in hazard and operability studies and as pressure relief scenarios whenever an exothermic reaction is used. Additional safety systems may need to be added to the process, and the safety integrity level (SIL) of safety instrumented systems associated with the reactor may need to be increased.

15.13.3 Venting and relief of reactive systems

The contents of a reactor can continue to react if containment is lost or if an overpressure event causes discharge of material into the pressure relief system. The presence of reaction substantially complicates the design of the pressure relief device and the downstream relief system.

In 1976 a consortium of 29 companies formed the Design Institute for Emergency Relief Systems (DIERS) under the auspices of the American Institute of Chemical Engineers (AIChE). The DIERS consortium developed design methods and software for designing relief systems for runaway reactions. At the time of writing, the DIERS methodology is maintained and updated by a users group drawn from 160 companies. Details of the DIERS consortium and their publications can be found via the AIChE website, www.aiche.org.

When a runaway reaction could occur, the design of the relief system should follow the guidelines specified in the DIERS methodology. The reactor design team must have a very good understanding of the reaction mechanism and kinetics, including the role of any trace compounds that may act as catalysts or free radical promoters or might otherwise accelerate the reaction. The experiments to collect kinetics data for exothermic reactions can also be hazardous and must be carefully planned and executed. They are often subcontracted to companies that specialize in this area. Additional information on designing relief systems for reactive mixtures is given in the articles by Fauske (2000) and Melhem and Powell (2005).

15.14 Capital cost of reactors

Most reactors are designed to be operated under conditions that cause the reaction vessel to be classified as a pressure vessel. Once the reactor dimensions have been determined following the procedure given in Section 15.2.1 and the other methods given in this chapter, the pressure vessel design methods given in Chapter 14 can be used to estimate the wall thickness and hence determine the capital cost. Additional costs may need to be added to cover the cost of reactor internals or other ancillary equipment such as agitators. This procedure was illustrated in Example 15.4.

The costs of jacketed stirred-tank reactors are not easily addressed using simple pressure vessel cost correlations. A substantial part of the cost is in the construction of the vessel jacket. In some cases, the jacket can cause the wall of the reaction vessel to be in compression, requiring a more complex analysis of the pressure vessel wall thickness. Instead, a commonly used method in preliminary cost estimating is to use a different correlation for jacketed stirred tanks operated at low pressures (below about 20 bar). Table 7.2 gives such correlations for plain and glass-lined stirred vessels.

15.15 References

- Aglave, R., & Eppinger, T. (2018). CFD Simulation in chemical reaction engineering. *Hydrological Processes*, 97(4), 41.
- Alford, J. S. (2009). Principles of bioprocess control. *Chemical Engineering Progress*, 105(11), 44.
- Al-Dahhan, M. H., & Dudukovic, M. P. (1994). Pressure drop and liquid holdup in high pressure trickle-bed reactors. *Chemical Engineering Science*, 49(24B), 5681.
- Bailey, J. F., & Ollis, D. F. (1986). *Biochemical engineering fundamentals* (2nd ed.). McGraw-Hill.
- Baker, J. R. (1991). Motionless mixtures stir up new uses. *Chemical Engineering Progress*, 87(6), 32.
- Basu, P. (2006). *Combustion and gasification in fluidized beds*. CRC Press.
- Benson, S. W. (1976). *Thermochemical kinetics* (2nd ed.). Wiley.
- Benson, S. W., Cruickshank, F. R., Golden, D. M., Haugen, G. R., O'Neal, H. E., Rogers, A. S., et al. (1969). Activity rules for the estimation of thermochemical properties. *Chemical Reviews*, 69, 279.

- Benz, G. T. (2019). Optimize aspect ratio in industrial fermenters. *Chemical Engineering Progress*, 115(4), 26.
- Bisio, A., & Kabel, R. L. (1985). *Scale up of chemical processes: Conversion from laboratory scale tests to successful commercial size design*. Wiley.
- Blanch, H. W., & Clark, D. S. (1996). *Biochemical engineering*. Marcel Dekker.
- Breck, D. (1974). *Zeolite molecular sieves: Structure, chemistry and uses*. Wiley.
- Bunn, G. P. (2019). *Good manufacturing practices for pharmaceuticals* (7th ed.). CRC Press.
- Cameron, I. T., & Raman, R. (2005). *Process systems risk management* (Vol. 6). Academic Press (Process systems engineering).
- CCPS. (2008). *Inherently safer chemical processes: A life cycle approach* (2nd ed.). Wiley.
- Chhabra, R. P., & Shankar, V. (Eds.). (2017). *Coulson and Richardson's chemical engineering: Volume 1A: Fluid flow: Fundamentals and applications* (7th ed.). Butterworth-Heinemann.
- Dickey, D. S. (2015). *Fundamentals of high-shear dispersers*, 122 (p. 40). New York: Chemical Engineering, 2.
- Douglas, J. M. (1988). *Conceptual design of chemical processes*. McGraw-Hill.
- Curtis, W. R., Carvalho, E. B., & Tescione, L. D. (2001). Advances and challenges in bioreactor design for the production of chemicals from plant tissue culture. *Acta Horticulturae*, 560, 247 (Proceedings of the 4th International Symposium on In Vitro Culture and Horticultural Breeding, 2000).
- Crowl, D. A., & Louvar, J. F. (2011). *Chemical process safety: Fundamentals with applications* (3rd ed.). Prentice-Hall.
- Domalski, E. S. (1972). Selected values of heats of combustion and heats of formation of organic compounds containing the elements C, H, N, O, P, and S. *Journal of Physical and Chemical Reference Data*, 1, 221.
- Ergun, S. (1952). Fluid flow through packed columns. *Chemical Engineering Progress*, 48, 89.
- Fair, J. R. (1967). Designing gas-sparged reactors. *Chemical Engineering*, 74(14), 67.
- Fasano, J. B. (2015). Select the right impeller. *Chemical Engineering Progress*, 111(6), 30.
- Fauske, H. K. (2000). Properly size vents for nonreactive and reactive chemicals. *Chemical Engineering Progress*, 96(2), 17.
- Fossett, H., & Prosser, L. E. (1949). The application of free jets to the mixing of fluids in tanks. *Proceedings of the Institution of Mechanical Engineers*, 160, 224.
- Froment, G. F., Bischoff, K. B., & De Wilde, J. (2010). *Chemical reactor analysis and design* (3rd ed.). Wiley.
- Frössling, N. (1938). Über die Verdunstung fallender Tropfen. *Gerlands Beitr Age Zur Geophysik*, 52, 170.
- Fuller, E. N., Schettler, P. D., & Giddings, J. C.. (1966). A new method for the prediction of gas-phase diffusion coefficients. *Industrial Engineering Chemistry*, 58, 19.
- Gambill, W. R. (1958). Predict diffusion coefficient. *D. Chemical Engineering*, 65(6), 125. New York.
- Grace, J. R., Knowlton, T. M., & Avidan, A. A. (Eds.). (1996). *Circulating fluidized beds*. Springer.
- Green, D. W., & Southard, M. Z. (Eds.). (2018). *Perry's chemical engineers' handbook* (9th ed.). McGraw-Hill.
- Grenville, R. K., Giacomelli, J. J., Padron, G., & Brown, D. A. R. (2017). Impeller performance in stirred tanks. *Chemical Engineering*, 124(8), 42. New York.
- Gupta, A. S., & Thodos, G. (1963). Direct analogy between heat and mass transfer in beds of spheres. *AIChE Journal*, 9, 751.
- Harnby, N., Edwards, M. F., & Nienow, A. W. (Eds.). (1997). *Mixing in the process industries* (2nd ed.). Butterworth-Heinemann.
- Harriott, P. (2002). *Chemical reactor design*. CRC Press.
- Heefner, D. L., Weaver, C. A., Yarus, M. J., & Burdzinski, L. A. (1992). US patent 5,164,303, issued 11.17.1992 to ZeaGen, Inc.
- Henkel, K.-D. (2005). Reactor types and their industrial applications. In F. Ullmann (Ed.), *Ullmann's chemical engineering and plant design*. Wiley VCH.
- Hicks, R. W. (1976). How to select turbine agitators for dispersing gas into liquids. *Chemical Engineering*, 83, 141. New York.
- Jung, J., Gidaspow, D., & Gamwo, I. K. (2009). *Design and understanding of fluidized-bed reactors: Application of CFD techniques to multiphase flows*. VDM Verlag.
- Kletz, T. A., & Amyotte, P. (2010). *Process plants: A handbook for inherently safer design* (2nd ed.). CRC Press.
- Krahe, M. (2005). Biochemical engineering. In F. Ullmann (Ed.), *Ullmann's chemical engineering and plant design*. Wiley VCH.
- Kokossis, A. C., & Floudas, C. A. (1990). Optimization of complex reactor networks – 1. Isothermal operation. *Chemical Engineering Science*, 45(3), 595.
- Korovessi, E., & Linniger, A. (2005). *Batch processes*. CRC.
- Kulprathipanja, S. (Ed.). (2010). *Zeolites in industrial separation and catalysis*. Wiley.
- Lavik, E., & Langer, R. (2004). Tissue engineering: Current state and perspectives. *Applied Microbiology and Biotechnology*, 65(1), 1.
- Lee, A. L. (2010). *AIChE Corporate Innovation Award lecture*. Salt Lake City, UT: 11.11.2010, Genentech: A rich history of chemical engineering innovation.
- Lee, J., & Brodkey, R. S. (1964). Turbulent motion and mixing in a pipe. *AIChE Journal*, 10, 187.
- Leskova, E., Kubikova, J., Kovacikova, E., Kosicka, M., Porubská, J., & Holcikova, K. (2006). Vitamin losses: Retention during heat treatment and continual changes expressed by mathematical models. *Journal of Food Composition and Analysis*, 19(4), 252.
- Leva, M. (1949). Fluid flow through packed beds. *Chemical Engineering*, 56(5), 115.
- Levenspiel, O. (1998). *Chemical reaction engineering* (3rd ed.). Wiley.
- Levin, M. (2011). *Pharmaceutical process scale-up* (3rd ed.). Informa Healthcare.
- Mannan, S. (Ed.). *Lees' loss prevention in the process industries* (4th ed.). (Vol. 3). Butterworth-Heinemann.
- Martin, Y., & Vermette, P. (2005). Bioreactors for tissue mass culture: Design, characterization, and recent advances. *Biomaterials*, 26(35), 7481.
- Mccabe, W. L., Smith, J. C., & Harriott, P. (2004). *Unit operations of chemical engineering* (7th ed.). McGraw-Hill.
- Mead, D. J., & Van Urk, H. (2000). US patent 6,150,133, issued 11.21.2000 to Delta Biotechnology Ltd.
- Melhem, G. A., & Howell, P. (2005). Designing emergency relief systems for runaway reactions. *Chemical Engineering Progress*, 101(9), 23.
- Meyers, R. A. (2016). *Handbook of petroleum refining processes* (4th ed.). McGraw-Hill.
- Mullin, R. (2016). Single-use becomes state-of-the-art in bioprocessing. *Chemical & Engineering News*, 94(28), 21.
- Nally, J. D. (2006). *Good manufacturing practices for pharmaceuticals* (6th ed.). Informa Healthcare.
- Oldshue, J. Y., Hirshland, H. E., & Gretton, A. T. (1956). Side-entering mixers. *Chemical Engineering Progress*, 52, 481.
- Paul, E. L., Atiemo-Obeng, V., & Kresta, S. M. (2003). *Handbook of industrial mixing: Science and practice*. Wiley.

- Penny, N. R. (1970). Guide to trouble free mixing. *Chemical Engineering*, 77, 171. New York.
- Post, T. (2010). Understand the real world of mixing. *Chemical Engineering Progress*, 106(3), 25.
- Previs, T., & Roche, L. (2018). Methods to achieve sustainable CIP. *Chemical Engineering*, 125(4), 53. New York.
- Ranade, V. V., Chaudhari, R., & Gunjal, P. R. (2011). *Trickle bed reactors: Reactor engineering and applications*. Elsevier.
- Rase, H. F. (1977). *Chemical reactor design for process plants* (Vol. 2). Wiley.
- Rase, H. F. (1990). *Fixed-bed reactor design and diagnostics*. Butterworths.
- Reid, R. C., Prausnitz, J. M., & Poling, B. E. (1987). *Properties of liquids and gases* (4th ed.). McGraw-Hill.
- Rosennzweig, M. D. (1977). Motionless mixers move into new processing roles. *Chemical Engineering*, 84, 95. New York.
- Rushton, J. H., Costich, E. W., & Everett, H. J. (1950). Power characteristics of mixing impellers. *Chemical Engineering Progress*, 46, 467.
- Sharratt, P. N. (1997). *Handbook of batch process design*. Springer.
- Shibuya, K., Haga, R., & Namba, M. (2010). U.S. patent application US20100081122A1: System and method for cultivating cells.
- Shuler, M. L., Kargi, F., & DeLisa, M. (2017). *Bioprocess engineering: Basic concepts* (3rd ed.). Prentice-Hall.
- Solomon, B. O., & Erickson, L. E. (1981). Biomass yields and maintenance requirements for growth on carbohydrates. *Process Biochemistry*, 16(2), 44.
- Stark, A., & Seddon, K. R. (2007). Ionic liquids. In *Kirk-Othmer encyclopedia of chemical technology*. Wiley.
- Stanier, R. Y., Ingraham, J. L., Wheelis, M. L., & Painter, P. R. (1986). *The microbial world* (5th ed.). Prentice-Hall.
- Stewart, W. E., & Caracotsios, M. (2008). *Computer-aided modeling of reactive systems*. Wiley.
- Storey, K. B., & Schafhauser-Smith, D. Y. (1994). Immobilization of polysaccharide-degrading enzymes. *Biotechnology and Genetic Engineering Reviews*, 12, 409.
- Tattersson, G. B. (1991). *Fluid mixing and gas dispersion in agitated tanks*. McGraw-Hill.
- Tattersson, G. B. (1993). *Scale-up and design of industrial mixing processes*. McGraw-Hill.
- Thayer, A. M. (2010). Outfitting the full disposable kit. *Chemical & Engineering News*, 88(50), 18.
- Uhl, W. W., & Gray, J. B. (Eds.). (1967). *Mixing, theory and practice* (Vol. 2). Academic Press.
- Van't Riet, K. (1979). Review of measuring methods and results in nonviscous gas-liquid mass transfer in stirred vessels. *Industrial Engineering Chemistry, Process, Design and Development*, 18(3), 357.
- Wilke, C. R., & Chang, P. (1955). Correlation of diffusion coefficients in dilute solutions. *AIChE Journal*, 1, 264.
- Wilkinson, W. L., & Edwards, M. F. (1972). Heat transfer in agitated vessels (in two parts). *London Chemical Engineering*, 264, 310, 265, 328.
- Willig, S. H., & Stoker, J. R. (1996). *Good manufacturing practices for pharmaceuticals: A plan for total quality control* (4th ed.). Marcel Dekker.
- Wolschlag, L. M., & Couch, K. A. (2010). Upgrade FFC performance - part 1. *Hydrocarbon Processing*, 89(9), 57.
- Xu, Z. C., Zhang, W. J., Li, H., Cui, L., Cen, L., Zhou, G. D., et al. (2008). Engineering of an elastic large muscular vessel wall with pulsatile stimulation in bioreactor. *Biomaterials*, 29(10), 1464.
- Zenz, F. A., & Othmer, D. F. (1960). *Fluidization and fluid-particle systems*. Reinhold Publishing.
- Zughi, H. D., Khokar, Z. H., & Sharna, R. H. (2003). Mixing in pipelines with side and opposed tees. *Industrial Engineering Chemistry Research*, 42, 2003.

Bibliography

Books on Reactor Design

- Aris, R. (1989). *Elementary chemical reactor analysis*. Butterworths.
- Carberry, J. J., & Varma, A. (1986). *Chemical reaction and reactor engineering*. CRC Press.
- Chen, N. H. (1983). *Process reactor design*. Allyn and Bacon.
- Doraiswamy, L. K., & Sharma, M. M. (1984). *Heterogeneous reactions: Analysis, examples, and reactor design*, (Vol. 2). Wiley.
- Fogler, H. S. (2016). *Elements of chemical reactor design* (5th ed.). Prentice Hall.
- Levenspiel, O. (1979). *The chemical reactor minibook*. OSU Book Center.
- Nauman, E. B. (2008). *Handbook of chemical reactor design, optimization and scaleup* (2nd ed.). McGraw-Hill.
- Rose, L. M. (1981). *Chemical reactor design in practice*. Elsevier.
- Smith, J. M. (1981). *Chemical engineering kinetics* (3rd ed.). McGraw-Hill.
- Westerterp, K. R., Van Swaaij, W. P. M., & Beenackers, A. A. C. M. (1984). *Chemical reactor design and operation* (2nd ed.). Wiley.

American and International Standards

- 21 CFR Part 210 – Current good manufacturing practice in manufacturing, processing, packing, or holding of drugs; general.
- 21 CFR Part 211 – Current good manufacturing practice for finished pharmaceuticals.
- WHO Technical Report Series 908: WHO Expert Committee on Specifications for Pharmaceutical Preparations. 37th report.(2003).

15.16 Nomenclature

		Dimensions in MLT
A	Area	L^2
a	Interfacial area per unit volume	L^{-1}
a_i	activity of component i	—
b	Index in equation 15.16	—
C_A	Concentration of A in the liquid phase	ML^{-3}
$C_{A,i}$	Concentration of A in the liquid at the interface	ML^{-3}
$C_{A,bulk}$	Concentration of A in the bulk liquid	ML^{-3}
C_B	Concentration of B in the liquid phase	ML^{-3}
C_p	Specific heat capacity	$L^2T^{-2}\theta^{-1}$
C_{pw}	Specific heat capacity of cooling water	$L^2T^{-2}\theta^{-1}$
C	Index in equation 15.16	—
D	Diffusivity	L^2T^{-1}
D	Impeller diameter (Fig. 15.15)	L
D_A	Diffusivity of component A	L^2T^{-1}
D_i	Vessel internal diameter	L
D_L	Liquid phase diffusivity	L^2T^{-1}
D_T	Tank diameter (Fig. 15.15)	L
D_v	Vapor phase diffusivity	L^2T^{-1}
d_a	Agitator diameter	L
d_p	Particle diameter	L
d_t	Tube diameter	L
E	Welded joint efficiency	—
ΔG	Change in Gibbs free energy	L^2T^{-2}
g	Gravitational acceleration	LT^{-2}
ΔH_{rT}	Heat of reaction at temperature T	L^2T^{-2}
$\Delta H_{react.}$	Enthalpy change to bring reactants to standard temperature	L^2T^{-2}
$\Delta H_{prod.}$	Enthalpy change to bring products to reaction temperature	L^2T^{-2}
h_i	Tube-side heat transfer coefficient for a packed tube	$MT^{-3}\theta^{-1}$
K	Reaction equilibrium constant	units vary
K	Constant in equation 15.16	—
k	Mass transfer coefficient	LT^{-1}
k_L	Mass transfer coefficient in liquid phase	LT^{-1}
k_1	First order rate constant	T^{-1}
k_2	Second order rate constant	$M^{-1}L^3T^{-1}$
L_b	Length of a packed bed of solids	L
M_i	Molecular mass of component i	M

		Dimensions in MLT
M_{in}	Molar flow rate of any component into the reactor	MT^{-1}
M_{out}	Molar flow rate of any component out of the reactor	MT^{-1}
M_w	Molecular mass of solvent	M
m	Mass flow rate	MT^{-1}
m_i	Rate of consumption of substrate i to maintain cell life	T^{-1}
m_L	Mass of liquid in vessel	M
m_w	Mass flow rate of cooling water	MT^{-1}
N	Agitator speed	T^{-1}
n	Total number of components	—
P	Pressure	$ML^{-1}T^{-2}$
P_a	Agitator power input	ML^2T^{-3}
P_i	Internal pressure	$ML^{-1}T^{-2}$
ΔP	Pressure differential (pressure drop)	$ML^{-1}T^{-2}$
P	Partial pressure of adsorbed component	$ML^{-1}T^{-2}$
P	Agitator blade pitch (Fig. 15.15)	L
Q	Gas volumetric flow rate	L^3T^{-1}
R	Universal gas constant	$L^2T^2\theta^{-1}$
$\mathcal{R}$	Rate of reaction	$ML^{-3}T^{-1}$
S	Maximum allowable stress	$ML^{-1}T^{-2}$
s_i	Concentration of substrate i	ML^{-3}
T	Temperature	θ
T_0	Initial temperature	θ
T_w	Wall temperature	θ
T_1	Cooling water inlet temperature	θ
T_2	Cooling water outlet temperature	θ
ΔT_{eff}	Effective temperature difference	θ
t	Time	T
t	Pressure vessel wall thickness (Equation 14.13)	L
U	Overall heat transfer coefficient	$MT^{-3}\theta^{-1}$
u	Superficial velocity	LT^{-1}
u_p	Particle velocity relative to the fluid	LT^{-1}
V	Reactor volume	L^3
V_{liq}	Liquid volume	L^3
V_m	Molar volume of the solute at its boiling point	$M^{-1}L^3$
v_i	Special diffusion volume coefficient in Equation 15.9	L^3
w	Impeller width (Fig. 15.16)	L
x	Concentration of cells	ML^{-3}
Y_i	Yield of new cells on substrate i	—
α_j	stoichiometric coefficient of component i	—
δ	Thickness of boundary layer	L

		Dimensions in MLT
ε	Bed void fraction	—
λ_f	Fluid thermal conductivity	$\text{MLT}^{-3}\theta^{-1}$
μ	Fluid viscosity	$\text{ML}^{-1}\text{T}^{-1}$
μ_g	Growth rate	T^{-1}
ρ	Liquid density	ML^{-3}
ρ_f	Fluid density	ML^{-3}
τ	Residence time	T
Φ	Solvent association factor	—
Dimensionless numbers		
Fr	Froude number	
Ha	Hatta number	
N_p	Power number, defined by Equation 15.16	
Pr	Prandtl number	
Re	Reynolds number	
Sc	Schmidt number	
Sh	Sherwood number	

15.17 Problems

- 15.1.** Acrolein ($\text{H}_2\text{C}=\text{CHCHO}$) is made by selective oxidation of propylene at 2 bar, 350 °C using a molybdenum, iron, and bismuth catalyst on a silica support. The reactor yields based on propylene are 85% acrolein, 10% acrylic acid, and 5% light by-products. The light by-products are mostly acetaldehyde, but for the purpose of this problem, it can be assumed that the yield is 85% acrolein and 15% acrylic acid. The feed to the reactor on a volume percent basis is propylene 6%, propane 28%, steam 6%, oxygen 11%, and balance nitrogen. Estimate the reactor cooling requirement for a plant that produces 20,000 metric tons per year (20 kt/y) of acrolein if the reactor is operated isothermally.
- 15.2.** Styrene can be made by catalytic dehydrogenation of ethylbenzene. The reaction is usually carried out in the presence of steam, which acts as a heat carrier and reduces coking on the catalyst. If a mixture of steam and ethylbenzene is fed to an adiabatic reactor at 2 bar, 640 °C, estimate the equilibrium conversion and outlet temperature as a function of the ratio of steam to ethylbenzene. What ratio would you recommend using in practice?
- 15.3.** Aniline is produced by the hydrogenation of nitrobenzene. The reaction takes place in a fluidized bed reactor operating at 270 °C and 20 bar. The excess heat of reaction is removed by a heat transfer fluid passing through tubes in the fluidized bed. Nitrobenzene vapor and hydrogen enter the reactor at a temperature of 260 °C. A typical reactor off-gas composition, mole percent, is: aniline 10.73, cyclo-hexylamine 0.11, water 21.68, nitrobenzene 0.45, hydrogen 63.67, and inerts (take as nitrogen) 3.66. Estimate the heat removed by the heat transfer fluid for a feed rate of nitrobenzene to the reactor of 2500 kg/h.
- 15.4.** Hydrogen chloride is produced by burning chlorine with an excess of hydrogen. The reaction is highly exothermic and reaches equilibrium quite rapidly. The equilibrium mixture contains approximately 4% free chlorine, but this is rapidly combined with the excess hydrogen as the mixture is cooled. Below 200 °C the conversion of chlorine is essentially complete. The burner is fitted with a cooling jacket, which cools the exit gases to 200 °C. The gases are further cooled to 50 °C in an external heat exchanger.

For a production rate of 10,000 tons per year of hydrogen chloride, calculate the heat removed by the burner jacket and the heat removed in the external cooler. Take the excess hydrogen as 1% over stoichiometric. The hydrogen supply contains 5% inerts (take as nitrogen) and is fed to the burner at 25 °C. The chlorine is essentially pure and is fed to the burner as a saturated vapor. The burner operates at 1.5 bar.

- 15.5. A 100-liter fed-batch fermenter was successfully operated for 200 h using an aeration rate of 14 l/s and a 5 W agitator. It is proposed to scale the fermentation up to 10,000 liters. What agitator power consumption is required?
- 15.6. Isophthalic acid (IPA, formula $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$) is made by oxidation of *m*-xylene with air. The reaction is carried out in acetic acid solvent at about 200 °C using a cobalt–manganese catalyst promoted by bromine. Acetic acid is boiled off, condensed, and returned to the reactor to remove the heat of reaction, and the reactor pressure is set to allow an adequate evaporation rate of acetic acid. IPA is not soluble in acetic acid, so the product crystallizes and a slurry can be removed from the base of the reactor for product recovery. Design and size a reactor system for production of 25,000 metric tons per year of IPA.
- 15.7. In the acrolein process introduced in Problem 15.1, the catalyst is packed in tubes and the reactor is cooled using a circulating molten salt. The molten salt rejects heat to boiling steam in a similar arrangement to that shown for phthalic anhydride in Fig. 15.25. The required gas hourly space velocity is 200 standard liters (at STP)/liter.h. Design and size a reactor to produce 20 kt/y of acrolein.
- 15.8. Acetylene is formed as a by-product in the manufacture of ethylene. It is removed by selective hydrogenation using noble metal catalysts (see, for example, US 7,453,017). A particular catalyst achieves 90% acetylene saturation with 50% hydrogen selectivity for acetylene at a weight hourly space velocity of 800 h^{-1} based on ethylene. Design a reactor to remove 1% acetylene from ethylene in a plant that produces 1.5 million metric tons per year of ethylene using this catalyst.
- 15.9. It is proposed to sterilize the growth medium fed to a continuous bioreactor by holding it at either 120 °C for 15 minutes or at 140 °C for 3 minutes. The feed must contain 12 mg/l of vitamin C and 0.3 mg/l of thiamin. Leskova et al. (2006) give the following first-order rate constants for thermal decomposition of these vitamins:

Vitamin	$k_o \text{ (s}^{-1}\text{)}$	$E_a \text{ (kJ/mol)}$
Ascorbic acid (vitamin C)	3.6×10^2	46
Thiamin (vitamin B1)	1.88×10^9	97

Where the first-order rate constant $k_1 = k_o \exp(-E_a/RT)$

Determine the amount of each vitamin that must be added to the feed before sterilization for each heat treatment. Which sterilization treatment do you recommend?

- 15.10. The reaction of Example 15.5 can be carried out in a heat exchanger reactor, using boiling water as coolant. Design and size a reactor to produce 40,000 metric tons per year of cyclohexane.
- 15.11. The fermentation of cane sugar to ethanol was introduced in Problem 4.3, and typical yields were given. Design the reactor system for a plant that produces 500,000 metric tons per year of ethanol using continuous fermentation.
- 15.12. The batch fermentation of glucose to glutamic acid using *Brevibacterium divaricatum* is described in detail in US patent 2,978,384. In Example I in the patent a 10.5 wt% glucose feed was fermented for 72 hours at 30 °C, giving 33.5 wt% glutamic acid yield on glucose. What annual production of glutamic acid can be achieved in a batch fermentation plant that has four 10,000-liter reactors? The reactors have jackets that can be used for steam heating or water cooling, but no internal coils.

16

Separation of fluids

KEY LEARNING OBJECTIVES

- How to separate mixtures of gases
- How to separate droplets of liquid from a gas or second liquid phase
- How to separate dissolved components from liquid mixtures
- Design of separations using adsorption, membranes, knockout pots, decanters, coalescers, ion exchange, reverse osmosis, and chromatography

16.1 Introduction

Separation processes play an important role in all of the process industries. In addition to recovering components from reactor products, separations are used to purify feeds, products, and recycle streams and to treat effluent streams to meet environmental regulations.

Most separation processes are based on the concept of partitioning a component between two phases that can be physically separated. In some cases, two or more phases are formed from the process stream, for example, in condensation or evaporation, whereas in other cases an additional separating agent is introduced, as in adsorption, chromatography, or liquid–liquid extraction.

This chapter and the following chapter cover fluid-phase separations. Chapter 16 discusses separation of gas mixtures, physical separation of vapor–liquid and liquid–liquid mixtures, and recovery of dissolved components from liquids. [Chapter 17](#) focuses on separations that use multistage contacting columns, particularly vapor–liquid separations such as distillation, absorption, and stripping, which are the most widely used separation techniques in the chemical process industries. Separations that involve the formation or recovery of a flowing solid phase are intrinsically linked to solids forming, sizing, and handling processes that are described in the chapter on solids processing; see [Chapter 18](#).

Many of the separation techniques described in this chapter require the use of proprietary separating agents or equipment, and so would normally be designed in consultation with a technology vendor. Sufficient information is given here for a design engineer to make a preliminary design and estimate the critical sizing parameters that are needed to generate a Class 4 or Class 5 cost estimate using commercial costing software; however, a design selection should not be made without obtaining a more realistic estimate from discussion with a vendor. Contact information for technology and equipment suppliers can easily be obtained by searching the Internet.

Several excellent textbooks provide an overview of separations processes, discuss the underlying physical phenomena, and derive the design equations. [Chhabra and Gurappa \(2019\)](#) and [McCabe et al. \(2004\)](#) are particularly recommended. Descriptions and illustrations of most of the equipment used can also be found in various handbooks:

Green and Southard (2018), Schweitzer (1997), and Couper et al. (2012). Other texts that focus on specific unit operations are referenced in the relevant sections. Equipment manufacturers' advertisements in the technical press and online should also be studied. Manufacturer's websites are usually easily located online and often provide details of equipment construction, standard sizes, available metallurgies, specification sheets, and performance information. It is worthwhile building up a personal file of vendors' websites that have good information to eliminate repeated searching.

16.2 Gas–gas separations

Separation of gaseous or vapor components from the gas phase is important in many industrial processes:

1. Preparation of high-purity industrial gases such as oxygen, nitrogen, argon, neon, hydrogen, etc.
2. Purification of natural gas to meet specifications for pipelines and liquefaction plants.
3. Treatment of process gases to prevent accumulation of contaminants in recycles or to protect catalysts.
4. Removal of pollutants from vent gases to meet legislative requirements.
5. Drying and purification of air for use in sterile dryers, clean rooms, as instrument air, etc.

Bulk removal of condensable vapor components from a gas can usually be accomplished by cooling and condensation; see Section 19.10; however, even refrigerated condensers seldom give adequate recovery of the vapor, and additional separation is usually needed to meet product gas or vent gas specifications.

16.2.1 Adsorption

Adsorption is probably the most widely used method in gas separation. Adsorption can be expensive for bulk separation of large quantities of gas, but it has the advantage of allowing very high gas purities or high recoveries of contaminants to be achieved.

The basic principle of adsorption is to exploit differences in the strength of interaction between different species in the gas phase and the surface of a solid known as a sorbent or adsorbent. If one species is more strongly adsorbed, then it accumulates on the sorbent as the gas passes through a bed of sorbent, and hence is removed from the gas phase. If the gas flows down through the sorbent bed, as shown in Fig. 16.1(a), then it continues to contact fresh sorbent and all of the adsorbed component is removed, giving a purified gas. As the sorbent accumulates the adsorbed component, it eventually becomes saturated and can adsorb no more. A concentration profile is thus established in the sorbent bed (see Fig. 16.1b). The concentration profile moves down through the bed until it reaches the bottom, at which point *breakthrough* of the adsorbed component occurs and the concentration in the exit gas begins to rise. Breakthrough occurs at the time labeled t_B in Fig. 16.1. The thermodynamics and kinetics of adsorption that lead to the formation of the concentration profiles shown in Fig. 16.1(b) are discussed in detail in the books by Ruthven (1984), Yang (1997), Chhabra and Gurappa (2019), and Ruthven et al. (1993). An introduction to adsorption design is given in the article by Gabelman (2017). Adsorption is also covered in the books by Suzuki (1990) and Crittenden and Thomas (1998).

Irreversible adsorption

When adsorption is used to remove a low level of a contaminant from a gas stream, a sorbent can be selected that reacts irreversibly with the contaminant. In this case, the process is termed irreversible adsorption, and the adsorption vessel is sometimes referred to as a guard bed. An example of irreversible adsorption is the use of zinc or copper oxide to remove trace amounts of hydrogen sulfide from natural gas and petrochemical processes.

When a bed of irreversible sorbent becomes saturated, it must be replaced. A common arrangement is to use two beds in parallel, with one in service and the other isolated to allow for sorbent change-out (Fig. 16.2). This scheme is simple, but can be wasteful of sorbent, as the bed must usually be replaced before breakthrough occurs.

An alternative is to use a *lead–lag* arrangement (Fig. 16.3). In this flow scheme, the gas normally flows through both beds in series, as illustrated in the figure. When the first (lead) bed becomes saturated and breakthrough occurs, then the second (lag) bed removes the contaminant, so there is no breakthrough to the product gas. The first bed is then removed from service for sorbent change-out while the feed gas flows through the second bed alone. The first bed is then placed back into service and becomes the lag bed, while the second bed now operates as lead bed. This arrangement allows the sorbent to be completely used up and reduces spent sorbent waste disposal costs.

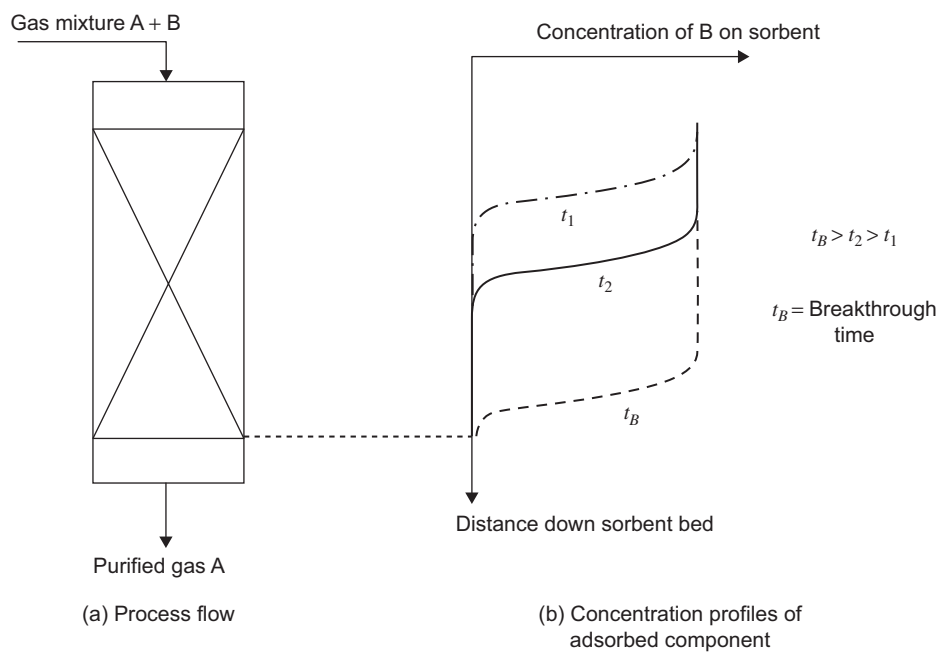


FIG. 16.1 Adsorption.

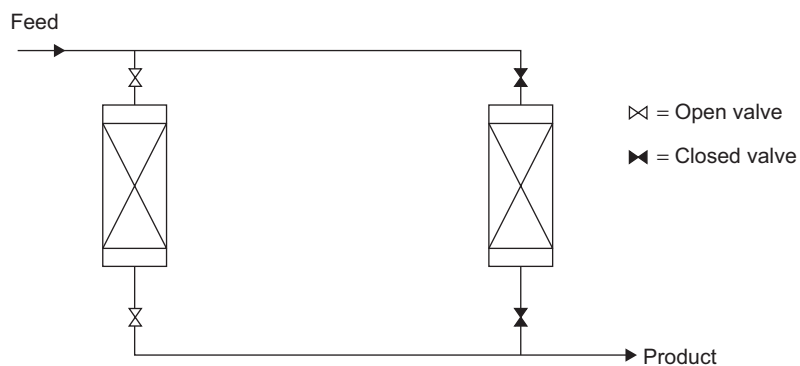


FIG. 16.2 Parallel flow through guard beds.

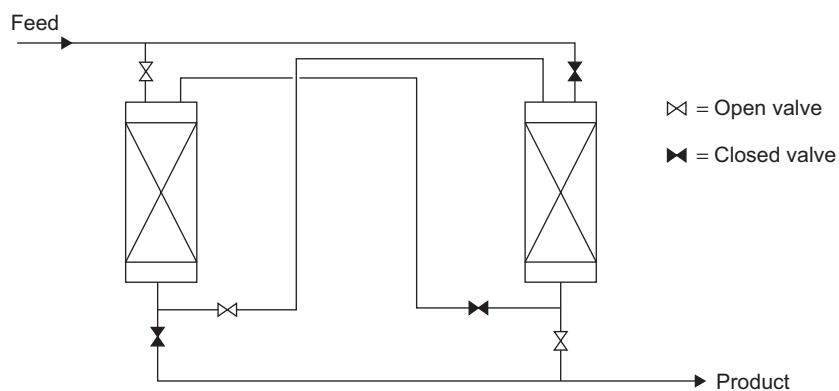


FIG. 16.3 Lead-lag guard beds.

The sorbents for irreversible adsorption are usually cheap, but the process can only be justified when the amount of contaminant removed is small, because the inconvenience and labor costs of changing out sorbent and the costs of waste disposal must also be taken into account.

Reversible adsorption

In most cases, irreversible adsorption is not practical, as the adsorbed component either has high flow rate or is desired as a product. When the bed reaches or nears breakthrough, it must be taken out of process service so that the sorbent can be regenerated and the adsorbed components recovered. Multiple beds are usually used, with an arrangement of isolation valves that allows the beds to be sequenced so that a fresh bed of sorbent is brought into service whenever a bed is switched to regeneration.

Regeneration of the sorbent is usually carried out by increasing temperature or reducing pressure to give a lower equilibrium concentration of the adsorbed component on the solid surface. The equilibrium between the partial pressure of a component in the vapor phase and the adsorbent loading, expressed as grams of adsorbed material per gram of adsorbent, can be plotted at constant temperature as an isotherm (Fig. 16.4). Various expressions can be used to model isotherms, but for design purposes they can usually be fitted with an equation of the form:

$$m = kp^n \quad (16.1)$$

where m = adsorbent loading, g/g adsorbent

k = equilibrium constant

p = partial pressure of adsorbed component

n = exponent

Fig. 16.4 shows isotherms at two temperatures, T_2 and T_1 , where T_2 is greater than T_1 . It can be seen from the figure that a change in adsorbent loading from m_1 to m_2 could be accomplished either by decreasing the partial pressure from p_1 to p_2 at constant temperature T_1 , or by increasing temperature from T_1 to T_2 at constant pressure p_1 . Cycling the bed over a range of temperature or pressure thus allows material to be adsorbed under process conditions and then recovered from the bed under regeneration conditions.

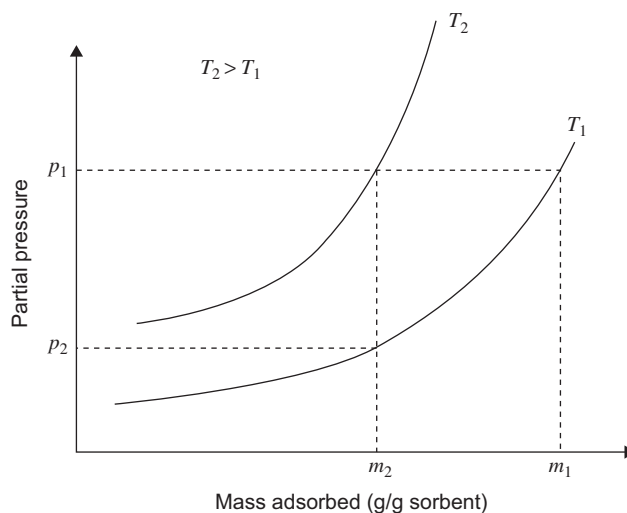


FIG. 16.4 Adsorption isotherms.

Pressure swing adsorption

When the bed is regenerated by reducing pressure, the process is known as pressure swing adsorption (PSA). Pressure swing adsorption processes are nominally isothermal, although heat of adsorption and desorption effects must be allowed for in the design. Because the process is not limited by heat transfer rates, very rapid cycles as short as a few minutes can be used. Rapid cycling gives efficient use of adsorbent and leads to smaller vessel sizes and lower capital cost.

If the process feed is under high pressure, then carrying out the regeneration at or near atmospheric pressure may be adequate. Alternatively, a vacuum pump can be used to generate a lower pressure for regeneration, in which case the process is known as vacuum swing adsorption (VSA or VPSA).

A small flow of sweep gas is usually passed through the bed during regeneration to promote desorption and clear away the desorbed components. The sweep gas is typically a slip stream from the purified product, although other gases are occasionally used. When the product gas is used as sweep gas, very high-purity product gas can be obtained, but the recovery of the product gas is typically less than 95%.

Typical industrial flowsheets for PSA use 4 to 12 adsorbent vessels. With multiple vessels, the designer can sequence the switching valves to compensate for heating and cooling effects due to heat of adsorption and desorption, and also to economize on losses of product gas during depressurization, regeneration, and repressurization. Examples of typical commercial configurations are given by [Ruthven et al. \(1993\)](#), [Cassidy and Doshi \(1984\)](#), and [Kumar et al. \(1994\)](#).

Pressure swing adsorption is used in many processes; for example, air separation, hydrogen manufacture by steam reforming, and dehydration of ethanol–water azeotropic mixtures (see Section 17.6.5).

Temperature swing adsorption

Regeneration of the adsorbent by raising temperature is known as temperature swing adsorption (TSA). In TSA processes, heat is required to raise the temperature of the adsorbent, the vessel and piping, and to provide the heat of desorption. This heat is usually provided by flowing a hot stripping gas over the adsorbent. The stripping gas is typically steam, dry air, nitrogen, or a slip stream of purified product. The stripping gas must be chosen carefully to ensure that flammable atmospheres are not formed and the adsorbent is not deactivated. The stripping gas is usually heated in a fired heater, but electric or steam heaters are used in smaller plants.

After desorption is completed, the bed is usually flushed with cold purified product to bring the temperature back down to the adsorption temperature. Temperature swing processes typically operate on longer time cycles than PSA, as the cycle time is usually governed by the maximum attainable rates of heating and cooling.

Temperature swing adsorption is widely used for drying air and for removing trace amounts of organic compounds from vent gases using activated carbon as adsorbent. The activated carbon can then be regenerated with steam.

Adsorbent selection

Adsorbents for a reversible process such as PSA or TSA are chosen to give a wide range of adsorbent loading over an acceptable range of either temperature or pressure. Many adsorbents are proprietary materials that have been designed to have high specific surface area and hence give high loadings; however, isotherm data for common adsorbents such as activated carbon, alumina, silica gel, and some of the more common zeolites are available in the open literature and in books such as [Breck \(1974\)](#), [Ruthven \(1984\)](#), and [Yang \(2003\)](#).

Care must be taken to ensure that the adsorbent is not irreversibly poisoned by any component of the gas phase. When multiple contaminants must be removed from a gas, several layers of different adsorbents with selectivity for different contaminants can be loaded in the same vessel so that each contaminant is removed in turn.

Prices for some of the more common adsorbents are listed in Aspen ICARUS and other costing programs. Current prices for commodity adsorbents such as activated carbon, silica gel, and alumina can be found online or by contacting manufacturers.

Adsorption equipment design

Adsorption plants are often purchased from an industrial gas company as packaged modular plants that include the adsorbent, vessels, and valve skid. For TSA plants, the heater may also be included in the packaged plant scope. The design and optimization of PSA and TSA processes require detailed understanding of the adsorbent properties and heat of adsorption effects and are best left to specialist suppliers. The simplified method that follows is suitable for undergraduate design projects and for generating preliminary estimates.

When a plant is designed from scratch, the first step is to estimate the amount of adsorbent required. The amount of adsorbent can be determined from the flow rate of the adsorbed species and the change in the bed loading during the cycle. A mass balance on the bed gives:

$$(F_1 y_1 - F_2 y_2) M_w t_a = 1000 (m_1 - m_2) M_a f_L \quad (16.2)$$

where F_1 = feed molar flow rate (mol/s)

F_2 = product molar flow rate (mol/s)

y_1 = feed mole fraction of adsorbed component

y_2 = product mole fraction of adsorbed component

M_w = molecular weight of adsorbed component (g/mol)

t_a = time the bed is in the adsorption stage of the cycle (s)

m_1 = maximum adsorbent loading (g/g adsorbent)

m_2 = minimum adsorbent loading (g/g adsorbent)

M_a = mass of adsorbent per bed (kg)

f_L = fraction of bed that is fully loaded at end of adsorption phase of cycle

The fraction of the adsorbent bed, f_L , that reaches loading m_1 at the end of the adsorption step of the cycle depends on the process arrangement and the number of beds used. For a simple two-bed system it will usually be less than 0.7 unless a very sharp front is formed or a lead–lag arrangement can be used. With four or more beds, the cycle can usually be designed to give fractional loading close to 1.0 unless multiple components are being removed.

The time that the bed spends in the adsorption phase of the cycle is less than the total cycle time, which also includes time for regeneration, depressurizing, heating, cooling, etc. The rates of these processes are governed by intrinsic rates of desorption, mass transfer, and heat transfer. The development of rate-based models of these processes is discussed by [Ruthven \(1984\)](#) and [Chhabra and Gurappa \(2019\)](#). For a preliminary analysis, a time in the range of 5 to 60 minutes can be assumed for PSA and 60 to 200 minutes for TSA. The total cycle time is equal to the time spent in adsorption multiplied by the number of beds in the sequence.

The volume of each adsorbent bed can be estimated from the mass of adsorbent and the adsorbent bulk density. Fixed beds of adsorbent are usually used to give a sharp adsorption concentration profile. The adsorption vessel can then be sized as a cylindrical pressure vessel that contains the adsorbent volume, using the pressure vessel design methods given in [Chapter 14](#). The head space is usually left empty, and up to 20% of the volume between the tangent lines of the vessel is packed with inert material to ensure that a uniform flow profile is established at the entry and exit of the bed and to prevent “fingering” of contaminant through the bed ([Fig. 16.5](#)). The aspect ratio of the bed is usually at least 3:1 to ensure that high fractional loading is obtained and the adsorbent is used efficiently.

The total plant cost will include at least two adsorbent vessels, the set of switching valves and any other equipment; such as blowers, vacuum pumps or heaters, that is required for regeneration.

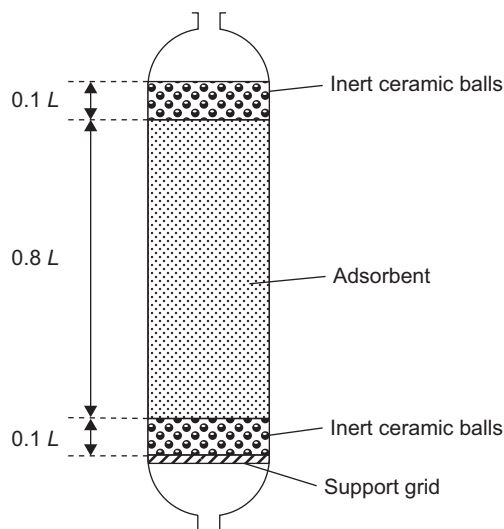


FIG. 16.5 Adsorption vessel internals.

Example 16.1

[Yang et al. \(1997\)](#) give isotherm data for adsorption of methane on zeolite 5A in the presence of hydrogen:

Pressure (atm)	0	1.5	2.5	4.5	6.5	9	12.5	16.5	19.5
Adsorbed amount (mmol/g)	0.0	0.4	0.7	1.0	1.25	1.45	1.6	1.7	1.75

It has been determined that a four-bed PSA plant using this adsorbent can be used to purify a stream containing 40% CH₄ in H₂ to give 90% recovery of hydrogen at purity 99.99 mol%. The adsorption time is 300s, and the beds are 85% loaded at the end of the adsorption step. If the feed gas is available at 25 atm and the off-gas can be discharged at 2 atm, estimate the amount of adsorbent per bed and the dimensions of each adsorbent vessel for a feed rate of 1000 kmol/h.

Data (Yang et al., 1997): Adsorbent bulk density = 795 kg/m³.

Solution

Feed flow rate = 1000 kmol/h

hydrogen in feed = $1000 \times 0.60 = 600$ kmol/h

methane in feed = $1000 \times 0.40 = 400$ kmol/h, $y_{\text{CH}_4, \text{feed}} = 0.4$

90% recovery at 99.99% purity:

hydrogen in product gas = $600 \times 0.9 = 540$ kmol/h

methane in product gas = $\frac{(1 - 0.9999)}{0.9999} \times 540 = 5.4 \times 10^{-2}$ kmol/h

hydrogen in off-gas = $600 - 540 = 60$ kmol/h

methane in off-gas ≈ 400 kmol/h

So, methane mole fraction in off-gas, $y_{\text{CH}_4, \text{off-gas}} = \frac{400}{460} = 0.870$

Partial pressures of methane:

in feed at 25 atm = $25 \times 0.4 = 10$ atm

in off-gas at 2 atm = $2 \times 0.87 = 1.74$ atm

By fitting an isotherm through the data, plotting or interpolating, we find:

At 10 atm, adsorbent loading = 1.5 mmol/g

At 1.74 atm, adsorbent loading = 0.5 mmol/g

Substituting into Equation 16.2 (noting that we do not need the molecular weight, as the adsorbent loadings were given in mmol/g = mol/kg):

$$\{(1000 \times 0.4) - (540 \times 0.0001)\} \times \frac{1000}{3600} \times 300 = (1.5 - 0.5) \times M_a \times 0.85 \text{ (mol/s)}$$

Hence, mass of adsorbent $M_a = \frac{400 \times 300}{3.6 \times 0.85 \times 1.0} = \underline{\underline{39,200 \text{ kg}}}$.

Volume of adsorbent bed = $\frac{39200}{795} = 49 \text{ m}^3$.

Allow 20% of total volume for inert packing for good flow distribution, so tangent-to-tangent volume = $49/0.8 = 61.7 \text{ m}^3$.

Assume 4:1 cylinder, volume = $\frac{\pi D_T^2 L_v}{4} = \pi D_T^3$

where D_T = vessel diameter

L_v = tangent-to-tangent length

So, $D_T^3 = 61.7/\pi$, vessel diameter = 2.70 m

Round up to nearest standard head size (see Chapter 14):

diameter = 2.74 m (9 ft)

tangent length = 11.0 m (36 ft)

We now have sufficient information to complete the pressure vessel design and costing (see Chapters 7 and 14). Note that four vessels will be needed and the total quantity of adsorbent is $4 \times 39,200 \text{ kg} = 156.8 \text{ t}$.

16.2.2 Membrane separation

A membrane is a thin layer of material that allows species to pass through by permeation. The material that flows through the membrane is known as the *permeate* and the material that does not is called the *retentate*. If some species

pass through the membrane faster than others, these species will accumulate in the permeate, and the membrane can be used to separate these components from the gas mixture.

The flux through a membrane is defined as the flow rate through a unit area of membrane per unit time. The flux is proportional to the applied partial pressure gradient:

$$M_i = \frac{P_i}{\delta} (p_{i,f} - p_{i,p}) \quad (16.3)$$

where M_i = molar flux of component i (mol/m².s)

P_i = permeability of membrane for component i (mol/m.s.bar)

δ = membrane thickness (m)

$p_{i,f}$ = local partial pressure of component i on feed side (bar)

$p_{i,p}$ = local partial pressure of component i on permeate side (bar)

The average flux across a long cylindrical membrane is given by:

$$M_{i,ave} = \frac{\int_0^{L_m} M_i dx}{L_m} \quad (16.4)$$

where $M_{i,ave}$ = average molar flux of component i (mol/m².s)

L_m = length of membrane (m)

x = length (m)

If the feed or permeate partial pressure varies substantially along the membrane length, an expression for the variation of partial pressure with length must be substituted into Equation 16.4 before the average flux can be found.

The ratio of permeability of two components is called the membrane selectivity, or ideal separation factor:

$$S_{ij} = \frac{P_i}{P_j} \quad (16.5)$$

where S_{ij} = selectivity of the membrane for component i over component j .

If the selectivity of the membrane for a component is very high, then that component can be recovered at high purity in the permeate.

At the retentate end of the membrane, the partial pressure of the permeating component is reduced; consequently, the driving force for permeation is lower and the flux is lower. If the partial pressure for the permeating species becomes too low, the flux becomes very low and an uneconomically large membrane area is required. Because of this effect, membrane processes for gas separation usually cannot achieve high purity in the retentate or high recovery of permeate. Techniques for maintaining an adequate flux at the retentate end are discussed later.

Membrane selection and construction

Membranes are described as *microporous* if they have an open pore structure and *dense* if there are no holes or pores. Microporous membranes perform essentially like sieves and allow small molecules to move through the pores while large molecules are blocked. The separation of ²³⁵UF₆ from ²³⁸UF₆ in uranium enrichment is the only commercial gas separation process that uses a microporous membrane. Most gas separations use *asymmetric* membranes, consisting of a very thin dense layer supported on a thick, porous layer that gives strength to the membrane (Fig. 16.6).

Gas separation membranes are usually made from elastomeric or glassy polymers, although very thin layers of metal or ceramic materials are used in some special applications. The use of polymers restricts membrane processes

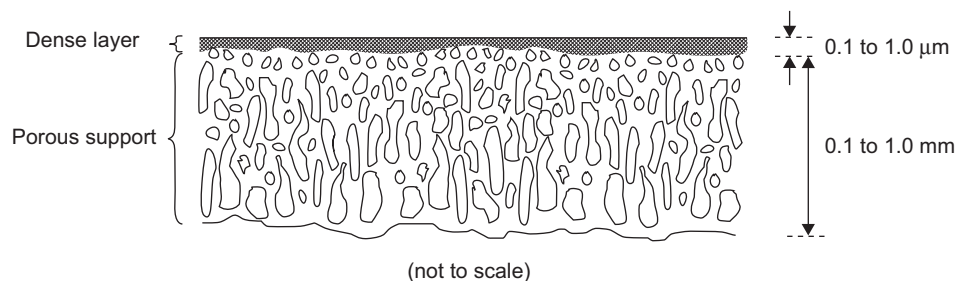


FIG. 16.6 Asymmetric membrane structure.

to operating at low temperatures; typically below 100 °C. For a dense membrane, the mechanism of permeation is for the gas to dissolve in the membrane on the feed side and then evaporate from the membrane on the permeate side. The membrane permeability and selectivity therefore depend on the solubility of different species in the membrane. With careful design and selection of membrane material, high selectivities can be obtained.

In addition to high selectivity, it is desirable to have high membrane permeability, as this will determine the area required, and hence the membrane cost. Table 16.1 gives permeability coefficients for some membrane materials. Other important parameters in membrane selection include mechanical stability, chemical resistance, thermal stability, ease of manufacture, and cost.

Gas separation membranes are usually manufactured as hollow fibers or else cast as flat sheets that are then spiral-wound into modular assemblies. Details of the manufacturing methods are given by Rautenbach and Albrecht (1989) and Scott and Hughes (1996). Hollow-fiber membranes are glued (“potted”) into a resin to form a closure similar to the tube sheet of a shell and tube exchanger (Fig. 16.7). In a spiral-wound membrane, two membrane sheets are placed back to back and sealed along three edges to form an envelope. The fourth edge is attached to a perforated tube, from which the permeate can be withdrawn. Several membrane envelopes can be

TABLE 16.1 Membrane permeability coefficients [$\text{cm}^3(\text{STP})\cdot\text{cm}/\text{cm}^2\cdot\text{s}\cdot\text{cmHg}$] $\times 10^{10}$

Membrane	Temperature (°C)	CO ₂	O ₂	N ₂
Natural rubber	25	99.6	17.7	6.12
Ethyl cellulose	25	113	15	3.0
Polystyrene	20	10.0	2.01	0.32
Polycarbonate	25	8.0	1.4	0.3
Poly(dimethylsiloxane)	25	3240	605	300
Poly(ethylene phthalate)	25	0.15	0.03	0.006
Poly(vinyl alcohol)	20	0.0005	0.00052	0.00045

From Osada and Nakagawa, 1992.

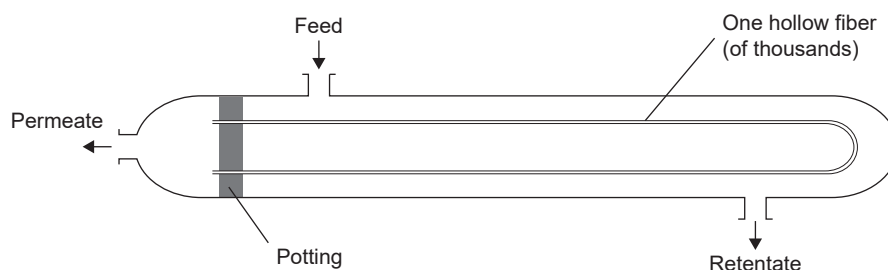


FIG. 16.7 Hollow-fiber membrane module.

attached to one tube, with netlike spacers between them. The entire assembly is then wrapped into a spiral and inserted into a tubular shell with a suitable head assembly through which the permeate can be withdrawn (Fig. 16.8).

Both the hollow-fiber and spiral-wound designs use dead-ended membranes that do not allow the use of a sweep gas on the permeate side. The permeate stream can be withdrawn from the feed end or retentate end, giving countercurrent or co-current flow (Fig. 16.9). With spiral-wound membranes, the permeate stream can also be withdrawn from both ends, giving an approximation of cross flow (see Fig. 16.9c). The choice of flow arrangement determines the boundary conditions for integrating Equation 16.4; see Chapter 5 of Scott and Hughes (1996).

In both the hollow-fiber and spiral-wound designs, the cost of the membrane itself is typically 80% to 90% of the total module cost.

Membrane process design

A typical membrane process uses a large number of membrane modules to achieve the overall separation. The modules can be arranged in various networks, depending on the processing objective and membrane performance. Some of the more common arrangements are illustrated in Fig. 16.10 and described here.

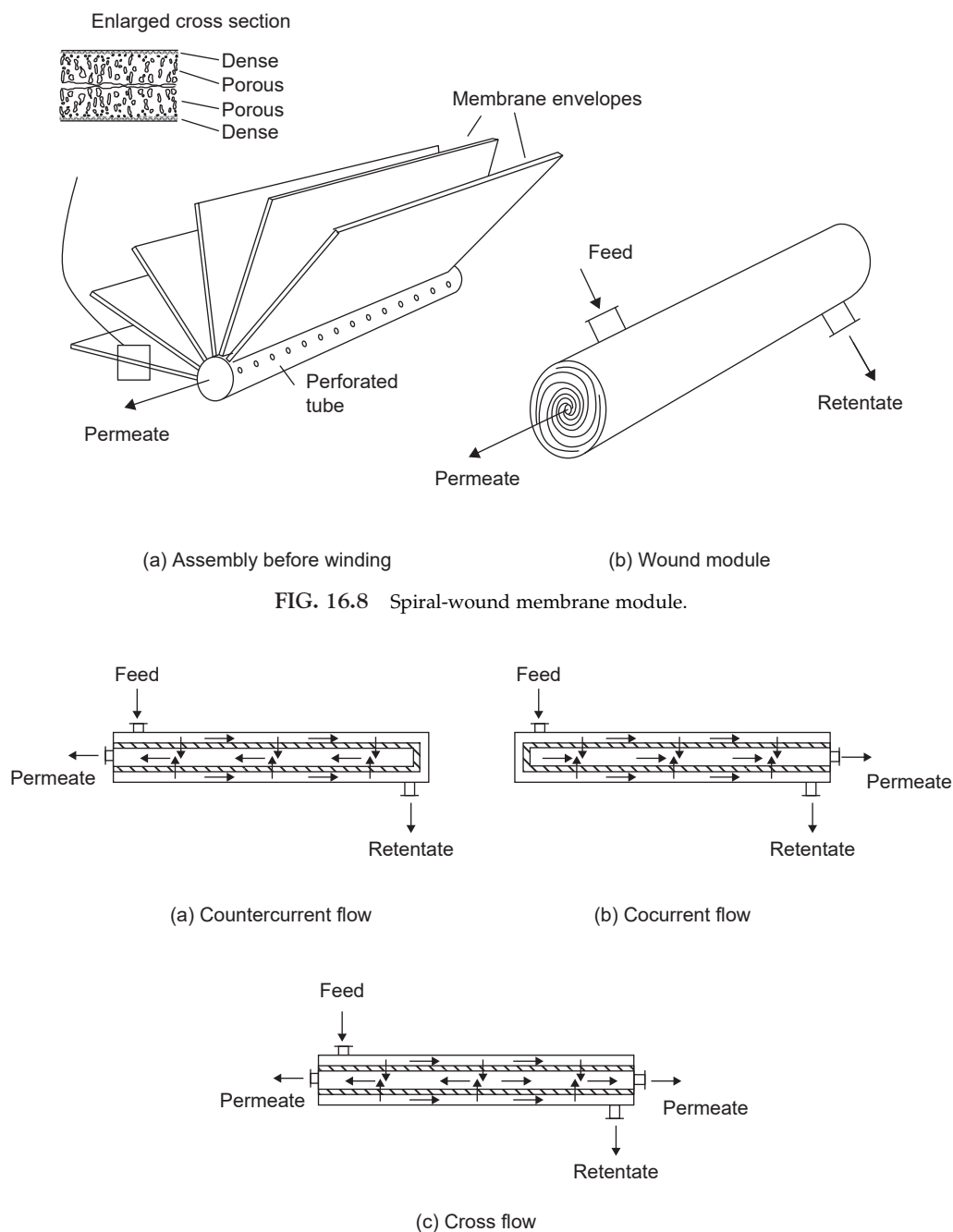


FIG. 16.8 Spiral-wound membrane module.

FIG. 16.9 Membrane flow patterns.

1. Tapered cascade (see Fig. 16.10a). This flow scheme is used when the final retentate flow is a small fraction of the feed flow rate. It ensures that the retentate side velocities are maintained in downstream modules as the flow rate decreases. Tapered cascades with retentate or permeate recycle can also be used.
2. Permeate recycle (see Fig. 16.10b). The use of a permeate recycle increases both the recovery and the purity of the permeate. The permeate stream from the downstream module is recycled to the feed of the upstream module. This increases the concentration and partial pressure of the permeating species in the first module, leading to better permeate purity and higher flux. The downstream module can be run to higher recovery, even if this causes the purity obtained in the downstream module to be lower.

3. Retentate recycle (see Fig. 16.10c). In this scheme, the permeate stream from a first module is sent to a second module. Retentate from the second module is recycled to the first module. This flow scheme is used to increase permeate purity when the membrane average selectivity is low. In some cases, several modules can be placed in series, with retentate recycle from the downstream module to the preceding module.

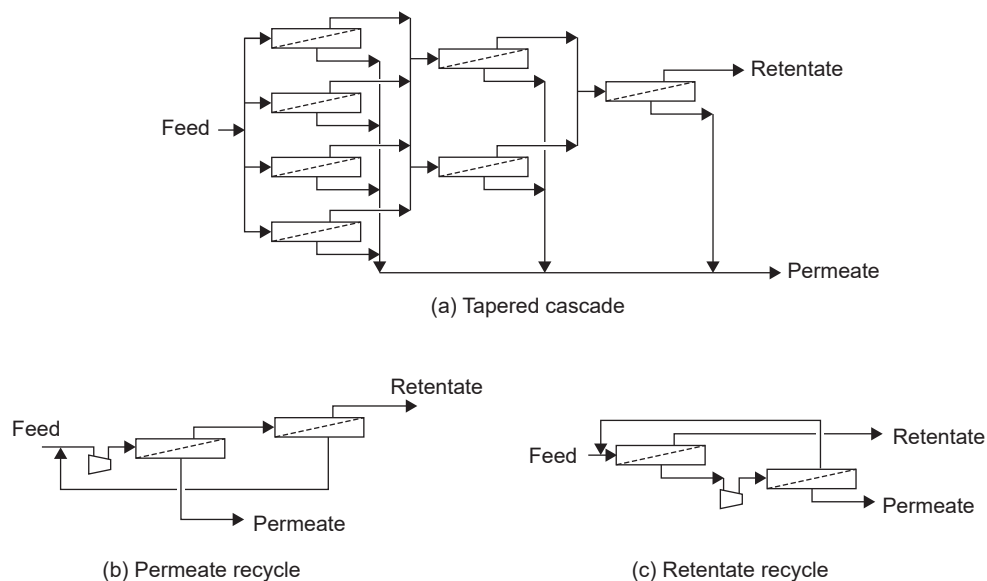


FIG. 16.10 Membrane flow schemes.

In addition to the membrane modules, membrane separation units often incorporate chillers, condensers, heaters, and guard beds. If the feed contains condensable components, these are usually knocked out by chilling and condensing upstream of the membrane so that they do not condense on the membrane and “blind” the surface. The gas is then reheated to give a comfortable margin of temperature above the dew point. When a network of membrane modules is used, intermediate chillers, condensers, and reheaters may be needed if the condensable components build up to the dew point concentration in either the permeate or the retentate of an intermediate module. Recompression between stages may also be necessary, but is avoided whenever possible, because of the expense of compression.

Sometimes, condensable or reactive species can poison or permanently damage the membrane. In such cases, these components are removed upstream of the membrane unit, usually by using an adsorption process.

The design of membrane separation plants is usually left to specialist vendors. Details of the membrane design and construction are usually proprietary, and optimal process design requires careful selection of flow pattern, flow scheme, and ancillary equipment. Accurate prediction of the overall purity and recovery requires a good knowledge of the permeability of all species present so that Equation 16.4 can be integrated for every species simultaneously. Design methods for membrane units are described in Scott and Hughes (1996), Noble and Stern (1995), Mulder (1996), Hoffman (2003), Sridhar and Moulik (2018), and Baker (2012). Short-cut design methods are given by Hogsett and Mazur (1983) and Fleming and Dupuis (1993), but both papers caution that final design should be left to experts. A simplified membrane design example is given in Example 16.6 in Section 16.5.4 for the purification of water by reverse osmosis. Gas separations are more complex in that there are usually multiple species that permeate through the membrane.

The proprietary nature of membrane technology for gas separations means that there is little reliable cost information in the open literature. The *Handbook of Gas Processing Processes* published biennially as a supplement to *Hydrocarbon Processing* usually provides some information from vendors that can be used to make preliminary estimates.

16.2.3 Cryogenic distillation

In cryogenic distillation, the feed gas is compressed to high pressure, cooled, and then chilled until the gas is partially liquefied and distillation can be carried out. The refrigeration load for chilling the gas may be met by expansion of the products or by using an external refrigeration cycle. The distillation column is designed to operate at very low temperatures and usually has many stages, but other than that is much like any conventional distillation; see Chapter 17.

The efficiency of cryogenic separations depends critically on heat recovery (cold recovery) during chilling of the feed. Very close temperature approaches of the order of 1 to 5 °C are used to minimize the amount of low-temperature refrigeration that is needed. Complex multilevel refrigeration cycles are also used to improve the coefficient of performance and reduce the compression work. The design of refrigeration cycles and heat recovery networks is covered in [Chapter 3](#).

If the pressure of a cryogenic process is increased, the process can be carried out closer to ambient temperature, giving a smaller temperature range for the refrigeration cycle and hence a more efficient refrigeration (see Section 3.2.6). It also becomes easier to generate the required refrigeration by expansion of the products; however, the feed compression becomes more costly and the distillation becomes more difficult, as relative volatility generally decreases with increased pressure. One of the main trade-offs in design of cryogenic processes is therefore between feed compression and refrigeration plant compression.

Cryogenic distillation is usually the lowest-cost process when handling large quantities of gases heavier than hydrogen. It is widely used industrially for air separation, natural gas liquids recovery, ethylene recovery, and propylene recovery; see [Flynn \(2004\)](#).

16.2.4 Absorption and stripping

In an absorption process, a component is removed from a gas by contacting the gas with a solvent that dissolves the component selectively. The solvent is then regenerated in a stripping process and recycled to the absorber.

Absorption can take place through physical dissolution of gas components in the liquid phase (*physisorption*) or chemical reaction between gas-phase components and the liquid phase (*chemisorption*). Absorption can be very cost-effective if a solvent can be found that has high affinity for the desired gas-phase components. Absorption is widely used for the removal of acid gases such as carbon dioxide, hydrogen sulfide, and sulfur dioxide, using basic solutions as solvent.

Absorption and stripping are usually carried out in vapor–liquid contacting columns; see Section 17.14.

16.2.5 Condensation

A condensable vapor can be recovered from a mixture with other gases that have much lower dew points by simple condensation. Any type of heat exchanger can be used as a condenser, although conventional shell and tube exchangers are most common. When one component is condensed from a non-condensing gas, the process is known as partial condensation, and care must be taken to include mass transfer limitations, as well as heat transfer in the exchanger design. The design of partial condensers is covered as a part of the treatment of heat exchanger design in Section 19.10.8.

When a partial condenser is used, it will often be followed by a settling drum to ensure that condensed liquid is not entrained into the gas flow. The design of vapor–liquid settling drums is discussed in Section 16.3.

The extent of recovery that can be accomplished by condensation depends on the temperature, pressure, and phase equilibrium properties of the mixture. The process performance is easily predicted using equilibrium flash models in commercial process simulation programs, as described in [Chapter 4](#).

Recovery of the condensed component can be increased by operating at higher pressure or lower temperature. In situations where the initial concentration of condensed component is high, it may be economical to use two or more stages of condensation, with the later stages operated at lower temperature or higher pressure. This obviates cooling all of the condensed liquid to the lowest temperature or compressing the entire vapor to the highest pressure.

16.3 Gas–liquid separators

The separation of entrained liquid droplets and mists from gas or vapor streams is possible whenever the temperature of a vapor is lowered below the dew point. Where the carryover of some fine droplets can be tolerated, it is often sufficient to rely on gravity settling in a vertical or horizontal separating vessel (known as a knockout pot). Knockout pots are often used immediately downstream of condensers or coolers to prevent entrainment of liquid and two-phase flow in downstream vapor lines. Knockout pots are also used before compressors and between intermediate stages of compressors to prevent liquid droplets from damaging the compressor blades.

Knitted mesh demisting pads are frequently used to improve the performance of separating vessels where the droplets are likely to be small, down to 1 μm , and where high separating efficiencies are required. Proprietary demister pads are available in a wide range of materials, metals and plastics; thickness and pad densities. For liquid separators, stainless steel pads around 100 mm thick and with a nominal density of 150 kg/m^3 would generally be used. Use of a demister pad allows a smaller vessel to be used. Separating efficiencies above 99% can be obtained with low-pressure drop. The design and specification of demister pads for gas–liquid separators are discussed by Pryce Bailey and Davies (1973). The design methods for horizontal separators given later are based on a procedure given by Gerunda (1981).

The separation of liquid droplets and mists from gas or vapor streams is analogous to the separation of solid particles from gas flows, and with the possible exception of filtration, the same techniques and equipment can be used; see Section 18.5. Cyclone separators are frequently used for gas–liquid separation. They can be designed using the same methods used for gas–solids cyclones. The inlet velocity should be kept below 30 m/s to avoid pick-up of liquid from the cyclone surfaces.

16.3.1 Settling velocity

Equation 16.6 can be used to estimate the settling velocity of the liquid droplets for the design of separating vessels.

$$u_t = 0.07[(\rho_L - \rho_v)/\rho_v]^{1/2} \quad (16.6)$$

where u_t = settling velocity, m/s

ρ_L = liquid density, kg/m^3

ρ_v = vapor density, kg/m^3

If a demister pad is not used, the value of u_t obtained from Equation 16.6 should be multiplied by a factor of 0.15 to provide a margin of safety and to allow for flow surges.

16.3.2 Vertical separators

The layout and typical proportions of a vertical gas–liquid separator are shown in Fig. 16.11.

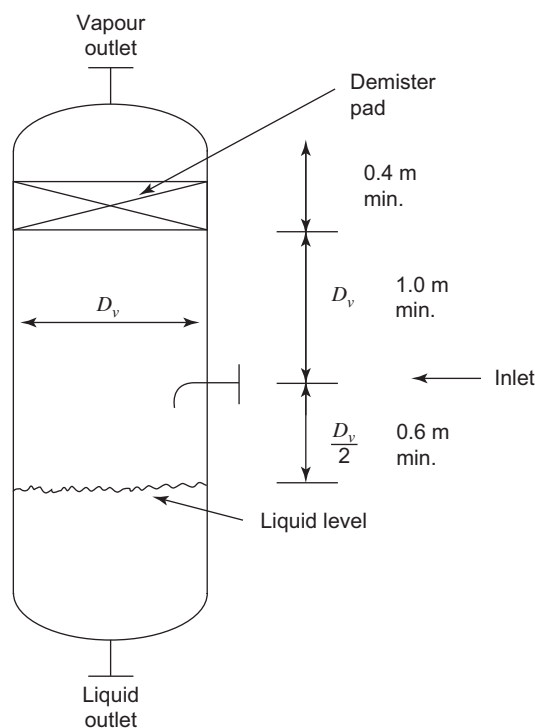


FIG. 16.11 Vertical liquid–vapor separator.

The diameter of the vessel must be large enough to slow the gas down to below the velocity at which the droplets will settle out. So the minimum allowable diameter will be given by:

$$D_v = \sqrt{\left(\frac{4V_v}{\pi u_s}\right)} \quad (16.7)$$

where D_v = minimum vessel diameter, m

V_v = gas or vapor volumetric flow rate, m^3/s

$u_s = u_t$ if a demister pad is used, and $0.15 u_t$ for a separator without a demister pad; u_t from Equation 16.6, m/s

The diameter is usually rounded up to the nearest standard vessel size so that standard vessel closures can be used; see Section 14.5.2.

The height of the vessel outlet above the gas inlet should be sufficient to allow for disengagement of the liquid drops. A height equal to the diameter of the vessel or 1 m, whichever is the greatest, should be used (see Fig. 16.11).

The liquid level will depend on the hold-up time necessary for smooth operation and control; typically 10 minutes of liquid hold-up would be allowed.

Example 16.2

Make a preliminary design for a separator to separate a mixture of steam and water; flow rates: steam 2000 kg/h, water 1000 kg/h; operating pressure 4 bar.

Solution

From steam tables, at 4 bar: saturation temperature 143.6°C , liquid density 926.4 kg/m^3 , and vapor density 2.16 kg/m^3 .

$$u_t = 0.07[(926.4 - 2.16)/2.16]^{\frac{1}{2}} = 1.45 \text{ m/s} \quad (16.6)$$

As the separation of condensate from steam is unlikely to be critical, a demister pad will not be specified.

$$\text{So, } u_t = 0.15 \times 1.45 = 0.218 \text{ m/s}$$

$$\text{Vapor volumetric flow rate} = \frac{2000}{3600 \times 2.16} = 0.257 \text{ m}^3/\text{s}$$

$$D_v = \sqrt{[(4 \times 0.257)/(\pi \times 0.218)]} = 1.23 \text{ m, round to nearest standard vessel size, 1.25 m (4 ft)} \quad (16.7)$$

$$\text{Liquid volumetric flow-rate} = \frac{1000}{3600 \times 926.14} = 3.0 \times 10^{-4} \text{ m}^3/\text{s}$$

Allow a minimum of 10 minutes hold-up.

$$\text{Volume held in vessel} = 3.0 \times 10^{-4} \times (10 \times 60) = 0.18 \text{ m}^3$$

$$\begin{aligned} \text{Liquid depth required, } h_L &= \frac{\text{volume held up}}{\text{vessel cross sectional area}} \\ &= \frac{0.18}{(\pi \times 1.25^2/4)} = 0.15 \text{ m} \end{aligned}$$

Increase to 0.3 m to allow space for positioning the level controller.

16.3.3 Horizontal separators

The layout of a typical horizontal separator is shown in Fig. 16.12. A horizontal separator would be selected when a longer liquid hold-up time is required, for example, to give better process control of the liquid flow.

In the design of a horizontal separator, the vessel diameter cannot be determined independently of its length, unlike for a vertical separator. The diameter and length, and the liquid level, must be chosen to give sufficient vapor residence time for the liquid droplets to settle out and for the required liquid hold-up time to be met.

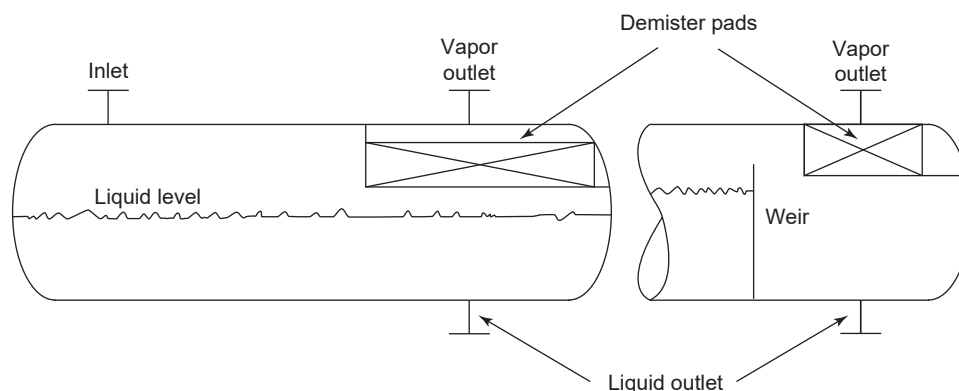


FIG. 16.12 Horizontal liquid–vapor separator.

The most economical length-to-diameter ratio will depend on the operating pressure (see [Chapter 14](#)). As a general guide, the following values can be used:

Operating pressure, bar	Length/diameter, L_v/D_v
0–20	3
20–35	4
>35	5

The relationship between the area for vapor flow, A_v , and the height above the liquid level, h_v , can be found from tables giving the dimensions of the segments of circles; see [Green and Southard \(2018\)](#) or Figs. 17.39 and 17.40 in [Chapter 17](#).

For preliminary designs, set the liquid height at half the vessel diameter,

$$h_v = D_v/2 \text{ and } f_v = 0.5$$

where f_v is the fraction of the total cross-sectional area occupied by the vapor.

The design procedure for horizontal separators is illustrated in the following example.

Example 16.3

Design a horizontal separator to separate 10,000 kg/h of liquid, density 962.0 kg/m³, from 12,500 kg/h of vapor, density 23.6 kg/m³. The vessel operating pressure will be 21 bar.

Solution

$$u_t = 0.07[(962.0 - 23.6)/23.6]^{1/2} = 0.44 \text{ m/s}$$

Try a separator without a demister pad.

$$u_s = 0.15 \times 0.44 = 0.066 \text{ m/s}$$

$$\text{Vapor volumetric flow rate} = \frac{12,500}{3600 \times 23.6} = 0.147 \text{ m}^3/\text{s}$$

Take $h_v = 0.5D_v$ and $L_v/D_v = 4$

$$\text{Cross-sectional area for vapor flow} = \frac{\pi D_v^2}{4} \times 0.5 = 0.393 D_v^2$$

$$\text{Vapor velocity, } u_v = \frac{0.147}{0.393 D_v^2} = 0.374 D_v^{-2}$$

Vapor residence time required for the droplets to settle to liquid surface:

$$h_v / u_s = 0.5D_v / 0.066 = 7.58D_v$$

Actual residence time = vessel length/vapor velocity:

$$= \frac{L_v}{u_v} = \frac{4D_v}{0.374D_v^{-2}} = 10.70D_v^3$$

For satisfactory separation, required residence time = actual:

$$\begin{aligned} \text{So, } 7.58D_v &= 10.70D_v^3 \\ D_v &= 0.84 \text{ m, say } 0.92 \text{ m (3 ft, standard pipe size)} \end{aligned}$$

Liquid hold-up time:

$$\text{liquid volumetric flow - rate} = \frac{10,000}{3600 \times 962.0} = 0.00289 \text{ m}^3/\text{s}$$

$$\text{liquid cross-sectional area} = \frac{\pi \times 0.92^2}{4} \times 0.5 = 0.332 \text{ m}^2$$

$$\text{Length, } L_v = 4 \times 0.92 = 3.7 \text{ m}$$

$$\text{Hold-up volume} = 0.332 \times 3.7 = 1.23 \text{ m}^3$$

$$\text{Hold-up time} = \text{liquid volume/liquid flow rate}$$

$$= 1.23/0.00289 = 426 \text{ s} = 7 \text{ minutes}$$

This is unsatisfactory, 10 minutes minimum required.

Need to increase the liquid volume. This is best done by increasing the vessel diameter. If the liquid height is kept at half the vessel diameter, the diameter must be increased by a factor of roughly $(10/7)0.5 = 1.2$.

$$\text{New } D_v = 0.92 \times 1.2 = 1.1 \text{ m}$$

Check liquid residence time:

$$\text{new liquid volume} = \frac{\pi \times 1.1^2}{4} \times 0.5 \times (4 \times 1.1) = 2.09 \text{ m}^3$$

$$\text{new residence time} = 2.09/0.00289 = 723 \text{ s} = 12 \text{ minutes, satisfactory}$$

Increasing the vessel diameter will have also changed the vapor velocity and the height above the liquid surface. The liquid separation will still be satisfactory, as the velocity, and hence the residence time, is inversely proportional to the diameter squared, whereas the distance the droplets have to fall is directly proportional to the diameter.

In practice, the distance traveled by the vapor will be less than the vessel length, L_v , as the vapor inlet and outlet nozzles will be set in from the ends. This could be allowed for in the design but will make little difference.

16.4 Liquid-liquid separation

Separation of two liquid phases, immiscible or partially miscible liquids, is a common requirement in the process industries. For example, in single-stage liquid-liquid extraction, the liquid-contacting step must be followed by a settling stage. It is also frequently necessary to separate small quantities of entrained water from organic-phase process streams. The simplest form of equipment used to separate liquid phases is the gravity settling tank, the decanter. Various types of proprietary equipment are also used to promote coalescence and improve separation in difficult systems or where emulsions are likely to form. Centrifuges and hydrocyclones are also used.

Liquid-liquid extraction can also be carried out as a multistage countercurrent process. This is described with other multistage separation processes in [Chapter 17](#), Section 17.16.